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स. 9(2)/DSR/CE/CSQ(E)/2025/E- 9178656/416

दिनांक 13/08/25

कार्यालय ज्ञापन

Sub: Release of Delhi Schedule of Rates (E&M)-2025.

It is hereby notified that Delhi Schedule of Rates (E&M) - 2025 has been released and is applicable with immediate effect. The cost Index base is 100 as on 01.04.2025 in the Delhi Schedule of Rates (E&M)-2025. The English Version of Delhi Schedule of Rates (E&M) -2025 is attached in PDF Format with this OM.

Errors or Omissions and suggestions for improvement, if any, may kindly be brought to the notice of the Superintending Engineer (E) TAS in the office of Chief Engineer CSQ (E) CPWD New Delhi-110011 (Tel No-011-23061418, Email: delceecsqa.cpwa@nic.in, delseetas.cpwa@nic.in).

This issues with the approval of the Competent Authority.



(Santosh Kumar Dhangar)
Superintending Engineer(E), TLQA

Copy to (Through CPWD Website): -
All Officers of CPWD, PWD and others.



Superintending Engineer(E), TLQA



भारत सरकार

GOVERNMENT OF INDIA

केंद्रीय लोक निर्माण विभाग



CENTRAL PUBLIC WORKS DEPARTMENT

DELHI SCHEDULE OF RATES (E&M)-2025



महानिदेशालय, के.लो.नि.वि.,निर्माण भवन, नई दिल्ली 110011

DIRECTORATE GENERAL, CPWD, NIRMAN BHAWAN, NEW DELHI 110011



GOVERNMENT OF INDIA
CENTRAL PUBLIC WORKS DEPARTMENT

DELHI SCHEDULE OF RATES
(E & M)
2025



PUBLISHED UNDER THE AUTHORITY OF
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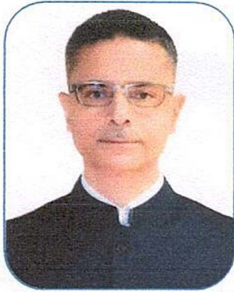
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Disclaimer

The Rates taken in the Delhi Schedule of Rates (E&M) 2025 are indicative and actual rates shall be governed by the market forces. Further, its use by the own risk and discretion, CPWD shall not be responsible for any ambiguity, discrepancy, dispute or financial loss arising directly or indirectly by using or following items of DSR (E&M) 2025 by Department/PSUs/Private Bodies/Individuals.

A GOVERNMENT OF INDIA PUBLICATION

Published by under the authority of
Director General
Central Public Works Department
Nirman Bhawan
New Delhi-110011



सतिंदर पाल सिंह
महानिदेशक
Satinder Pal Singh
Director General



भारत सरकार
Government of India



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FOREWORD

CPWD is the benchmarking organisation in the building construction sector and its documents are also made use of by other Government Departments, Undertakings and Autonomous Bodies for their works. This Delhi Schedule of Rates (E&M) is a comprehensive document which may be widely followed by these organisations.

The Delhi Schedule of Rates (E&M) 2025 is a revision and amalgamation of CPWD Delhi Schedule of Rates (E&M) 2022, DSR (E&M) for Solar Photovoltaic Power Plant – 2019, DSR (E&M) for Facade Lighting -2019 and Delhi Schedule of Rates (E&M) (Energy efficient materials) Vol.-II- 2025. It takes into account the increased labour and material cost, including the effect of GST on Works Contracts. In addition, new chapters for Low Voltage E&M services (CCTV Camera System, EPABX /IP-PBX, LAN, Access Control System & BMS), Fountains and Lifts have also been added.

I acknowledge the hard work and sincere effort put in by Sh. Mohd. Kamal Ahmad, SDG (HQ), Sh. Chita Ranjan Nanda, ADG (Tech.), Sh. Ram Raj Meena, CE CSQ (E), Sh. Ramayan Prasad Gupta, SE TAS(E), Sh. Santosh Kumar Dhangar, SE TLQA (E) and other staff of CSQ (E) unit, whose names are not mentioned here for the sake of brevity.

I hope that the Delhi Schedule of Rates (E&M) 2025 will be useful to all Engineers of CPWD and to other engineering organizations of Central/State Governments, practicing Architects and E&M Consultants.

(Satinder Pal Singh)
Director General

Place: New Delhi
Date: July, 2025



Mohd. Kamal Ahmad
Spl. Director General (HQ)



भारत सरकार
Government of India



केन्द्रीय लोक निर्माण विभाग
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MESSAGE

CPWD is the major Central Government organization setting norms in building construction and its documents are used by various government departments and organizations. CPWD aims to lead in executing, maintaining, and standardizing India's built environment while supporting sustainable development. To achieve this goal, the CPWD continuously updates its Delhi Schedule of Rates by incorporating the New Items of E&M services, Energy Efficient Materials, Building Security System and Building Management Systems.

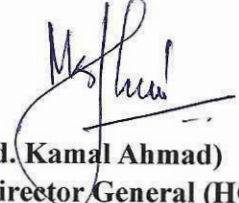
The DSR (E&M) – 2025 is a updated version of CPWD Delhi Schedule of Rates (E&M) 2022, DSR (E&M) for Solar Photovoltaic Power Plant – 2019, DSR (E&M) for Facade Lighting -2019 and Delhi Schedule of Rates (E&M) (Energy efficient materials) Vol.-II- 2025 and incorporation of New Chapters related to Low Voltage E&M services (CCTV Camera System, EPABX /IP-PBX, LAN, Access Control System & BMS), Fountain and Lifts.

Credit is given to individuals who contributed to the preparation of this schedule specially to Shri Chita Ranjan Nanda, ADG(Tech), Shri Ram Raj Meena, CE, CSQ (E), Sh. Ramayan Prasad Gupta, SE TAS(E), Sh. Santosh Kumar Dhangar, SE TLQA (E) and others.

This edition incorporates updated items related to various E&M services which are energy efficient and leads to sustainable development. To ensure significant energy savings compared to conventional buildings the items of higher level of energy performance parameters and Low Voltage E&M Services related to Building Management, Security and Communication have been added.

Sincere gratitude is expressed to Shri Satinder Pal Singh, Director General, CPWD for his trust and support in the preparation of this document.

Place: New Delhi
Date: July, 2025


(Mohd. Kamal Ahmad)
Special Director General (HQ)



Chita Ranjan Nanda
Additional Director General



भारत सरकार
Government of India

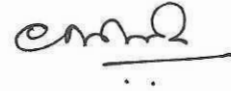


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PREFACE

1. The Delhi Schedule of Rates (E&M) usually called as DSR (E&M) is updated as DSR (E&M) 2025 which derived from Delhi Schedule of Rates (E&M) 2022, DSR (E&M) for Solar Photovoltaic Power Plant – 2019, DSR (E&M) for Facade Lighting -2019 and Delhi Schedule of Rates (E&M) (Energy efficient materials) Vol.-II- 2025 & new specialized E&M services chapters in the Construction Industry i.e. CCTV- Camera System, EPABX /IP-PBX, LAN, Access Control System, BMS, Fountain and Lifts.
2. DSR (E&M) 2025 is based on the prevailing market rates of materials in Delhi during the last six months. The labour rates adopted are as per minimum wages applicable with effect from 01.04.2025 issued by Chief Labour Commissioner, Govt. of India/Commissioner (Labour), Govt. of Delhi, whichever is higher.
3. The cost index for DSR (E&M) 2025 is 100 (as on 1.04.2025) with reference to CPWD Plinth Area Rates 2025.
4. This Delhi Schedule of Rates (E&M) 2025 shall be read along with relevant CPWD Specifications of Electrical and Mechanical Services. The DSR (E&M) 2025 hereby replace the Delhi Schedule of Rates (E&M) 2022, DSR (E&M) for Solar Photovoltaic Power Plant – 2019, DSR (E&M) for Facade Lighting -2019, Items for VRF/VRV Air-Conditioning System – 2019, Wet Riser and Sprinkler System - 2019 and Delhi Schedule of Rates (E&M) Vol.-II- 2025 Energy efficient materials.
5. The salient Features of DSR (E&M) 2025 are as follows:
 - a. The Items of Works contained herein are to be executed as per CPWD General Specifications for Electrical Works Part-I Internal 2023, Part-II External 2023, Part – III Lift & Escalators 2003, Part- IV Sub-Station 2013, Part – V Wet Riser and Sprinkler System 2020, Part – VI Fire Detection and Alarm System 2018, Part – VII DG Set 2013, CPWD General Specification for HVAC Works 2024, with up-to-date amendments/correction slips and relevant standards/codes of Bureau of Indian Standards.
 - b. The rates in this DSR (E&M) are for working height up to 4.5 meter from floor level wherever height not specified. The rates of materials are for reputed brands, factory fabricated and tested wherever available from manufacturers as submitted by field units and in confirmation with Bureau of Indian Standard Codes and Guidelines.
 - c. Effect of GST on works contracts and Labour Cess have been incorporated in all items.
6. Wholehearted commitment and considerable efforts have been put by team of CSQ (E) unit for preparation of this edition of Delhi Schedule of Rates (E&M) 2025. I convey my deep appreciation and sincere thanks to Sh. Ram Raj Meena, CE CSQ (E), Sh. Ramayan Prasad Gupta, SE TAS(E), Sh. Santosh Kumar Dhangar, SE TLQA (E), Sh. Himanshu Phulwaria, EE (E), Sh. Shashank Gaur, AEE (E)TAS, Sh. Sandeep Kumar Das, AE(E)TAS-I, Sh. Harjeet Singh, AE (E) TLQA-I, Sh. Sundeep Kumar, AE (E) TLQA-II, Ms. Preeti Verma, AE (E) TLQA – III and other staff of CSQ (E) unit.

7. Although, due care has been taken in bringing out DSR (E&M) 2025, still there is possibility that some errors may remain inadvertently. Errors & omission may be brought to the notice of Superintending Engineer (TAS) (E) O/o CE CSQ (E), CPWD, Room No. 224, A-wing, Nirman Bhawan, New Delhi-110011 Telephone No. 011-23061418/ 1697 and Email: delseetas.cpwd@nic.in / delceecsq.cpwd@nic.in.



Place: New Delhi
Date: July, 2025

(Chita Ranjan Nanda)
Additional Director General (Tech)

GENERAL NOTE

The Delhi Schedule of Rate (E&M) 2025 has been prepared on the following

1. The items of works are to be executed as per CPWD General Specifications for Electrical Works Part-I Internal 2023 , Part-II External 2023, Part-III-Lift & Escalators 2003, Part-IV Sub-Station 2013, Part V Wet Riser & Sprinkler Systems 2020, Part VI Fire Detection and Alarm System 2018, Part VII D.G. Sets 2013, Part VIII Gas Based Fire Extinguishing System 2013 and CPWD General Specifications for HVAC Works 2024, with up to date amendments/correction slips, the Bureau of Indian Standards Codes as amendment upto date and latest ENS 2024/ ECSBC 2024 whichever is more stringent.
2. The Basic Rates of labour and materials have been provided in Appendix-I & II. The effect of GST on works contract has been incorporated in all the items.
3. The rates are complete i.e. including all material, labour, taxes & duties (in Delhi), T&P etc. and overhead & profit as per orders of the department applicable as on date.
4. The rates in this DSR (E&M) 2025 and DAR (E&M) 2025 are for working height upto 4.5m from floor level wherever height is not specified.
5. The Rating of APFC Panel shall be selected by the NIT approving authority so as achieve average power factor ideally up to unity including considering the depreciation. Hybrid type of APFC panel shall be used where requirement of APFC and building has complex type load using electronics ballast/VFD Drives etc.
6. The rates of materials are from reputed manufacturers and market trends, factory fabricated and tested where ever available from manufactures.
7. The Delhi Schedule of Rates (E&M) 2025 is supersede the CPWD Delhi Schedule of Rates (E&M) 2022, DSR (E&M) for Solar Photovoltaic Power Plant – 2019, DSR (E&M) for Facade Lighting -2019 and Delhi Schedule of Rates (E&M) (Energy efficient materials) Vol.-II- 2025. In addition, new chapters for Low Voltage E&M services (CCTV Camera System, EPABX /IP-PBX, LAN, Access Control System & BMS), Fountains and Lifts have also been added.
8. This edition also aims to incorporate latest items to achieve Energy Efficiency in Buildings by implementation of Energy conservation Sustainable building code-2024 & Eco-Niwas Samhita (ENS) 2024 developed by BEE.
9. To adopt modern technology and Energy Efficient material for faster, ecofriendly and quality Contribution to buildings the NIT approving authority have to analyze for any change in parameter.

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CHAPTER-1 WIRING

Item No.	Description	Unit	Rate(₹)
WIRING IN STEEL & PVC CONDUIT			
1.1	Wiring for light point/ fan point/ exhaust fan point/ call bell point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable in surface / recessed steel conduit, with piano type switch, phenolic laminated sheet, suitable size MS box and earthing the point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable etc. as required.		
1.1.1	Group A	Point	1406
1.1.2	Group B	Point	1661
1.1.3	Group C	Point	2078
1.2	Wiring for twin control light point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable in surface / recessed steel conduit, 2way piano type switch, phenolic laminated sheet, suitable size MS box and earthing the point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable etc. as required.	Point	2090
1.3	Wiring for light point/ fan point/ exhaust fan point/ call bell point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable in surface/ recessed steel conduit, with modular switch, modular plate, suitable GI box and earthing the point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable etc. as required.		
1.3.1	Group A	Point	1429
1.3.2	Group B	Point	1684
1.3.3	Group C	Point	2101
1.4	Wiring for twin control light point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable in surface / recessed steel conduit, 2 way modular switch, modular plate, suitable GI box and earthing the point with 1.5 sq.mm. FRLS/HFFR PVC insulated copper conductor single core cable etc .as required.	Point	2242

1.5	Wiring for light/ power plug with 2X4 sq. mm FRLS/HFFR PVC insulated copper conductor single core cable in surface/ recessed steel conduit alongwith 1 No. 4 sq. mm FRLS PVC insulated copper conductor single core cable for loop earthing as required.	Metre	480
1.6	Wiring for light/ power plug with 4X4 sq. mm FRLS/HFFR PVC insulated copper conductor single core cable in surface/ recessed steel conduit alongwith 2 Nos. 4 sq. mm FRLS/HFFR PVC insulated copper conductor single core cable for loop earthing as required.	Metre	724
1.7	Wiring for circuit/ submain wiring alongwith earth wire with the following sizes of FRLS/HFFR PVC insulated copper conductor, single core cable in surface/ recessed steel conduit as required.		
1.7.1	2 X 1.5 sq. mm + 1 X 1.5 sq. mm earth wire	Metre	370
1.7.2	2 X 2.5 sq. mm + 1 X 2.5 sq. mm earth wire	Metre	414
1.7.3	2 X 4 sq. mm + 1 X 4 sq. mm earth wire	Metre	476
1.7.4	2 X 6 sq. mm + 1 X 6 sq. mm earth wire	Metre	652
1.7.5	2 X 10 sq. mm + 1 X 6 sq. mm earth wire	Metre	778
1.7.6	2 X 16 sq. mm + 1 X 6 sq. mm earth wire	Metre	1020
1.7.7	4 X 2.5 sq. mm + 2 X 2.5 sq. mm earth wire	Metre	621
1.7.8	4 X 4 sq. mm + 2 X 4 sq. mm earth wire	Metre	744
1.7.9	4 X 6 sq. mm + 2 X 6 sq. mm earth wire	Metre	1001
1.7.10	4 X 10 sq. mm + 2 X 6 sq. mm earth wire	Metre	1253
1.7.11	4 X 16 sq. mm + 2 X 6 sq. mm earth wire	Metre	1760
1.8	Wiring for light point/ fan point/ exhaust fan point/ call bell point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable in surface / recessed medium class PVC conduit, with piano type switch, phenolic laminated sheet, suitable size M.S. box and earthing the point with 1.5 sq.mm. FRLS/HFFR PVC insulated copper conductor single core cable etc. as required.		

1.8.1	Group A	Point	1166
1.8.2	Group B	Point	1339
1.8.3	Group C	Point	1680
1.9	Wiring for twin control light point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable in surface / recessed medium class PVC conduit, 2 way piano type switch, phenolic laminated sheet, suitable size MS box and earthing the point with 1.5 sq.mm. FRLS/HFFR PVC insulated copper conductor single core cable etc. as required.	Point	1693
1.10	Wiring for light point/ fan point/ exhaust fan point/ call bell point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable in surface / recessed medium class PVC conduit, with modular switch, modular plate, suitable GI box and earthing the point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable etc. as required.		
1.10.1	Group A	Point	1189
1.10.2	Group B	Point	1362
1.10.3	Group C	Point	1704
1.11	Wiring for twin control light point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable in surface / recessed medium class PVC conduit, 2 way modular switch, modular plate, suitable GI box and earthing the point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable etc. as required.	Point	1803
1.12	Wiring for light/ power plug with 2X4 sq. mm FRLS/HFFR PVC insulated copper conductor single core cable in surface/ recessed medium class PVC conduit alongwith 1 No. 4 sq. mm FRLS/HFFR PVC insulated copper conductor single core cable for loop earthing as required.	Metre	373
1.13	Wiring for light/ power plug with 4X4 sq. mm FRLS/HFFR PVC insulated copper conductor single core cable in surface/ recessed medium class PVC conduit alongwith 2 Nos. 4 sq. mm FRLS/HFFR PVC insulated copper conductor single core cable for loop earthing as required.	Metre	587

1.14	Wiring for circuit/ submain wiring alongwith earth wire with the following sizes of FRLS/HFFR PVC insulated copper conductor, single core cable in surface/ recessed medium class PVC conduit as required.		
1.14.1	2 X 1.5 sq. mm + 1 X 1.5 sq. mm earth wire	Metre	267
1.14.2	2 X 2.5 sq. mm + 1 X 2.5 sq. mm earth wire	Metre	311
1.14.3	2 X 4 sq. mm + 1 X 4 sq. mm earth wire	Metre	4508
1.14.4	2 X 6 sq. mm + 1 X 6 sq. mm earth wire	Metre	4525
1.14.5	2 X 10 sq. mm + 1 X 6 sq. mm earth wire	Metre	4279
1.14.6	2 X 16 sq. mm + 1 X 6 sq. mm earth wire	Metre	832
1.14.7	4 X 2.5 sq. mm + 2 X 2.5 sq. mm earth wire	Metre	452
1.14.8	4 X 4 sq. mm + 2 X 4 sq. mm earth wire	Metre	587
1.14.9	4 X 6 sq. mm + 2 X 6 sq. mm earth wire	Metre	813
1.14.10	4 X 10 sq. mm + 2 X 6 sq. mm earth wire	Metre	1078
1.14.11	4 X 16 sq. mm + 2 X 6 sq. mm earth wire	Metre	1454
1.15	Rewiring for light point/ fan point/ exhaust fan point/ call bell point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable and 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable as earth wire in existing surface/ recessed steel/PVC conduit including dismantling as required.		
1.15.1	Group A	Point	687
1.15.2	Group B	Point	752
1.15.3	Group C	Point	1064
1.16	Rewiring for twin control light point with 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable and 1.5 sq.mm FRLS/HFFR PVC insulated copper conductor single core cable as earth wire in existing surface/ recessed steel/PVC conduit including dismantling as required.	Point	1064

1.17	Supplying and drawing following sizes of FRLS/HFFR PVC insulated copper conductor, single core cable in the existing surface/ recessed steel/ PVC conduit as required.		
1.17.1	1 x 1.5 sq. mm	Metre	51
1.17.2	2 x 1.5 sq. mm	Metre	77
1.17.3	3 x 1.5 sq. mm	Metre	104
1.17.4	4 x 1.5 sq. mm	Metre	130
1.17.5	5 x 1.5 sq. mm	Metre	169
1.17.6	6 x 1.5 sq. mm	Metre	195
1.17.7	7 x 1.5 sq. mm	Metre	234
1.17.8	8 x 1.5 sq. mm	Metre	260
1.17.9	9 x 1.5 sq. mm	Metre	311
1.17.10	10 x 1.5 sq. mm	Metre	338
1.17.11	2 x 2.5 sq. mm	Metre	107
1.17.12	3 x 2.5 sq. mm	Metre	148
1.17.13	4 x 2.5 sq. mm	Metre	189
1.17.14	5 x 2.5 sq. mm	Metre	242
1.17.15	6 x 2.5 sq. mm	Metre	283
1.17.16	7 x 2.5 sq. mm	Metre	337
1.17.17	8 x 2.5 sq. mm	Metre	378
1.17.18	9 x 2.5 sq. mm	Metre	443
1.17.19	10 x 2.5 sq. mm	Metre	484
1.17.20	2 x 4 sq. mm	Metre	160
1.17.21	3 x 4 sq. mm	Metre	4357

1.17.22	4 x 4 sq. mm	Metre	296
1.17.23	5 x 4 sq. mm	Metre	362
1.17.24	6 x 4 sq. mm	Metre	424
1.17.25	7 x 4 sq. mm	Metre	488
1.17.26	8 x 4 sq. mm	Metre	550
1.17.27	9 x 4 sq. mm	Metre	616
1.17.28	10 x 4 sq. mm	Metre	690
1.17.29	2 x 6 sq. mm	Metre	234
1.17.30	3 x 6 sq. mm	Metre	326
1.17.31	4 x 6 sq. mm	Metre	431
1.17.32	5 x 6 sq. mm	Metre	524
1.17.33	6 x 6 sq. mm	Metre	641
1.17.34	7 x 6 sq. mm	Metre	733
1.17.35	8 x 6 sq. mm	Metre	838
1.17.36	9 x 6 sq. mm	Metre	930
1.17.37	10 x 6 sq. mm	Metre	1029
1.18	Supplying and drawing following pair 0.5 mm dia FRLS/HFFR PVC insulated annealed copper conductor, unarmored telephone cable in the existing surface/ recessed steel/ PVC conduit as required.		
1.18.1	1 Pair	Metre	33
1.18.2	2 Pair	Metre	39
1.18.3	4 Pair	Metre	53
1.19	Supplying and drawing co-axial TV cable RG-6 grade, 0.7 mm solid copper conductor PE insulated, shielded with fine tinned copper braid and protected with PVC sheath in the existing surface/ recessed steel/ PVC conduit as required.		
		Metre	54

1.20	Supplying and fixing of following sizes of steel conduit along with accessories in surface/recess including painting in case of surface conduit, or cutting the wall and making good the same in case of recessed conduit as required.		
1.20.1	20 mm	Metre	254
1.20.2	25 mm	Metre	289
1.20.3	32 mm	Metre	379
1.20.4	40 mm	Metre	515
1.20.5	50 mm	Metre	634
1.21	Supplying and fixing of following sizes of medium class PVC conduit along with accessories in surface/recess including cutting the wall and making good the same in case of recessed conduit as required.		
1.21.1	20 mm	Metre	151
1.21.2	25 mm	Metre	168
1.21.3	32 mm	Metre	209
1.21.4	40 mm	Metre	257
1.21.5	50 mm	Metre	329
1.22	Supplying and fixing metal box of following sizes (nominal size) on surface or in recess with suitable size of phenolic laminated sheet cover in front including painting etc. as required.		
1.22.1	75 mm X 75 mm X 60 mm deep	Each	229
1.22.2	100 mm X 100 mm X 60 mm deep	Each	245
1.22.3	150 mm X 75 mm X 60 mm deep	Each	252
1.22.4	150 mm X 150 mm X 60 mm deep	Each	360
1.22.5	180 mm X 100 mm X 60 mm deep	Each	282
1.22.6	200 mm X 125 mm X 60 mm deep	Each	372
1.22.7	200 mm X 150 mm X 60 mm deep	Each	406

1.22.8	200 mm X 150 mm X 75 mm deep	Each	414
1.22.9	200 mm X 250 mm X 60 mm deep	Each	528
1.22.10	200 mm X 250 mm X 75 mm deep	Each	549
1.22.11	200 mm X 150 mm X 100 mm deep	Each	469
1.22.12	200 mm X 250 mm X 100 mm deep	Each	568
1.22.13	200 mm X 300 mm X 60 mm deep	Each	587
1.22.14	200 mm X 300 mm X 100 mm deep	Each	625
1.22.15	250 mm X 300 mm X 60 mm deep	Each	673
1.22.16	250 mm X 300 mm X 100 mm deep	Each	721
1.23	Supplying and fixing following piano type switch/ socket on the existing switch box/ cover including connections etc. as required.		
1.23.1	5/6 amps switch	Each	63
1.23.2	2 way 5/6 A switch	Each	75
1.23.3	15/16 A switch	Each	140
1.23.4	3 pin 5/6 A socket outlet	Each	81
1.23.5	6 pin 15/16 A socket outlet	Each	156
1.23.6	Telephone socket outlet	Each	127
1.23.7	TV antenna socket outlet	Each	112
1.23.8	Bell push	Each	100
1.24	Supplying and fixing following modular switch/ socket on the existing modular plate & switch box including connections but excluding modular plate etc. as required.		
1.24.1	5/6 A switch	Each	121
1.24.2	2 way 5/6 A switch	Each	168
1.24.3	15/16 A switch	Each	176

1.24.4	3 pin 5/6 A socket outlet	Each	136
1.24.5	6 pin 15/16 A socket outlet	Each	219
1.24.6	Telephone socket outlet	Each	168
1.24.7	TV antenna socket outlet	Each	168
1.24.8	Bell push	Each	159
1.25	Supplying and fixing two module stepped type electronic fan regulator on the existing modular plate switch box including connections but excluding modular plate etc. as required.	Each	402
1.26	Supplying and fixing modular blanking plate on the existing modular plate & switch box excluding modular plate as required.	Each	47
1.27	Supplying and fixing following size/ modules, GI box alongwith modular base & cover plate for modular switches in recess etc. as required.		
1.27.1	1 or 2 Module (75mmX75mm)	Each	354
1.27.2	3 Module (100mmX75mm)	Each	384
1.27.3	4 Module (125mmX75mm)	Each	401
1.27.4	6 Module (200mmX75mm)	Each	462
1.27.5	8 Module (125mmX125mm)	Each	517
1.27.6	12 Module (200mmX150mm)	Each	614
1.28	Supplying and fixing following Modular base & cover plate on existing modular metal boxes etc. as required.		
1.28.1	1 or 2 Module	Each	156
1.28.2	3 Module	Each	175
1.28.3	4 Module	Each	182
1.28.4	6 Module	Each	207
1.28.5	8 Module	Each	238
1.28.6	12 Module	Each	301

1.29	Supplying and fixing metal box of 150mm X 75mm X 60mm deep (nominal size) on surface or in recess with suitable size of phenolic laminated sheet cover in front including providing and fixing 3 pin 5/6 A socket outlet and 5/6 A piano type switch, connections, painting etc. as required.	Each	423
1.30	Supplying and fixing metal box of 180mm X 100mm X 60mm deep (nominal size) on surface or in recess with suitable size of phenolic laminated sheet cover in front including providing and fixing 6 pin 5/6 & 15/16 A socket outlet and 15/16 A piano type switch, connections, painting etc. as required.	Each	558
1.31	Supplying and fixing suitable size GI box with modular plate and cover in front on surface or in recess, including providing and fixing 3 pin 5/6 A modular socket outlet and 5/6 A modular switch, connections etc. as required.	Each	545
1.32	Supplying and fixing suitable size GI box with modular plate and cover in front on surface or in recess, including providing and fixing 6 pin 5/6 & 15/16 A modular socket outlet and 15/16 A modular switch, connections etc. as required.	Each	659
1.33	Supplying and fixing 3 pin, 5 A ceiling rose on the existing junction box/ wooden block including connections etc. as required.	Each	104
1.34	Supplying and fixing brass batten/ angle holder including connection etc. as required.	Each	154
1.35	Installation ,Testing, Commissioning of wall bracket /ceiling fittings of all sizes and shapes containing upto two GLS/CFL/LED lamps per fitting, complete with all accessories including connections etc. as required.	Each	151
1.36	Supplying and fixing stiff pendent with 300 mm long, 20 mm dia X 1.6 mm thick steel conduit, aluminium cast back plate and brass holder complete, including wiring the down rod with 1.5 sq. mm FRLS/HFFR PVC insulated, copper conductor, single core cable and painting etc. as required.	Each	347
1.37	DELETED		
1.38	Supplying and fixing call bell/ buzzer suitable for single phase, 230 V, complete as required.	Each	112
1.39	Providing and fixing plain 16/0.20mm (0.50sqmm) twin flat flexible, FRLS/HFFR PVC insulated, copper conductor cable, in PVC sleeve of suitable size on the floor/ wall, or side of the table/ door etc. as required.	Metre	47

1.40	Providing and fixing plain 16/0.20mm (0.50sqmm) twin circular flexible FRLS/HFFR PVC insulated, PVC sheathed copper conductor cable direct on the wall with PVC clips etc. as required.	Metre	40
1.41	Installation, testing and commissioning of pre-wired, fluorescent fitting / compact fluorescent/LED fitting of all types, complete with all accessories and tube/lamp etc. directly on ceiling/ wall, including connections with 1.5 sq. mm FRLS/HFFR PVC insulated, copper conductor, single core cable and earthing etc. as required.	Each	259
1.42	Installation, testing and commissioning of pre-wired, fluorescent fitting / compact fluorescent/LED fitting of all types, complete with all accessories and tube/lamp etc., including supplying and fixing ball and socket arrangement, 2 Nos. down rods of 20 mm dia X 1.6 mm thick steel conduit upto 30 cm length, painting and wiring the down rods and connections with 1.5 sq. mm FRLS/HFFR PVC insulated, copper conductor, single core cable and earthing etc. as required.	Each	556
1.43	Providing and fixing extra conduit down rod of 20 mm dia, 2 X 10 cm length, wiring with 2 X 1.5 sq. mm FRLS/HFFR PVC insulated, copper conductor, single core cable including painting etc. as required. (Note : More than 5 cm length shall be rounded to the nearest 10 cm and 5 cm or less shall be ignored)	Each	64
1.44	Installation, testing and commissioning of ceiling fan, including wiring the down rods of standard length (upto 30 cm) with 1.5 sq. mm FRLS/HFFR PVC insulated, copper conductor, single core cable etc. as required.	Each	266
1.45	Installation, testing and commissioning of ceiling fan, including wiring the down rods of standard length (upto 30 cm) with 1.5 sq. mm FRLS/HFFR PVC insulated, copper conductor, single core cable, including providing and fixing phenolic laminated sheet cover on the fan box etc. as required.	Each	332
1.46	DELETED		
1.47	Supplying and fixing extra down rod of 10 cm length G.I. pipe, 15 mm dia, heavy gauge including painting etc. as required. (Note: More than 5 cm length shall be rounded to the nearest 10 cm and 5 cm or less shall be ignored)	Each	53
1.48	Supplying and fixing extra conduit down rod of 20 cm length G.I. pipe 15 mm dia, heavy gauge including painting etc. as required. (Note : More than 5 cm length shall be rounded to the nearest 10 cm and 5 cm or less shall be ignored)	Each	59

1.49	Numbering of ceiling fan/ exhaust fan/ fluorescent fittings as required.	Each	72
1.50	Installation of exhaust fan in the existing opening, including making good the damage, connection, testing, commissioning etc. as required.		
1.50.1	Upto 450 mm sweep	Each	556
1.50.2	510 mm sweep	Each	800
1.51	Extra for fixing the louvers/ shutters complete with frame for a exhaust fan of all sizes.	Each	260
1.52	Painting of ceiling fan in installed position with one or more coats of spray painting with synthetic enamel paint of approved brand and manufacture to give an even shade, including cleaning of surface with detergent etc. as required.	Each	186
1.53	Deleted		
1.54	Wiring for group controlled (looped) light point/fan point/exhaust fan point/ call bell point (without independent switch etc.) with 1.5 sq. mm FRLS/HFFR PVC insulated copper conductor single core cable in surface/ recessed steel conduit, and earthing the point with 1.5 sq. mm FRLS/HFFR PVC insulated copper conductor single core cable etc. as required.		
1.54.1	Group A	Point	914
1.54.2	Group B	Point	1083
1.54.3	Group C	Point	1246
1.55	Wiring for group controlled (looped) light point/fan point/exhaust fan point/ call bell point (without independent switch etc.) with 1.5 sq. mm FRLS/HFFR PVC insulated copper conductor single core cable in surface/ recessed PVC conduit, and earthing the point with 1.5 sq. mm FRLS/HFFR PVC insulated copper conductor single core cable etc. as required.		
1.55.1	Group A	Point	776
1.55.2	Group B	Point	885
1.55.3	Group C	Point	994

1.56	Supplying and fixing suitable size GI box with modular plate and cover in front on surface or in recess, including providing and fixing 2 nos. 3 pin 5/6 A modular socket outlet and 2 nos. 5/6 A modular switch, connections etc. as required. (For light plugs to be used in non residential buildings).	Each	874
1.57	Supplying & fixing suitable size GI box with modular plate and cover in front on surface or in recess including providing and fixing 25 A modular socket outlet and 25 A modular SP MCB, "C" curve including connections, painting etc. as required.	Each	807
1.58	Supplying and fixing PVC batten/ angle holder including connections etc. as required.	Each	132
1.59	Dismantling of ceiling fan and painting the same with with one or more coats of spray painting with synthetic enamel paint of approved brand and manufacture to give an even shade, including cleaning of surface with detergent and replacing the damaged rubber reel, nuts and bolts with washers and safety pins, reinstalling the same as required.	Each	324

CHAPTER-2 MCCB, MCB & DB'S

Item No.	Description	Unit	Rate(₹)
2.1	Providing and fixing following capacity TP&N disconnecter fuse switch unit inside the existing panel board with ISI marked HRC fuses including drilling holes in cubicle panel, making connections, etc. as required.		
2.1.1	32 A TP&N	Each	2292
2.1.2	63 A TP&N	Each	3342
2.1.3	100 A TP&N	Each	6414
2.1.4	125 A TP&N	Each	7577
2.1.5	160 A TP&N	Each	8546
2.1.6	200 A TP&N	Each	10198
2.1.7	315 A TP&N	Each	15308
2.1.8	400 A TP&N	Each	18722
2.2	Providing and fixing following rating and breaking capacity and pole MCCB with thermomagnetic release and terminal spreaders in existing cubicle panel board including drilling holes in cubicle panel, making connections, Ics=100% Icu and Operational Voltage 690V etc. as required.		
2.2.1	100 A, 16 KA,TPMCCB	Each	4003
2.2.2	125 A, 16 KA,TPMCCB	Each	5103
2.2.3	150 A, 16 KA,TPMCCB	Each	6326
2.2.4	200 A, 16 KA,TPMCCB	Each	8111
2.2.5	200 A, 25 KA,TPMCCB	Each	11586
2.2.6	250 A, 25 KA,TPMCCB	Each	13647
2.2.7	250 A, 35 KA,TPMCCB	Each	14584
2.2.8	315 A, 35 KA,TPMCCB	Each	22644

2.2.9	400 A, 35 KA,TPMCCB	Each	22644
2.2.10	500 A, 35 KA,TPMCCB	Each	26812
2.2.11	630 A, 50 KA,TPMCCB	Each	29222
2.2.12	800 A, 50 KA,TPMCCB	Each	37895
2.2.13	100 A,30KA,FPMCCB	Each	8127
2.2.14	125 A,36KA,FPMCCB	Each	8644
2.2.15	200 A,36KA,FPMCCB	Each	15956
2.2.16	250 A,36KA,FPMCCB	Each	18372
2.2.17	250 A,50KA,FPMCCB	Each	19313
2.2.18	400 A,50KA,FPMCCB	Each	44663
2.2.19	630 A,50KA,FPMCCB	Each	45108
2.3	Supplying and fixing following way, single pole and neutral, sheet steel, MCB distribution board, 240 V, on surface/ recess, complete with tinned copper bus bar, neutral bus bar, earth bar, din bar, interconnections, powder painted including earthing etc. as required. (But without MCB/RCCB/Isolator)		
2.3.1	6 way, Double door	Each	2390
2.3.2	8 way, Double door	Each	2773
2.3.3	12 way, Double door	Each	2503
2.3.4	16 way, Double door	Each	3367
2.4	Supplying and fixing following way, horizontal type three pole and neutral, sheet steel, MCB distribution board, 415 V, on surface/ recess, complete with tinned copper bus bar, neutral bus bar, earth bar, din bar, interconnections, powder painted including earthing etc. as required. (But without MCB/RCCB/Isolator)		
2.4.1	4 way (4 + 12), Double door	Each	4377
2.4.2	6 way (4 + 18), Double door	Each	5299

2.4.3	8 way (4 + 24), Double door	Each	19477
2.5	Supplying and fixing of following ways surface/ recess mounting, vertical type, 415 V, TPN MCB distribution board of sheet steel, dust protected, duly powder painted, inclusive of 200 A tinned copper bus bar, common neutral link, earth bar, din bar for mounting MCBs (but without MCBs and incomer) as required . (Note : Vertical type MCB TPDB is normally used where 3 phase outlets are required.)		
2.5.1	4 way (4 + 12), Double door	Each	7951
2.5.2	8 way (4 + 24), Double door	Each	10724
2.5.3	12 way (4 + 36), Double door	Each	13512
2.6	DELETED		
2.7	DELETED		
2.8	DELETED		
2.9	DELETED		
2.10	Supplying and fixing 5 A to 32 A rating, 240/415 V, 10 kA, "C" curve, miniature circuit breaker suitable for inductive load of following poles in the existing MCB DB complete with connections, testing and commissioning etc. as required.		
2.10.1	Single pole	Each	285
2.10.2	Single pole and neutral	Each	285
2.10.3	Double pole	Each	696
2.10.4	Triple pole	Each	1070
2.10.5	Triple pole and neutral	Each	1301
2.11	Supplying and fixing single pole blanking plate in the existing MCB DB complete etc. as required.	Each	14
2.12	Supplying and fixing following rating, double pole, 240 V, isolator in the existing MCB DB complete with connections, testing and commissioning etc. as required.		
2.12.1	40 A	Each	472
2.12.2	63 A	Each	569

2.13	Supplying and fixing following rating, four pole, 415 V, isolator in the existing MCB DB complete with connections, testing and commissioning etc. as required.		
2.13.1	40 A	Each	1049
2.13.2	63 A	Each	1115
2.13.3	100 A	Each	1317
2.14	Supplying and fixing following rating, double pole, (single phase and neutral), 240 V, residual current circuit breaker (RCCB), having a sensitivity current 30 mA in the existing MCB DB complete with connections, testing and commissioning etc. as required.		
2.14.1	25 A	Each	2141
2.14.2	40 A	Each	2512
2.14.3	63 A	Each	2866
2.15	Supplying and fixing following rating, four pole, (three phase and neutral), 415 volts, residual current circuit breaker (RCCB), having a sensitivity current 30 mA in the existing MCB DB complete with connections, testing and commissioning etc. as required.		
2.15.1	25 A	Each	2737
2.15.2	40 A	Each	3366
2.15.3	63 A	Each	3036
2.16	Supplying and fixing DP sheet steel enclosure on surface/ recess along with 25/32 A 240 V "C" curve DP MCB complete with connections, testing and commissioning etc. as required.	Each	1258
2.17	Supplying and fixing TP sheet steel enclosure on surface/ recess along with 16/25/32 A 415 V "C" curve TP MCB complete with connections, testing and commissioning etc. as required.	Each	2352
2.18	Supplying and fixing 20 A, 240 V, SPN Industrial type socket outlet, with 2 pole and earth, metal enclosed plug top alongwith 20 A "C" curve, SP, MCB, in sheet steel enclosure, on surface or in recess, with chained metal cover for the socket out let and complete with connections, testing and commissioning etc. as required.	Each	1806

2.19	Supplying and fixing 20 A, 415 V, TPN Industrial type socket outlet, with 4 pole and earth, metal enclosed plug top alongwith 20 A "C" curve, TPMCB, in sheet steel enclosure, on surface or in recess, with chained metal cover for the socket out let and complete with connections, testing and commissioning etc. as required.	Each	2773
2.20	Supplying and fixing 30 A, 415 V, TPN Industrial type socket outlet, with 4 pole and earth, metal enclosed plug top alongwith 30 A "C" curve, TPMCB, in sheet steel enclosure, on surface or in recess, with chained metal cover for the socket out let and complete with connections, testing and commissioning etc. as required.	Each	4358
2.21	Providing and fixing M.V. danger notice plate of 200 mm X 150 mm, made of mild steel, at least 2 mm thick, and vitreous enameled white on both sides, and with inscription in single red colour on front side as required.	Each	315
2.22	Providing and fixing H.T. danger notice plate of 250 mm X 200 mm, made of mild steel, at least 2 mm thick, and vitreous enameled white on both sides, and with inscription in single red colour on front side as required.	Each	340
2.23	Supplying and fixing Cable End Box (Loose Wire Box) suitable for following single pole and neutral, sheet steel, MCB distribution board, 240 Volts, on surface/ recess, complete with testing and commissioning etc. as required.		
2.23.1	For 6 way, Double door SPN MCBDB	Each	834
2.23.2	For 8 way, Double door SPN MCBDB	Each	918
2.23.3	For 10 way, Double door SPN MCBDB	Each	895
2.23.4	For 14 way, Double door SPN MCBDB	Each	990
2.24	Supplying and fixing Cable End Box (Loose Wire Box) suitable for following triple pole and neutral, sheet steel, MCB distribution board, 415 Volts, on surface/ recess, complete with testing and commissioning etc.as required.		
2.24.1	For 4 way, Double door TPN MCBDB	Each	1177
2.24.2	For 6 way, Double door TPN MCBDB	Each	1223
2.24.3	For 8 way, Double door TPN MCBDB	Each	1448
2.25	Supplying and fixing Cable End Box (Loose Wire Box) suitable for triple pole and neutral, sheet steel, Vertical MCB distribution board, 415 Volts, on surface/ recess, complete with testing and commissioning etc. as required.	Each	1271

CHAPTER 3

RISING MAINS & BUS TRUNKING

Item No.	Description	Unit	Rate(₹)
3.1	Supplying, installing on wall, testing and commissioning of following capacity rising mains made of 1.6mm thick IP 42 sheet steel enclosure duly painted with powder coating, wall straps, fully PVC insulated 4 Nos aluminium bus bars having current density of 130 A/ sq cm at nominal current rating in convenient sections and suitable for 415 V, 3 phase, 4 wire, 50 Hz, A.C. supply with extension joints, fire proof barriers, expansion joints, thrust pads including jointing and earthing with 2 runs of galvanised iron strips etc. as required. (Rising mains confirming to IS 8623, TEC 439)		
3.1.1	200 A (S.C. rating for 1 sec - 15 kA)	Metre	7778
3.1.2	300 A (S.C. rating for 1 sec - 20 kA)	Metre	8097
3.1.3	400 A (S.C. rating for 1 sec - 25 kA)	Metre	9907
3.1.4	600 A (S.C. rating for 1 sec - 45 kA)	Metre	11033
3.1.5	800 A (S.C. rating for 1 sec - 50 kA)	Metre	13241
3.2	Supplying, installing, testing and commissioning of following capacity TPN tap off box made of 1.6mm thick sheet steel enclosure duly painted with powder coating on existing rising mains complete with TPN disconnecter FSU and HRC fuses, connections, earthing etc. as required.		
3.2.1	16 A TPN	Each	8550
3.2.2	32 A TPN	Each	10711
3.2.3	63 A TPN	Each	12116
3.2.4	100 A TPN	Each	14902
3.2.5	200 A TPN	Each	19804
3.2.6	315 A TPN	Each	26889
3.2.7	400 A TPN	Each	30594
3.3	Supplying, installing, testing and commissioning of following capacity TPN distribution tap off box made of 1.6mm thick		

	sheet steel enclosure duly painted with powder coating on existing rising mains complete with HRC fuses, interconnections, earthing etc. as required.		
3.3.1	16 A TPN, 2 way	Each	12817
3.3.2	16 A TPN, 4 way	Each	28498
3.3.3	16 A TPN, 6 way	Each	40364
3.3.4	16 A TPN, 8 way	Each	55866
3.3.5	32 A TPN, 2 way	Each	15772
3.3.6	32 A TPN, 4 way	Each	32134
3.3.7	32 A TPN, 6 way	Each	42181
3.3.8	32 A TPN, 8 way	Each	58592
3.3.9	63 A TPN, 2 way	Each	17953
3.3.10	63 A TPN, 4 way	Each	30315
3.3.11	63 A TPN, 6 way	Each	32727
3.3.12	63 A TPN, 8 way	Each	56230
3.4	Supplying, installing, testing and commissioning of following capacity End Feed Unit made of 1.6mm thick sheet steel enclosure duly painted with powder coating to existing rising mains complete with TPN disconnecter FSU and HRC fuses, mounting stands, cable end box, brass compression gland, connections, earthing etc. as required.		
3.4.1	200 A TPN	Each	25637
3.4.2	300 A TPN	Each	28194
3.4.3	400 A TPN	Each	31265
3.4.4	600 A TPN	Each	33791
3.4.5	800 A TPN	Each	38140
3.5	Supplying, installation, including suspension, testing and commissioning of following capacity bus trunking with aluminium bus bars having current density of 130 A/ sq cm at		

	nominal current rating in 1.6mm thick IP 42 sheet steel enclosure in convenient sections for use on, 3 phase, 4 wire, 415 V, 50 Hz, A.C. supply including jointing of sections, flexible joints, expansion joints, bends and earthing with 2 runs of galvanised iron strips, suspenders, angle iron bracket, steel fasteners, connecting to earthing system etc. as required.		
3.5.1	800 A	Metre	14713
3.5.2	1000 A	Metre	17350
3.5.3	1250 A	Metre	22339
3.5.4	1400 A	Metre	27793
3.5.5	1600 A	Metre	27906
3.6	Supplying, installation, testing and commissioning of following capacity overhead distribution bus trunking with aluminium bus bars having current density of 130 A/ sq cm at nominal current rating in 1.6mm thick IP 42 sheet steel enclosure in convenient sections with provision of tapping points for use on, 3 phase, 4 wire, 415 volts, 50 Hz, A.C. supply including jointing of sections, bends, earthing with 2 runs of galvanised iron strips, all installation accessories etc. as required.		
3.6.1	200 A	Metre	8623
3.6.2	400 A	Metre	10986
3.7	Supplying, installing, testing and commissioning of following capacity TPN tap off box / plug-in box made of 1.6mm thick sheet steel enclosure duly painted with powder coating on existing overhead bus bar system complete with TPN disconnecter FSU and HRC fuses, connections, earthing etc. as required.		
3.7.1	32 A	Each	10711
3.7.2	63 A	Each	12116
3.7.3	100 A	Each	14902
3.8	Supplying, installing on wall, testing and commissioning of following capacity Air Insulated Compact Type Rising Mains for use on 3 phase 4 wire 415 volts, 50Hz A.C. supply with enclosure having IP-54 rating after fixing the tap off boxes and all accessories, made of 1.6mm thick steel sheet duly powder coated in convenient sections complete with 4 Nos aluminium		

bus bars having current density of 130 A/ sq cm at nominal current rating, necessary joints & expansion joints, fire barrier at each floor, provision of tapping at every metre, continuous earthing with 2 Nos aluminium strip of suitable size (one on each side) including, G.I. clamping brackets, angle iron bracket, steel fasteners, connecting to earthing system etc. as required.

3.8.1	200 A, I _{sc} = 15kA for 1 second	Metre	8445
3.8.2	315 A, I _{sc} = 25kA for 1 second	Metre	9849
3.8.3	400 A, I _{sc} = 30kA for 1 second	Metre	11717
3.8.4	500 A, I _{sc} = 35kA for 1 second	Metre	11779
3.8.5	630 A, I _{sc} = 50kA for 1 second	Metre	13584
3.8.6	800 A, I _{sc} = 50kA for 1 second	Metre	19714
3.8.7	1000 A, I _{sc} = 50kA for 1 second	Metre	20411
3.8.8	1250 A, I _{sc} = 50kA for 1 second	Metre	24174
3.9	Supplying, installing on wall or suspension on ceiling, testing and commissioning of following capacity Air Insulated Compact Type Bus Trunking for use on 3 phase 4 wire 415 volts, 50Hz A.C. supply with metal clad enclosure having IP-54 rating after fixing the tap off boxes and all accessories, made of 1.6mm thick steel sheet duly powder coated in convenient sections complete with 4 Nos aluminium bus bars having current density of 130 A/ sq cm at nominal current rating, necessary joints, elbow joints & expansion joints, fire barrier at each floor, continuous earthing with 2 Nos aluminium strip of suitable size (one on each side) including, G.I. clamping brackets, suspenders, angle iron bracket, steel fasteners, connecting to earthing system etc. as required.		
3.9.1	200 A, I _{sc} = 15kA for 1 second	Metre	8522
3.9.2	315 A, I _{sc} = 25kA for 1 second	Metre	9940
3.9.3	400 A, I _{sc} = 30kA for 1 second	Metre	11827
3.9.4	500 A, I _{sc} = 35kA for 1 second	Metre	11888
3.9.5	630 A, I _{sc} = 50kA for 1 second	Metre	13711

3.9.6	800 A, I _{sc} = 50kA for 1 second	Metre	19902
3.9.7	1000 A, I _{sc} = 50kA for 1 second	Metre	19902
3.9.8	1250 A, I _{sc} = 50kA for 1 second	Metre	23098
3.10	DELETED		
3.11	Supplying, installation, testing & commissioning of following capacity End Feed Unit for the existing Air Insulated Compact Type bus trunking/ rising mains for use on 3 phase 4 wire 415 volts, 50Hz A.C. supply made with 1.6mm thick steel sheet enclosure (IP54) duly powder coated with provision of MCCB/ACB (but without MCCB/ACB) complete with necessary joints including clamping brackets, angle iron bracket, steel fasteners, connecting to earthing system etc. as required		
3.11.1	200 A, I _{sc} = 15kA for 1 sec	Each	8899
3.11.2	315 A, I _{sc} = 25kA for 1 sec	Each	10718
3.11.3	400 A, I _{sc} = 30kA for 1 sec	Each	11888
3.11.4	500 A, I _{sc} = 35kA for 1 sec	Each	11888
3.11.5	630 A, I _{sc} = 50kA for 1 sec	Each	14120
3.11.6	800 A, I _{sc} = 50kA for 1 sec	Each	18772
3.11.7	1000 A, I _{sc} = 50kA for 1 sec	Each	18772
3.11.8	1250 A, I _{sc} = 50kA for 1 sec	Each	20873
3.12	Supplying, installation, testing & commissioning of following capacity Plug In/ tap off box on the existing Air Insulated Compact Type bus trunking/ rising mains for use on 3 phase 4 wire 415 volts, 50Hz A.C. supply made with 1.6mm thick sheet steel enclosure (IP54) duly powder coated with provision of MCCB/ACB (but without MCCB/ACB) complete etc. as required		
3.12.1	125 A, I _{sc} = 15kA for 1 sec	Each	9104
3.12.2	200 A, I _{sc} = 25kA for 1 sec	Each	9104
3.12.3	250 A, I _{sc} = 30kA for 1 sec	Each	11171

3.12.4	400 A, Isc= 35kA for 1 sec	Each	11171
3.12.5	500 A, Isc= 50kA for 1 sec	Each	16806
3.12.6	630 A, Isc= 50kA for 1 sec	Each	16806
3.13	Supplying, installing by suspension on ceiling, testing and commissioning of following capacity Sandwich Type Rising Mains for use on 3 phase 4 wire 415 volts, 50Hz A.C. supply with metal clad enclosure having IP-54 rating after fixing the tap off boxes and all accessories, made of 1.6mm thick steel sheet duly powder coated in convenient sections complete with 4 Nos aluminium bus bars having current density of 130 A/ sq cm at nominal current rating, necessary joints, elbow joints & expansion joints and bends, fire barrier at each floor, provision of tapping at every metre, adopter box and copper flexible for joints, continuous earthing with 2 Nos aluminium strip of suitable size (one on each side) including, G.I. clamping brackets, suspenders, angle iron bracket, steel fasteners, connecting to earthing system etc. as required		
3.13.1	400 A, Isc= 25kA for 1 sec	Meter	16196
3.13.2	500 A, Isc= 30kA for 1 sec	Meter	17152
3.13.3	630 A, Isc= 50kA for 1 sec	Meter	20225
3.13.4	800 A, Isc= 50kA for 1 sec	Meter	22841
3.13.5	1000 A, Isc= 50kA for 1 sec	Meter	25416
3.13.6	1250 A, Isc= 50kA for 1 sec	Meter	27733
3.13.7	1600 A, Isc= 50kA for 1 sec	Meter	32974
3.13.8	2000 A, Isc= 50kA for 1 sec	Meter	42110
3.13.9	2500 A, Isc= 50kA for 1 sec	Meter	46745
3.13.10	3200 A, Isc= 50kA for 1 sec	Meter	58373
3.13.11	4000 A, Isc= 50kA for 1 sec	Meter	98975

CHAPTER 4 CABLE TRAYS

Item No.	Description	Unit	Rate(₹)
4.1	Supplying and installing following size of perforated painted with powder coating M.S. cable trays with perforation not more than 17.5%, in convenient sections, joined with connectors, suspended from the ceiling with two numbers MS/GI Suspenders i/c base of suitable size MS/GI angle, bolts & nuts, painting suspenders, fastner etc as required.		
4.1.1	100 mm width X 50 mm depth X 1.6 mm thickness	Meter	623
4.1.2	150 mm width X 50 mm depth X 1.6 mm thickness	Meter	686
4.1.3	200 mm width X 50 mm depth X 1.6 mm thickness	Meter	819
4.1.4	300 mm width X 50 mm depth X 1.6 mm thickness	Meter	866
4.1.5	375 mm width X 50 mm depth X 2.0 mm thickness	Meter	1116
4.1.6	450 mm width X 50 mm depth X 2.0 mm thickness	Meter	1243
4.1.7	600 mm width X 50 mm depth X 2.0 mm thickness	Meter	1526
4.1.8	300 mm width X 62.5 mm depth X 2.0 mm thickness	Meter	1025
4.1.9	375 mm width X 62.5 mm depth X 2.0 mm thickness	Meter	1160
4.1.10	450 mm width X 62.5 mm depth X 2.0 mm thickness	Meter	1348
4.1.11	600 mm width X 62.5 mm depth X 2.0 mm thickness	Meter	1597
4.1.12	750 mm width X 62.5 mm depth X 2.0 mm thickness	Meter	1830
4.1.13	900 mm width X 62.5 mm depth X 2.0 mm thickness	Meter	2109
4.1.14	600 mm width X 75 mm depth X 2.0 mm thickness	Meter	1633
4.1.15	750 mm width X 75 mm depth X 2.0 mm thickness	Meter	1844
4.1.16	900 mm width X 75 mm depth X 2.0 mm thickness	Meter	2145
4.2	Supplying and installing following size of perforated painted with powder coating M.S. cable trays bends with perforation not more than 17.5%, joined with connectors, suspended from the		

	ceiling with two numbers MS/GI Suspenders i/c base of suitable size MS/GI angle, bolts & nuts, painting suspenders, fastner etc as required.		
4.2.1	100 mm width X 50 mm depth X 1.6 mm thickness	Each	948
4.2.2	150 mm width X 50 mm depth X 1.6 mm thickness	Each	1105
4.2.3	200 mm width X 50 mm depth X 1.6 mm thickness	Each	1323
4.2.4	300 mm width X 50 mm depth X 1.6 mm thickness	Each	1544
4.2.5	375 mm width X 50 mm depth X 2.0 mm thickness	Each	2138
4.2.6	450 mm width X 50 mm depth X 2.0 mm thickness	Each	2437
4.2.7	600 mm width X 50 mm depth X 2.0 mm thickness	Each	3009
4.2.8	300 mm width X 62.5 mm depth X 2.0 mm thickness	Each	1930
4.2.9	375 mm width X 62.5 mm depth X 2.0 mm thickness	Each	2244
4.2.10	450 mm width X 62.5 mm depth X 2.0 mm thickness	Each	2657
4.2.11	600 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3244
4.2.12	750 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3805
4.2.13	900 mm width X 62.5 mm depth X 2.0 mm thickness	Each	4403
4.2.14	600 mm width X 75 mm depth X 2.0 mm thickness	Each	3332
4.2.15	750 mm width X 75 mm depth X 2.0 mm thickness	Each	813
4.2.16	900 mm width X 75 mm depth X 2.0 mm thickness	Each	4493
4.3	Supplying and installing following size of perforated painted with powder coating M.S. cable trays Tee with perforation not more than 17.5%, joined with connectors, suspended from the ceiling with two numbers MS/GI Suspenders i/c base of suitable size MS/GI angle, bolts & nuts, painting suspenders, fastner etc as required.		
4.3.1	100 mm width X 50 mm depth X 1.6 mm thickness	Each	1089
4.3.2	150 mm width X 50 mm depth X 1.6 mm thickness	Each	1281

4.3.3	225 mm width X 50 mm depth X 1.6 mm thickness	Each	1552
4.3.4	300 mm width X 50 mm depth X 1.6 mm thickness	Each	1826
4.3.5	375 mm width X 50 mm depth X 2.0 mm thickness	Each	2558
4.3.6	450 mm width X 50 mm depth X 2.0 mm thickness	Each	2923
4.3.7	600 mm width X 50 mm depth X 2.0 mm thickness	Each	3628
4.3.8	300 mm width X 62.5 mm depth X 2.0 mm thickness	Each	2305
4.3.9	375 mm width X 62.5 mm depth X 2.0 mm thickness	Each	2686
4.3.10	450 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3165
4.3.11	600 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3885
4.3.12	750 mm width X 62.5 mm depth X 2.0 mm thickness	Each	4578
4.3.13	900 mm width X 62.5 mm depth X 2.0 mm thickness	Each	5309
4.3.14	600 mm width X 75 mm depth X 2.0 mm thickness	Each	3995
4.3.15	750 mm width X 75 mm depth X 2.0 mm thickness	Each	4691
4.3.16	900 mm width X 75 mm depth X 2.0 mm thickness	Each	5422
4.4	Supplying and installing following size of perforated painted with powder coating M.S. cable trays Cross Member with perforation not more than 17.5%, joined with connectors, suspended from the ceiling with two numbers MS/GI Suspenders i/c base of suitable size MS/GI angle, bolts & nuts, painting suspenders, fastner etc as required.		
4.4.1	100 mm width X 50 mm depth X 1.6 mm thickness	Each	1231
4.4.2	150 mm width X 50 mm depth X 1.6 mm thickness	Each	1498
4.4.3	225 mm width X 50 mm depth X 1.6 mm thickness	Each	1783
4.4.4	300 mm width X 50 mm depth X 1.6 mm thickness	Each	2110
4.4.5	375 mm width X 50 mm depth X 2.0 mm thickness	Each	2978
4.4.6	450 mm width X 50 mm depth X 2.0 mm thickness	Each	3410

4.4.7	600 mm width X 50 mm depth X 2.0 mm thickness	Each	4247
4.4.8	300 mm width X 62.5 mm depth X 2.0 mm thickness	Each	2681
4.4.9	375 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3129
4.4.10	450 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3674
4.4.11	600 mm width X 62.5 mm depth X 2.0 mm thickness	Each	4526
4.4.12	750 mm width X 62.5 mm depth X 2.0 mm thickness	Each	5352
4.4.13	900 mm width X 62.5 mm depth X 2.0 mm thickness	Each	6215
4.4.14	600 mm width X 75 mm depth X 2.0 mm thickness	Each	4658
4.4.15	750 mm width X 75 mm depth X 2.0 mm thickness	Each	5487
4.4.16	900 mm width X 75 mm depth X 2.0 mm thickness	Each	6350
4.5	Supplying and installing following size of perforated painted with powder coating M.S. cable trays Reducer with perforation not more than 17.5%, joined with connectors, suspended from the ceiling with two numbers MS/GI Suspenders i/c base of suitable size MS/GI angle, bolts & nuts, painting suspenders, fastner etc as required.		
4.5.1	100 mm width X 50 mm depth X 1.6 mm thickness	Each	1089
4.5.2	150 mm width X 50 mm depth X 1.6 mm thickness	Each	1281
4.5.3	225 mm width X 50 mm depth X 1.6 mm thickness	Each	1562
4.5.4	300 mm width X 50 mm depth X 1.6 mm thickness	Each	1876
4.5.5	375 mm width X 50 mm depth X 2.0 mm thickness	Each	2568
4.5.6	450 mm width X 50 mm depth X 2.0 mm thickness	Each	2933
4.5.7	600 mm width X 50 mm depth X 2.0 mm thickness	Each	3628
4.5.8	300 mm width X 62.5 mm depth X 2.0 mm thickness	Each	2305
4.5.9	375 mm width X 62.5 mm depth X 2.0 mm thickness	Each	2686
4.5.10	450 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3165

4.5.11	600 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3885
4.5.12	750 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3880
4.5.13	900 mm width X 62.5 mm depth X 2.0 mm thickness	Each	5309
4.5.14	600 mm width X 75 mm depth X 2.0 mm thickness	Each	3995
4.5.15	750 mm width X 75 mm depth X 2.0 mm thickness	Each	4691
4.5.16	900 mm width X 75 mm depth X 2.0 mm thickness	Each	5422
4.6	Hot Dipped Galvanized Iron Cable Tray Supplying and installing following size of perforated Hot Dipped Galvanised Iron cable tray (Galvanisation thickness not less than 50 microns) with perforation not more than 17.5%, in convenient sections, joined with connectors, suspended from the ceiling with two numbers GI Suspenders i/c base of suitable size GI angle, GI bolts & nuts, fastner etc as required.		
4.6.1	100 mm width X 50 mm depth X 1.6 mm thickness	Metre	754
4.6.2	150 mm width X 50 mm depth X 1.6 mm thickness	Metre	803
4.6.3	225 mm width X 50 mm depth X 1.6 mm thickness	Metre	991
4.6.4	300 mm width X 50 mm depth X 1.6 mm thickness	Metre	1062
4.6.5	375 mm width X 50 mm depth X 2.0 mm thickness	Metre	1322
4.6.6	450 mm width X 50 mm depth X 2.0 mm thickness	Metre	1439
4.6.7	600 mm width X 50 mm depth X 2.0 mm thickness	Metre	1925
4.6.8	300 mm width X 62.5 mm depth X 2.0 mm thickness	Metre	1299
4.6.9	375 mm width X 62.5 mm depth X 2.0 mm thickness	Metre	1477
4.6.10	450 mm width X 62.5 mm depth X 2.0 mm thickness	Metre	1713
4.6.11	600 mm width X 62.5 mm depth X 2.0 mm thickness	Metre	2064
4.6.12	750 mm width X 62.5 mm depth X 2.0 mm thickness	Metre	2428
4.6.13	900 mm width X 62.5 mm depth X 2.0 mm thickness	Metre	2793

4.6.14	600 mm width X 75 mm depth X 2.0 mm thickness	Metre	2111
4.6.15	750 mm width X 75 mm depth X 2.0 mm thickness	Metre	2441
4.6.16	900 mm width X 75 mm depth X 2.0 mm thickness	Metre	2799
4.7	Supplying and installing following size of perforated Hot Dipped Galvanised Iron cable tray "bends" (galvanisation not less than 50 microns) with perforation not more than 17.5%, in convenient sections, joined with connectors, suspended from the ceiling with two numbers GI Suspenders i/c base of suitable size GI angle, GI bolts & nuts, fastner etc as required.		
4.7.1	100 mm width X 50 mm depth X 1.6 mm thickness	Each	1252
4.7.2	150 mm width X 50 mm depth X 1.6 mm thickness	Each	1368
4.7.3	225 mm width X 50 mm depth X 1.6 mm thickness	Each	1811
4.7.4	300 mm width X 50 mm depth X 1.6 mm thickness	Each	1977
4.7.5	375 mm width X 50 mm depth X 2.0 mm thickness	Each	2580
4.7.6	450 mm width X 50 mm depth X 2.0 mm thickness	Each	2844
4.7.7	600 mm width X 50 mm depth X 2.0 mm thickness	Each	3985
4.7.8	300 mm width X 62.5 mm depth X 2.0 mm thickness	Each	2537
4.7.9	375 mm width X 62.5 mm depth X 2.0 mm thickness	Each	2946
4.7.10	450 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3460
4.7.11	600 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3966
4.7.12	750 mm width X 62.5 mm depth X 2.0 mm thickness	Each	5122
4.7.13	900 mm width X 62.5 mm depth X 2.0 mm thickness	Each	5932
4.7.14	600 mm width X 75 mm depth X 2.0 mm thickness	Each	4383
4.7.15	750 mm width X 75 mm depth X 2.0 mm thickness	Each	5156
4.7.16	900 mm width X 75 mm depth X 2.0 mm thickness	Each	5957
4.8	Supplying and installing following size of perforated Hot Dipped Galvanised Iron cable tray "Tee" (galvanisation not less than		

50 microns) with perforation not more than 17.5%, in convenient sections, joined with connectors, suspended from the ceiling with two numbers GI Suspenders i/c base of suitable size GI angle, GI bolts & nuts, fastner etc as required.

4.8.1	100 mm width X 50 mm depth X 1.6 mm thickness	Each	1427
4.8.2	150 mm width X 50 mm depth X 1.6 mm thickness	Each	1486
4.8.3	225 mm width X 50 mm depth X 1.6 mm thickness	Each	2097
4.8.4	300 mm width X 50 mm depth X 1.6 mm thickness	Each	2294
4.8.5	375 mm width X 50 mm depth X 2.0 mm thickness	Each	3012
4.8.6	450 mm width X 50 mm depth X 2.0 mm thickness	Each	3324
4.8.7	600 mm width X 50 mm depth X 2.0 mm thickness	Each	4687
4.8.8	300 mm width X 62.5 mm depth X 2.0 mm thickness	Each	2965
4.8.9	375 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3505
4.8.10	450 mm width X 62.5 mm depth X 2.0 mm thickness	Each	4038
4.8.11	600 mm width X 62.5 mm depth X 2.0 mm thickness	Each	5084
4.8.12	750 mm width X 62.5 mm depth X 2.0 mm thickness	Each	6014
4.8.13	900 mm width X 62.5 mm depth X 2.0 mm thickness	Each	6984
4.8.14	600 mm width X 75 mm depth X 2.0 mm thickness	Each	5137
4.8.15	750 mm width X 75 mm depth X 2.0 mm thickness	Each	6061
4.8.16	900 mm width X 75 mm depth X 2.0 mm thickness	Each	7013
4.9	Supplying and installing following size of perforated Hot Dipped Galvanised Iron cable tray "Cross member" (galvanisation not less than 50 microns) with perforation not more than 17.5%, in convenient sections, joined with connectors, suspended from the ceiling with two numbers GI Suspenders i/c base of suitable size GI angle, GI bolts & nuts, fastner etc as required.		
4.9.1	100 mm width X 50 mm depth X 1.6 mm thickness	Each	1427

4.9.2	150 mm width X 50 mm depth X 1.6 mm thickness	Each	1486
4.9.3	225 mm width X 50 mm depth X 1.6 mm thickness	Each	2117
4.9.4	300 mm width X 50 mm depth X 1.6 mm thickness	Each	2513
4.9.5	375 mm width X 50 mm depth X 2.0 mm thickness	Each	3289
4.9.6	450 mm width X 50 mm depth X 2.0 mm thickness	Each	3770
4.9.7	600 mm width X 50 mm depth X 2.0 mm thickness	Each	4702
4.9.8	300 mm width X 62.5 mm depth X 2.0 mm thickness	Each	2972
4.9.9	375 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3448
4.9.10	450 mm width X 62.5 mm depth X 2.0 mm thickness	Each	4044
4.9.11	600 mm width X 62.5 mm depth X 2.0 mm thickness	Each	5000
4.9.12	750mm width X 62.5 mm depth X 2 mm thickness	Each	6014
4.9.13	900 mm width X 62.5 mm depth X 2.0 mm thickness	Each	6984
4.9.14	600 mm width X 75 mm depth X 2.0 mm thickness	Each	5140
4.9.15	750 mm width X 75 mm depth X 2.0 mm thickness	Each	4505
4.9.16	900 mm width X 75 mm depth X 2.0 mm thickness	Each	5197
4.10	Supplying and installing following size of perforated Hot Dipped Galvanised Iron cable tray "Reducer" (galvanisation not less than 50 microns) with perforation not more than 17.5%, in convenient sections, joined with connectors, suspended from the ceiling with two numbers GI Suspenders i/c base of suitable size GI angle, GI bolts & nuts, fastner etc as required.		
4.10.1	100 mm width X 50 mm depth X 1.6 mm thickness	Each	1771
4.10.2	150 mm width X 50 mm depth X 1.6 mm thickness	Each	2049
4.10.3	225 mm width X 50 mm depth X 1.6 mm thickness	Each	2572
4.10.4	300 mm width X 50 mm depth X 1.6 mm thickness	Each	2991
4.10.5	375 mm width X 50 mm depth X 2.0 mm thickness	Each	3624

4.10.6	450 mm width X 50 mm depth X 2.0 mm thickness	Each	3929
4.10.7	600 mm width X 50 mm depth X 2.0 mm thickness	Each	5418
4.10.8	300 mm width X 62.5 mm depth X 2.0 mm thickness	Each	3583
4.10.9	375 mm width X 62.5 mm depth X 2.0 mm thickness	Each	4178
4.10.10	450 mm width X 62.5 mm depth X 2.0 mm thickness	Each	5000
4.10.11	600 mm width X 62.5 mm depth X 2.0 mm thickness	Each	6291
4.10.12	750mm width X 62.5 mm depth X 2 mm thickness	Each	6758
4.10.13	900 mm width X 62.5 mm depth X 2.0 mm thickness	Each	7826
4.10.14	600 mm width X 75 mm depth X 2.0 mm thickness	Each	5567
4.10.15	750 mm width X 75 mm depth X 2.0 mm thickness	Each	6842
4.10.16	900 mm width X 75 mm depth X 2.0 mm thickness	Each	7912

CHAPTER-5 EARTHING

Item No.	Description	Unit	Rate(₹)
5.1	Earthing with G.I. earth pipe 4.5 metre long, 40 mm dia including accessories, and providing masonry enclosure with cover plate having locking arrangement and watering pipe etc. (but without charcoal/ coke and salt) as required.	Set	5958
5.2	Earthing with G.I. earth pipe 4.5 metre long, 40 mm dia including accessories, and providing masonry enclosure with cover plate having locking arrangement and watering pipe etc. with charcoal/ coke and salt as required.	Each	7658
5.3	Earthing with G.I. earth plate 600 mm X 600 mm X 6 mm thick including accessories, and providing masonry enclosure with cover plate having locking arrangement and watering pipe of 2.7 metre long etc. (but without charcoal/ coke and salt) as required.	Set	7080
5.4	Earthing with G.I. earth plate 600 mm X 600 mm X 6 mm thick including accessories, and providing masonry enclosure with cover plate having locking arrangement and watering pipe of 2.7 metre long etc. with charcoal/ coke and salt as required.	Set	8351
5.5	Earthing with copper earth plate 600 mm X 600 mm X 3 mm thick including accessories, and providing masonry enclosure with cover plate having locking arrangement and watering pipe of 2.7 metre long etc. (but without charcoal/ coke and salt) as required.	Set	13733
5.6	Earthing with copper earth plate 600 mm X 600 mm X 3 mm thick including accessories, and providing masonry enclosure with cover plate having locking arrangement and watering pipe of 2.7 metre long etc. with charcoal/ coke and salt as required.	Set	15004
5.7	Supplying and laying 6 SWG G.I. wire at 0.50 metre below ground level for conductor earth electrode, including connection/ termination with GI thimble etc. as required.	Metre	62
5.8	Supplying and laying 25 mm X 5 mm copper strip at 0.50 metre below ground as strip earth electrode, including connection/ terminating with nut, bolt, spring, washer etc. as required. (Jointing shall be done by overlapping and with 2 sets of brass nut bolt & spring washer spaced at 50mm)	Metre	1024

5.9	Supplying and laying 25 mm X 5 mm G.I strip at 0.50 metre below ground as strip earth electrode, including connection/terminating with G.I. nut, bolt, spring, washer etc. as required. (Jointing shall be done by overlapping and with 2 sets of G.I. nut bolt & spring washer spaced at 50mm)	Metre	160
5.10	Providing and fixing 25 mm X 5 mm copper strip in 40 mm dia G.I. pipe from earth electrode including connection with brass nut, bolt, spring, washer excavation and re-filling etc. as required.	Metre	1638
5.11	Providing and fixing 25 mm X 5 mm G.I. strip in 40 mm dia G.I. pipe from earth electrode including connection with G.I. nut, bolt, spring, washer excavation and re-filling etc. as required.	Metre	755
5.12	Providing and laying earth connection from earth electrode with 6 SWG dia G.I. Wire in 15 mm dia G.I. pipe from earth electrode including connection with G.I. thimble excavation and re-filling as required.	Metre	313
5.13	Providing and laying earth connection from earth electrode with 4.00 mm dia copper wire in 15 mm dia G.I. pipe from earth electrode including connection with copper thimble excavation and re-filling as required.	Metre	400
5.14	Providing and fixing 25 mm X 5 mm copper strip on surface or in recess for connections etc. as required.	Metre	1246
5.15	Providing and fixing 25 mm X 5 mm G.I. strip on surface or in recess for connections etc. as required.	Metre	287
5.16	Providing and fixing 6 SWG dia G.I. wire on surface or in recess for loop earthing as required.	Metre	84
5.17	Providing and fixing 4.00 mm dia copper wire on surface or in recess for loop earthing as required.	Metre	173
5.18	Providing and fixing 6 SWG dia G.I. wire on surface or in recess for loop earthing along with existing surface/ recessed conduit/ submain wiring/ cable as required.	Metre	50
5.19	Providing and fixing 4.00 mm dia copper wire on surface or in recess for loop earthing along with existing surface/ recessed conduit/ submain wiring/ cable as required.	Metre	139
5.20	Providing and fixing earth bus of 50 mm X 5 mm copper strip on surface for connections etc. as required.	Metre	2245

CHAPTER 6 LIGHTNING CONDUCTOR

Item No.	Description	Unit	Rate(₹)
6.1	DELETED		
6.2	Providing and fixing of lightning conductor finial, made of 25 mm dia 300 mm long, G.I. tube, having single prong at top, with 85 mm dia 6 mm thick G.I. base plate including holes etc. complete as required.	Each	609
6.3	Fixing of lightning conductor finial (single prong) with base plate including holes etc. complete as required.	Each	422
6.4	Jointing copper / G.I. tape (with another copper/ G I tape, base of the finial or any other metallic object) by riveting / nut bolting/ sweating and soldering etc as required.	Each	139
6.5	DELETED		
6.6	DELETED		
6.7	Providing and fixing G.I. tape 20 mm X 3 mm thick on parapet or surface of wall for lightning conductor complete as required.(For horizontal run)	Metre	148
6.8	Providing and fixing G.I. tape 20 mm X 3 mm thick on parapet or surface of wall for lightning conductor complete as required.(For vertical run)	Metre	238
6.9	Fixing of copper/ G.I. tape 20 mm X 3 mm thick on parapet or surface of wall for lightning conductor complete as required.(For horizontal run)	Metre	100
6.10	Fixing of copper/ G.I. tape 20 mm X 3 mm thick on parapet or surface of wall for lightning conductor complete as required.(For vertical run)	Metre	281
6.11	DELETED		
6.12	Providing and fixing testing joint, made of 20 mm X 3 mm thick G.I. strip, 125 mm long, with 4 nos. of G.I. bolts, nuts, chuck nuts and spring washers etc. complete as required.	Each	3614
6.13	DELETED		
6.14	Providing and laying G.I. tape 32 mm X 6 mm from earth electrode directly in ground as required.	Metre	221
6.15	Laying copper/ G.I. tape 32 mm X 6 mm from earth electrode directly in ground as required.	Metre	96

CHAPTER 7 LIGHTING CONTROLS

Item No.	Description	Unit	Rate(₹)
7.1	Supplying,installation,testing and commissioning of Passive Infrared(PIR) technology based occupancy sensor having high preformance, non regulating programmable type, suitable for connected load upto 10Amp , for mounting height up to 2.8 mtr and for 5 m diameter coverage area along with necessary fixing arrangements i/c programming at site etc. complete as required.	Each	4896
7.2	Supplying,installation,testing and commissioning of Passive Infrared(PIR) technology based occupancy sensor with day light dimming(lighting level shall be regulated as per availability of natural day light in an area along with occupancy detection.) having high preformance, regulating programmable type, suitable for connected load upto 10Amp , for mounting height up to 2.8 mtr and for 5 m diameter coverage area along with necessary fixing arrangements i/c programming at site etc. complete as required.	Each	9670
7.3	Supplying,installation,testing and commissioning of Microwave technology based occupancy sensor having high preformance, non regulating programmable type, suitable for connected load upto 10Amp , for mounting height up to 2.6 mtr and for 5m X 20m coverage area along with necessary fixing arrangements i/c programming at site etc. complete as required.	Each	12515
7.4	Supplying,installation,testing and commissioning of Astronomical time switch of following configuration to be mounted in feeder pillars / Lighting DBs for automatic switching On & OFF of street lights at sun set & sun rise or twilight(Auto ON, Auto OFF, Auto modes) with manual override facility with 12/24 hour display format with suitable battery and indication for relay status i/c programming at site complete as required.		
7.4.1	1 output per phase and suitable for single phase supply	Each	4673
7.4.2	2 output per phase and suitable for single phase supply	Each	9029
7.4.3	3output(1output per phase) and suitable for three phase supply	Each	11161

CHAPTER -8 LED LIGHT

Item No.	Description	Unit	Rate(₹)
8.1	LED Down lighter (SMD Type) (System lumen efficacy $\geq 105 < 120\text{lm/Watt}$) Supplying, installation, Testing & Commissioning of LED Recessed / surface Down lighter (Round/ square/ Rectangular) SMD type of following body material with PMMA and prismatic diffuser and construction as per IS : 10322 with driver as per the requirement with Driver efficiency $>85\%$, Operating voltage AC 140-270 Volt, frequency 50/60 hz, Operating temp range -5 deg to 40 deg centigrade, internal surge protection of 2.5 KV with Short & Open circuit protection ,THD $< 10\%$, P. F.≥ 0.95, IP20, CRI ≥ 80 , UGR (Unified Glare Rating) ≤ 19, Flicker free (flicker should be below 5%), life time (LED,Driver & electrical circuitary), of minimum 50000 Burning Hours with 70% of initial Lumen maintained till life ends , CCT 3000°K / 4000°K / 5700°K / 6500°K (As per ANSI Bin), SDCM (Standard Deviation Color Matching) <3, Maximum power consumption should not more than the specified rating and Fixture shall be confirming to relevant BIS standards and trade mark certificate (T.C.). Manufactures Word Mark/ Name Engraved/ Embossing/ Screen printing on housing Complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy ≥ 105 and $<120\text{lm/Watt}$ output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted.. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C) Powder coated die cast /Extruded aluminium Body including trim		
8.1.1	5 - 7 watt	Each	542
8.1.2	8 - 10 watt	Each	624
8.1.3	12 -15 watt	Each	715
8.1.4	18 watt	Each	920
8.1.5	22 watt	Each	942

8.1.6	30 watt	Each	1439
8.2	LED Down lighter (SMD Type) (System lumen efficacy ≥ 120 <135lm/Watt) Supplying, installation, Testing & Commissioning of LED Recessed /surface Down lighter (Round / square/ Rectangular) SMD type of following body material with PMMA and prismatic diffuser and construction as per IS : 10322 with driver as per the requirement with Driver efficiency >85%, Operating voltage AC 140-270 Volt, freq 50/60 hz, Operating temp range -5 deg to 40 deg centigrade, internal surge protection of 2.5 KV with Short & Open circuit protection ,THD < 10% , P. F.≥ 0.95, IP20, CRI ≥ 80 , UGR (Unified Glare Rating) ≤ 19, Flicker free (flicker should be below 5 %), life time (LED,Driver & electrical circuitary), life time of minimum 50000 Burning Hours with , 70% of initial Lumen maintained till life ends , CCT 3000°K / 4000°K / 5700°K /6500°K (As per ANSI Bin), SDCM(Standard Deviation Color Matching) <3, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard and trade mark certificate (T.C.). Manufactures Word Mark/ Name Engraved/ Embossing/ Screen printing on housing. OEM must have its own in house NABL lab setup for all testing facilities for LED fixtures. "complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy ≥ 120 <135 lm/Watt output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted.. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C). Powder coated die cast /Extruded aluminium Body including trim .		
8.2.1	5 - 7 watt	Each	542
8.2.2	8 - 10 watt	Each	624
8.2.3	12 -15 watt	Each	715
8.2.4	18 watt	Each	942
8.2.5	22 watt	Each	967

8.2.6	30 watt	Each	1539
8.3	LED Down lighter (SMD Type) (System lumen efficacy >135 lm/Watt) Supplying, installation, Testing & Commissioning of LED Recessed/surface Down lighter (Round /square/Rectangular) SMD type of following body material with PMMA and prismatic diffuser and construction as per IS : 10322 with driver as per the requirement with Driver efficiency >85%, Operating voltage AC 140-270 Volt, freq 50/60 hz, Operating temp range - 5 deg to 40 deg centigrade, internal surge protection of 2.5 KV with Short & Open circuit protection ,THD < 10% , P. F.≥0.95, IP20, CRI <u>>80</u>, UGR (Unified Glare Rating) <u>≤</u> 19, Flicker free, (flicker should be below 5 %), life time (LED,Driver & electrical circuitary), life time of minimum 50000 Burning Hours with, 70% of initial Lumen maintained till life ends as per LM80 extrapolation IES TM-21-11 report, CCT 3000°K / 4000°K / 5700°K / 6500°K (As per ANSI Bin) , SDCM(Standard Deviation Color Matching) <3, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard and trade mark certificate (T.C.). Manufactures Word Mark/ Name Engraved/ Embossing/ Screen printing on housing. OEM must have its own in house NABL lab setup for all testing facilities for LED fixtures. complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy >135 lm/Watt output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C). Powder coated die cast /Extruded aluminium Body including trim .		
8.3.1	5 - 7 watt	Each	567
8.3.2	8 - 10 watt	Each	657
8.3.3	12 -15 watt	Each	756
8.3.4	18 watt	Each	1004
8.3.5	22 watt	Each	1029

8.3.6	30 watt	Each	1579
8.4	LED Down lighter (COB Type) (System lumen efficacy $\geq 105 < 120\text{lm/Watt}$) Supplying, installation, Testing & Commissioning of LED Recessed/ surface Down lighter (Round/ square/ Rectangular) COB Type of following body material and construction as per IS : 10322 with driver as per the requirement with Driver efficiency $>85\%$, Operating voltage AC 140-270 Volt, frequency 50/60 hz, Operating temp range -5 deg to 40 deg centigrade, internal surge protection of 2.5 KV with Short & Open circuit protection ,THD $< 10\%$, P. F.≥ 0.95, IP20, CRI ≥ 80 , UGR (Unified Glare Rating) ≤ 19, Flicker free (flicker should be below 5%), life time (LED,Driver & electrical circuitary), of minimum 50000 Burning Hours with 70% of initial Lumen maintained till life ends ,CCT 3000°K / 4000°K / 5700°K / 6500°K (As per ANSI Bin), SDCM (Standard Deviation Color Matching) <3, Maximum power consumption should not more than the specified rating and Fixture shall be confirming to relevant BIS standards and trade mark certificate (T.C.). Manufactures Word Mark/ Name Engraved/ Embossing/ Screen printing on housing Complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy ≥ 105 and $<120\text{ lm/Watt}$ output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C) Powder coated die cast /Extruded aluminium Body including trim with Aluminium Reflector		
8.4.1	5 - 7 watt	Each	826
8.4.2	8 - 10 watt	Each	878
8.4.3	12 -15 watt	Each	1042
8.4.4	18 watt	Each	1181
8.4.5	22 watt	Each	1330
8.4.6	30 watt	Each	1654
8.5	LED Down lighter (COB Type) (System lumen efficacy ≥ 120		

lm/Watt)

Supplying, installation, Testing & Commissioning of LED Recessed/surface Down lighter (Round / square/ Rectangular) COB Type of following body material and construction as per IS : 10322 with driver as per the requirement with Driver efficiency >85%, Operating voltage AC 140-270 Volt, freq 50/60 hz, Operating temp range -5 deg to 40 deg centigrade, internal surge protection of 2.5 KV with Short & Open circuit protection ,THD < 10% , P. F.≥0.95, IP20, CRI ≥80 , UGR (Unified Glare Rating) ≤ 19, Flicker free (flicker should be below 5 %), life time (LED,Driver & electrical circuitary), life time of minimum 50000 Burning Hours with , 70% of initial Lumen maintained till life ends , CCT 3000°K / 4000°K / 5700°K /6500°K (As per ANSI Bin), SDCM(Standard Deviation Color Matching) <3, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard and trade mark certificate (T.C.). Manufactures Word Mark/ Name Engraved/ Embossing/ Screen printing on housing. OEM must have its own in house NABL lab setup for all testing facilities for LED fixtures. "complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy ≥120 <135 lm/Watt output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C).

Powder coated die cast /Extruded aluminium Body including trim with Aluminium Reflector

8.5.1	5 - 7 watt	Each	926
8.5.2	8 - 10 watt	Each	945
8.5.3	12 -15 watt	Each	1187
8.5.4	18 watt	Each	1470
8.5.5	22 watt	Each	1506
8.5.6	30 watt	Each	2052
8.6	LED Panel light 2x2 ft. (System lumen efficacy ≥105 <120 lm/Watt)		

Supplying, installation, Testing & Commissioning of Panel light 2x2 ft., of following body material and construction as per IS : 10322 with driver as per the requirement with Driver efficiency >85%, Operating voltage AC 140-270 Volt, freq 50/60 hz, Operating temp range -5 deg to 40 deg centigrade, internal surge protection of 2.5 KV with Short & Open circuit protection, THD < 10% , P. F.≥0.95, IP20, CRI ≥80, UGR (Unified Glare Rating) ≤ 19, Flicker free, (flicker should be below 5 %), life time (LED,Driver & electrical circuitary), of minimum 50000 Burning Hours with, 70% of initial Lumen maintained till life ends as per LM80 extrapolation IES TM-21-11 report, CCT 3000°K / 4000°K/ 5700°K /6500°K (As per ANSI Bin), SDCM(Standard Deviation Color Matching) <3, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard and trade mark certificate (T.C.). Manufactures Word Mark/ Name Engraved/ Embossing/ Screen printing on housing. complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy ≥105 <120 lm/Watt output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C).

Powder coated die cast /Extruded aluminium Body (Thickness ≥ 1.20 mm)

8.6.1	15 watt	Each	1508
8.6.2	18 watt	Each	2060
8.6.3	22 watt	Each	2370
8.6.4	36 watt	Each	2508
8.6.5	40 watt	Each	2681
8.6.6	45 watt	Each	3166
Powder Coated CRCA Sheet Body (Thickness ≥ 0.50 mm)			
8.6.7	15 watt	Each	1388
8.6.8	18 watt	Each	1920

8.6.9	22 watt	Each	1948
8.6.10	36 watt	Each	2059
8.6.11	40 watt	Each	2197
8.6.12	45 watt	Each	2527
8.7	LED Panel light 2x2 ft. (System lumen efficacy ≥ 120 <135 lm/Watt) Supplying, installation, Testing & Commissioning of Panel light 2x2 ft., of following body material and construction as per IS : 10322 with driver as per the requirement with Driver efficiency >85%, Operating voltage AC 140-270 Volt, freq 50/60 hz, Operating temp range -5 deg to 40 deg centigrade, internal surge protection of 2.5 KV with Short & Open circuit protection ,THD < 10% , P. F.≥ 0.95, IP20, CRI ≥ 80, UGR (Unified Glare Rating) ≤ 19, Flicker free, (flicker should be below 5 %), life time (LED,Driver & electrical circuitary), of minimum 50000 Burning Hours with , 70% of initial Lumen maintained till life ends,CCT 3000°K / 4000°K / 5700°K /6500°K (As per ANSI Bin), SDCM (Standard Deviation Color Matching) <3, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard and trade mark certificate (T.C.). Manufactures Word Mark/ Name Engraved/ Embossing/ Screen printing on housing. complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy ≥ 120 <135 lm/Watt output .LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C). Powder coated die cast /Extruded aluminium Body (Thickness ≥ 1.20 mm)		
8.7.1	15 watt	Each	1633
8.7.2	18 watt	Each	2240
8.7.3	22 watt	Each	2695
8.7.4	36 watt	Each	2962

8.7.5	40 watt	Each	3685
8.7.6	45 watt	Each	3913
	Powder Coated CRCA Sheet Body (Thickness $\geq$ 0.50 mm)		
8.7.7	15 watt	Each	1549
8.7.8	18 watt	Each	1920
8.7.9	22 watt	Each	2031
8.7.10	36 watt	Each	2225
8.7.11	40 watt	Each	2527
8.7.12	45 watt	Each	2584
8.8	LED Panel light 2x2 ft., (System lumen efficacy >135 lm/Watt) Supplying, installation, Testing & Commissioning of Panel light 2x2 ft., of following body material and construction as per IS : 10322 with driver as per the requirement with Driver efficiency >85%, Operating voltage AC 140-270 Volt, freq 50/60 hz, Operating temp range -5 deg to 40 deg centigrade, internal surge protection of 2.5 KV with Short & Open circuit protection, THD < 10% , P. F. $\geq$ 0.95, IP20, CRI $\geq$ 80, UGR (Unified Glare Rating) $\leq$ 19, Flicker free, (flicker should be below 5 %), life time (LED, Driver & electrical circuitary), of minimum 50000 Burning Hours with , 70% of initial Lumen maintained till life ends , CCT 3000°K / 4000°K / 5700°K / 6500°K (As per ANSI Bin), SDCM (Standard Deviation Color Matching) < 3, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard and trade mark certificate (T.C.). Manufactures Word Mark/ Name Engraved/ Embossing/ Screen printing on housing. complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy >135 lm/Watt output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C). Powder coated die cast /Extruded aluminium Body (Thickness $\geq$ 1.20 mm)		

8.8.1	15 watt	Each	1758
8.8.2	18 watt	Each	2420
8.8.3	22 watt	Each	3581
8.8.4	36 watt	Each	3997
8.8.5	40 watt	Each	4246
8.8.6	45 watt	Each	4827
Powder Coated CRCA Sheet Body (Thickness $\geq$ 0.50 mm)			
8.8.7	15 watt	Each	1646
8.8.8	18 watt	Each	2024
8.8.9	22 watt	Each	2793
8.8.10	36 watt	Each	2846
8.8.11	40 watt	Each	2867
8.8.12	45 watt	Each	2924

8.9	LED Batten light (System lumen efficacy ≥ 105 <120 lm/Watt) Supplying, installation, Testing & Commissioning of LED surface mounted Batten light of following body material and construction as per IS : 10322 with driver (Replaceable) as per the requirement with Driver efficiency >85% ,Operating voltage AC 140-270 Volt, freq 50/60 hz, Operating temp range -5 deg to 40 deg centigrade, internal surge protection of 2.5 KV with Short & Open circuit protection ,THD < 10% , P. F.≥ 0.95, IP20, CRI ≥ 80, Flicker free, (flicker should be below 5 %), life time (LED,Driver & electrical circuitary), of minimum 50000 Burning Hours with , 70% of initial Lumen maintained till life ends, CCT 3000°K / 4000°K / 5700°K /6500°K (As per ANSI Bin), SDCM(Standard Deviation Color Matching) <3, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard and trade mark certificate (T.C.). Manufactures Word Mark/ Name Engraved/ Embossing/ Screen printing on housing. complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy ≥ 105 <120 lm/Watt output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C). Powder coated die cast /Extruded aluminium Body (Thickness ≥ 1.20 mm)		
8.9.1	18- 22 Watt	Each	681
8.9.2	24 -26watt	Each	692
8.9.3	36 watt	Each	678
8.9.4	40 watt	Each	721
	Powder Coated CRCA Sheet Body (Thickness ≥ 0.50 mm)		
8.9.5	18- 22 Watt	Each	629
8.9.6	24 -26watt	Each	639
8.9.7	36 watt	Each	649
8.9.8	40 watt	Each	663

8.10	LED Batten light (System lumen efficacy $\geq 120 < 135$ lm/Watt) Supplying, installation, Testing & Commissioning of LED surface mounted Batten light of following body material and construction as per IS : 10322 with driver (Replaceable) as per the requirement with Driver efficiency $> 85\%$, Operating voltage AC 140-270 Volt, freq 50/60 hz, Operating temp range -5 deg to 40 deg centigrade, internal surge protection of 2.5 KV with Short & Open circuit protection ,THD $< 10\%$, P. F.≥ 0.95, IP20, CRI ≥ 80 , Flicker free (flicker should be below 5 %), life time (LED,Driver & electrical circuitary), of minimum 50000 Burning Hours with , 70% of initial Lumen maintained till life ends,CCT 3000°K / 4000°K / 5700°K /6500°K (As per ANSI Bin), SDCM(Standard Deviation Color Matching) < 3, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard and trade mark certificate (T.C.). Manufactures Word Mark/ Name Engraved/ Embossing/ Screen printing on housing. OEM must have its own in house NABL lab setup for all testing facilities for LED fixtures. complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy $\geq 120 < 135$ lm/Watt output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C). Powder coated die cast /Extruded aluminium Body (Thickness ≥ 1.20 mm)		
8.10.1	18- 22 Watt	Each	734
8.10.2	24 -26watt	Each	745
8.10.3	36 watt	Each	760
8.10.4	40 watt	Each	779
	Powder Coated CRCA Sheet Body (Thickness ≥ 0.50 mm)		
8.10.5	18- 22 Watt	Each	681
8.10.6	24 -26watt	Each	692
8.10.7	36 watt	Each	705

8.10.8	40 watt	Each	721
8.11	LED Batten light (System lumen efficacy >135 lm/Watt) Supplying, installation, Testing & Commissioning of LED surface mounted Batten light of following body material and construction as per IS : 10322 with driver (Replaceable) as per the requirement with Driver efficiency >85% ,Operating voltage AC 140-270 Volt, freq 50/60 hz, Operating temp range -5 deg to 40 deg centigrade, internal surge protection of 2.5 KV with Short & Open circuit protection ,THD < 10% , P. F.≥0.95, IP20, CRI ≥80 ,Flicker free (flicker should be below 5 %), life time (LED,Driver & electrical circuitary), of minimum 50000 Burning Hours with , 70% of initial Lumen maintained till life ends, CCT 3000°K / 4000°K / 5700°K /6500°K (As per ANSI Bin), SDCM(Standard Deviation Color Matching) <3, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard and trade mark certificate (T.C.). Manufactures Word Mark/ Name Engraved/ Embossing/ Screen printing on housing. OEM must have its own in house NABL lab setup for all testing facilities for LED fixtures. complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy >135 lm/Watt output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted.. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C). Powder coated die cast /Extruded aluminium Body (Thickness ≥ 1.20 mm)		
8.11.1	18- 22 Watt	Each	787
8.11.2	24 -26watt	Each	797
8.11.3	36 watt	Each	816
8.11.4	40 watt	Each	837
	Powder Coated CRCA Sheet Body (Thickness ≥ 0.50 mm)		
8.11.5	18- 22 Watt	Each	734
8.11.6	24 -26watt	Each	745

8.11.7	36 watt	Each	760
8.11.8	40 watt	Each	779
8.12	LED Street light fixture, powder coated pressure die cast aluminium body (System lumen efficacy $\geq 105 < 120$ lm/Watt) Supplying, installation, Testing & Commissioning of Street light LED fixture powder coated pressure die cast aluminium body with driver as per the requirement with Driver efficiency $> 85\%$, Input voltage: 140-270 Volt AC, freq 50/60 hz, Operating temp range -5 deg to 50 deg centigrade, internal surge protection of 5 KV L,N,E as per IEC 61000-4-5, Driver efficiency $> 85\%$, THD $< 10\%$ as per IEC 61000-3-2, P.F. ≥ 0.95, IP-66, IK-10, CRI ≥ 70, under voltage and over voltage protection, EMI-EMC As per CISPR -15, lenses for beam angle as per IESNA type I/II/III as per the width of the road and the project requirement., suitable to fit in up to 65mm dia pipe, life time (LED, Driver & electrical circuitary) of minimum 50000 Burning Hours with 70% of initial Lumen maintained till life ends as per LM80 extrapolation IES TM-21-11 report, CCT 3000°K/4000°K/5700°K/6500°K (As per ANS Bin), SDCM (Standard Deviation Color Matching) < 5, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy $\geq 105 < 120$ lm/Watt output .LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C).		
8.12.1	10 watt	Each	778
8.12.2	14 watt	Each	917
8.12.3	15 watt	Each	930
8.12.4	18 watt	Each	973
8.12.5	20 watt	Each	982
8.12.6	24 watt	Each	986

8.12.7	25 watt	Each	1000
8.12.8	30 watt	Each	1020
8.12.9	36 watt	Each	1228
8.12.10	40 watt	Each	1297
8.12.11	45 watt	Each	1366
8.12.12	50 watt	Each	1782
8.12.13	72 watt	Each	1920
8.12.14	90 watt	Each	2162
8.12.15	100 watt	Each	2543
8.12.16	120 watt	Each	2612
8.12.17	150 watt	Each	3374
8.12.18	180 watt	Each	4550
8.12.19	200 watt	Each	4689
8.13	LED Street light fixture, powder coated pressure die cast aluminium body (System lumen efficacy $\geq 120 < 135$ lm/Watt) Supplying, installation, Testing & Commissioning of Street light LED fixture powder coated pressure die cast aluminium body with driver as per the requirement with Driver efficiency $> 85\%$, Input voltage: 140-270 Volt AC, freq 50/60 hz, Operating temp range -5 deg to 50 deg centigrade, internal surge protection of 5 KV L,N,E as per IEC 61000-4-5, Driver efficiency $> 85\%$, THD $< 10\%$ as per IEC 61000-3-2, P.F. ≥ 0.95 , IP-66, IK-10, CRI ≥ 70 , under voltage and over voltage protection, EMI-EMC As per CISPR 15, lenses for beam angle as per IESNA type I/II/III as per the width of the road and the project requirement., suitable to fit in up to 65mm dia pipe, life time (LED, Driver & electrical circuitary) of minimum 50000 Burning Hours with 70% of initial Lumen maintained till life ends as per LM80 extrapolation IES TM-21-11 report, CCT 3000°K / 4000°K / 5700°K/6500°K(As per ANSI Bin), SDCM(Standard Deviation Color Matching) < 5 , Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard complete in all		

respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy $\geq 120 < 135$ lm/Watt output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C).

8.13.1	10 watt	Each	830
8.13.2	14 watt	Each	982
8.13.3	15 watt	Each	1011
8.13.4	18 watt	Each	1173
8.13.5	20 watt	Each	1297
8.13.6	24 watt	Each	1477
8.13.7	25 watt	Each	1526
8.13.8	30 watt	Each	1553
8.13.9	36 watt	Each	1629
8.13.10	40 watt	Each	1643
8.13.11	45 watt	Each	1744
8.13.12	50 watt	Each	1934
8.13.13	72 watt	Each	2128
8.13.14	90 watt	Each	2335
8.13.15	100 watt	Each	2681
8.13.16	120 watt	Each	2958
8.13.17	150 watt	Each	3443
8.13.18	180 watt	Each	4827
8.13.19	200 watt	Each	5035

8.14 LED Street light fixture, powder coated pressure die cast aluminium body (System lumen efficacy > 135 lm/Watt) Supplying, installation, Testing & Commissioning of Street light LED fixture, powder coated pressure die cast aluminium body with built in or separate driver as per the requirement with Driver efficiency $> 85\%$, Input voltage: 140-270 Volt AC, freq 50/60 hz, Operating temp range -5 deg to 50 deg centigrade, internal surge protection of 5 KV L,N,E as per IEC 61000-4-5, Driver efficiency $> 85\%$, THD $< 10\%$ as per IEC 61000-3-2 , P. F. ≥ 0.95 , IP-66, IK-10, CRI ≥ 70 , under voltage and over voltage protection, EMI-EMC as per CISPR-15, lenses for beam angle as per IESNA type I/II/III as per the width of the

road and the project requirement., suitable to fit in up to 65mm dia pipe, life time (LED,Driver & electrical circuitary) of minimum 50000 Burning Hours with 70% of initial Lumen maintained till life ends as per LM80 extrapolation IES TM-21-11 report , CCT 3000°K / 4000°K / 5700°K /6500°K (As per ANSI Bin) , SDCM(Standard Deviation Color Matching) <5, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy >135 lm/Watt output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted.. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C).

8.14.1	10 watt	Each	916
8.14.2	14 watt	Each	1055
8.14.3	15 watt	Each	1087
8.14.4	18 watt	Each	1193
8.14.5	20 watt	Each	1228
8.14.6	24 watt	Each	1539
8.14.7	25 watt	Each	1591
8.14.8	30 watt	Each	1678
8.14.9	36 watt	Each	1747
8.14.10	40 watt	Each	1886
8.14.11	45 watt	Each	1966
8.14.12	50 watt	Each	2185
8.14.13	72 watt	Each	2578
8.14.14	90 watt	Each	2820

8.14.15	100 watt	Each	3374
8.14.16	120 watt	Each	4066
8.14.17	150 watt	Each	5090
8.14.18	180 watt	Each	5450
8.14.19	200 watt	Each	6142
8.15	LED Flood Light, powder coated pressure die cast aluminium (System lumen efficacy 105 <120 lm/Watt) Supplying, installation, Testing & Commissioning of Flood Light, powder coated pressure die cast aluminium body with built in or separate driver as per the requirement with Driver efficiency >85%, Input voltage: Input voltage: 140-270 Volt AC, freq 50/60 hz, Operating temp range -5 deg to 50 deg centigrade, internal surge protection of 5 KV L,N,E as per IEC 61000-4-5, Driver efficiency >85%,THD < 10% as per IEC 61000-3-2 , P. F.≥0.95, IP-66,IK-10, CRI ≥70, under voltage and over voltage protection,EMI-EMC as per CISPR -15, lenses for beam angle 30 deg-120deg as per the application and the project requirementdeg., suitable tilt able fitting, life time (LED,Driver & electrical circuitary) of minimum 50000 Burning Hours with 70% of initial Lumen maintained till life ends as per LM80 extrapolation IES TM-21-11 report, CCT 3000°K / 4000°K / 5700°K /6500°K (As per ANSI Bin) , SDCM(Standard Deviation Color Matching) <5, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy 105 <120 lm/Watt output .LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted.. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C).		
8.15.1	50 watt	Each	1436
8.15.2	70 watt	Each	2114
8.15.3	100 watt	Each	2266

8.15.4	150 watt	Each	3166
8.15.5	200 watt	Each	4550
8.15.6	250 watt	Each	5450
8.16	LED Flood Light, powder coated pressure die cast aluminium (System lumen efficacy ≥ 120 and < 135 lm/Watt) Supplying, installation, Testing & Commissioning of Flood Light, powder coated pressure die cast aluminium body with built in or separate driver as per the requirement with Driver efficiency $> 85\%$, Input voltage: 140-270 Volt AC, freq 50/60 hz, Operating temp range -5 deg to 50 deg centigrade, internal surge protection of 5 KV L,N,E as per IEC 61000-4-5, Driver efficiency $> 85\%$, THD $< 10\%$ as per IEC 61000-3-2, P. F. ≥ 0.95, IP-66, IK-10, CRI ≥ 70, under voltage and over voltage protection, EMI-EMC as per CISPR-15, lenses for beam angle 30 deg-120deg as per the application and the project requirement deg, suitable tilt able fitting, life time (LED, Driver & electrical circuitry) of minimum 50000 Burning Hours with 70% of initial Lumen maintained till life ends as per LM80 extrapolation IES TM-21-11 report, CCT 3000°K / 4000°K / 5700°K / 6500°K (As per ANSI Bin), SDCM (Standard Deviation Color Matching) < 5, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy ≥ 120 and < 135 lm/Watt output. LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted.. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C).		
8.16.1	50 watt	Each	2009
8.16.2	70 watt	Each	2299
8.16.3	100 watt	Each	2467
8.16.4	150 watt	Each	3457
8.16.5	200 watt	Each	4689

8.16.6	250 watt	Each	5969
8.17	LED Flood Light, powder coated pressure die cast aluminium (System lumen efficacy >135 lm/Watt) Supplying, installation, Testing & Commissioning of Flood Light, powder coated pressure die cast aluminium body with built in or separate driver as per the requirement with Driver efficiency >85%, Input voltage: Input voltage: 140-270 Volt AC, freq 50/60 hz, Operating temp range -5 deg to 50 deg centigrade, internal surge protection of 5 KV L,N,E as per IEC 61000-4-5, Driver efficiency >85%,THD < 10% as per IEC 61000-3-2 , P. F.≥0.95, IP-66,IK-10, CRI ≥70, under voltage and over voltage protection,EMI-EMC as per CISPR-15, lenses for beam angle 30 deg-120deg as per the application and the project requirementdeg., suitable tilt able fitting, life time (LED,Driver & electrical circuitary) of minimum 50000 Burning Hours with 70% of initial Lumen maintained till life ends as per LM80 extrapolation IES TM-21-11 report, CCT 3000°K / 4000°K / 5700°K /6500°K (As per ANSI Bin), SDCM(Standard Deviation Color Matching) <5, Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy >135 lm/Watt output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted. . Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C).		
8.17.1	50 watt	Each	2089
8.17.2	70 watt	Each	2392
8.17.3	100 watt	Each	2567
8.17.4	150 watt	Each	3602
8.17.5	200 watt	Each	5504
8.17.6	250 watt	Each	6229
8.18	LED Smart Street light fixture, powder coated pressure die cast aluminium (System. System lumen efficacy ≥105 and <120		

lm/Watt)

Supplying, installation, Testing & Commissioning of Smart Street light LED fixture, powder coated pressure die cast aluminium body with built in or separate driver as per the requirement (< 700ma), Input voltage: 140-270 Volt AC, freq 50/60 hz, Operating temp range -15 deg to 50 deg centigrade, internal surge protection of 5 KV L,N,E as per IEC 61000-4-5, Driver efficiency >85%,THD < 10% as per IEC 61000-3-2, P. F.≥0.95, IP-66,IK-08, CRI ≥70 , under voltage and over voltage protection,EMI- EMC as per CISPR-15, lenses for beam angle as per IESNA type I/II/III as per the width of the road and the project requirement., suitable to fit in up to 65mm dia pipe, life time of minimum 50000 Burning Hours with 70% of initial Lumen maintained till life ends as per LM80 extrapolation IES TM-21-11 report, CCT 3000°K / 4000°K / 5700°K /6500°K (As per ANSI Bin), Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard complete in all respect i/c external connections with 1.5 sq mm FRLS, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy ≥105 and <120 lm/Watt output . LM79 & LM80 Test report from NABL lab for all testing required for LED fixtures as per BIS shall be submitted. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C). Smart inbuilt controller shall have following features.

- 1.Control and monitor LED luminaries with bi directional control (Status, Fault, Alarm, dimming level, wattage, energy)
2. to measure voltage, current, power, power factor, apparent energy, active energy, operating hours.
3. Inbuilt ambient light sensor, motion sensor based on Passive Infra Red (PIR).
4. Wi-Fi LoRA/Zigbee/Powerline with ethernet network based IOT feature as per site requirement or engineer in charge.
5. should be controlled through auto/ manual
6. Programmable level of not less than 48 different light intensity settings,
7. inbuilt repeater & relay signals function to other controllers

8.18.1	45 watt	Each	5081
8.18.2	50 watt	Each	5300

8.18.3	72 watt	Each	5825
8.18.4	90 watt	Each	6088
8.18.5	100 watt	Each	7226
8.18.6	120 watt	Each	7576
8.18.7	150 watt	Each	9485
8.18.8	180 watt	Each	9660
8.18.9	200 watt	Each	11341

8.19 LED Smart Street light fixture, powder coated pressure die cast aluminium (System. System lumen efficacy ≥ 120 and < 135 lm/Watt)

Supplying, installation, Testing & Commissioning of Smart Street light LED fixture, powder coated pressure die cast aluminium body with built in or separate driver as per the requirement ($< 700\text{ma}$), Input voltage: 140-270 Volt AC, freq 50/60 hz, Operating temp range -5 deg to 50 deg centigrade, internal surge protection of 5 KV L,N,E as per IEC 61000-4-5, Driver efficiency $> 85\%$, THD $< 10\%$ as per IEC 61000-3-2, P. F. ≥ 0.95 , IP-66, IK-08, CRI ≥ 70 , under voltage and over voltage protection, EMI- EMC as per CISPR-15, lenses for beam angle as per IESNA type I/II/III as per the width of the road and the project requirement., suitable to fit in up to 65mm dia pipe, life time of minimum 50000 Burning Hours with 70% of initial Lumen maintained till life ends as per LM80 extrapolation IES TM-21-11 report , CCT 3000°K / 4000°K / 5700°K / 6500°K (As per ANSI Bin) , Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy ≥ 120 and < 135 lm/Watt output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted.. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C).

Smart inbuilt controller shall have following features.

1. Control and monitor LED luminaries with bi directional control(Status, Fault, Alarm, dimming level, wattage, energy)
2. To measure voltage, current, power, power factor, apparent energy, active energy, operating hours.
3. Inbuilt ambient light sensor, motion sensor based on Passive Infra Red (PIR).
4. Wi-Fi LoRA/Zigbee/Powerline with ethernet network based IOT feature as per site requirement or engineer in charge.
5. Should be controlled through auto/ manual
6. Programmable level of not less than 48 different light intensity settings,
7. Inbuilt repeater & relay signals function to other controllers

8.19.1	45 watt	Each	5563
8.19.2	50 watt	Each	5804
8.19.3	72 watt	Each	6382
8.19.4	90 watt	Each	6671
8.19.5	100 watt	Each	7923
8.19.6	120 watt	Each	8308
8.19.7	150 watt	Each	10408
8.19.8	180 watt	Each	10600
8.19.9	200 watt	Each	12449

8.20 LED Smart Street light fixture, powder coated pressure die cast aluminium (System. System lumen efficacy >135 lm/Watt) Supplying, installation, Testing & Commissioning of Smart Street light LED fixture, powder coated pressure die cast aluminium body with built in or separate driver as per the requirement (< 700ma), Input voltage: 140-270 Volt AC, freq 50/60 hz, Operating temp range -5 deg to 50 deg centigrade, internal surge protection of 5 KV L,N,E as per IEC 61000-4-5, Driver efficiency >85%,THD < 10% as per IEC 61000-3-2, P. F.≥0.95, IP-66,IK-08, CRI ≥70 , under voltage and over voltage protection,EMI- EMC as per CISPR-15, lenses for beam angle as per IESNA type I/II/III as per the width of the road and the project requirement., suitable to fit in up to 65mm dia pipe, life time of minimum 50000 Burning Hours with 70% of initial Lumen maintained till life ends as per LM80 extrapolation IES TM-21-11 report, CCT 3000°K / 4000°K / 5700°K /6500°K (As per ANSI Bin), Maximum power consumption should not more than the specified rating and Fixture shall be of relevant BIS standard complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required with Minimum 5 year OEM warranty. System lumen efficacy >135 lm/Watt output . LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted.. Shape size and CCT shall be as approved by Engineer-in-Charge as per requirement. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise above 90°C). Smart inbuilt controller shall have following features.

- 1.Control and monitor LED luminaries with bi directional control(Status, Fault, Alarm, dimming level, wattage, energy)
2. to measure voltage, current, power, power factor, apparent energy, active energy, operating hours
3. Inbuilt ambient light sensor, motion sensor based on Passive Infra Red (PIR).
4. Wi-Fi LoRA/Zigbee/Powerline with ethernet network based IOT feature as per site requirement or engineer in charge.
5. should be controlled through auto/ manual
6. Programmable level of not less than 48 different light intensity settings,
7. inbuilt repeater & relay signals function to other controllers

8.20.1	45 watt	Each	5804
8.20.2	50 watt	Each	6056
8.20.3	72 watt	Each	6660

8.20.4	90 watt	Each	6962	
8.20.5	100 watt	Each	8271	
8.20.6	120 watt	Each	8674	
8.20.7	150 watt	Each	10869	
8.20.8	180 watt	Each	11070	
8.20.9	200 watt	Each	13003	
8.21	Solar outdoor LED light			
8.21.1	Supplying, installation, Testing & Commissioning of the integrated type solar PV lighting system on the existing pole structure, comprising of 20 watt, 6V Mono Passivated Emitter and Rear Contact (PERC) Solar Panel (minimum efficiency 21%),, Pulse with modulation (PWM)/Maximum Power point tracking (MPPT) Charge Controller in the box with a sleek appearance and a sturdy structure, is weather-proof, and is simple to install, With Lithium- Iron Phosphate Battery (LiFePO4) 3.2Volt (Cell) 24 AH battery, charging time 6-8 hours, Battery backup time 12 hours (minimum), LED fixture 20 watt, Input voltage: 12V DC , Operating temp range -5 deg to 50 deg centigrade, internal surge protection of 5 KV L,N,E as per IEC 61000-4-5, Driver efficiency >85%, P. F.≥0.95, IP-66,IK-08, CRI ≥70 , under voltage and over voltage protection, Electro Magnetic Interference (EMI) Electro Magnetic Compatibility (EMC) As per CISPR 15 , lenses for beam angle as per Illuminating Engineering Society of North America (IESNA) type I/II/III as per the width of the road and the project requirement, Correlated Colour Temperature (CCT) 5700°K / (As per American National Standard Institute (ANSI Bin)), life time (LED,Driver & electrical circuitary) of 50K hours lamp burning hours till the 70 % of initial Lumen maintained as per LM80 extrapolation IES TM-21-11 report, automatic switch on/off, Alliuminium or Acrylonitrile Butadiene Styrene (ABS body), can be installed on a pole or wall. System lumen efficacy >120 lm/Watt output. LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted.. All as per pre approved by Engineer in-charge complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required. (Part 1), for fixtures up to 60 watt. LED light Complete with mounting structure for the battery and accessories and wind storm withstand capacity as per the zone. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise aboven 90°C) 20 Watt (System lumen efficacy >120 lm/Watt)		Each	18601
8.21.2	Supplying, installation, Testing & Commissioning of the			

integrated type solar PV lighting system on the existing pole structure, comprising of 30 watt, 6V Mono Passivated Emitter and Rear Contact (PERC) Solar Panel (minimum efficiency 21%), Pulse with modulation (PWM)/Maximum Power point tracking (MPPT) charge Controller in the box with a sleek appearance and a sturdy structure, is weather-proof, and is simple to install, With Lithium- Iron Phosphate Battery (LiFePO4) 3.2Volt (Cell) 30 AH battery, charging time 8-10 hours, Battery backup time 12 hours (minimum), LED fixture watt 30 watt, Input voltage: 12V DC , Operating temp range - 5 deg to 50 deg centigrade, internal surge protection of 5 KV L,N,E as per IEC 61000-4-5, Driver efficiency >85%, P. F.≥0.95, IP-66,IK-08, CRI >70 , under voltage and over voltage protection, Electro Magnetic Interference (EMI) Electro Magnetic Compatibility (EMC) As per CISPER 15 A, lenses for beam angle as per Illuminating Engineering Society of North America (IESNA) type I/II/III as per the width of the road and the project requirement, Correlated Colour Temperature (CCT) 5700°K (As per American National Standard Institute (ANSI Bin)) , life time (LED,Driver & electrical circuitary) of 50K hours lamp burning hours till the 70 % of initial Lumen maintained as per LM80 extrapolation IES TM-21-11 report, automatic switch on/off, Alluminium or Acrylonitrile Butadiene Styrene (ABS body), can be installed on a pole or wall. System lumen efficacy >120 lm/Watt output. LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted.. All as per pre approved by Engineer in-charge complete in all respect i/c external connections with 1.5 sq mm FRLS/HFFR, PVC insulated copper conductor single core cable and earthing etc. as required. (Part 1), for fixtures up to 60 watt. LED light Complete with mounting structure for the battery and accessories and wind storm withstand capacity as per the zone. (Thermal management: heat sink of aluminium housing such that LED junction temperature do'nt rise aboven 90°C). 30 Watt (System lumen efficacy >120 lm/Watt)

Nos 23100

8.21.3 Supplying, installation, Testing & Commissioning of the integrated type solar PV lighting system on the existing pole structure, comprising of 35 watt, 6V Mono Passivated Emitter and Rear Contact (PERC) Solar Panel (minimum efficiency 21%), Pulse with modulation (PWM)/Maximum Power point tracking (MPPT) Charge Controller in the box with a sleek appearance and a sturdy structure, is weather-proof, and is simple to install, with Lithium-Iron Phosphate Battery (LiFePO4) 3.2Volt (Cell) 35 AH battery, charging time 8-10 hours, Battery backup time 12 hours (minimum), LED fixture watt 35 watt, Input voltage: 12V DC , Operating temp range -5 deg to 50

deg centigrade, internal surge protection of 5 KV L,N,E as per IEC 61000-4-5, Driver efficiency >85%, P. F.≥0.95, IP-66,IK-08, CRI >70 , under voltage and over voltage protection, Electro Magnetic Interference (EMI) Electro Magnetic Compatibility (EMC) As per CISPR 15 , lenses for beam angle as per Illuminating Engineering Society of North America (IESNA) type I/II/III as per the width of the road and the project requirement, Correlated Colour Temperature (CCT) 5700°K (As per American National Standard Institute (ANSI Bin)), life time (LED,Driver & electrical circuitary) of 50K hours lamp buring hours till the 70 % of initial Lumen maintained as per LM80 extrapolation IES TM-21-11 report, automatic swich on/off, Alluminium or Acrylonitrile Butadiene Styrene (ABS body), can be installed on a pole or wall. System lumen efficacy >120 lm/Watt output. LM79 & LM80 Test report and all testing required for LED fixtures as per BIS shall be submitted.. All as per pre approved by Engineer in-charge complete in all respect i/c external connections with 1.5 sq mm FRLS, PVC insulated copper conductor single core cable and earthing etc. as required. (Part 1), for fixtures up to 60 watt. LED light Complete with mounting structure for the battery and accessories and wind storm withstand capacity as per the zone. (Thermal management: heat sink of aluminium housing such that LED junction temperature shall not rise aboven90°C) 35 Watt (System lumen efficacy >120 lm/Watt)

Each 29329

CHAPTER-9 BLDC FAN

Item No.	Description	Unit	Rate(₹)
9.1	Brush Less Direct Current (BLDC) Motor Supply, Installation, Testing and Commissioning of ceiling fan with Brush Less Direct Current (BLDC) Motor, class of insulation: B, 3 nos. metal(Aluminum alloy) blades, 30 cm long down rod, 2 nos. canopies, shackle kit, safety rope, copper winding, steel/Al bodyPower Factor not less than 0.9, Service Value (CM/M/W) minimum as below, 350 RPM (tolerance as per IS : 374-2019), THD (Total Harmonic Distortion) less than 10%, remote (prefreably mobile app based) for speed control and all remaining accessories including safety pin, nut bolts, washers, temperature rise=75 degree C (max.), insulation resistance more than 2 mega ohm, suitable for 230 V, 50 Hz, single phase AC Ceiling Fan compliant to IS 374:2019 fan Supply, earthing etc. complete as req.		
9.1.1	900mm, service value ≥ 5.1 CM/Min/Watt, air delivery 130 CM/Min (Minimum)	Each	2737
9.1.2	1050mm, service value ≥ 5.1 CM/Min/Watt, air delivery 150 CM/Min (Minimum)	Each	2766
9.1.3	1200mm, service value ≥ 6.0 CM/Min/Watt, air delivery 210 CM/Min (Minimum)	Each	2882
9.1.4	1400mm, service value ≥ 6.0 CM/Min/Watt, air delivery 245 CM/Min (Minimum)	Each	2913
9.2	Brush Less Direct Current (BLDC) Motor Supply, Installation, Testing and Commissioning of ceiling fan with Brush Less Direct Current (BLDC) Motor, class of insulation: B, 3 nos. metal(Aluminum alloy) blades, 30 cm long down rod, 2 nos. canopies, shackle kit, safety rope, copper winding, steel/Al bodyPower Factor not less than 0.9, Service Value (CM/M/W) minimum as below, 350 RPM (tolerance as per IS : 374-2019), THD (Total Harmonic Distortion) less than 10%, suitable for operation with regulator for speed control and all remaining accessories including safety pin, nut bolts, washers, temperature rise=75 degree C (max.), insulation resistance more than 2 mega ohm, suitable for 230 V, 50 Hz, single phase AC Ceiling Fan compliant to IS 374:2019 fan Supply, earthing etc. complete as req.		
9.2.1	900mm, service value ≥ 5.1 CM/Min/Watt, air delivery 130 CM/Min (Minimum)	Each	2560
9.2.2	1050mm, service value ≥ 5.1 CM/Min/Watt, air delivery 150 CM/Min (Minimum)	Each	2587
9.2.3	1200mm, service value ≥ 6.0 CM/Min/Watt, air delivery 210	Each	2695

	CM/Min (Minimum)		
9.2.4	1400mm, service value ≥ 6.0 CM/Min/Watt, air delivery 245 CM/Min (Minimum)	Each	2737

CHAPTER 10

MV CABLE LAYING

Item No.	Description	Unit	Rate(₹)
10.1	Laying of one number PVC insulated and PVC sheathed / XLPE power cable of 1.1 KV grade of following size direct in ground including excavation, sand cushioning, protective covering and refilling the trench etc as required.		
10.1.1	Upto 35 sq. mm	Metre	484
10.1.2	Above 35 sq. mm and upto 95 sq. mm	Metre	475
10.1.3	Above 95 sq. mm and upto 185 sq. mm	Metre	497
10.1.4	Above 185 sq. mm and upto 400 sq. mm	Metre	564
10.2	Laying of one number additional PVC insulated and PVC sheathed / XLPE power cable of 1.1 KV grade of following size direct in ground in the same trench in one tier horizontal formation including excavation, sand cushioning, protective covering and refilling the trench etc as required.		
10.2.1	Upto 35 sq. mm	Metre	313
10.2.2	Above 35 sq. mm and upto 95 sq. mm	Metre	335
10.2.3	Above 95 sq. mm and upto 185 sq. mm	Metre	357
10.2.4	Above 185 sq. mm and upto 400 sq. mm	Metre	423
10.3	Laying of one number PVC insulated and PVC sheathed / XLPE power cable of 1.1 KV grade of following size direct in ground including excavation and refilling the trench etc as required, but excluding sand cushioning and protective covering.		
10.3.1	Upto 35 sq. mm	Metre	253
10.3.2	Above 35 sq. mm and upto 95 sq. mm	Metre	276
10.3.3	Above 95 sq. mm and upto 185 sq. mm	Metre	298
10.3.4	Above 185 sq. mm and upto 400 sq. mm	Metre	364
10.4	Laying of one number additional PVC insulated and PVC sheathed / XLPE power cable of 1.1 KV grade of following size direct in ground in the same trench in one tier horizontal formation including excavation and refilling the trench etc as		

	required, but excluding sand cushioning and protective covering.		
10.4.1	Upto 35 sq. mm	Metre	160
10.4.2	Above 35 sq. mm and upto 95 sq. mm	Metre	182
10.4.3	Above 95 sq. mm and upto 185 sq. mm	Metre	204
10.4.4	Above 185 sq. mm and upto 400 sq. mm	Metre	270
10.5	Laying of one number PVC insulated and PVC sheathed / XLPE power cable of 1.1 KV grade of following size in the existing RCC/ HUME/ METAL pipe as required.		
10.5.1	Upto 35 sq. mm	Metre	47
10.5.2	Above 35 sq. mm and upto 95 sq. mm	Metre	72
10.5.3	Above 95 sq. mm and upto 185 sq. mm	Metre	97
10.5.4	Above 185 sq. mm and upto 400 sq. mm	Metre	170
10.6	Laying of one number PVC insulated and PVC sheathed / XLPE power cable of 1.1 KV grade of following size in the existing masonry open duct as required.		
10.6.1	Upto 35 sq. mm	Metre	36
10.6.2	Above 35 sq. mm and upto 95 sq. mm	Metre	58
10.6.3	Above 95 sq. mm and upto 185 sq. mm	Metre	80
10.6.4	Above 185 sq. mm and upto 400 sq. mm	Metre	146
10.7	Laying and fixing of one number PVC insulated and PVC sheathed / XLPE power cable of 1.1 KV grade of following size on wall surface as required.		
10.7.1	Upto 35 sq. mm (clamped with 1mm thick saddle)	Metre	64
10.7.2	Above 35 sq. mm and upto 95 sq. mm (clamped with 25x3mm MS flat clamp)	Metre	151
10.7.3	Above 95 sq. mm and upto 185 sq. mm (clamped with 25/40x3mm MS flat clamp)	Metre	180
10.7.4	Above 185 sq. mm and upto 400 sq. mm (clamped with 40x3mm MS flat clamp)	Metre	271
10.8	Laying and fixing of one number PVC insulated and PVC sheathed / XLPE power cable of 1.1 KV grade of following size on cable tray as required.		

10.8.1	Upto 35 sq. mm (clamped with 1mm thick saddle)	Metre	53
10.8.2	Above 35 sq. mm and upto 95 sq. mm (clamped with 25x3mm MS flat clamp)	Metre	109
10.8.3	Above 95 sq. mm and upto 185 sq. mm (clamped with 25/40x3mm MS flat clamp)	Metre	138
10.8.4	Above 185 sq. mm and upto 400 sq. mm (clamped with 40x3mm MS flat clamp)	Metre	226
10.9	Supplying and making cable route marker with cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size) of size 60 cm X 60 cm at the bottom and 50 cm X 50 cm at the top with a thickness of 10cm including inscription duly engraved as required.	Each	683
10.10	Supplying and fixing cable route marker with 10 cm X 10 cm X 5 mm thick G.I. plate with inscription there on, bolted /welded to 35 mm X 35 mm X 6 mm angle iron, 60 cm long and fixing the same in ground as required.	Each	598

CHAPTER 11

MV CABLE JOINTING & END TERMINATION

Item No.	Description	Unit	Rate(₹)
11.1	Supplying and making end termination with brass compression gland and aluminium lugs for following size of PVC insulated and PVC sheathed / XLPE aluminium conductor cable of 1.1 KV grade as required.		
11.1.1	2 X 6 sq. mm (19mm)	Each	294
11.1.2	2 X 10 sq. mm (19mm)	Each	294
11.1.3	2 X 16 sq. mm (22mm)	Each	298
11.1.4	2 X 25 sq. mm (22mm)	Each	314
11.1.5	2 X 35 sq. mm (25mm)	Each	366
11.1.6	2 X 50 sq. mm (28mm)	Each	383
11.1.7	3 X 10 sq. mm (22mm)	Each	358
11.1.8	3 X 16 sq. mm (25mm)	Each	315
11.1.9	3 X 25 sq. mm (25mm)	Each	330
11.1.10	3 X 35 sq. mm (28mm)	Each	383
11.1.11	3 X 50 sq. mm (32mm)	Each	442
11.1.12	3 X 70 sq. mm (35mm)	Each	495
11.1.13	3 X 95 sq. mm (38mm)	Each	636
11.1.14	3 X 120 sq. mm (45mm)	Each	694
11.1.15	3 X 150 sq. mm (50mm)	Each	781
11.1.16	3 X 185 sq. mm (57mm)	Each	984
11.1.17	3 X 225 sq. mm (62mm)	Each	1102
11.1.18	3 X 240 sq. mm (62mm)	Each	1136

11.1.19	3 X 300 sq. mm (70mm)	Each	1305
11.1.20	3½ X 25 sq. mm (28mm)	Each	371
11.1.21	3½ X 35 sq. mm (32mm)	Each	437
11.1.22	3½ X 50 sq. mm (35mm)	Each	484
11.1.23	3½ X 70 sq. mm (38mm)	Each	542
11.1.24	3½ X 95 sq. mm (45mm)	Each	684
11.1.25	3½ X 120 sq. mm (45mm)	Each	710
11.1.26	3½ X 150 sq. mm (50mm)	Each	798
11.1.27	3½ X 185 sq. mm (57mm)	Each	1001
11.1.28	3½ X 225 sq. mm (62mm)	Each	1126
11.1.29	3½ X 240 sq. mm (62mm)	Each	1160
11.1.30	3½ X 300 sq. mm (70mm)	Each	1336
11.1.31	3½ X 400 sq. mm (82mm)	Each	1713
11.1.32	4 X 10 sq. mm (25mm)	Each	325
11.1.33	4 X 16 sq. mm (28mm)	Each	366
11.1.34	4 X 25 sq. mm (28mm)	Each	373
11.1.34	4 X 35 sq. mm (32mm)	Each	438
11.1.35	4 X 50 sq. mm (35mm)	Each	489
11.2	Supplying and making outdoor end termination with cast resin compound including aluminium lugs and other jointing materials for following size of PVC insulated and PVC sheathed / XLPE aluminium conductor cable of 1.1 KV grade as required.		
11.2.1	2 X 16 sq. mm	Each	1477
11.2.2	2 X 25 sq. mm	Each	1582
11.2.3	2 X 35 sq. mm	Each	1582

11.2.4	2 X 50 sq. mm	Each	1582
11.2.5	3 X 16 sq. mm	Each	1582
11.2.6	3 X 25 sq. mm	Each	1582
11.2.7	3 X 35 sq. mm	Each	1582
11.2.8	3 X 50 sq. mm	Each	1679
11.2.9	3 X 70 sq. mm	Each	1679
11.2.10	3 X 95 sq. mm	Each	2023
11.2.11	3 X 120 sq. mm	Each	2023
11.2.12	3 X 150 sq. mm	Each	2023
11.2.13	3 X 185 sq. mm	Each	2643
11.2.14	3 X 225 sq. mm	Each	2643
11.2.15	3 X 240 sq. mm	Each	3028
11.2.16	3 X 300 sq. mm	Each	3028
11.2.17	3½ X 25 sq. mm	Each	1582
11.2.18	3½ X 35 sq. mm	Each	1679
11.2.19	3½ X 50 sq. mm	Each	1679
11.2.20	3½ X 70 sq. mm	Each	1827
11.2.21	3½ X 95 sq. mm	Each	2023
11.2.22	3½ X 120 sq. mm	Each	2023
11.2.23	3½ X 150 sq. mm	Each	2643
11.2.24	3½ X 185 sq. mm	Each	2117
11.2.25	3½ X 225 sq. mm	Each	3028
11.2.26	3½ X 240 sq. mm	Each	3028

11.2.27	3½ X 300 sq. mm	Each	3444
11.2.28	3½ X 400 sq. mm	Each	3897
11.2.29	4 X 16 sq. mm	Each	1582
11.2.30	4 X 25 sq. mm	Each	1582
11.2.31	4 X 35 sq. mm	Each	1679
11.2.32	4 X 50 sq. mm	Each	1679
11.3	Supplying and making straight through joint with cast resin compound including ferrules and other jointing materials for following size of PVC insulated and PVC sheathed / XLPE aluminium conductor cable of 1.1 KV grade as required.		
11.3.1	2 X 16 sq. mm	Each	2902
11.3.2	2 X 25 sq. mm	Each	2902
11.3.3	2 X 35 sq. mm	Each	2902
11.3.4	2 X 50 sq. mm	Each	3285
11.3.5	3 X 16 sq. mm	Each	3285
11.3.6	3 X 25 sq. mm	Each	2902
11.3.7	3 X 35 sq. mm	Each	2902
11.3.8	3 X 50 sq. mm	Each	3285
11.3.9	3 X 70 sq. mm	Each	3810
11.3.10	3 X 95 sq. mm	Each	4093
11.3.11	3 X 120 sq. mm	Each	4565
11.3.12	3 X 150 sq. mm	Each	4565
11.3.13	3 X 185 sq. mm	Each	5386
11.3.14	3 X 225 sq. mm	Each	6067
11.3.15	3 X 240 sq. mm	Each	6067

11.3.16	3 X 300 sq. mm	Each	7845
11.3.17	3½ X 25 sq. mm	Each	3285
11.3.18	3½ X 35 sq. mm	Each	3285
11.3.19	3½ X 50 sq. mm	Each	3810
11.3.20	3½ X 70 sq. mm	Each	3841
11.3.21	3½ X 95 sq. mm	Each	4565
11.3.22	3½ X 120 sq. mm	Each	5386
11.3.23	3½ X 150 sq. mm	Each	5386
11.3.24	3½ X 185 sq. mm	Each	6067
11.3.25	3½ X 225 sq. mm	Each	6172
11.3.26	3½ X 240 sq. mm	Each	7429
11.3.27	3½ X 300 sq. mm	Each	9156
11.3.28	3½ X 400 sq. mm	Each	5482
11.3.29	4 X 16 sq. mm	Each	2902
11.3.30	4 X 25 sq. mm	Each	3285
11.3.31	4 X 35 sq. mm	Each	3285
11.3.32	4 X 50 sq. mm	Each	3810
11.4	Supplying and making straight through joint with heat shrinkable kit including ferrules and other jointing materials for following size of PVC insulated and PVC sheathed / XLPE aluminium conductor cable of 1.1 KV grade as required.		
11.4.1	2 X 16 sq. mm	Each	2374
11.4.2	2 X 25 sq. mm	Each	2698
11.4.3	2 X 35 sq. mm	Each	2698
11.4.4	2 X 50 sq. mm	Each	2698

11.4.5	3 X 16 sq. mm	Each	2374
11.4.6	3 X 25 sq. mm	Each	2698
11.4.7	3 X 35 sq. mm	Each	2698
11.4.8	3 X 50 sq. mm	Each	2698
11.4.9	3 X 70 sq. mm	Each	3623
11.4.10	3 X 95 sq. mm	Each	3819
11.4.11	3 X 120 sq. mm	Each	4886
11.4.12	3 X 150 sq. mm	Each	4886
11.4.13	3 X 185 sq. mm	Each	4886
11.4.14	3 X 225 sq. mm	Each	5283
11.4.15	3 X 240 sq. mm	Each	5283
11.4.16	3 X 300 sq. mm	Each	6599
11.4.17	3½ X 25 sq. mm	Each	2698
11.4.18	3½ X 35 sq. mm	Each	2698
11.4.19	3½ X 50 sq. mm	Each	2698
11.4.20	3½ X 70 sq. mm	Each	3623
11.4.21	3½ X 95 sq. mm	Each	3819
11.4.22	3½ X 120 sq. mm	Each	4886
11.4.23	3½ X 150 sq. mm	Each	4886
11.4.24	3½ X 185 sq. mm	Each	4886
11.4.25	3½ X 225 sq. mm	Each	5283
11.4.26	3½ X 240 sq. mm	Each	5283
11.4.27	3½ X 300 sq. mm	Each	6599

11.4.28	3½ X 400 sq. mm	Each	8541
11.4.29	4 X 16 sq. mm	Each	4210
11.4.30	4 X 25 sq. mm	Each	2698
11.4.31	4 X 35 sq. mm	Each	3623
11.4.32	4 X 50 sq. mm	Each	3623

CHAPTER 12 POLE ERECTION

Item No.	Description	Unit	Rate(₹)
12.1	Erection of RCC/ PCC pole of following length in brick ballast and ramming the foundation, finishing with 150mm thick cement concrete (1:3:6) layer on top with including excavation and refilling etc as required.		
12.1.1	Above 4.5 metre and upto 6.5 metre	Each	6847
12.1.2	Above 6.5 metre and upto 8.0 metre	Each	7594
12.1.3	Above 8.0 metre and upto 11.0 metre	Each	9135
12.1.4	Above 11.00 metre and upto 13.00 metre	Each	10016
12.2	Erection of RCC/ PCC pole strut in brick ballast and ramming the foundation including excavation and refilling and secured with holding clamps, bolts, nuts, etc. as required.	Each	11859
12.3	Erection of metallic pole of following length in cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) foundation including excavation and refilling etc. as required.		
12.3.1	Above 4.5 metre and upto 6.5 metre	Each	6123
12.3.2	Above 6.5 metre and upto 8.0 metre	Each	7178
12.3.3	Above 8.0 metre and upto 10.0 metre	Each	8300
12.3.4	Above 10.00 metre and upto 12.00 metre	Each	9517
12.4	Erection of steel tubular or rail pole strut in cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) foundation including excavation and refilling and secured with holding clamps, bolts, nuts, etc. as required.	Each	10363
12.5	Providing and making steel pole collar with cement concrete (1 cement : 3 coarse sand : 6 stone aggregate 20mm) of specified size and shape including form work, plastering if required, curing etc as required. (volume of pole/ pipe not to be deducted)	cum	8834
12.6	Supplying and embedding following dia G.I. pipe (medium class) in pole collar/ foundation (during casting) for cable entry including bending the pipe to the required shape complete as		

required.

12.6.1	32 mm dia	Metre	559
12.6.2	40 mm dia	Metre	626

CHAPTER 13

MV OVER HEAD LINE WORK

Item No.	Description	Unit	Rate(₹)
13.1	Supplying and erection of stay set complete (galvanised) with 19/20 mm dia X 1.8 metres long stay rod, anchor plate of size 45 cm X 45 cm X 7.5 mm, thimble, stay clamps, turn buckle (20 mm X 60 cm), 7/ 4.00 mm dia G.I. stay wire and strain insulator etc in cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) foundation including excavation and refilling etc. as required.	Each	6098
13.2	Supplying and erection of stay set complete (galvanised) with 19/20 mm dia X 1.8 metres long stay rod, anchor plate of size 45 cm X 45 cm X 7.5 mm, thimble, stay clamps, turn buckle (20 mm X 60 cm), 7/ 3.15 mm dia G.I. stay wire and strain insulator etc in cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) foundation including excavation and refilling etc. as required.	Each	5961
13.3	Supplying and erection of stay set complete (galvanised) with 19/20 mm dia X 1.8 metres long stay rod, anchor plate of size 45 cm X 45 cm X 7.5 mm, thimble, stay clamps, bow tightener, 7/ 4.00 mm dia G.I. stay wire and strain insulator etc in cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) foundation including excavation and refilling etc. as required.	Each	5901
13.4	Supplying and erection of stay set complete (galvanised) with 19/20 mm dia X 1.8 metres long stay rod, anchor plate of size 45 cm X 45 cm X 7.5 mm, thimble, stay clamps, bow tightener, 7/ 3.15 mm dia G.I. stay wire and strain insulator etc in cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) foundation including excavation and refilling etc. as required.	Each	5759
13.5	Supplying and erection of bow stay set complete (galvanised) with 19/20 mm dia X 1.8 metres long stay rod, anchor plate of size 45 cm X 45 cm X 7.5 mm, thimble, stay clamps, turn buckle (20 mm X 60 cm), 7/ 4.00 mm dia G.I. stay wire and strain insulator etc in cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) foundation including excavation and refilling and also with 0.6 m long brace of size 50 mm X 50 mm X 6 mm angle iron with 50 mm dia pulley fixed at one end of the brace as required.	Each	6494

13.6	Supplying and erection of bow stay set complete (galvanised) with 19/20 mm dia X 1.8 metres long stay rod, anchor plate of size 45 cm X 45 cm X 7.5 mm, thimble, stay clamps, turn buckle (20 mm X 60 cm), 7/ 3.15 mm dia G.I. stay wire and strain insulator etc in cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) foundation including excavation and refilling and also with 0.6 m long brace of size 50 mm X 50 mm X 6 mm angle iron with 50 mm dia pulley fixed at one end of the brace as required.	Each	6260
13.7	Supplying and erection of bow stay set complete (galvanised) with 19/20 mm dia X 1.8 metres long stay rod, anchor plate of size 45 cm X 45 cm X 7.5 mm, thimble, stay clamps, bow tightener), 7/ 4.00 mm dia G.I. stay wire and strain insulator etc in cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) foundation including excavation and refilling and also with 0.6 m long brace of size 50 mm X 50 mm X 6 mm angle iron with 50 mm dia pulley fixed at one end of the brace as required.	Each	6297
13.8	Supplying and erection of bow stay set complete (galvanised) with 19/20 mm dia X 1.8 metres long stay rod, anchor plate of size 45 cm X 45 cm X 7.5 mm, thimble, stay clamps, bow tightener), 7/ 3.15mm dia G.I. stay wire and strain insulator etc in cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) foundation including excavation and refilling and also with 0.6 m long brace of size 50 mm X 50 mm X 6 mm angle iron with 50 mm dia pulley fixed at one end of the brace as required.	Each	6155
13.9	Erection of stay set complete (galvanised) in cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) foundation including excavation and refilling etc. as required.	Each	4775
13.10	Supplying of channel iron 75 mm X 40 mm X 6 mm (7.14 kg per metre) cross arm for 2 wire over head line complete with clamps, bolts, nuts and washer etc including drilling of holes for insulator pins etc (as per drawing) and painting with primer and finished paint as required .	Set	885
13.11	Supplying of channel iron 75 mm X 40 mm X 6 mm (7.14 kg per metre) cross arm for 4 wire over head line complete with clamps, bolts, nuts and washer etc including drilling of holes for insulator pins etc (as per drawing) and painting with primer and finished paint as required .	Set	1444

13.12	Erection of angle iron/ channel iron cross arm on wood/ RCC/ PCC/ steel tubular/ rail pole etc. as required.	Set	391
13.13	Supplying and erection of galvanised 'D' iron clamps complete with shackle insulator (75 mm X 90 mm), G. I. bolts, nuts and washers, coach screws etc. as required.	Set	316
13.14	Supplying and erection of galvanised 'D' iron clamps complete with shackle insulator (100 mm X 110 mm), G.I. bolts, nuts and washers, coach screws etc. as required.	Set	345
13.15	Erection of galvanised 'D' iron clamps and insulator on pole as required.	Set	98
13.16	Supplying and erection of 75 mm X 90 mm shackle insulator with G. I. Bolt, nuts and straps etc. as required .	Set	206
13.17	Supplying and erection of 100 mm X 110 mm shackle insulator with G. I. Bolt, nuts and straps etc. as required .	Set	236
13.18	Supplying and erection of 100 mm X 65 mm pin insulator complete with G. I. Spindle and nuts etc. as required .	Set	176
13.19	Supplying and erection of 100 mm X 80 mm pin insulator complete with G. I. Spindle and nuts etc. as required .	Set	176
13.20	Erection of LV/MV shackle/pin insulator etc. as required.	Set	107
13.21	Erection of ACSR conductor of 7/2.11 mm to 7/3.00 mm diameter including binding etc. as required.	Kg	258
13.22	Erection of ACSR conductor of 7/3.35 mm to 7/4.00 mm diameter and above including binding etc. as required.	Kg	340
13.23	Erection of all aluminium conductor of 7/1.96 mm to 7/3.10 mm diameter including binding etc. as required.	Kg	340
13.24	Erection of all aluminium conductor of 7/3.40 mm to 7/4.00 mm diameter and above including binding etc. as required.	Kg	260
13.25	Erection of G.I. Wire No. 8 SWG including binding etc. as required.	Kg	52
13.26	Erection of hexagonal type guard as required.	Each	279

13.27	Erection of ring type guard as required.	Each	93
13.28	Erection of cradle guard as required.	Each	186
13.29	Supplying and erection of 15 A aerial fuse complete as required.	Each	160
13.30	Supplying and erection of 30 A aerial fuse complete as required.	Each	170
13.31	Erection of aerial fuse as required.	Each	142
13.32	Supplying and fixing MV horn gap lightening arrestor as required.	Each	323
13.33	Fixing of MV lightning arrestor as required.	Each	213
13.34	Supplying and fixing of 32 mm dia X 2.00 metres long G.I. pipe (medium class) bracket for mounting of fluorescent / HPMV / HPSV street light fitting on pole including bending the pipe to the required shape, 2 nos 40 mm X 3 mm flat iron clamps with nuts, bolts and washer, painting the flat iron with primer and finish paint etc. as required.	Each	1647
13.35	Providing and fixing 50 mm dia X 3.2 metres G.I. pipe (including accessories) complete with 50 mm X 50 mm X 6 mm angle iron bracket on wall and 75 mm X 90 mm shackle insulator with straps for house service connection including painting the angle and flat iron with primer and finish paint etc. as required.	Each	3357
13.36	Providing and fixing 50 mm dia X 5.6 metres G.I. pipe (including accessories) complete with 50 mm X 50 mm X 6 mm angle iron cross arm and 40 mm X 3mm M.S. flat iron clamps bends for guard wire, 75 mm X 90 mm shackle insulator and straps, 7/ 3.15 mm G.I. wire stay set for house service connection including painting the angle and flat iron with primer and finish paint etc. as required.	Each	5225
13.37	Supplying and fixing of light class G.I. pipe of 50 mm dia. (nominal) 3 metres length along the pole for protection of under ground cable as required.	Each	2783
13.38	Supplying and fixing of light class G.I. pipe of 80 mm dia. (nominal) 3 metres length along the pole for protection of under ground cable as required.	Each	3871

13.39	Supplying and fixing of light class G.I. pipe of 100 mm dia. (nominal) 3 metres length along the pole for protection of under ground cable as required.	Each	5231
13.40	Dismantling of over head lines comprising of copper/ aluminium over head conductor, G.I. wire, cross arms, insulators etc. as required.	Kg	111
13.41	Dismantling of pole/ street light standard/ strut embedded in brick ballast foundation etc. as required.	Each	2543
13.42	Dismantling of pole/ street light standard/ strut embedded in cement concrete foundation etc. as required.	Each	3378

CHAPTER 14

HV OVER HEAD LINE WORK

Item No.	Description	Unit	Rate(₹)
14.1	Supplying and erection of galvanised stay set for 11 KV over head lines complete with 19/ 20 mm dia X 1.8 metres long stay rod, anchor plate of size 45 cm X 45 cm X 7.5 mm thick, thimble, stay clamps, turn buckle (20 mm X 600 mm), 7/ 4.00 mm dia G.I. stay wire and 11 KV strain insulator etc in cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) foundation including excavation and refilling etc. as required.	Set	6130
14.2	Supplying of channel iron 75 mm X 40 mm X 6 mm (7.14 kg per metre) pole top bracket/ cross arm for single 11 KV over head line conductor complete with 50 mm X 6mm flat iron clamp, bolts, nuts and washers including drilling holes for insulator pins, bolts and nuts etc and painting with primer and finish paint as required.	Each	693
14.3	Supplying of 50 mm X 8 mm M.S. flat iron pole top bracket/ cross arm for single 11/22/33 KV over head line conductor complete with fixing clamps, bolts, nuts and washers drilling holes for insulator pins, bolts and nuts etc and painting with primer and finish paint as required.	Each	756
14.4	Supplying of channel iron 75 mm X 40 mm X 6 mm (7.14 kg per metre) cross arm for two 11 KV over head line conductors complete with 50 mm X 50 mm X 6 mm (angle iron bracket welded to the channel iron and complete with 50 mm X 6mm M.S. flat iron clamps, bolts and nuts including drilling holes for insulator pins, bolts, nuts and washers etc (as per drawing) and painting with primer and finish paint as required.	Each	2604
14.5	Supplying of channel iron 75 mm X 40 mm X 6 mm (7.14 kg per metre) V shape cross arm for two 11 KV over head line conductors complete with 50 mm X 6mm M.S. flat iron clamps, bolts and nuts including drilling holes for insulator pins, bolts, nuts and washers etc (as per drawing) and painting with primer and finish paint as required.	Each	2310
14.6	Erection of pole top/ straight two wire/ V shape two wire, angle iron/ channel iron, cross arm on steel tubular/ rail/ PCC pole for 11/22/33 KV as required.	Set	710

14.7	Supplying of two lengths of channel iron 75 mm X 40 mm X 6 mm (7.14 kg per metre) double pole cross arm for three wire 11KV over head line conductors complete with through bolts and nuts for clamping to the poles, 50 mm X 6 mm M.S. flats welded on one side to the channel iron and with bolts and nuts on the other side for tying the cross arms together, including drilling holes for insulator pins/ fittings, bolts, nuts and washers etc (as per drawing) and painting with primer and finish paint as required.	Set	5789
14.8	Supplying and erection of a set of cross bracing FRL Same work for 11 KV over head line double pole structure having four members fabricated out of 50 mm X 50 mm X 6 mm angle iron to form a rectangle of minimum size 1400 mm width X 2500 mm height , complete with 50 mm X 6mm M.S. flat iron clamps, bolts and nuts including drilling holes for insulator pins, bolts and nuts etc (as per drawing) and painting with primer and finish paint as required.	Set	9084
14.9	Erection of double pole 3 wire cross arm for 11KV/ 22KV/ 33 KV over head lines as required.	Each	1629
14.10	Supplying and erection of 11 KV pin insulator complete with large steel head G.I. pin, nuts, washers etc. as required.	Set	370
14.11	Supplying and erection of 11 KV disc insulator for 11 KV over head lines with galvanised insulator fittings, ball and socket type and complete with galvanised strain clamps, bolts, nuts, washers etc. as required.	Set	1086
14.12	Erection of disc/ pin insulator for 11 KV over head lines as required.	Set	213
14.13	Supplying and erection of three piece nonlinear resistor type lightning arrestor suitable for 3 wire, 11 KV overhead lines with rated voltage 9 KV (rms) with a nominal discharge current rating of 5 KA and complete with galvanised clamping arrangement, G.I. bolts, nuts, washers etc. as required.	Set	2854
14.14	Supplying and erection of galvanised stay set for 33 KV over head lines complete with 19/ 20 mm dia X 1.8 metres long stay rod, anchor plate of size 45 cm X 45 cm X 7.5 mm thick, thimble, stay clamps, turn buckle (20 mm X 600 mm), 7/ 4.00 mm dia G.I. stay wire and 33 KV strain insulator etc in cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) foundation including excavation and refilling		

	etc. as required.	Set	6142
14.15	Supplying of channel iron 75 mm X 40 mm X 6 mm (7.14 kg per metre) pole top bracket/ cross arm for single 33 KV over head line conductor complete with 50 mm X 6mm flat iron clamps, bolts and nuts including drilling holes for insulator pins, bolts, nuts, washers etc and painting with primer and finish paint as required.	Each	721
14.16	Supplying of channel iron 75 mm X 40 mm X 6 mm (7.14 kg per metre) cross arm for two 33 KV over head line conductors complete with 50 mm X 50 mm X 6 mm angle iron bracket welded to the channel iron and complete with 50 mm X 6mm M.S. flat iron clamps, bolts and nuts including drilling holes for insulator pins, bolts, nuts, washers etc and painting with primer and finish paint as required.	Each	3195
14.17	Supplying of channel iron 75 mm X 40 mm X 6 mm (7.14 kg per metre) V shape cross arm for two 33 KV over head line conductors complete with 50 mm X 6mm M.S. flat iron clamps, bolts and nuts including drilling holes for insulator pins, bolts and nuts etc and painting with primer and finish paint as required.	Each	2565
14.18	Supplying of channel iron 100 mm X 50 mm X 6 mm (9.56 kg per metre), pole top bracket/ cross arm for single 33 KV over head line conductor complete with 50 mm X 6 mm M.S. flat iron clamps, bolts and nuts including drilling holes for insulator pins, bolts and nuts etc and painting with primer and finish paint as required.	Each	926
14.19	Supplying of channel iron 100 mm X 50 mm X 6 mm (9.56 kg per metre) cross arm for two wire 33 KV over head line conductors complete with 50 mm X 50 mm X 6 mm angle iron bracket welded to the channel iron and complete with 50 mm X 6mm M.S. flat iron clamps, bolts and nuts including drilling holes for insulator pins/ fittings, bolts, nuts, washers etc and painting with primer and finish paint as required .	Each	3919
14.20	Supplying of channel iron 100 mm X 50 mm X 6 mm (9.56 kg per metre) V shape cross arm for two 33 KV over head line conductors complete with 50 mm X 6mm M.S. flat iron clamps, bolts and nuts including drilling holes for insulator pins, bolts, nuts, washers etc and painting with primer and finish paint as required.	Each	3043

14.21	Supplying of two lengths of channel iron 100 mm X 50 mm X 6 mm (9.56 kg per metre) double pole cross arm for three wire 33 KV over head line conductors complete with through bolts and nuts for clamping to the poles, 50 mm X 6 mm M.S. flats welded on one side to the channel iron and with bolts and nuts on the other side for tying the cross arms together, including drilling holes for insulator pins/ fittings, bolts, nuts, washers etc and painting with primer and finish paint as required .	Set	8164
14.22	Supplying and erection of a set of cross bracing FRLSame work for 33 KV over head line double pole structure having four members fabricated out of 65 mm X 65 mm X 6 mm angle iron to form a rectangle of minimum size 2400 mm width X 2800 mm height complete with 50 mm X 6mm M.S. flat iron clamps, bolts and nuts including drilling holes for insulator pins, bolts, nuts, washers etc and painting with primer and finish paint as required .	Set	12444
14.23	Supplying and erection of 33 KV pin insulator complete with large steel head G.I. pin, nuts, washers etc. as required.	Set	928
14.24	Supplying and erection of a set of three 11 KV disc insulator for 33 KV over head lines with galvanised insulator fittings, ball and socket type and complete with galvanised strain clamps, bolts, nuts, washers etc. as required.	Set	3122
14.25	Supplying and erection of single piece nonlinear resistor type lightning arrestor suitable for 3 wire, 33 KV overhead lines with rated voltage 30 KV (rms) with a nominal discharge current rating of 10 KA (station class) and complete with galvanised clamping arrangement, G.I. bolts, nuts, washers etc. as required.	Set	18176

CHAPTER 15 MISC.CIVIL ITEMS

Item No.	Description	Unit	Rate(₹)
15.1	Excavation for foundation in soft soil including dressing of sides and ramming of bottoms, lift upto 1.5 m including getting out the excavated soil and disposal of surplus excavated soil as directed, within a lead of 50 metres.	cum	Rate to be taken from Civil DSR
15.2	Excavation for cable trenches of depth upto 1.2 m in soft soil including getting out the excavated soil and disposal of surplus excavated soil as directed within a lead of 50 metres.	cum	Rate to be taken from Civil DSR
15.3	Filling available excavated earth (excluding rock) in trenches, sides of foundations etc. in layers not exceeding 20cm in depth, consolidating each deposited layer by ramming and watering, lead up to 50 metres.	cum	323
15.4	Excavation for cable trenches in soft soil, depth upto 1.2 m including dressing of sides lift upto 1.5 m, including getting out the excavated soil, refilling with sand and or good soil after laying of cable/ pipe etc in layers of 20 cm, ramming, watering and disposal of surplus excavated soil as directed, within a lead of 50 metres.	cum	830
15.5	Deduct for not consolidation by ramming and watering while re-filling with sand and or good soil in cable/ pipe laying.	cum	184
15.6	Excavation of the cable trenches in hard rock not exceeding 1.5 metres in width, and lift upto 1.5 metres, including getting out the excavated soil and disposal of excavated soil as directed within a lead of 50 metres.	cum	1543
15.7	Providing and laying in position cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size) in foundation of pump, DG set etc including form work etc as required.	cum	Rate to be taken from Civil DSR
15.8	Providing and laying in position cement concrete 1:3:6 (1 cement : 2 coarse sand : 6 graded stone aggregate 20 mm nominal size) in foundation of pump, DG set etc including form work etc as required.	cum	Rate to be taken from Civil DSR
15.9	Providing and laying in position cement concrete 1:3:6 (1 cement : 2 coarse sand : 6 graded stone aggregate 40 mm nominal size) in foundation of pump, DG set etc including form work etc as required.	cum	Rate to be taken from Civil DSR

15.10	Providing and laying in position reinforced cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size) in foundation of pump, DG set etc including form work etc as required (excluding reinforcement).	cum	Rate to be taken from Civil DSR
15.11	Providing brick work (in width 225 mm or more) with F.P.S. bricks of class designation 7.5 in cement mortar 1:4 (1 cement : 4 coarse sand) at all levels.	cum	Rate to be taken from Civil DSR
15.12	Providing 15mm thick cement plaster of mix 1:4 (1 cement : 4 fine sand) at all levels.	Sq.m	Rate to be taken from Civil DSR
15.13	Providing, laying and fixing following dia G.I. pipe (medium class) in ground complete with G.I. fittings including trenching (75 cm deep) and re-filling etc as required		
15.13.1	50 mm dia	Metre	Rate to be taken from Civil DSR
15.13.2	80 mm dia	Metre	Rate to be taken from Civil DSR
15.13.3	100 mm dia	Metre	1918
15.13.4	150 mm dia	Metre	2894
15.14	Providing, laying and fixing following dia RCC pipe NP2 class (light duty) in ground complete with RCC collars, jointing with cement mortar 1:2 (1 cement : 2 fine sand) including trenching (75 cm deep) and refilling etc as required.		
15.14.1	100 mm dia	Metre	771
15.14.2	150 mm dia	Metre	846
15.14.3	250 mm dia	Metre	1110
15.14.4	300 mm dia	Metre	1293
15.15	Supplying and laying of following size DWC HDPE pipe ISI marked along with all accessories like socket, bend, couplers etc. conforming to IS 14930, Part II complete with fitting and cutting, jointing etc. in the existing trench, complete as required.		
15.15.1	63 mm dia (OD-63 mm & ID-51 mm nominal)	Metre	136
15.15.2	90 mm dia (OD-90 mm & ID-76 mm nominal)	Metre	181
15.15.3	120 mm dia (OD-120 mm & ID-103 mm nominal)	Metre	264

15.15.4	160 mm dia (OD-160 mm & ID-135 mm nominal)	Metre	385
15.15.5	200 mm dia (OD-200 mm & ID-175 mm nominal)	Metre	580
15.16	Supplying and laying of following size DWC HDPE pipe ISI marked along with all accessories like socket, bend, couplers etc. conforming to IS 14930, Part II complete with fitting and cutting, jointing etc.direct in ground (75 cm below ground level) including excavation and refilling the trench but excluding sand cushioning and protective covering etc., complete as required.		
15.16.1	63 mm dia (OD-63 mm & ID-51 mm nominal)	Metre	289
15.16.2	90 mm dia (OD-90 mm & ID-76 mm nominal)	Metre	334
15.16.3	120 mm dia (OD-120 mm & ID-103 mm nominal)	Metre	417
15.16.4	160 mm dia (OD-160 mm & ID-135 mm nominal)	Metre	537
15.16.5	200 mm dia (OD-200 mm & ID-175 mm nominal)	Metre	733

ADDITIONAL CIVIL ITEMS INTRODUCED

Note:-	There are few items which are contained in Civil DSR which are required to be used in exclusive electrical works. Field units should use these items for such purposes.		
i)	Providing and fixing M.S. Fan clamps of 16 mm dia M.S. bar, bent to shape with Hooked ends in R.C.C. slabs or beams during laying,including painting the exposed portion of loop, all as per standard design complete (DSR Civil-2014 item no. 10.17)	Rate to be taken from Civil DSR	
ii)	Providing and fixing circular/Hexagonal G.I. Box for ceiling fan clamps, of internal dia 140 mm, 73 mm height, top LID of 1.5 mm thick G.I. sheet with its top surface hacked for proper bonding, top LID shall be screwed into the G.I. box by means of 3.3 mm dia rounded screws, one lock at the corners. Clamps shall be made of 12 mm dia M.S. bar bent to shap as per standard drawing. (DSR Civil-2014 item no. 10.18)	Rate to be taken from Civil DSR	
iii)	Making hole for installation of exhaust fan of size upto 450mm sweep by demolishing brick work munually/by machnical means including disposal of malba within 50 meters lead and making good the wall by plastering etc. as required. (DSR Civil-2014 item no. 15.7.4 and 13.1.2)	Rate to be taken from Civil DSR	

iv) Painting with one or more coat of synthetic enamel paint of approved brand and manufacture of required colour to give an even shade. (DSR Civil-2014 item no. 14.54)	Rate to be taken from Civil DSR
v) Painting with one or more coat of aluminium paint of approved brand and manufacture to give an even shade. (DSR Civil-2014 item no. 14.55)	Rate to be taken from Civil DSR
vi) Painting with one or more coat of black anticorrosive bituminous paint of approved brand & manufacturer to give an even shade (DSR civil 2014- Item no. 14.56)	Rate to be taken from Civil DSR
vii) Re-lettering with black Japan paint of approved brand and manufacture. (DSR Civil-2014 item no. 14.60)	Rate to be taken from Civil DSR

CHAPTER- 16

SOLAR WATER HEATING SYSTEM

Item No.	Description	Unit	Rate(₹)
	SOLAR WATER HEATING SYSTEM (Evacuated Tube Collector)		
16.1	Supplying, installation, Testing, Commissioning of following capacity Evacuated Tube Collector (ETC) Solar Water Heating System comprising of all glass ETC tube absorber. The inner layer of absorber shall be of solar selected absorbing coated tube, Vacuum jacket, cover glass tube, getter and getter mirror surface, as per IS 16543. The system shall have temperature gauges, strainer, 2 nos. water meters, Suitable capacity cold and hot water tank, all MS structure for installation including suitable electric control panel complete with control and power wiring, necessary plumbing including piping for cold and hot water line between tank and solar water system, water heater and thermostat including non-return valve, float valve and other valve etc. as required. The various component shall have following specification. 1. The Absorber area i.e. the number, dimension and thickness of solar evacuated tube as per IS: 16544 clause 5.4 and IS: 16543 clause 4.2 2. Boro Silicate Glass 3.3 for cover plat as per ISO: 3585 3. The material for three target coating shall be aluminum nitrate, aluminum nitrate stainless steel and copper multi layer selecting coating as per IS: 16543. 4. Manifold shall be of Mild steel section with PP coating and Inner material shall be of SS 304. 5. Recommended operating pressure:10 Bars. 6. The capacity of hot water tank shall be minimum 1.5 times the rated capacity system. Inner Material shall be Stainless Steel SS 316 b) as per IS 1730 grade SS304-2B (22SWG). The hot water tank shall be insulated with high density injected PUF insulation 50 mm thickness between inner and outer tank. Tank stand shall be of mild steel and shall be design to withstand wind velocity of 100km/hours (minimum) or more as per site. 7. Suitable nos. ISI Marked electrical heaters along control panel, MCCB, with all protections, and all safety provisions so as to achive 60 0C temperature rise in an hour. The range of thermostat shall be upto 80 0C.		
16.1.1	100 LPD	Each	16477
16.1.2	200 LPD	Each	32378

16.1.3	300 LPD	Each	48351
16.1.4	500 LPD	Each	65835

SOLAR WATER HEATER (FLAT PLATE TYPE COLLECTOR)

- 16.2 Supply, Installation, Testing & Commissioning of Flat Plate collector (FPC) Solar Water Heating System comprising of solar flat plate collector ISI Marked made of copper sheet/copper tube, absorber toughened glass cover and aluminum extruded channel confirm to IS: 12933 (Part 1,2,3&5). The system shall have temperature gauges, strainer, water meter 2 nos., cold and hot water tank. The system shall have suitable electric backup complete with control and power wiring etc., as following.
1. Cover plate: cover plate shall be toughened glass and thickness of 4.0 mm (min) conforming to section -1 of IS: 12933(pt-2)/2003 the solar transmittances of the cover plate shall be minimum 82 percent at near normal incidence.
 2. Collector box: collector box shall be made of aluminum sections. The type grade, size, and finish of the material used shall be as per section-2 of IS: 12933 (pt-2)/2003: the minimum thickness of aluminum shall be as under:
 1. Channel section for sides 1.6 mm
 2. Sheet for bottom 0.7 mm
 3. Support for glass retaining 1.2 mm
 4. Sheet for entire body 1.0 mm

The insulation of collector box shall be minimum 0.96 m² OC/W for back insulation and minimum 0.48 m square degree c/w for side insulation conform to sec. 4 of IS 12933 (pt – 2) / 2003. (b) Gaskets and grommets: gaskets and grommets shall conform to Sec. 5 of IS 12933 (pt-2)/ 2003.
 3. Absorber Shall Consist of riser, Header and Sheet for absorber. The Diameter of header shall be 25.4 + /-0.5mm and thickness 0.71mm. The Diameter of riser shall be 12.7 + /-0.5 mm and thickness 0.56mm and made of copper only. The distance between the risers from center to center shall be 120mm. type grade, size , workmanship and finish of the material used shall be as per section- 3 of IS : 12933 (pt -2) /2003 the sheet for absorber shall be of copper sheet 34 gauge/copper tube (at least 10 nos.)
 4. Riser and header assembly designed for working pressure up 245 k pa (2.5 kg/ cm square) shall be tested for leakage at a minimum hydraulic pressure of 490 k pa (5 kg/ cm square). Sheet for absorber shall be made of copper only. Type Grade, size, workmanship and finis of the material used shall be per Section -3 of IS: 12933 (pt -2) /2003.
 5. HDPE/LDPE cold water tank and hot water tank shall be

dully elected on MS angle /channel duly painted with dual coats of enamel paint. The overall structure of solar collector plate module shall be design to with stand wind velocity of 100 kms /hr (minimum) or more as per site.

6.Hot water tank : The tank capacity shall be minimum 1.25 time the rated capacity of system. Inner tank material shall be stainless steel SS 316, as per IS 1730 GRADE SS304-2B. Hot water shall be insulated with high density injected PUF insulation: 50 mm, of 50 mm thickness between inner and outer tank ensures maximum heat rotenone ever season.

7. Suitable nos. ISI Marked electrical heaters along control panel, MCCB, with all protections, and all safety provisions so as to achive 60⁰C temperature rise in an hour. The range of thermostat shall be upto 80⁰C.

16.2.1	100 LPD	Each	20879
16.2.2	200 LPD	Each	41759
16.2.3	250 LPD	Each	48641
16.2.4	300 LPD	Each	62638
16.2.5	500 LPD	Each	104397

CHAPTER- 17

SOLAR PV POWER GENERATION

Item No.	Description	Unit	Rate(₹)
	Solar Photovoltaic Power Plant		
17.1	Supply, Installation, Testing and Commissioning of on-grid Solar Photovoltaic Power Plant conforming to various applicable standards BIS, IEC, MNRE guidelines, the Central Electricity Authority Regulations and CPWD Specifications as amended up to date, consisting of Mono/Poly Crystalline silicon solar cells module, net metering facility, necessary control, protections, earthing, cabling, mounting structure, junction boxes, power conditioning units, Real time online web interfaced Data Monitoring System, Distribution panels, grid connecting arrangement, conduits, pipes, cable trays and other accessories etc. as required. a) High Energy Efficiency Solar Photovoltaic Module of capacity 330 Wp or above, manufactured in India, conforming to IS 14286/IEC 61215, IS/IEC 61730-Part-1, IS/IEC 61730-Part-2. Solar Photovoltaic Module conversion efficiency shall not be less than 23% at STC with temperature coefficient of Pmax better than -0.30% per degree Celsius. PV modules used in solar power plants/ systems must be warranted for their output peak watt capacity, which should not be less than 90% at the end of 10 years and 80% at the end of 25 years. Solar Modules shall be designed to operate in relative humidity upto 100% with temperature between -10 Deg C and +85 Deg C. Further, each PV module used in any solar power project must have Radio frequency identification tag with information such as name of manufacturer, month and year of manufacturing, country of origin (separately for Solar cell and module), I-V curve, Unique Serial No and Model No of the module, Wattage, Im, Vm and FF, name of test lab issuing IEC certificate. b)Power Conditioning Unit (PCU) of 350-800 V DC Input voltage range and 415 V AC, three phase, 4 wire, 50Hz +/- 2.5 Hz, output voltage suitable to generate AC with a variation of 10% at nominal voltage. Power with efficiency not less than 97%, total harmonic distortion less than 3% and suitable for ambient temperature from 0 to 50 degree C , Minimum IP-65 for outdoor and Minimum IP 21 for indoor, Built-in meter and data logger, MPPT, switching devices IGBT/MOSFETs and controller Microprocessor /DSP . PCU/inverter shall be capable of complete automatic operation including wake-up, synchronization & shutdown. The PCU shall be able to withstand unbalanced load conforming to IEC standard with shutdown/standby mode.It must be provided with grid islanding		

along with manual disconnect pole isolation switch besides automatic disconnection. Minimum protections: Mains Under / Over Voltage, Over current, Over/Under grid frequency, Over temperature, Surge voltage induced at output due to external source, Shortcircuit, Lightening, Anti Islanding (for grid synch. Mode) and other protections as per applicable standards. LCD/LED display of minimum parameters: DC input voltage, DC current, AC Voltage and current (all 3 phases, in case of 3 phase), Instantaneous & cumulative AC output power, Daily DC energy produced and other parameters applicable standard. Communication interface RS 485 / RS 232.

c) Module mounting structure: The roof top solar plant generation units shall be installed by using supporting Aluminium/Galvanized MS structure (mass of zinc coating shall be as per IS4759) having minimum head room clearance of 2.4 meter above the terrace level / ground level. The mounting structure would be designed to sustain wind load and seismic parameter of the site of installation. All the structure shall be design as per applicable BIS code and the material shall also confirm the applicable BIS Code. Structural material shall be corrosion resistant and electrolytically compatible with the materials used in the module frame, its fasteners, nuts and bolts. The suitable arrangements for maintenance and cleaning shall be provided.

d) Real time online web interfaced Data Monitoring System complete with accessories for various parameters such as Solar Irradiance, temperature, AC Output Voltage and current, Output Power, Power factor, DC Input Voltage and Current, Time Active, Time disabled, Time Idle, Power produced and other parameters as per standard practices.

e) Array junction box & Main junction box with IP 65 protection and termination arrangement for incoming and outgoing cable along with glands, lugs and other accessories etc. as required. Each junction box shall be made of GRP/FRP/Powder Coated Aluminum /cast aluminum alloy with full dust, water & vermin proof arrangement with High quality Suitable capacity Metal Oxide Varistors(MOVs) (semiconductor diode with resistant applied voltage)/ surge arrestors and suitable Reverse Blocking Diodes, isolation switches isolate the DC input to Inverter, copper bus bar etc.

f) Lightning ,surge voltage protection, earthing protection and grid islanding.

g) Cables: Connections & Interconnections by required size IR/UV protected XLPE insulated copper conductor 1.1 kV grade armored power and control cables (ISI Marked) along with supplying & fixing of necessary channel/conduit, GI cable trays, supports, lugs, thimble and other accessories etc. as required.

h) DC Distribution Board And AC Distribution Panel Board: IP65, free standing, metal clad, having copper bus bar, having required protection and control gears, connection-interconnection, etc. as required.

kWp 54280

CHAPTER 18

ELECTRICAL VEHICLE CHARGER

Item No.	Description	Unit	Rate(₹)
18.1	Supply, Installation, Testing and commissioning of EV charging station As per specifications and in Compliance to relevant IS codes etc.		
18.1.1	Light EV AC Charger (Mode-3) Power : 7 kW, Input power supply: 1phase 230 $\pm$10% Volt, output supply: 230 Volt AC, Frequency:50 Hz $\pm$3%, Operational temprature range : -25 to 55 degree C (outdoor), -5 to 55 degree C(Indoor)., RH upto 95%, Charging Device as per IS-17017-22-1 EV-EVSE Communication: as per relevent IS Codes, Bluetooth Low Energy, one Charge Point Plug/ Socket as per IS-60309 and IS-17017-2, Vehicle Inlet/ Connector As per EV manufacturer, suitable for 2 Wheelers and 4 wheelers.Indoor use: at least IP41; Outdoor use: at least IP44. Mechanical Strength :protection of the external enclosure against mechanical impact shall be IK08 according to IEC 62262.O/L,S/C protection. Insulation Resistance > 1 M Ω. Cable Length: 7.5 m. RCD having a rated residual operating current not exceeding 30 mA; Seprate RCD for multiple outputs. Telecommunication port of the EV supply equipment according to IS 13252 (Part 1) : 2010. OCPP(Open charge point protocol) 1.6J upgradble to ocpp 2.0. Device Should follow 17017 series of IS codes in general and the installation of the system shall comply with relevent IS Codes.	Each	23385
18.1.2	Light EV DC Charger (Mode 4) Power Level 1: Up to 7 kW, Input power supply: 1phase 230 $\pm$10% Volt/3phase 415 Volt, Frequency: 50 Hz $\pm$5%, output supply: 12/24 Volt DC. Operational temprature range : -25 to 55 degree C (outdoor), -5 to 55 degree C (Indoor), RH upto 95%, Charging Device as per IS-17017-25, EV-EVSE Communication:IS-17017-25, one Charge Point Plug/ Socket as per IS-60309 and IS-17017-2, Vehicle Inlet/ Connector As per EV manufacturer, suitable for 2 Wheelers and 4 wheelers. Indoor use: at least IP41; Outdoor use: at least IP44.Mechanical Strength :protection of the external enclosure against mechanical impact shall be IK08 according to IEC 62262.O/L,S/C protection. Insulation Resistance > 1 M Ω. Cable Length: 7.5 m. RCD having a rated residual operating current not exceeding 30 mA; Seprate RCD for multiple outputs.Telecommunication port of the EV supply equipment according to IS 13252 (Part 1) : 2010. OCPP(Open charge		

point protocol) 1.6J upgradable to ocpp 2.0. Device Should follow 17017 series of IS codes in general and the installation of the system shall comply with relevant IS Codes.

Each 350914

18.1.3 Parkbay AC Charger (Mode -3)

Power Level 2: Normal Power ~11kW/ 22 kW, 3 phase 415VAC(-40% to +20%), Frequency:50 Hz +/-5%, output supply: 240 Volt AC, Operational temperature range : -25 to 55 degree C (outdoor), -5 to 55 degree C(Indoor), RH upto 95%, Charging Device as per IS-17017-1 EV-EVSE ISO-15118 for Smart Charging, Infrastructure Socket as per IS-17017-2-2, Vehicle Connector as per IS-17017-2-2 Vehicle Inlet/ Connector As per EV manufacturer, suitable for 4 wheelers. Indoor use: at least IP41; Outdoor use: at least IP44. Mechanical Strength :protection of the external enclosure against mechanical impact shall be IK08 according to IEC 62262. O/L,S/C protection. Insulation Resistance > 1 M Ω . Cable Length: 7.5 m. RCD having a rated residual operating current not exceeding 30 mA; Separate RCD for multiple outputs. Telecommunication port of the EV supply equipment according to IS 13252 (Part 1) : 2010. OCPP(Open charge point protocol) 1.6J upgradable to ocpp 2.0. Device Should follow 17017 series of IS codes in general and the installation of the system shall comply with relevant IS Codes.

Each 84219

18.1.4 Parkbay DC Charger (Mode-3)

Power Level 2: Normal Power ~24KW and above, 3 phase 415VAC(-40% to +20%), Frequency:50 Hz +/-5%, output supply: DC 12/24 Volt, Operational temperature range : -25 to 55 degree C (outdoor), -5 to 55 degree C(Indoor), RH upto 95%, Charging Device as per Device/protocol: IS-17017-23, EV-EVSE Communication as per IS-17017-24 ,ISO-15118, Infrastructure Socket as per IS-17017-2-2/3, Vehicle Connector as per IS-17017-2-3 Vehicle Inlet/ Connector As per EV manufacturer, suitable for 4 wheelers. Indoor use: at least IP41; Outdoor use: at least IP44. Mechanical Strength :protection of the external enclosure against mechanical impact shall be IK08 according to IEC 62262. O/L,S/C protection. Insulation Resistance > 1 M Ω . Cable Length: 7.5 m. RCD having a rated residual operating current not exceeding 30 mA; Separate RCD for multiple outputs. Telecommunication port of the EV supply equipment according to IS 13252 (Part 1) : 2010. OCPP(Open charge point protocol) 1.6J upgradable to ocpp 2.0. Device Should follow 17017 series of IS codes in general and the installation of the system shall comply with relevant IS Codes.

Each 888982

CHAPTER-19 FAÇADE LIGHTING

Item No.	Description	Unit	Rate(₹)
19.1	LED floodlight Supply, Installation, Testing and Commissioning of Circular/ Rectangular/square LED RGB/RGBWW/RGBNW/RGBA/RGBW (as per site requirement) Facade Floodlight/ Projector luminaire for facade lighting with Pressure Die-cast aluminium housing, powder-coated finish, with inbuilt driver of following wattage as given below. The luminaire bracket shall be made of cold-rolled steel & shall be so designed that the luminaire can be tilted to $\pm 90^\circ$, the scale shall be provided for rotation degrees. The luminaire shall be IP66 protection also having silicon gasket between optical cover & LED chamber to ensure proper IP66 protection and impact resistance of IK 07. CRI>80, SDCM<5, THD<10%. as per IEC 61000-3-2, under voltage, over voltage protection, driver inbuilt surge protection of 4KV and external surge protection (SPD) of 10 KV common mode/differential mode, Driver Efficiency>85%, EMI/EMC as per CISPR-15. There should be options of beam angles of 5/10/15/20/25/30/40/45/60 degree. Luminaire should have efficacy of minimum 35 lm/W for RGB/RGBNW/RGBWW/TNW and 65 lm/W for mono colour options. Luminaire shall be capable of producing dynamic color changing light for with 16 million colors by DMX/Ethernet based control. The beam angle of the fitting shall be narrow, medium, wide or asymmetric as per the requirement. Luminaire shall be complete with replaceable driver and capable of operating at line voltage without any separate power supply from 100-270 V AC, 50 Hz with power factor > 0.9. Fixture shall be suitable to operate at an ambient temperature range of -10°C to $+50^\circ\text{C}$. Lifetime of complete luminaire(LED, Driver, Electrical & control circuitary) shall be at least L70:50000 burning hours at 50°C. Luminaire shall be conforming to BIS: (IS 10322) with duly R Number marked and shall be complete with all necessary accessories required for proper working of fixture including connection/inter connection with weather proof ISI Marked copper conductor of minimum size 1.5sqmm connection cables and accessories, Integral male/female water proof connectors IP 68 etc. as required. The successful bidder shall submit the following certifications/reports for approval before supply of materials: product data sheets, LM79 report from NABL accredited lab, illuminance diagram, polar curve & LM80 report issued by LED manufacturer, RoHS		

Compliance.

19.1.1	25W	Each	36027
19.1.2	40W	Each	42765
19.1.3	50W	Each	43888
19.1.4	60W	Each	45759
19.1.5	100W	Each	48627
19.1.6	120W	Each	80196
19.1.7	150W	Each	86372
19.1.8	200W	Each	94933

19.2 LED Linear Profile (AC)

Supply, Installation, Testing and Commissioning of LED RGB/RGBWW/RGBNW/ RGBA/ RGBW (as per site requirement) Linear Profile / Wall Washer luminaire for Facade lighting with aluminium extruded/ Die Cast housing with inbuilt driver of following wattage as given below. The luminaire bracket shall be made of cold-rolled steel & shall be so designed that the luminaire can be tilted to $\pm 90^\circ$, having the rotation degrees scale shall be provided. The luminaire shall be IP66 protection also having silicon gasket between optical cover & LED chamber to ensure proper IP66 protection. CRI>80, SDCM<5, THD<10%. as per IEC 61000-3-2, under voltage , over voltage protection, driver inbuilt surge protection of 4KV and external surge protection (SPD) of 10 KV common mode/differential mode , Driver Efficiency>85%, EMI/EMC as per CISPR-15. There should be options of beam angles of 10/20/30/40/60, asymmetric degree. Luminaire should have efficacy of minimum 40 lm/W for RGB/RGBNW/RGBWW/TNW and 65 lm/W for mono colour options. Luminaire shall be capable of producing dynamic color changing light for with 16 million colors by DMX/Ethernet based control. The beam angle of the fitting shall be narrow, medium, wide or asymmetric as per the requirement. Luminaire shall be complete with replaceable driver and capable of operating at line voltage without any separate power supply from 100-270 V AC, 50 Hz with power factor > 0.9 . Fixture shall be suitable to operate at an ambient temperature range of - - 10°C to $+50^\circ\text{C}$ and shall have impact resistance of IK 07. Lifetime of complete luminaire (LED ,Driver , Electrical & control circuitary) shall be at least L70:50000 burning hours at 50°C .

Luminaire shall be conforming to BIS (IS 10322) with dully R Number marked and shall be complete with all necessary accessories required for proper working of fixture including connection/inter connection with weather proof ISI Marked copper conductor of minimum size 1.5sqmm connection cables and accesories , Integral male/female water proof connectors of IP68 etc. as required.The successful bidder shall submit the following certifications/reports for approval before supply of materials: product data sheets, LM79 report from NABL accredited lab, illuminance diagram, polar curve & LM80 report issued by LED manufacturer,RoHS Compliance.

Wattage per meter

19.2.1	120W	Each	65878
19.2.2	100W	Each	56408
19.2.3	80W	Each	42783
19.2.4	60W	Each	42307
19.2.5	50W	Each	26613
19.2.6	40W	Each	25631
19.2.7	25W	Each	19715

19.3 LED Linear Profile (DC)

Supply, Installation, Testing and Commissioning of LED RGB/RGBWW/RG BNW/ RGBA/ RGBW (as per site requirement) Linear Profile / Wall Washer luminaire for Facade lighting with aluminium extruded/ Die Cast housing with inbuilt driver of following wattage as given below.The luminaire bracket shall be made of cold-rolled steel & shall be so designed that the luminaire can be tilted to $\pm 90^\circ$, having the rotation degrees scale shall be provided. The luminaire shall be IP66 protection also having silicon gasket between optical cover & LED chamber to ensure proper IP66 protection. CRI>80, SDCM<5, THD<10%. as per IEC 61000-3-2, under voltage , over volatge protection, driver inbuilt surge protection of 4KV and external surge prootection (SPD) of 10 KV common mode/differential mode , Driver Efficiency>85%, EMI/EMC as per CISPR-15. There should be options of beam angles of 10/20/30/40/60, asymmetric degree.Luminaire should have efficacy of minimum 40 lm/W for RGB/RGBNW/RGBWW/TNW and 65 lm/W for mono colour options. Luminaire shall be capable of producing dynamic color changing light for with 16 million colors by

DMX/Ethernet based control. The beam angle of the fitting shall be narrow, medium, wide or asymmetric as per the requirement. Luminaire shall be complete with replaceable driver and capable of operating at line voltage without any separate power supply from 24 V DC. Fixture shall be suitable to operate at an ambient temperature range of -10°C to +50°C and shall have impact resistance of IK 07. Lifetime of complete luminaire (LED ,Driver , Electrical & control circuitary) shall be at least L70:50000 burning hours at 50°C. Luminaire shall be conforming to BIS (IS 10322) with dully R Number marked and shall be complete with all necessary accessories required for proper working of fixture including connection/inter connection with weather proof ISI Marked copper conductor of minimum size 1.5sqmm connection cables and accesories , Integral male/female water proof connectors of IP68 etc. as required. The successful bidder shall submit the following certifications/reports for approval before supply of materials: product data sheets, LM79 report from NABL accredited lab, illuminance diagram, polar curve & LM80 report issued by LED manufacturer, RoHS Compliance.

Wattage per meter

19.3.1	100W	Each	24162
19.3.2	50W	Each	16362
19.3.3	40W	Each	16507
19.3.4	30W	Each	12012
19.3.5	15W	Each	3426

19.4 RGB LED Strip

Supply, installation, testing and commissioning of LED RGB/RGBWW/ RGBNW/ RGBA/RGBW suitable for exterior use having IP68 rating (as per site requirement). Flexible light strip of 5 mtr standard length with 12-15 Watt per meter. Luminaire shall be capable of producing dynamic color changing light for with 16 million colors by DMX/Ethernet based control. Minimum System lumens 450 lumens/meter and System Lumen Efficacy > 30lm/w. Luminaire light beam spread - 100-110°, The Strip should have IP 68 protection and is offered IP 68 rated connectors to join the LED Strips incontinuity. The luminaire must be offered with Aluminum channel for mounting the strip lights. The product should be complete with all accessories and Driver. Lifetime of complete luminaire (LED ,Driver , Electrical & control circuitary) shall be at least L70:50000 burning hours at 50°C. Luminaire shall be conforming to BIS (IS 10322) with dully

R Number marked. Luminaire shall be complete with replaceable driver and capable of operating at line voltage without any separate power supply from 24 V DC, Fixture shall be suitable to operate at an ambient temperature range of - 10°C to +50°C. The successful bidder shall submit the following certifications/reports for approval before supply of materials: product data sheets, LM79 report from NABL accredited lab, illuminance diagram, polar curve & LM80 report issued by LED manufacturer & RoHS Compliance along with CE certificate.

Each 9689

19.5 Controller

19.5.1 Supply, installation, testing and commissioning of DMX Controller for the system including address writer and programming the DMX controller with the suitable designs as decided by the Engineer-in-charge/client department (not less than 10 nos designs) and all other accessories required for the running of the system Controller should have Input Power 5-5.5V DC 0.6A and Output Protocol DMX512 with 4 universe, 2 universe should work in Stand Alone Mode and 2 universe should work in live mode with laptop/ tablet / computer. The universe should be expandable from 2 to 4 DMX512 universes. It should be compatible for Programmability with PC, Mac, Tablet, Smartphone by using USB & Ethernet . Dmx controller should have network controllability feature to control the scene remotely through LAN and triggered via UDP packets on port 2430. It's must have Connections Type C USB, XLR3, Ethernet, battery. There should be provision for Memory microSD card. Device should have Intuitive backlit keyboard with 10 presents and also have provision of keypad on device for changing scene manually from device. Device should be capable to work on Extended triggering possibilities (TCA) & also have Smart Upgrade Technology (SUT). It is an IP 20 unit and has an Operating Temperature of 0°C to 50°C.OS Requirements for the controller is Mac OS X 10.8-10.12, Windows 7/8/10/11. Standards should be EC, EMC, ROHS. The controller should have a variety of features including 2048 DMX channels, clock/calendar, iPhone/iPad /Android remote control, Ethernet facilities, dry contact port triggering and multi-zone SD card memory. Network synchronization of one area across hundreds of universes with the possibility of 99 scenes across 5 areas with auto clock triggering for sunrise and sunset and as per the festivals for specific occasions. Device should be equipped with meta clamp to mount on wall with data enablers and splitters as required. The Facade Lighting products and DMX controller should be of same / single make. Note: DMX Controller Should be in-house. To be provided with IP 66 enclosure if installed in

	external environment.	Each	65891
19.6	Splitter		
19.6.1	Supply, installation testing and commissioning of DMX Splitter with Input Voltage range : 190-240V AC, Power Consumption 5W max. Housing is made of CRCA with powder coated finish Mounting, 19 inch rack mountable, Operation Temperature - 30°C to +55°C, 8 Output with 3pin XLR terminal sockets, Ingress Protection Rating IP20. This unit takes the DMX Signal from DMX Controller & split the signal into 8 separate output channels. The output terminals of Splitters are optically isolated from the input. Splitters should be capable to amplify the DMX signal & allowing the data link to be extended by max. 300 m (1044 ft.) using AWG 22 DMX cable or 150 m (500 ft.) using AWG 24 DMX cable or maximum 20-25 fixture per output channel. To be provided with IP enclosure if installed in external environment etc., complete as required.	Each	6224
19.6.2	Supply, Installation, Testing and Commissioning of RDM-6Wall Splitter, should be able to provide 6- output DMX/RDM splitter with DMX/RDM pass- through supporting bi-directional communications for discovery, addressing and DMX control of Ethernet/DMX products.This unit takes the incoming DMX/RDM signal and splits the signal into six separate output channels allowing for expanding the number of devices controlled from 32 to 6x32. The output terminals are electronically isolated from the input & RDM-through port and share a common ground reference. All six output ports have an independent output driver to boost the DMX/RDM signal. Surface or DIN-rail mountable,Isolated DMX/RDM Input/Outputs,Automatic switching between DMX and RDM mode for auto-discovery, Environment-IP20, Interior applications, Finish/Color-Polycarbonate, Certification: CE Certified. DMX Splitter/Booster etc. as required.	Each	15142
19.7	Signal Amplifier		
19.7.1	Supply, installation testing and commissioning of Signal Amplifier to boost the Signal, IP 65 rated module to enhance the depleting DMX Signals, should have IP 66/67 rates inbuilt connectors compatible with Luminaires. It should be offered with inbuilt electronic circuit for functioning. No external LED driver needed etc., complete as required.	Each	2519
19.8	DMX decoder		

19.8.1	Supplying, installation, testing and commissioning of high quality DMX decoder, Power Input : DC 24V etc complete as required.	Each	5399
19.9	Connector		
19.9.1	Supplying and Fixing of IP65 "I" power connectors having 3/4 pole cable connector, chuck type strain relief, Current rating 20 A rms continuous, Easy and extremely precise locking system "Quick Lock" High impact materials - long-lasting and reliable etc complete as required.	Each	649
19.9.2	Supplying and Fixing of IP65 "I" DMX connectors having 3/5 pole cable connector, Male/Female connector with improved latch, cage type contacts, "solder stop" for easy soldering, and additional ground contact for best contact integrity between chassis and cable connector. More durable housing for maximum reliability, etc complete as required.	Each	649
19.9.3	Supplying and Fixing of 5 pin XLR Male/Female connectors having 5 pole cable connector, Male/Female connector with improved latch, cage type contacts, "solder stop" for easy soldering, and additional ground contact for best contact integrity between chassis and cable connector. More durable housing for maximum reliability etc as required.	Each	1958
19.10	Processor		
19.10.1	SITC of push button type keypad with 10 buttons. Each button is programmable up to 16 scenerios. Also has got two dry-contact input to trigger any preset scene, etc as required	Each	10473

CHAPTER- 20 FOUNTAIN

Item No.	Description	Unit	Rate(₹)
20.1	Supply of Installation Testing Commissioning of Nozzle made with gunmetal Grade Geyser jet 100mm with adjustable nozzle for height adjustment of the fountain etc., Complete as required.	Each	10473
20.2	Supply of Installation Testing Commissioning of Fixing of 40 mm dia Bubbler Jet Nozzles in SS 304 Gr to produce water effect complete as required.	Each	6481
20.3	Supply of Installation Testing and Commissioning of fixing of 40 mm dia SS Difuser Nozzles in SS 304 Gr to produce water effect complete as required.	Each	2954
20.4	Supply of Installation Testing and Commissioning of Stainless steel 304 gr. Comet effect nozzle with compatible Digitally controlled pump to produce desirable effects with 40 mm dia and 20 mm orifice nozzle as per direction given by engineer in charge.	Each	10335
20.5	SITC of Single Jet Crown Nozzles Complete with ball joint for 15 Degree pivoting made of brass with Nickle plating, flow straightner in following size . Height of water stream upto 10 mtrs.		
20.5.1	15 mm internal threaded with 6 mm orifice.	Each	2322
20.5.2	15 mm internal threaded with 8 mm orifice.	Each	2538
20.5.3	15 mm internal threaded with 10 mm orifice.	Each	2764
20.5.4	25 mm internal threaded with 12 mm orifice.	Each	3078
20.5.5	25 mm internal threaded with 14 mm orifice.	Each	3726
20.5.6	40 mm internal threaded with 17 mm orifice.	Each	3078
20.5.7	40 mm internal threaded with 20 mm orifice.	Each	7829
20.6	SITC of Eco Cluster Nozzle made of stainless steel with inlet dia 40 mm and 21 nos Single jets to create 38 mm dia full jet for height 10 mtr water level independed.	Each	37251
20.7	SITC of Eco Cluster Nozzle made of stainless steel with inlet dia 80 mm and 64 nos Single jets to Create 67 mm dia full jet for fountain height 10 mtr water level independed.	Each	63380

20.8	SITC of Central hollow jet 65 mm dia for achieving maximum height of 20 mtrs to produce clear full jet stable in wind water level independent	Each	50208
20.9	SITC of Lava bell nozzle for producing transparent water pattern with adjustable ball dia made out of SS 304 Gr./Brass of following Size.		
20.9.1	15 mm Dia inlet 290 mm long	Each	4158
20.9.2	25 mm Dia inlet 330 mm long	Each	5237
20.9.3	25 mm Dia inlet 488 mm long	Each	6317
20.9.4	25 mm Dia inlet 636 mm long	Each	7289
20.10	SITC of Foam jet High contrast for producing high contrast/foam effect to extensively wind stable, water level independent moc Brass /SS 316L of following size.		
20.10.1	25 mm Dia inlet 32 mm orifice 160 mm long.	Each	17816
20.10.2	25 mm Dia inlet 50 mm orifice 180 mm long.	Each	29801
20.10.3	40 mm Dia inlet 50 mm orifice 180 mm long.	Each	41570
20.11	Supply of Installation Testing Commissioning of Providing of 20 mm nylon rope all the accessories Floater Grit Clamping both sides etc., Complete as required.	Mtrs.	69
20.12	Supply of Installation Testing Commissioning of fixing of Suction Screen strainer made of ABS to be fitted in suction to protect the pump from entry of external materials.	Each	1971
20.13	Supply Installation Testing and Commissioning of 100 mm Dia Debris Collection Strainer in Stainless steel 304 gr Complete as reqd. All fixing to be done as per direction instruction and entire satisfaction of Engineer-in-charge.	Each	7685
20.14	Supply Installation Testing and Commissioning of 400 mm Dia wide SS Overflow Skimmer in Stainless steel 304 gr with front openable fitted debris trapping mesh with outlet size of 65mm Dia for fixing in wall. Complete as reqd. All fixing to be done as per direction instruction and entire satisfaction of Engineer-in-charge.	Each	10420
20.15	Supply and Fixing of 300x300 mm Dia Main Drain in SS 304 Gr with perforations to disallow foreign material with top openable for maintenance.	Each	7745

20.16	Supply, installation, testing and commissioning of ISI Marked openwell submersible pump set contain with stainless steel shaft and cast iron/SS body, the pump shall be designed to prevent over loading and having dynamically balanced rotating parts etc. the pump capable of delivering following capacities discharge at suitable head and lowering the pumps in water bodies and suitable for operation on 415V(+/-10%), 3-phase AC supply including providing required no's of heavy duty M.S clamps made of flat iron and suitable size top cover made of suitable thick MS sheet including connction, interconnection ,supports etc. as required. suitable for 900LPM at 25 Mtrs Head	Each	49514
20.17	Supply, installation, testing and commissioning of ISI Marked vertical Multi Stage submersible pump set contain with stainless steel shaft and cast iron/SS body, the pump shall be designed to prevent over loading and having dynamically balanced rotating parts etc. the pump capable of delivering following capacities discharge at suitable head and lowering the pumps in water bodies and suitable for operation on 415V(+/-10%), 3- phase AC supply including providing required no's of heavy duty M.S clamps made of flat iron and suitable size top cover made of suitable thick MS sheet as required. suitable for 1200LPM at14 mtr. head.	Each	67966
20.18	Supply of Installation Testing and Commissioning of three phase openwell ISI Marked submersible pumpset of 1.5H.P. capacity suitable for operation on 415V(+/-10%), 3- phase AC supply compaitable for operations on dynamic variable frequency / speed having IP 68 Protection including providing required no's of heavy duty M.S clamps made of flat iron and suitable size top cover made of suitable thick MS sheet as required.and making of flexible IP68 cable connections etc. complete of suitable head and discharge as aproved by engineer incharge as reqd.	Each	28535
20.19	Supply of Installation Testing and Commissioning of three phase openwell ISI Marked submersible pumpset of 7.5 H.P. capacity suitable for operation on 415V(+/-10%), 3- phase AC supply compaitable for operations on dynamic variable frequency / speed having IP 68 Protection including providing required no's of heavy duty M.S clamps made of flat iron and suitable size top cover made of suitable thick MS sheet as required. and making of flexible IP68 cable connections etc complete as reqd.	Each	56470
20.20	Supply of Installation Testing and Commissioning of fixing of following dia. Gun metal gate valve (Screwed type) on the		

	existing pipeline, including testing & Commissioning etc. as required.		
20.20.1	80 mm dia	Each	17486
20.20.2	50 mm dia	Each	7881
20.21	Supply of Installation Testing and Commissioning of fixing of following dia. Gun metal Horizontal Non Return valve (screwed type) on the existing pipeline, including testing & Commissioning etc. As required.		
20.21.1	100 mm dia	Each	36832
20.21.2	50 mm dia	Each	7881
20.22	Supplying, Installation, Testing and Commissioning of underwater Luminaries surface mounted RGB LED spot light 6 x 3W with IP 68 protection, SS/ aluminium dia casting body with power coated paint, soft rubber gasket along with 2 years free replacement guaranty etc. as required.	Each	13659
20.23	Supplying, Installation, Testing and Commissioning of underwater Luminaries surface mounted RGB LED spot light 3 x 3W with IP 68 protection, SS/ aluminium dia casting body with power coated paint, soft rubber gasket along with 2 years free replacement guaranty etc. as required.	Each	7073
20.24	Supplying, Installation, Testing and Commissioning of water proof (IP 68) power supply unit of not less than 200W i/c connection along with 2 years free replacement guaranty etc. as required.	Each	7410
20.25	Supplying, Installation, Testing and Commissioning of RGB controller with remote Kit wattage not less than 200 watt i/c connection along with 2 years free replacement guaranty etc. as required.	Each	107061
20.26	Supplying, Installation, Testing and Commissioning of Signal Amplifier of RGB signal as require i/c connection along with 2 years free replacement guaranty etc. as required.	Each	16242
20.27	Supply of Installation Testing and Commissioning of fixing of Power Supply/SMPS for suitable to run LED Lights, Body-Metal input Power-220 V AC/ Output Power-12 V DC, complete as per the Direction & Entire Satisfaction of engineer in charge	Each	7210

20.28	Supplying & fixing PVC insulated PVC sheathed copper conductor submersible flat cable ISI Marked in existing metal/ HDPE / PVC/ pipe/ in bore well/ in sump including fixing the cable to GI/HDPE pipe with suitable cable tags of following sizes etc. as required.		
20.28.1	3 Core 1.5 Sq mm Flat Cable	Meter	76.00
20.28.2	3 Core 2.5 Sq mm Flat Cable	Meter	124.00
20.28.3	3 Core 4.0 Sq mm Flat Cable	Meter	186.00
20.28.4	3 Core 6 Sq mm Flat Cable	Meter	266.00
20.28.5	3 Core 10 Sq mm Flat Cable	Meter	461.00

CHAPTER -21 LIFT

Item No.	Description	Unit	Rate(₹)
	Passenger Lifts		
21.1	Supplying, Installation, Testing & Commissioning of following capacity passenger Machine room less (MRL) lift with regenerative drive as per CPWD General Specification for Electrical Works (Part-III Lift & Escalator) 2003, BIS Codes, NBC 2016 as amended upto date having provision for barrier free access as per Harmonised Guidelines & Standards for Universal Accessibility in India - 2021 of MoHUA, serving different floors in the lift shaft as per detailed specifications enclosed and as under including all other equipment and accessories complete as required :- (i) Speed-1 MPS (ii) Floors-5 (G+4) (iii) Travel -17 Meters (approx.) (iv) Stops & opening- 5 stops and 5 openings (v) Controller: A.C. variable voltage & variable frequency (vi) Automatic rescue device (ARD) complete with dry maintenance free batteries as required. (vii) Operation: Microprocessor based single automatic push button/simplex selective collective/duplex collective selective with/without attendant (viii) Power-415 Volts$\pm$10%, 3 phase, 50 Hz, 4 wires system (ix) Car Enclosure: Stainless steel scratch proof (Moon Rock/Honeycomb/ Hairline Finish) on all sides and stainless steel decorative ceiling with fans (indirect throw) and LED fitting. (x) The car flooring should be smooth and antiskid as approved by engineer-in-charge. (xi) Type of doors a) Car: Power operated, centre opening horizontal sliding stainless steel scratch proof (Moon Rock/Honeycomb/ Hairline Finish). (b) Landing doors: stainless steel scratch proof . (xii) A hand rail not less than 600mm long at 900mm above floor level to be fixed adjacent to control panel in the lift car. (xiii) Voice announcement system in the car to announce the position of the elevator in the hoist way as the car passes or stops at floor served by the elevator.		

(xiv) Closed loop Control System, protection against power fluctuation, self diagnostic control, Landing doors 2 hours Fire rated, Overload detection and protection, Overspeed Governor, Dot Matrix LED D & P Indicators In Car & Landings, BMS Compatibility, tamper proof infrared curtain covering almost the entire height of the lift car door, Alarm horn, Anti Nuisance Operation, CCTV camera system with 4 MP IP camera, suitable NVR and HDD for 30 days storage backup, Firemen switch, 2/3 way intercom, Emergency Stop Switch with battery backup for fan and light fitting for 1 hour, pit ladder.

(xv) The lift doors shall have a vision panel to enable persons with hearing impairment to signal for help or assistance in the event of an emergency.

(xvi) An appropriate technological support be provided (Through emergency messaging |services or alarms etc.) to respond to the emergency requirements of person with hearing impairment or deafness.

(xvii) In case of failure of ARD or other electronic devices, a provision of manual rescue be provided.

21.1.1	13 passenger (884kg)	Each	1684226
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21.1.2	16 passenger (1088kg)	Each	2348212
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Goods Lifts

21.2	Supplying, Installation, Testing & Commissioning of 2000kg Machine room less (MRL) goods lift with regenerative drive as per CPWD General Specification for Electrical Works (Part-III Lift & Escalator) 2003, BIS Codes, NBC 2016 as amended upto date, serving different floors in the lift shaft as per detailed specifications enclosed and as under including all other equipment and accessories complete as required :-
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(i) Speed - 0.5 MPS

(ii) Floors - 5(G+4)

(iii) Travel - 17 metres(approx)

(iv) Stops & opening : 5 stops and 5 openings

(v) Controller: A.C. variable voltage & variable frequency

(vi) Automatic rescue device (ARD) complete with dry maintenance free batteries as required.

(vii) Operation: Microprocessor based single automatic push button/simplex selective collective/duplex collective selective with/without attendant

(viii) Power-415 Volts $\pm$ 10%, 3 phase, 50 Hz, 4 wires system

(ix) Car Enclosure: Stainless steel scratch proof (Moon Rock/Honeycomb/ Hairline Finish) on all sides and stainless steel

decorative ceiling with fans (indirect throw) and LED fitting.

(x) The car flooring should be strong to take the rated load without any deformation or damage and antiskid as approved by engineer-in-charge.

(xi) Type of doors:-

(a) Car: Power operated, centre opening horizontal sliding stainless steel scratch proof (Moon Rock/Honeycomb/ Hairline Finish).

(b) Landing doors: stainless steel scratch proof .

(xii) A hand rail not less than 600mm long at 900mm above floor level to be fixed adjacent to control panel in the lift car.

(xiii) Voice announcement system in the car to announce the position of the elevator in the hoist way as the car passes or stops at floor served by the elevator.

(xiv) Closed loop Control System, protection against power fluctuation, self diagnostic control, Landing doors 2 hours Fire rated, Overload detection and protection, Overspeed Governor, Dot Matrix LED D & P Indicators In Car & Landings, BMS Compatibility, tamper proof infrared curtain covering almost the entire height of the lift car door, Alarm horn, Anti Nuisance Operation, CCTV camera system with 4 MP IP camera, suitable NVR and HDD for 30 days storage backup , Firemen switch, 2/3 way intercom, Emergency Stop Switch with battery backup for fan and light fitting for 1 hour, pit ladder.

(xv) In case of failure of ARD or other electronic devices, a provision of manual rescue be provided.

21.2.1	2000 kg	Each	3391945
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Hospital Lifts (20 passenger (1360 kg))

21.3 Supplying, Installation, Testing & Commissioning of following capacity passenger cum bed lift Machine room less (MRL) Bed lift with regenerative drive as per CPWD General Specification for Electrical Works (Part-III Lift & Escalator) 2003, BIS Codes, NBC 2016 as amended upto date having provision for barrier free access as per Harmonised Guidelines & Standards for Universal Accessibility in India - 2021 of MoHUA, serving different floors in the lift shaft as per detailed specifications enclosed and as under including all other equipment and accessories complete as required :-

- (i) Speed-1.0 MPS
- (ii) Floors-5(G+4)
- (iii) Travel-17 metres(approx)
- (iv) Stops & opening : 5 stops and 5 openings
- (v) Controller: A.C. variable voltage & variable frequency

- (vi) Automatic rescue device (ARD) complete with dry maintenance free batteries as required.
- (vii) Operation: Microprocessor based single automatic push button/simplex selective collective/duplex collective selective with/without attendant
- (viii) Power-415 Volts $\pm$ 10%, 3 phase, 50 Hz, 4 wires system
- (ix) Car Enclosure: Stainless steel scratch proof (Moon Rock/Honeycomb/ Hairline Finish) on all sides and stainless steel decorative ceiling with fans (indirect throw) and LED fitting.
- (x) The car flooring should be smooth and antiskid as approved by engineer-in-charge.
- (xi) Type of doors:-
 - (a) Car: Power operated, centre opening horizontal sliding stainless steel scratch proof (Moon Rock/Honeycomb/ Hairline Finish).
 - (b) Landing doors: stainless steel scratch proof .
 - (xii) A hand rail not less than 600mm long at 900mm above floor level to be fixed adjacent to control panel in the lift car.
 - (xii) Voice announcement system in the car to announce the position of the elevator in the hoist way as the car passes or stops at floor served by the elevator.
 - (xiv) Closed loop Control System, protection against power fluctuation, self diagnostic control, Landing doors 2 hours Fire rated, Overload detection and protection, Overspeed Governor, Dot Matrix LED D & P Indicators In Car & Landings, BMS Compatibility, tamper proof infrared curtain covering almost the entire height of the lift car door, Alarm horn, Anti Nuisance Operation, CCTV camera system with 4 MP IP camera, suitable NVR and HDD for 30 days storage backup , Firemen switch, 2/3 way intercom, Emergency Stop Switch with battery backup for fan and light fitting for 1 hour, pit ladder.
 - (xv) The lift doors shall have a vision panel to enable persons with hearing impairment to signal for help or assistance in the event of an emergency.
 - (xvi) An appropriate technological support be provided (Through emergency messaging |services or alarms etc.) to respond to the emergency requirements of person with hearing impairment or deafness.
 - (xvii) In case of failure of ARD or other electronic devices, a provision of manual rescue be provided.

21.3.1

20 passenger (1360kg)

Each 2581451

CHAPTER 22 HT CABLE LAYING

Item No.	Description	Unit	Rate(₹)
22.1	Laying of one number PVC insulated and PVC sheathed / XLPE power cable of 11 KV grade of following size direct in ground including excavation, sand cushioning, protective covering and refilling the trench etc as required.		
22.1.1	Upto 120 sq. mm	Metre	628
22.1.2	Above 120 sq. mm and upto 400 sq. mm	Metre	694
22.2	Laying of one number additional PVC insulated and PVC sheathed / XLPE power cable of 11 KV grade of following size direct in ground in the same trench in one tier horizontal formation including excavation, sand cushioning, protective covering and refilling the trench etc as required.		
22.2.1	Upto 120 sq. mm	Metre	432
22.2.2	Above 120 sq. mm and upto 400 sq. mm	Metre	498
22.3	Laying of one number PVC insulated and PVC sheathed / XLPE power cable of 11 KV grade of following size in the existing RCC/ HUME/ METAL pipe as required.		
22.3.1	Upto 120 sq. mm	Metre	98
22.3.2	Above 120 sq. mm and upto 400 sq. mm	Metre	170
22.4	Laying of one number PVC insulated and PVC sheathed / XLPE power cable of 11 KV grade of following size in the existing masonry open duct as required.		
22.4.1	Upto 120 sq. mm	Metre	80
22.4.2	Above 120 sq. mm and upto 400 sq. mm	Metre	146
22.5	Laying of one number XLPE power cable of 33 KV grade of following size direct in ground including excavation, sand cushioning, protective covering and refilling the trench etc as required.		
22.5.1	Upto 120 sq. mm	Metre	635
22.5.2	Above 120 sq. mm and upto 400 sq. mm	Metre	701

22.6	Laying of one number additional XLPE power cable of 33 KV grade of following size direct in ground in the same trench in one tier horizontal formation including excavation, sand cushioning, protective covering and refilling the trench etc as required.		
22.6.1	Upto 120 sq. mm	Metre	439
22.6.2	Above 120 sq. mm and upto 400 sq. mm	Metre	505
22.7	Laying of one number XLPE power cable of 33 KV grade of following size in the existing RCC/ HUME/ METAL pipe as required.		
22.7.1	Upto 120 sq. mm	Metre	104
22.7.2	Above 120 sq. mm and upto 400 sq. mm	Metre	177
22.8	Laying of one number XLPE power cable of 33 KV grade of following size in the existing masonry open duct as required.		
22.8.1	Upto 120 sq. mm	Metre	86
22.8.2	Above 120 sq. mm and upto 400 sq. mm	Metre	153

CHAPTER 23

HV CABLE JOINTING & END TERMINATION

Item No.	Description	Unit	Rate(₹)
23.1	Supplying and making indoor cable end jointing with cast resin compound, including lugs and other jointing materials, for following size of 3 core, XLPE aluminium conductor cable of 11 KV grade as required :		
23.1.1	70 sq. mm	Each	2385
23.1.2	120 sq. mm	Each	3120
23.1.3	240 sq. mm	Each	4144
23.1.4	300 sq. mm	Each	4206
23.2	Supplying and making outdoor cable end jointing with cast resin compound, including lugs and other jointing materials, for following size of 3 core, XLPE aluminium conductor cable of 11 KV grade as required :		
23.2.1	70 sq. mm	Each	5201
23.2.2	120 sq. mm	Each	5988
23.2.3	240 sq. mm	Each	7010
23.2.4	300 sq. mm	Each	7010
23.3	Supplying and making straight through cable jointing with cast resin compound, including ferrule and other jointing materials, for following size of 3 core, XLPE aluminium conductor cable of 11 KV grade as required :		
23.3.1	70 sq. mm	Each	5077
23.3.2	120 sq. mm	Each	5715
23.3.3	240 sq. mm	Each	8392
23.3.4	300 sq. mm	Each	9491
23.4	supplying and making indoor cable end termination with heat shrinkable jointing kit complete with all accessories including lugs suitable for following size of 3 core, XLPE aluminium conductor cable of 11 KV grade as required :		

23.3	70 sq. mm	Each	12946
23.3.1	120 sq. mm	Each	15989
23.3.2	240 sq. mm	Each	17414
23.3.3	300 sq. mm	Each	17414
23.5	Supplying and making outdoor cable end termination with heat shrinkable jointing kit complete with all accessories including lugs suitable for following size of 3 core, XLPE aluminium conductor cable of 11 KV grade as required :		
23.5.1	70 sq. mm	Each	19286
23.5.2	120 sq. mm	Each	21351
23.5.3	240 sq. mm	Each	24255
23.5.4	300 sq. mm	Each	24255
23.6	Supplying and making straight through cable jointing with heat shrinkable jointing kit complete with all accessories including ferrules suitable for following size of 3 core, XLPE aluminium conductor cable of 11 KV grade as required :		
23.6.1	70 sq. mm	Each	30799
23.6.2	120 sq. mm	Each	40144
23.6.3	240 sq. mm	Each	44211
23.6.4	300 sq. mm	Each	44211
23.7	Supplying and making indoor cable end termination with heat shrinkable jointing kit complete with all accessories including lugs suitable for following size of 3 core, XLPE aluminium conductor cable of 33 KV grade as required :		
23.7.1	70 sq. mm	Each	20950
23.7.2	120 sq. mm	Each	31787
23.7.3	240 sq. mm	Each	36190
23.8	Supplying and making outdoor cable end termination with heat shrinkable jointing kit complete with all accessories including lugs suitable for following size of 3 core, XLPE aluminium conductor cable of 33 KV grade as required :		

23.8.1	70 sq. mm	Each	33200
23.8.2	120 sq. mm	Each	43405
23.8.3	240 sq. mm	Each	75301
23.9	Supplying and making straight through cable jointing with heat shrinkable jointing kit complete with all accessories including ferrules suitable for following size of 3 core, XLPE aluminium conductor cable of 33 KV grade as required :		
23.9.1	70 sq. mm	Each	57525
23.9.2	120 sq. mm	Each	73107
23.9.3	240 sq. mm	Each	103493

CHAPTER 24 TRANSFORMER

Item No.	Description Oil Type	Unit	Rate(₹)
	33/0.433 KV, 3 Phase, 50 Hz outdoor/ indoor mounting (Mineral oil filled)		
24.1	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 33/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better, suitable for out door/indoor applications with On Load Tap Changer (OLTC) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x400 sqmm XLPE cable of 33 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS Code including first filling of filtered dehydrated oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 35 dg.C for oil and 40 dg. C up to 200 KVA and 40 dg.C for oil and 45 dg. C for above 200 KVA for winding. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred to for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b)		

Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.1.1

500 KVA

Each 1328495

24.1.2	630KVA	Each	1673904
24.1.3	1000KVA	Each	2008307
24.1.4	1250 KVA	Each	2510384
24.1.5	1600KVA	Each	3213291
24.1.6	2000KVA	Each	4016614
24.1.7	2500KVA	Each	5020768
24.2	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 33/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better, suitable for out door/indoor applications with On Load Tap Changer (OLTC) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x400 sqmm XLPE cable of 33 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS Code including first filling of filtered dehydrated oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 35 dg.C for oil and 40 dg. C up to 200 KVA and 40 dg.C for oil and 45 dg. C for above 200 KVA for winding. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall		

be referred to for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth);

o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level

indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.2.1	500 KVA	Each	1594194
24.2.2	630KVA	Each	2008685
24.2.3	1000KVA	Each	2409969
24.2.4	1250 KVA	Each	3012461
24.2.5	1600KVA	Each	3855950
24.2.6	2000KVA	Each	4819937
24.2.7	2500KVA	Each	6024922
24.3	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 33/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better, suitable for out door/indoor applications with On Load Tap Changer (OLTC) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x400 sqmm XLPE cable of 33 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS Code including first filling of filtered dehydrated oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed		

35 dg.C for oil and 40 dg. C up to 200 KVA and 40 dg.C for oil and 45 dg. C for above 200 KVA for winding. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred to for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level

alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.3.1	500KVA	Each	1859893
24.3.2	630KVA	Each	2343466
24.3.3	1000KVA	Each	2811630
24.3.4	1250 KVA	Each	3514538
24.3.5	1600KVA	Each	4498608
24.3.6	2000KVA	Each	5623260
24.3.7	2500 KVA	Each	7029075

11/0.433 KV, 3 Phase, 50 Hz outdoor/ indoor mounting
(Mineral oil filled)

- 24.4 Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better, suitable for out door/indoor applications with On Load Tap Changer (OLTC) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS Code including first filling of filtered dehydrated oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026

(Part 1 to Part 5), IS : 1180 and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 35 dg.C for oil and 40 dg. C up to 200 KVA and 40 dg.C for oil and 45 dg. C for above 200 KVA for winding. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred to for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); o) Buchholz relay for transformers above 800 kVA. p) Arcing

horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.4.1	1000KVA	Each	1749171
24.4.2	1250 KVA	Each	2186463
24.4.3	1600KVA	Each	2798673
24.4.4	2000KVA	Each	3498342
24.5	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better, suitable for out door/indoor applications with On Load Tap Changer (OLTC) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS Code including first filling of filtered dehydrated oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 and duly ISI Marked and as per CPWD specifications complete in all respects etc as required		

at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 35 dg.C for oil and 40 dg. C up to 200 KVA and 40 dg.C for oil and 45 dg. C for above 200 KVA for winding. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred to for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge.

Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc.

Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding

temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.5.1	1000KVA	Each	2099005
24.5.2	1250 KVA	Each	2623756
24.5.3	1600KVA	Each	3358408
24.5.4	2000KVA	Each	4198010

24.6 Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better, suitable for out door/indoor applications with On Load Tap Changer (OLTC) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS Code including first filling of filtered dehydrated oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux

density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 35 dg.C for oil and 40 dg. C up to 200 KVA and 40 dg.C for oil and 45 dg. C for above 200 KVA for winding. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred to for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking

pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.6.1	1000KVA	Each	2448839
24.6.2	1250 KVA	Each	3061049
24.6.3	1600KVA	Each	3918143
24.6.4	2000KVA	Each	4897678
24.7	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better, suitable for out door/indoor applications with Offload Tap Changer/ On Load Tap Changer (OLTC) (below 500 KVA offload and above 500 KVA On Load Tap Changer) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS Code including first filling of filtered dehydrated oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux		

density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 35 dg.C for oil and 40 dg. C up to 200 KVA and 40 dg.C for oil and 45 dg. C for above 200 KVA for winding. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred to for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth);

o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking

pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.7.1	200KVA	Each	336877
24.7.2	250KVA	Each	421097
24.7.3	315KVA	Each	530582
24.7.4	400KVA	Each	673755
24.7.5	500KVA	Each	842193
24.7.6	630KVA	Each	1061164

24.8 Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better, suitable for out door/indoor applications with Offload Tap Changer/ On Load Tap Changer (OLTC) (below 500 KVA offload and above 500 KVA On Load Tap Changer) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS Code including first filling of filtered dehydrated oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026

(Part 1 to Part 5), IS : 1180 and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 35 dg.C for oil and 40 dg. C up to 200 KVA and 40 dg.C for oil and 45 dg. C for above 200 KVA for winding. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred to for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); o) Buchholz relay for transformers above 800 kVA. p) Arcing

horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.8.1	200KVA	Each	377303
24.8.2	250KVA	Each	471628
24.8.3	315KVA	Each	594252
24.8.4	400KVA	Each	754605
24.8.5	500KVA	Each	943257
24.8.6	630KVA	Each	1188503
24.9	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better, suitable for out door/indoor applications with Offload Tap Changer/ On Load Tap Changer (OLTC) (below 500 KVA offload and above 500 KVA On Load Tap Changer) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete		

with all accessories and safety provisions as per relevant IS Code including first filling of filtered dehydrated oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 35 dg.C for oil and 40 dg. C up to 200 KVA and 40 dg.C for oil and 45 dg. C for above 200 KVA for winding. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred to for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for

transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth);

o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.9.1	200KVA	Each	497830
24.9.2	250KVA	Each	622287
24.9.3	315KVA	Each	784082
24.9.4	400KVA	Each	995660
24.9.5	500KVA	Each	1244575
24.8.6	630KVA	Each	1568164
24.10	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better, suitable for out door/indoor applications with Offload Tap Changer in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS Code including first filling of filtered dehydrated oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum		

flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 35 dg.C for oil and 40 dg. C up to 200 KVA and 40 dg.C for oil and 45 dg. C for above 200 KVA for winding. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred to for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding

temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.10.1	63KVA	Each	112012
24.10.2	100KVA	Each	177796
24.10.3	160KVA	Each	284474
24.11	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better, suitable for out door/indoor applications with Offload Tap Changer in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS Code including first filling of filtered dehydrated oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 35 dg.C for oil and 40 dg. C up to 200 KVA and 40 dg.C for oil and 45 dg. C for above 200 KVA for winding. Inside of tank shall be painted with varnish or liquid resistant paint. For external		

surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred to for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top

and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.11.1	63KVA	Nos	125453
24.11.2	100KVA	Nos	199132
24.11.3	160KVA	Nos	318611
24.12	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better, suitable for out door/indoor applications with Offload Tap Changer in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS Code including first filling of filtered dehydrated oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site . The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 35 dg.C for oil and 40 dg. C up to 200 KVA and 40 dg.C for oil and 45 dg. C for above 200 KVA for winding. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred to for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b)		

Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth);

o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.12.1 63KVA

Each 156816

24.12.2	100KVA	Each	248915
24.12.3	160KVA	Each	398264

33/0.433 KV, 3 Phase, 50 Hz outdoor/ indoor mounting
(Synthetic organic Ester oil filled)

- 24.13 Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 33/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better), Dielectric material shall be type -A, suitable for out door/indoor applications with On Load Tap Changer (OLTC) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x400 sqmm XLPE cable of 33 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS 1180 (Part-3) including first filling of filtered dehydrated Synthetic organic Ester oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 (Part-3) and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site . The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with +12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 40 dg.C for oil and 45 dg. C for winding up to 200 KVA and 45 dg.C for oil and 50 dg. C for winding for above 200 KVA. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air

temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth);

o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.13.1	500 KVA	Each	1476106
24.13.2	630KVA	Each	1859893
24.13.3	1000KVA	Each	2231452
24.13.4	1250 KVA	Each	2789316
24.13.5	1600KVA	Each	3570324
24.13.6	2000KVA	Each	4462905
24.13.7	2500KVA	Each	5578631
24.14	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 33/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better), Dielectric material shall be type -A, suitable for out door/indoor applications with On Load Tap Changer (OLTC) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x400 sqmm XLPE cable of 33 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS 1180 (Part-3) including first filling of filtered dehydrated Synthetic organic Ester oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 (Part-3) and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with +12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 40 dg.C for oil and 45 dg. C for winding up to 200 KVA and 45 dg.C for oil and 50 dg. C for winding for above 200 KVA. Inside of tank		

shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge.

Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc.

Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram

Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.14.1	500 KVA	Each	1771327
24.14.2	630KVA	Each	2231872
24.14.3	1000KVA	Each	2677743
24.14.4	1250 KVA	Each	3347179
24.14.5	1600KVA	Each	4284389
24.14.6	2000KVA	Each	5355486
24.14.7	2500KVA	Each	6694357
24.15	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 33/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better), Dielectric material shall be type -A, suitable for out door/indoor applications with On Load Tap Changer (OLTC) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x400 sqmm XLPE cable of 33 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS 1180 (Part-3) including first filling of filtered dehydrated Synthetic organic Ester oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 (Part-3) and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at		

rated voltage and frequency shall be such that the flux density with +12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 40 dg.C for oil and 45 dg. C for winding up to 200 KVA and 45 dg.C for oil and 50 dg. C for winding for above 200 KVA. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding

temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.15.1	500KVA	Each	2066548
24.15.2	630KVA	Each	2603851
24.15.3	1000KVA	Each	3124033
24.15.4	1250 KVA	Each	3905042
24.15.5	1600KVA	Each	4998453
24.15.6	2000KVA	Each	6248067
24.15.7	2500 KVA	Each	7810083

11/0.433 KV, 3 Phase, 50 Hz outdoor/ indoor mounting (Synthetic organic Ester oil filled)

24.16	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better),Dielectric material shall be type -A, suitable for out door/indoor applications with On Load Tap Changer (OLTC) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor		
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multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS 1180 (Part-3) including first filling of filtered dehydrated Synthetic organic Ester oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 (Part-3) and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with +12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 40 dg.C for oil and 45 dg. C for winding up to 200 KVA and 45 dg.C for oil and 50 dg. C for winding for above 200 KVA. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for

transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth);

o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos.

q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.16.1	1000KVA	Each	1943523
24.16.2	1250 KVA	Each	2429404
24.16.3	1600KVA	Each	3109637
24.16.4	2000KVA	Each	3887046
24.17	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better), Dielectric material shall be type -A, suitable for out door/indoor applications with On Load Tap Changer (OLTC) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor		

multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS 1180 (Part-3) including first filling of filtered dehydrated Synthetic organic Ester oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 (Part-3) and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with +12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 40 dg.C for oil and 45 dg. C for winding up to 200 KVA and 45 dg.C for oil and 50 dg. C for winding for above 200 KVA. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for

transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth);

o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos.

q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.17.1	1000KVA	Each	2332228
24.17.2	1250 KVA	Each	2915285
24.17.3	1600KVA	Each	3731564
24.17.4	2000KVA	Each	4664455
24.18	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better), Dielectric material shall be type -A, suitable for out door/indoor applications with On Load Tap Changer (OLTC) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor		

multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS 1180 (Part-3) including first filling of filtered dehydrated Synthetic organic Ester oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 (Part-3) and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with +12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 40 dg.C for oil and 45 dg. C for winding up to 200 KVA and 45 dg.C for oil and 50 dg. C for winding for above 200 KVA. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for

transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth);

o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos.

q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.18.1	1000KVA	Each	2720932
24.18.2	1250 KVA	Each	3401165
24.18.3	1600KVA	Each	4353492
24.18.4	2000KVA	Each	5441865
24.19	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better), Dielectric material shall be type -A, suitable for out door/indoor applications with Offload Tap Changer/ On Load Tap Changer (OLTC) (below 500 KVA offload and above 500 KVA On Load Tap Changer) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including		

bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS 1180 (Part-3) including first filling of filtered dehydrated Synthetic organic Ester oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 (Part-3) and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with +12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 40 dg.C for oil and 45 dg. C for winding up to 200 KVA and 45 dg.C for oil and 50 dg. C for winding for above 200 KVA. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding

assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth);

o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.19.1	200KVA	Each	374308
24.19.2	250KVA	Each	467885
24.19.3	315KVA	Each	589535
24.19.4	400KVA	Each	748616
24.19.5	500KVA	Each	935770
24.19.6	630KVA	Each	1179071
24.20	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better),Dielectric material shall be type -A, suitable for out door/indoor applications with Offload Tap Changer/ On Load Tap Changer (OLTC) (below 500 KVA offload and above 500 KVA On Load		

Tap Changer) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS 1180 (Part-3) including first filling of filtered dehydrated Synthetic organic Ester oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be conforming to IS : 2026 (Part 1 to Part 5), IS : 1180 (Part-3) and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with +12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 40 dg.C for oil and 45 dg. C for winding up to 200 KVA and 45 dg.C for oil and 50 dg. C for winding for above 200 KVA. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked. Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating

and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.20.1	200KVA	Each	419225
24.20.2	250KVA	Each	524031
24.20.3	315KVA	Each	660280
24.20.4	400KVA	Each	838450
24.20.5	500KVA	Each	1048063
24.20.6	630KVA	Each	1320559
24.21	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11		

vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better), Dielectric material shall be type -A, suitable for out door/indoor applications with Offload Tap Changer/ On Load Tap Changer (OLTC) (below 500 KVA offload and above 500 KVA On Load Tap Changer) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS 1180 (Part-3) including first filling of filtered dehydrated Synthetic organic Ester oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 (Part-3) and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with +12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 40 dg.C for oil and 45 dg. C for winding up to 200 KVA and 45 dg.C for oil and 50 dg. C for winding for above 200 KVA. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition: a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely

fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be

provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth);

o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.21.1	200KVA	Each	524031
24.21.2	250KVA	Each	655039
24.21.3	315KVA	Each	825349
24.21.4	400KVA	Each	1048063
24.21.5	500KVA	Each	1310079

24.21.6	630KVA	Each	1650699
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24.22 Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better), Dielectric material shall be type -A, suitable for out door/indoor applications with Offload Tap Changer in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS 1180 (Part-3) including first filling of filtered dehydrated Synthetic organic Ester oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 (Part-3) and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with +12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 40 dg.C for oil and 45 dg. C for winding up to 200 KVA and 45 dg.C for oil and 50 dg. C for winding for above 200 KVA. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate

made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth);

o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.22.1	63KVA	Nos	117907
24.22.2	100KVA	Nos	187154
24.22.3	160KVA	Nos	299447
24.23	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt		

of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better), Dielectric material shall be type -A, suitable for out door/indoor applications with Offload Tap Changer in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS 1180 (Part-3) including first filling of filtered dehydrated Synthetic organic Ester oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 (Part-3) and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with +12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 40 dg.C for oil and 45 dg. C for winding up to 200 KVA and 45 dg.C for oil and 50 dg. C for winding for above 200 KVA. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked. Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol

b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having ($1\frac{1}{4}$ " nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.23.1	63KVA	Each	132056
24.23.2	100KVA	Each	209613
24.23.3	160KVA	Each	335380
24.24	Supply, installation, testing and commissioning of following capacity (continuous loading) BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better),Dielectric		

material shall be type -A, suitable for out door/indoor applications with Offload Tap Changer in range of +5% to -15% in steps of 2.5%, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade, including bus trunking arrangement on LV side including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/panel for safety tripping, complete with all accessories and safety provisions as per relevant IS 1180 (Part-3) including first filling of filtered dehydrated Synthetic organic Ester oil, i/c supplying and grouting of suitable M.S. Channel with all accessories and transformer shall be confirming to IS : 2026 (Part 1 to Part 5), IS : 1180 (Part-3) and duly ISI Marked and as per CPWD specifications complete in all respects etc as required at site. The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with +12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla. The permissible temperature-rise shall not exceed 40 dg.C for oil and 45 dg. C for winding up to 200 KVA and 45 dg.C for oil and 50 dg. C for winding for above 200 KVA. Inside of tank shall be painted with varnish or liquid resistant paint. For external surfaces one coat of thermo setting powder paint or one coat of epoxy primer followed by two coats of polyurethane base paint shall be used. IS: 1180 (Part 3) shall be referred for paint thickness for normal to medium corrosive atmosphere. For highly polluted atmosphere and special application external paint work shall be as per direction of Engineer-in-Charge. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc. Marking Each transformer shall be provided with rating plate made of anodized aluminum/ stainless steel material securely fixed on the outer body, easily accessible, as per IS: 1180 Part-3. The entries on the rating plate shall be indelibly marked.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Oil level gauge indicating oil level at minimum, 30°C and maximum operating temperature; c) Air release device (for non-sealed type transformers) d) Rating and terminal marking plates; e) Silica gel breather f) Drain-cum-sampling valve ($\frac{3}{4}$ " nominal size thread, IS 554) preferably steel with plug for three

phase transformers; g) Thermometer pocket with cap; h) Oil filling holes having (1¼" nominal size thread) with cover (for sealed type transformers without conservator); i) Lifting lugs for the complete transformer as well as for core and winding assembly; j) Pressure relief device or explosion vent above 200 kVA; k) One filter valve on the upper side of the tank (for transformers above 200 kVA); l) Unidirectional flat rollers (for transformers above 200 kVA); m) Inspection hole (for transformers above 200 kVA); n) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth);

o) Buchholz relay for transformers above 800 kVA. p) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. q) Bird guard; r) Oil temperature indicator and winding temperature indicators for transformers above 200 kVA with suitable tripping mechanism above permissible limit s) Jacking pads (for transformer above 1 600 kVA); t) Additional Neutral separately brought out on bushing for earthing. u) Magnetic oil level gauge (for transformer above 1600 kVA) with low oil level alarm contact; v) Non return valve (for conducting pressure test); w) Pressure relief device or explosion vent x). Monogram Plate y) Inspection cover z). Detachable type radiators with top and bottom shutoff valve. aa) Oil Conservator with Oil level indicator, minimum level marking and drain plug for all transformers of capacity 50 KVA and above. bb) Necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers.

24.24.1	63KVA	Nos	165070
24.24.2	100KVA	Nos	262016
24.24.3	160KVA	Nos	419225

DRY Type

33/0.433 KV, 3 Phase, 50 Hz Indoor mounting

- 24.25 Supply, installation, testing and commissioning of following capacity (continuous loading) 33/0.433 KV Delta/Star, step down, 3 Phase, 50 Hz, Dyn 11 vector group, Cast Resin / VPI (vacuum pressure impregnated) Dry Type, copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better) AN (air natural) cooled transformer suitable for indoor applications with On Load Tap Changer (OLTC) on HV side having AVS relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic

operation of OLTC and facility for remote/ manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, insulation class F (minimum), suitable for environment conditions class E4, suitable for fire behaviour class F1, climate class-C1, having cable end boxes on HV side suitable for 3x400 sqmm XLPE cable of 33 KV grade with necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers, bus trunking arrangement on LV side complete with all accessories and safety provisions as per relevant IS Code ,The transformer shall be provided with standard fittings/accessories as per relevant IS and mentioned below, protection alarm/trip protection, 3 nos. of Polymeric Zinc Oxide surge Arrestors on HV Side. Winding Temperature scanner (Digital) with alarm/Trip contacts with RTD Sensors per LV winding and space for mounting differential protection CT's in LV chamber with neutral brought out separately including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/HT Panel for safety tripping, complete as confirming to IS-2026 Part-11, i/c supplying and grouting of suitable M.S. Channel with all accessories ,complete in all respects as required at site as per CPWD specifications. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location, rating plate as per relevant IS Code etc. All testing shall as per relevant IS Code. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Rating and terminal marking plates; c) Thermometer pocket with cap; d) Lifting lugs for the complete transformer as well as for core and winding assembly; e) Bi-directional flat rollers (for transformers above 200 kVA); f) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); g) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. for transformers up to 200 kVA; h) Bird guard; i) Jacking pads (for transformer above 1 600 kVA); j) Name Rating & Diagram Plate. k) Monogram Plate.i) Additional neutral seperately brought out on bushing for earthing.

Note : The permissible total losses value shall not exceed by 15% the losses as mentioned below.

Level 3

24.25.1	1000 KVA (losses at 50% loading < 3000watt, losses at 100% loading < 9000watt)	Each	2530179
24.25.2	1250 KVA (losses at 50% loading < 3600watt, losses at 100% loading < 10750watt)	Each	3162724
24.25.3	1600 KVA (losses at 50% loading < 4500watt, losses at 100% loading < 13500watt)	Each	4048287
24.25.4	2000 KVA (losses at 50% loading < 5400watt, losses at 100% loading < 17000watt)	Each	5060358
24.25.5	2500 KVA (losses at 50% loading < 6500watt, losses at 100% loading < 20000watt)	Each	6325448
Level 4			
24.25.6	1000 KVA (losses at 50% loading < 2790watt, losses at 100% loading < 7700watt)	Each	2909706
24.25.7	1250 KVA (losses at 50% loading < 3300watt, losses at 100% loading < 9200watt)	Each	3637132
24.25.8	1600 KVA (losses at 50% loading < 4200watt, losses at 100% loading < 11800watt)	Each	4655530
24.25.9	2000 KVA (losses at 50% loading < 5050watt, losses at 100% loading < 15000watt)	Each	5819412
24.25.10	2500 KVA (losses at 50% loading < 6150watt, losses at 100% loading < 18500watt)	Each	7274265
Level 5			
24.25.11	1000 KVA (losses at 50% loading < 2620watt, losses at 100% loading < 7000watt)	Each	3289233
24.25.12	1250 KVA (losses at 50% loading < 3220watt, losses at 100% loading < 8400watt)	Each	4111541
24.25.13	1600 KVA (losses at 50% loading < 3970watt, losses at 100% loading < 11300watt)	Each	5262773
24.25.14	2000 KVA (losses at 50% loading < 4790watt, losses at 100% loading < 14100watt)	Each	6578466
24.25.15	2500 KVA (losses at 50% loading < 5900watt, losses at 100% loading < 17500watt)	Each	8223082
33/0.433 KV, 3 Phase, 50 Hz Outdoor mounting			
24.26	Supply, installation, testing and commissioning of following capacity (continuous loading) 33/0.433 KV Delta/Star, step down, 3 Phase, 50 Hz, Dyn 11 vector group, Cast Resin / VPI (vacuum pressure impregnated) Dry Type, copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better) AN (air natural) cooled transformer suitable for Outdoor applications with enclosure, On Load Tap Changer (OLTC) on HV side having AVS relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote/manual operation of OLTC HV side in range of +5% to -15% in		

steps of 2.5%, insulation class F (minimum), suitable for environment conditions class E4, suitable for fire behaviour class F1, climate class-C1, having cable end boxes on HV side suitable for 3x400 sqmm XLPE cable of 33 KV grade with necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers, bus trunking arrangement on LV side complete with all accessories and safety provisions as per relevant IS Code ,The transformer shall be provided with standard fittings/accessories as per relevant IS and mentioned below, protection alarm/trip protection, 3 nos. of Polymeric Zinc Oxide surge Arrestors on HV Side. Winding Temperature scanner (Digital) with alarm/Trip contacts with RTD Sensors per LV winding and space for mounting differential protection CT's in LV chamber with neutral brought out separately including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/HT Panel for safety tripping, complete as confirming to IS-2026 Part-11, i/c supplying and grouting of suitable M.S. Channel with all accessories ,complete in all respects as required at site as per CPWD specifications . Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location, rating plate as per relevant IS Code etc. All testing shall as per relevant IS Code. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The enclosure shall also have Welded Door handle , Danger plate on HV and LV side doors, caution plate for tap links for HT doors, Door limit switch on both HV and LV side doors to be wired up to WTI box terminal for tripping the transformer in case door is opened with the enclosure transformer energized, Phase marking plates on HV and LV doors.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Rating and terminal marking plates; c) Thermometer pocket with cap; d) Lifting lugs for the complete transformer as well as for core and winding assembly; e) Bi-directional flat rollers (for transformers above 200 kVA); f) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); g) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. for transformers up to 200 kVA; h) Bird guard; i) Jacking pads (for transformer above 1 600 kVA); j) Name Rating & Diagram Plate. k) Monogram Plate.i) Additional neutral seperately brought out on bushing for earthing. Note :

The permissible total losses value shall not exceed by 15% the losses as mentioned below.

Level 3

24.26.1	1000 KVA (losses at 50% loading < 3000watt, losses at 100% loading < 9000watt)	Each	2663346
24.26.2	1250 KVA (losses at 50% loading < 3600watt, losses at 100% loading < 10750watt)	Each	3329183
24.26.3	1600 KVA (losses at 50% loading < 4500watt, losses at 100% loading < 13500watt)	Each	4261354
24.26.4	2000 KVA (losses at 50% loading < 5400watt, losses at 100% loading < 17000watt)	Each	5326693
24.26.5	2500 KVA (losses at 50% loading < 6500watt, losses at 100% loading < 20000watt)	Each	6658366

Level 4

24.26.6	1000 KVA (losses at 50% loading < 2790watt, losses at 100% loading < 7700watt)	Each	3062848
24.26.7	1250 KVA (losses at 50% loading < 3300watt, losses at 100% loading < 9200watt)	Each	3828561
24.26.8	1600 KVA (losses at 50% loading < 4200watt, losses at 100% loading < 11800watt)	Each	4900557
24.26.9	2000 KVA (losses at 50% loading < 5050watt, losses at 100% loading < 15000watt)	Each	6125697
24.26.10	2500 KVA (losses at 50% loading < 6150watt, losses at 100% loading < 18500watt)	Each	7657121

Level 5

24.26.11	1000 KVA (losses at 50% loading < 2620watt, losses at 100% loading < 7000watt)	Each	3462350
24.26.12	1250 KVA (losses at 50% loading < 3220watt, losses at 100% loading < 8400watt)	Each	4327938
24.26.13	1600 KVA (losses at 50% loading < 3970watt, losses at 100% loading < 11300watt)	Each	5539761
24.26.14	2000 KVA (losses at 50% loading < 4790watt, losses at 100% loading < 14100watt)	Each	6924701
24.26.15	2500 KVA (losses at 50% loading < 5900watt, losses at 100% loading < 17500watt)	Each	8655876

11/0.433 KV, 3 Phase, 50 Hz Indoor mounting

- 24.27 Supply, installation, testing and commissioning of following capacity (continuous loading) 11/0.433 KV Delta/Star, step down, 3 Phase, 50 Hz, Dyn 11 vector group, Cast Resin / VPI (vacuum pressure impregnated) Dry Type, copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better) AN (air natural) cooled transformer suitable for indoor applications with Offload Tap Changer/ On Load Tap Changer (OLTC) (below 500 KVA offload and above 500 KVA

On Load Tap Changer) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, insulation class F (minimum), suitable for environment conditions class E4, suitable for fire behaviour class F1, climate class-C1, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade with necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers, bus trunking arrangement on LV side complete with all accessories and safety provisions as per relevant IS Code ,The transformer shall be provided with standard fittings/accessories as per relevant IS and mentioned below, protection alarm/trip protection, 3 nos. of Polymeric Zinc Oxide surge Arrestors on HV Side. Winding Temperature scanner (Digital) with alarm/Trip contacts with RTD Sensors per LV winding and space for mounting differential protection CT's in LV chamber with neutral brought out separately including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/HT Panel for safety tripping, complete as confirming to IS-2026 Part-11, i/c supplying and grouting of suitable M.S. Channel with all accessories ,complete in all respects as required at site as per CPWD specifications . Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. The transformer should have QR code which should contain drawing, test report OEM manual,Geo- Tag of manufacturing location, rating plate as per relevant IS Code etc. All testing shall as per relevant IS Code. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Rating and terminal marking plates; c) Thermometer pocket with cap; d) Lifting lugs for the complete transformer as well as for core and winding assembly; e) Bi-directional flat rollers (for transformers above 200 kVA); f) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); g) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. for transformers up to 200 kVA; h) Bird guard; i) Jacking pads (for transformer above 1 600 kVA); j) Name Rating & Diagram Plate. k) Monogram Plate.i) Additional neutral seperately brought out on bushing for earthing.

Level 3

24.27.1	63 KVA (losses at 50% loading < 380watt, losses at 100% loading < 1250watt)	Each	146477
24.27.2	100 KVA (losses at 50% loading < 520watt, losses at 100% loading < 1800watt)	Each	232503
24.27.3	160 KVA (losses at 50% loading < 770watt, losses at 100% loading < 2200watt)	Each	372005
24.27.4	200 KVA (losses at 50% loading < 890watt, losses at 100% loading < 2700watt)	Each	465006
24.27.5	250 KVA (losses at 50% loading < 1050watt, losses at 100% loading < 3150watt)	Each	581257
24.27.6	315 KVA (losses at 50% loading < 1100watt, losses at 100% loading < 3275watt)	Each	732384
24.27.7	400 KVA (losses at 50% loading < 1300watt, losses at 100% loading < 3875watt)	Each	930012
24.27.8	500 KVA (losses at 50% loading < 1600watt, losses at 100% loading < 4750watt)	Each	1162515
24.27.9	630 KVA (losses at 50% loading < 2000watt, losses at 100% loading < 5855watt)	Each	1464769
24.27.10	1000 KVA (losses at 50% loading < 3000watt, losses at 100% loading < 9000watt)	Each	2325029
24.27.11	1250 KVA (losses at 50% loading < 3600watt, losses at 100% loading < 10750watt)	Each	2906287
24.27.12	1600 KVA (losses at 50% loading < 4500watt, losses at 100% loading < 13500watt)	Each	3720047
24.27.13	2000 KVA (losses at 50% loading < 5400watt, losses at 100% loading < 17000watt)	Each	4650059
24.27.14	2500 KVA (losses at 50% loading < 6500watt, losses at 100% loading < 20000watt)	Each	5812574
	Level 4		
24.27.15	63 KVA (losses at 50% loading < 340watt, losses at 100% loading < 1140watt)	Each	168448
24.27.16	100 KVA (losses at 50% loading < 475watt, losses at 100% loading < 1650watt)	Each	267378
24.27.17	160 KVA (losses at 50% loading < 670watt, losses at 100% loading < 1950watt)	Each	427805
24.27.18	200 KVA (losses at 50% loading < 780watt, losses at 100% loading < 2300watt)	Each	534757
24.27.19	250 KVA (losses at 50% loading < 980watt, losses at 100% loading < 2930watt)	Each	668446
24.27.20	315 KVA (losses at 50% loading < 1025watt, losses at 100% loading < 3100watt)	Each	842242
24.27.21	400 KVA (losses at 50% loading < 1225watt, losses at 100% loading < 3450watt)	Each	1069514
24.27.22	500 KVA (losses at 50% loading < 1510watt, losses at 100% loading < 4300watt)	Each	1336892
24.27.23	630 KVA (losses at 50% loading < 1860watt, losses at 100%	Each	1684484

	loading < 5300watt)			
24.27.24	1000 KVA (losses at 50% loading < 2790watt, losses at 100% loading < 7700watt)	Each	2673784	
24.27.25	1250 KVA (losses at 50% loading < 3300watt, losses at 100% loading < 9200watt)	Each	3342230	
24.27.26	1600 KVA (losses at 50% loading < 4200watt, losses at 100% loading < 11800watt)	Each	4278054	
24.27.27	2000 KVA (losses at 50% loading < 5050watt, losses at 100% loading < 15000watt)	Each	5347568	
24.27.28	2500 KVA (losses at 50% loading < 6150watt, losses at 100% loading < 18500watt)	Each	6684460	
	Level 5			
24.27.29	63 KVA (losses at 50% loading < 300watt, losses at 100% loading < 1050watt)	Each	190420	
24.27.30	100 KVA (losses at 50% loading < 435watt, losses at 100% loading < 1500watt)	Each	302254	
24.27.31	160 KVA (losses at 50% loading < 570watt, losses at 100% loading < 1700watt)	Each	483606	
24.27.32	200 KVA (losses at 50% loading < 670watt, losses at 100% loading < 2100watt)	Each	604508	
24.27.33	250 KVA (losses at 50% loading < 920watt, losses at 100% loading < 2700watt)	Each	755635	
24.27.34	315 KVA (losses at 50% loading < 955watt, losses at 100% loading < 2750watt)	Each	952100	
24.27.35	400 KVA (losses at 50% loading < 1150watt, losses at 100% loading < 3330watt)	Each	1209015	
24.27.36	500 KVA (losses at 50% loading < 1430watt, losses at 100% loading < 4100watt)	Each	1511269	
24.27.37	630 KVA (losses at 50% loading < 1745watt, losses at 100% loading < 4850watt)	Each	1904199	
24.27.38	1000 KVA (losses at 50% loading < 2620watt, losses at 100% loading < 7000watt)	Each	3022538	
24.27.39	1250 KVA (losses at 50% loading < 3220watt, losses at 100% loading < 8400watt)	Each	3778173	
24.27.40	1600 KVA (losses at 50% loading < 3970watt, losses at 100% loading < 11300watt)	Each	4836061	
24.27.41	2000 KVA (losses at 50% loading < 4790watt, losses at 100% loading < 14100watt)	Each	6045077	
24.27.42	2500 KVA (losses at 50% loading < 5900watt, losses at 100% loading < 17500watt)	Each	7556346	
	11/0.433 KV, 3 Phase, 50 Hz Outdoor mounting			

- 24.28 Supply, installation, testing and commissioning of following capacity (continuous loading) 11/0.433 KV Delta/Star, step down, 3 Phase, 50 Hz, Dyn 11 vector group, Cast Resin / VPI (vacuum pressure impregnated) Dry Type, copper wound transformer (Electrolytic grade 99.9% pure copper, Core made of first grade Cold Rolled Grain Oriented (CRGO) Core grade MOH or better) AN (air natural) cooled transformer suitable for Outdoor applications with Offload Tap Changer/ On Load Tap Changer (OLTC) (below 500 KVA offload and above 500 KVA On Load Tap Changer) on HV side having AVR relay and Remote Tap Changer Control (RTCC) for automatic sensing of incoming voltage, automatic operation of OLTC and facility for remote and manual operation of OLTC HV side in range of +5% to -15% in steps of 2.5%, insulation class F (minimum), suitable for environment conditions Class E-O-3, suitable for fire behaviour class F1, climate class-C1, having cable end boxes on HV side suitable for 3x300 sqmm XLPE cable of 11 KV grade with necessary hardware, clamps, lugs etc. for termination on HV/MV etc. for all transformers, bus trunking arrangement on LV side complete with all accessories and safety provisions as per relevant IS Code ,The transformer shall be provided with standard fittings/accessories as per relevant IS and mentioned below, protection alarm/trip protection, 3 nos. of Polymeric Zinc Oxide surge Arrestors on HV Side. Winding Temperature scanner (Digital) with alarm/Trip contacts with RTD Sensors per LV winding and space for mounting differential protection CT's in LV chamber with neutral brought out separately including supplying and laying of copper conductor multicore control cable from transformer to HT breaker/HT Panel for safety tripping, complete as confirming to IS-2026 Part-11, i/c supplying and grouting of suitable M.S. Channel with all accessories ,complete in all respects as required at site as per CPWD specifications. Design ambient condition : a) air temperature 50 deg C, b) Relative Humidity 90 % Max, c) Seismic Zone as per location of site, d) Altitude as per location/site. The transformer should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location, rating plate as per relevant IS Code etc. All testing shall as per relevant IS Code. Noise level Shall not exceed limits as per NEMA TR-1 with all accessories running measured as per IEC 551 / NEMA standard. The enclosure shall also have Welded Door handle , Danger plate on HV and LV side doors, caution plate for tap links for HT doors, Door limit switch on both HV and LV side doors to be wired up to WTI box terminal for tripping the transformer in case door is

opened with the enclosure transformer energized, Phase marking plates on HV and LV doors.

Fitting and Accessories : The following fittings shall be provided:- a) Two earthing terminals with the earthing symbol b) Rating and terminal marking plates; c) Thermometer pocket with cap; d) Lifting lugs for the complete transformer as well as for core and winding assembly; e) Bi-directional flat rollers (for transformers above 200 kVA); f) HV side neutral grounding strip (where one of the HV bushing terminal is connected to earth); g) Arcing horns or suitable rating lightning arrestors for HT side – 3 Nos. for transformers up to 200 kVA; h) Bird guard; i) Jacking pads (for transformer above 1 600 kVA); j) Name Rating & Diagram Plate. k) Monogram Plate.i) Additional neutral seperately brought out on bushing for earthing.

Level 3

24.28.1	63 KVA (losses at 50% loading < 380watt, losses at 100% loading < 1250watt)	Each	154186
24.28.2	100 KVA (losses at 50% loading < 520watt, losses at 100% loading < 1800watt)	Each	244740
24.28.3	160 KVA (losses at 50% loading < 770watt, losses at 100% loading < 2200watt)	Each	391584
24.28.4	200 KVA (losses at 50% loading < 890watt, losses at 100% loading < 2700watt)	Each	489480
24.28.5	250 KVA (losses at 50% loading < 1050watt, losses at 100% loading < 3150watt)	Each	611850
24.28.6	315 KVA (losses at 50% loading < 1100watt, losses at 100% loading < 3275watt)	Each	770931
24.28.7	400 KVA (losses at 50% loading < 1300watt, losses at 100% loading < 3875watt)	Each	978960
24.28.8	500 KVA (losses at 50% loading < 1600watt, losses at 100% loading < 4750watt)	Each	1223700
24.28.9	630 KVA (losses at 50% loading < 2000watt, losses at 100% loading < 5855watt)	Each	1541862
24.28.10	1000 KVA (losses at 50% loading < 3000watt, losses at 100% loading < 9000watt)	Each	2447399
24.28.11	1250 KVA (losses at 50% loading < 3600watt, losses at 100% loading < 10750watt)	Each	3059249
24.28.12	1600 KVA (losses at 50% loading < 4500watt, losses at 100% loading < 13500watt)	Each	3915839
24.28.13	2000 KVA (losses at 50% loading < 5400watt, losses at 100% loading < 17000watt)	Each	4894799
24.28.14	2500 KVA (losses at 50% loading < 6500watt, losses at 100% loading < 20000watt)	Each	6118499

Level 4

24.28.15	63 KVA (losses at 50% loading < 340watt, losses at 100% loading < 1140watt)	Each	177314
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24.28.16	100 KVA (losses at 50% loading <475watt, losses at 100% loading <1650watt)	Each	281451
24.28.17	160 KVA (losses at 50% loading < 670watt, losses at 100% loading < 1950watt)	Each	450321
24.28.18	200 KVA (losses at 50% loading < 780watt, losses at 100% loading <2300watt)	Each	562902
24.28.19	250 KVA (losses at 50% loading < 980watt, losses at 100% loading < 2930watt)	Each	703627
24.28.20	315 KVA (losses at 50% loading < 1025watt, losses at 100% loading < 3100watt)	Each	886570
24.28.21	400 KVA (losses at 50% loading < 1225watt, losses at 100% loading < 3450watt)	Each	1125804
24.28.22	500 KVA (losses at 50% loading <1510watt, losses at 100% loading < 4300watt)	Each	1407255
24.28.23	630 KVA (losses at 50% loading < 1860watt, losses at 100% loading < 5300watt)	Each	1773141
24.28.24	1000 KVA (losses at 50% loading < 2790watt, losses at 100% loading < 7700watt)	Each	2814509
24.28.25	1250 KVA (losses at 50% loading < 3300watt, losses at 100% loading < 9200watt)	Each	3518137
24.28.26	1600 KVA (losses at 50% loading < 4200watt, losses at 100% loading < 11800watt)	Each	4503215
24.28.27	2000 KVA (losses at 50% loading < 5050watt, losses at 100% loading < 15000watt)	Each	5629019
24.28.28	2500 KVA (losses at 50% loading < 6150watt, losses at 100% loading < 18500watt) Level 5	Each	7036273
24.28.29	63 KVA (losses at 50% loading < 300watt, losses at 100% loading <1050watt)	Each	200442
24.28.30	100 KVA (losses at 50% loading < 435watt, losses at 100% loading < 1500watt)	Each	318162
24.28.31	160 KVA (losses at 50% loading < 570watt, losses at 100% loading <1700watt)	Each	509059
24.28.32	200 KVA(losses at 50% loading <670watt, losses at 100% loading < 2100watt)	Each	636324
24.28.33	250 KVA(losses at 50% loading < 920watt, losses at 100% loading < 2700watt)	Each	795405
24.28.34	315 KVA (losses at 50% loading < 955watt, losses at 100% loading < 2750watt)	Each	1002210
24.28.35	400 KVA (losses at 50% loading < 1150watt, losses at 100% loading < 3330watt)	Each	1272648
24.28.36	500 KVA (losses at 50% loading < 1430watt, losses at 100% loading < 4100watt)	Each	1590810
24.28.37	630 KVA (losses at 50% loading < 1745watt, losses at 100% loading < 4850watt)	Each	2004420
24.28.38	1000 KVA (losses at 50% loading < 2620watt, losses at 100%	Each	3181619

	loading < 7000watt)			
24.28.39	1250 KVA (losses at 50% loading < 3220watt, losses at 100% loading < 8400watt)	Each	3977024	
24.28.40	1600 KVA (losses at 50% loading < 3970watt, losses at 100% loading < 11300watt)	Each	5090591	
24.28.41	2000 KVA (losses at 50% loading < 4790watt, losses at 100% loading < 14100watt)	Each	6363239	
24.28.42	2500 KVA (losses at 50% loading < 5900watt, losses at 100% loading < 17500watt)	Each	7954048	

CHAPTER 25

AUTOMATIC POWER FACTOR CORRECTION (APFC) PANEL

Item No.	Description	Unit	Rate(₹)
25.1	Automatic Power Factor Correction (APFC) System		

Supply, Installation, testing and commissioning of Automatic Power Factor Correction (APFC) panel, indoor type floor mounted free standing totally enclosed, extendable, IP 42, of following capacity for 3 phase, 415 V + 10 %, 50 Hz AC System for Ambient temperature -5°C to +40°C, fabricated in compartmentalised designed made of CRCA sheet steel of 2.0mm thick for framework & covers, 3 mm thick for gland plate i/c cleaning & finishing complete with 9 tank process for powder coated of approved shade (RAL 7032-Siemens gray or as approved by Engineer-in-Charge), having front section (switch gear and control accessories) and rear section capacitor and reactor, front and rear access, having suitable current carrying capacity, extensible TPN Aluminium alloy bus bar of high conductivity, DMC/SMC bus bar supports, bottom base channel of MS Section, fabrication shall be done in transportable section, entire panel shall have common copper earth bar of minimum size of 25mm x 5mm with 2 nos. earth studs, the earth terminals provided on the body of capacitor bank shall also be bonded to the main capacitor panel earth bus with 2 nos. 8 SWG or 6 SWG GI earth wires/ equivalent size of copper conductor cable, forced ventilation for maintaining temperature rise not more than 5°C from ambient, interconnections, connections with 14% detuned reactor and heavy duty 525 V ISI marked Impregnated MPP(Metalized Polypropylene) Capacitor (IS 13340 Part -1 & 2) APFC Panel shall be in compliance with IS :16636 & CPWD Specifications etc. as per below details

(A) Incomers

Suitable capacity MCCB Microprocessor base with O/C, S/C, E/L release of TPN 50KA breaking capacity (Ics=Icu), ON, OFF, Trip, R, Y, B - LED Indicating Lamp set alongwith required Instruments and accessories with extended rotary handel and door interlocking arrangment. Current rating of the Incomer in ampere shall be APFC Panel rating in KVAR x 1.4 x 1.5 or Nearest higher standards rating.

(B) Instruments & Indications

i) 3-Phase current sensing APFC microprocessor relay/controller , advance 8/12 stages (8 stages for capacity below 100 KVAR and 12 stages 100 KVAR & above) with Communication Ethernet/RS485/SNMP port open protocol for BMS integration as per approved by Engineering in charge and

having display of Phase wise V, A, PF, Cos-Phi, Kw, KVA, KVAR, THD-V, THD-I, harmonics up to 31 level. 3 nos of dual core CT's accuracy class 1, 15VA at incomer of PCC Panel for APFC relay.

ii) Auto Manual Selector switch, auxiliary contactors with timer for delay in manual mode.

iii) Digital Multi function meter with LED Display for V, A, PF, KW, KVA, KVAR, THD-V & I, Frequency.

iv) Suitable rating control transformer shall be provided for control and indication circuit.

v) All components like control transformer, meter, relay and indicating lamp shall be protected by using suitable rating individual MCB's.

vi) Wiring of the control circuit shall be done by using 2.5 sq mm, FRLS/HFFR 1100 V grade, PVC insulated multi stranded copper control wire.

(C) Bus Bars

1.3 Amp per Sq.mm, TPN, Electrolytic grade Aluminium bus bar of capacity 1.25 times of incomer rating as per CPWD specification.

(D) Outgoings (APFC Section)

Selection of the capacitors combinations shall be for continuous rating and each capacitor bank shall have suitable capacity Heavy Duty ISI Marked Capacitor, capacitor duty contactor, the capacitor shall be mounted on channel with base of perforated M S Powder coated sheet, connections inter connections etc. and other features as per CPWD Specifications and relevant IS Code having following:

(i) Capacitor bank ratings & stages shall be as per the technical specifications sheet of NIT.

(ii) Capacitor will be MPP self healing type with discharge resistor, pressure release mechanism.

(iii) Since Capacitor Voltage is 525 Volts, thus higher KVAR has to be considered to get rated output at 415 Volts.

(iv) 14% Detuned Reactor of class H insulation & 150% linearity in series with Capacitor.

(Note: Technical specifications sheet for selection of the capacitors combinations shall be provided by the NIT Approving Authority with due consideration of number of capacitors i.e. 1 KVAR, 2 KVAR, 3 KVAR, 5 KVAR, 10 KVAR.....for smooth correction).

25.1.1	50 KVAR	Each	219906
25.1.2	75 KVAR	Each	271373

25.1.3	100 KVAR	Each	297575
25.1.4	125 KVAR	Each	320033
25.1.5	150 KVAR	Each	374776
25.1.6	175 KVAR	Each	402381
25.1.7	200 KVAR	Each	428583

25.2 HYBRID Power Factor Correction System

Supply, Installation, testing and commissioning of HYBRID APFC Panel, 3 phase 4 wire, 415 V, 50 Hz AC System for Ambient temperature -5°C to +40°C of following capacity with passive solution of 60% capacity and active solution of 40% capacity, 3Phase 4 wire Hybrid Power Factor Correction Solution (with arrangement for neutral current balance) to achieve >0.99 lag and TDDI/THDV values within IEEE recommended limits. APFC should be designed as per IS 16636 Or IEC 61921. The active section and passive section shall work in sync to give optimized output. The degree of protection of passive section should be IP 42, and of active section should be minimum IP 21. The switching device for APFC passive section should be through capacitor duty contactor and for the active compensation system shall be IGBT based with 3 level topology having 12 IGBT in inverter circuit. The active compensation system should filter harmonics from 2nd to 50th individual harmonic order and shall be selectable for the entire range. The active compensation system should have feature to improve PF correction and harmonic filtration having response time <25Micro second. The hybrid panel shall be indoor type floor mounted free standing totally enclosed, extensible, fabricated in compartmentalised designed made of CRCA sheet steel of 2.0mm thick for framework & covers, 3 mm thick for gland plate i/c cleaning & finishing complete with 9 tank process for powder coated of approved shade (RAL 7032-Siemens gray or as approved by Engineer-in-Charge), having front section (switch gear and control accessories) and rear section (capacitor and reactor),

front and rear access, having suitable current carrying capacity, extensible TPN Aluminium alloy bus bar of high conductivity, DMC/SMC bus bar supports, bottom base channel of MS Section, fabrication shall be done in transportable section, entire panel shall have common copper earth bar of minimum size of 25mm x 5mm with 2 nos. earth studs, the earth terminals

provided on the body of capacitor bank shall also be bonded to the main capacitor panel earth bus with 2 nos. 8 SWG or 6 SWG GI earth wires/ equivalent size of copper conductor cable, forced ventilation for maintaining temperature rise not more than 5°C from ambient, interconnections, connections with 14% detuned reactor and heavy duty 525 V ISI marked Impregnated MPP(Metalized Polypropylene) Capacitor (IS 13340 Part -1 & 2) APFC Panel shall be in compliance with IS :16636 & CPWD Specifications etc. as per below details.

(A) Incomers

Suitable capacity MCCB/ACB (Upto 300 KVAR, MCCB and above 300 KVAR, ACB) Microprocessor base with O/C, S/C, E/L release of TPN 50KA breaking capacity (Ics=Icu), ON, OFF, Trip, R, Y, B - LED Indicating Lamp set alongwith required Instruments and accessories with extended rotary handel and door interlocking arrangment. Current rating of the Incomer in ampere shall be APFC Panel rating in KVAR x 1.4 x 1.5 or Nearest higher standards rating.

25.2.1	250 KVAR	Each	1365289
25.2.2	300 KVAR	Each	1624497
25.2.3	350 KVAR	Each	1818202
25.2.4	400 KVAR	Each	2183152
25.2.5	450 KVAR	Each	2350655
25.2.6	500 KVAR	Each	2662267
25.2.7	550 KVAR	Each	2853164
25.2.8	600 KVAR	Each	3132023
25.2.9	650KVAR	Each	3314499
25.2.10	700KVAR	Each	3566221
25.2.11	750KVAR	Each	3876897
25.2.12	800KVAR	Each	4067794
25.2.13	850KVAR	Each	4214710

25.2.14	900KVAR	Each	4482340
25.2.15	950KVAR	Each	4741548
25.2.16	1000KVAR	Each	4992335

CHAPTER 26

UNINTERRUPTED POWER SUPPLY (UPS)

Item No.	Description	Unit	Rate(₹)
26.1	<u>Online UPS - Input supply: Single Phase, Output Supply : Single Phase</u> Supplying, installation, Testing & Commissioning of following capacity at full load (Unity Power Factor) ON LINE Uninterrupted Power Supply (UPS) system suitable for Single Phase input, Single Phase output AC Supply. The UPS shall include a Rectifier, inverter, battery bank suitable for 30 minutes back up (Battery VAH capacity shall not be less than 1600 VAH per KVA of UPS rating per Hour backup time) on full load (Battery shall be VRLA, SMF in ABS Container) and Static Bypass switch alongwith provision for manual bypass, suitable isolation transformer for additional protection against neutral faults etc. The UPS systems shall be Microprocessor based Digital Control, using Insulated Gate Bipolar Transistor (IGBT)'s both for the rectifier & inverter with PWM (Pulse Width Modulation) Technology. The quality of design, manufacturing and inspection process should confirm to the relevant Inter-national standards such as IEC/EN/VDE. The operating efficiency of the UPS systems shall be $\geq 95\%$ at 100% non-linear loads. Current total harmonic distortion (ITHD)/ total demand distortion (TDD) on the input grid shall be $\leq 5\%$ at 100 %load. (The required LC filters shall be included in UPS cost), extreme power factor kit to be include to limit the input pf to 0.99 and output power factor shall be unity (i.e. kw rating of the UPS shall be kva rating x 1) however UPS shall be suitable to take load at 0.7 lagging to 0.7 leading power factor loads. UPS shall be suitable for incoming supply AC single phase 160-270V 50 Hz and delivering output AC supply true sine wave single phase 220/230/240 Volt, 50 Hz +/- 0.2Hz, Overload capacity of 120% for 10 mins and 150% for 1 minute. Operating temperature 0 to 40 deg C, Relative humidity 0-95% non condensing, noise level less than 60db at 1 meter distance, Protection for Under voltage, over voltage, abnormal output voltage, battery over charging, output over current, short circuit, battery deep discharge , 10 KV surge. Display for watt/VA, Amp and Voltage power parameters etc. UPS shall comply with low voltage electromagnetic compatibility (EMC) achieved as per EN 6204, EN6204 Part I and Part 2, it shall be a Voltage and Frequency Independent (VFI) type UPS (as per standard IEC 62040-1, 2 & 3) . The UPS should be with IGBT Based Inverter Technology, Communication		

RS232/RS485/SNMP port open protocol for BMS integration as per approved by Engineering in charge. Required battery racks, DC breaker of suitable rating and interconnecting copper conductor cable of suitable size and connectors and all required accessories are inclusive in the cost. The UPS should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location etc

26.1.1	2KVA	Each	45894
26.1.2	3KVA	Each	55233
26.1.3	6KVA	Each	109793
26.1.4	10KVA	Each	168506

26.2 Online UPS- Input supply: Three Phase, Output supply: Three Phase

Supplying, installation, Testing & Commissioning of of following capacity at full load (Unity Power Factor) at operating temperature 0 to 40 deg C, Relative humidity 0 to 95%, Online double conversion true sine wave Uninterrupted hot swappable (allow for the replacement or addition of battery modules without shutting down the entire system) modular Power Supply (UPS) system with N+1 modules (N denotes total number of modules required for rated capacity). The UPS shall include a Rectifier, inverter, battery bank suitable for 30 minutes back up (Battery VAH capacity shall not be less than 1600 VAH per KVA of UPS rating per Hour backup time) on full load (Battery shall be VRLA, SMF in ABS Container) and Static Bypass switch alongwith provision for manual bypass, suitable isolation transformer for additional protection against neutral faults etc. UPS shall have inbuilt phase sequence correction. The UPS systems offered are to be of the latest technology with Digital Control Microprocessor based for reliable operation using Insulated Gate Bipolar Transistor (IGBT)'s both for the rectifier & inverter (3 Level) with PWM (Pulse Width Modulation). The quality of design, manufacturing and inspection process should confirm to the relevant Inter-national standards such as IEC/EN/VDE. The operating efficiency of the UPS systems shall be $\geq 96\%$ while operating on battery mode and delivering quality power to the 100% non-linear loads. Current total harmonic effect (ITHD) on the input grid shall be $< 5\%$ at 50 %load. (The required LC (inductor (L) and a capacitor (C)) filters shall be included in UPS cost), extreme power factor kit to be included to limit the input power factor (PF) to 0.99 and output power factor shall be unity (i.e. kw rating of the UPS shall be kva rating x 1), however UPS

shall be suitable to take load at 0.7 lagging to 0.7 leading power factor loads. UPS shall be suitable for incoming supply AC : 3Phase 400V +/-20%, 50 Hz +/-5 Hz, AC Output voltage: 3Phase 415 Volt, 50 Hz +/- 0.2Hz, Overload capacity of 120% for 10 mins, Sine wave output. Non condensing, noise level less than 60db at 1 meter distance, protections: Input Under voltage over voltage, abnormal out voltage, battery over charging, output over current, short circuit protection, battery deep discharge protection, 10KV surge. UPS must comply with low voltage electromagnetic compatibility (EMC) achieved as per EN 6204, EN6204 Part I and Part 2, it shall be a Voltage and Frequency Independent (VFI)-type UPS. . Communication RS232/RS485/SNMP port open protocol for BMS integration, all hardware & software for IoT Communication as per approved by Engineering in charge. Required battery racks and interconnecting copper conductor cables of suitable size and connectors and all required accessories are inclusive of the cost). This system must provide a means for logging and alarming of all monitored points plus email notification. Forced air-cooling with integral inbuilt fans with redundancy (if one fan fail UPS should be able to handle at least 80% of the load, Noise Level 65 DB at 1 meter distance. The system shall be in compliance IEC 62040- 1,2 & 3, IS: 16242 and CPWD Specification. Display Panel (minimum) (In-build 5 inch or more LC Display / LED) to display : a) Input: Voltage, current, Frequency. b) Bypass: Voltage, Frequency. c) Output: Voltage, frequency, Current. d) Battery: Voltage, Capacity. e) Load: KVA, KW, Percentage. f)Temperature: STS, Inverter, PFC. g) Event Logging & Statistical Data (On LCD/LED): UPS should capture and display up to 3000 events like: Over temperature / DC Bus Fail / Fan Fail / Fuse Fail / Overload / Short-circuit / Device Fail / Inverter Fail / Rectifier Fail / Bypass Fail, etc. h) Statistical Data: No. of power failures / Transfers to Bypass / Total Running time, etc. i) Mains Mode of Operation /Battery Mode of Operation / Bypass feeding the load / UPS Fault /Battery charging and discharging, overload, battery voltage and battery capacity. j) Audible Alarms : Mains Failure, Battery Low Alarm, UPS Overload, Fault, Shutdown, Input Over, Under Voltage, Output Over, Under Voltage, Battery Over, Under Voltage, Over Load and short circuit, Over Temperature. The UPS should have QR code which should contain drawing, test report OEM manual,Geo-Tag of manufacturing location etc

26.2.1	10KVA (Each Power module shall be $\leq$ 10 KVA)	Each	208893
26.2.2	20KVA (Each Power module shall be $\leq$ 10 KVA)	Each	341772

26.2.3	30KVA (Each Power module shall be $\leq$ 10 KVA)	Each	505748
26.2.4	40KVA (Each Power module shall be $\leq$ 10 KVA)	Each	599469
26.2.5	60KVA (Each Power module shall be $\leq$ 10 KVA)	Each	913600
26.2.6	80KVA (Each Power module shall be $\leq$ 25 KVA)	Each	1124076
26.2.7	100KVA (Each Power module shall be $\leq$ 25 KVA)	Each	1276967
26.2.8	120KVA (Each Power module shall be $\leq$ 25 KVA)	Each	1464409
26.2.9	160KVA (Each Power module shall be $\leq$ 25 KVA)	Each	1958639
26.2.10	200KVA (Each Power module shall be $\leq$ 50 KVA)	Each	2375417
26.2.11	300KVA (Each Power module shall be $\leq$ 50 KVA)	Each	3963347
26.2.12	400KVA (Each Power module shall be $\leq$ 50 KVA)	Each	5130901
26.2.13	500KVA (Each Power module shall be $\leq$ 50 KVA)	Each	6183283

CHAPTER 27

WET RISER & SPRINKLER SYSTEM

Item No.	Description	Unit	Rate(₹)
27.1	(a) Supplying, installation, testing and commissioning of Electric driven Main Fire Pump conforming to IS 12469: 2019 suitable for automatic operation and consisting of following, complete in all respects, as required : (b) Horizontal type, multistage, centrifugal, split casing pump of cast iron body & bronze impeller with stainless steel shaft, mechanical seal as required. (c) Suitable HP Squirrel cage induction motor, TEFC, synchronous speed 1500 RPM, suitable for operation on 415 volts, 3 phase 50 Hz, AC supply with IP 55 protection for enclosure, horizontal foot mounted type with Class-'F' insulation, conforming to IS-12615: 2018. (d) M.S. fabricated Common base plate, coupling, coupling guard, foundation bolts etc. as required. (e) Suitable cement concrete foundation duly plastered with anti vibration pads.		
27.1.1	2850 lpm at 88 m Head	Each	475390
27.1.2	2850 lpm at 70 m Head	Each	425556
27.1.3	2850 lpm at 56 m Head	Each	424172
27.1.4	2280 lpm at 88 m Head	Each	424172
27.1.5	2280 lpm at 70 m Head	Each	388181
27.1.6	2280 lpm at 56 m Head	Each	349421
27.1.7	1620 lpm at 88 m Head	Each	388181
27.1.8	1620 lpm at 70 m Head	Each	349421
27.1.9	1620 lpm at 56 m Head	Each	299587
	Note: *The head of the pump is selected in a manner so as to give a minimum 3.5kgf/cm² pressure at the highest/farthest point.		
27.2	(a) Supplying, installation, testing and commissioning of diesel engine driven main fire pump conforming to IS 12469: 2019 suitable for automatic operation and consisting of following, complete in all respects, as required : (Diesel Driven Pump)		

(b) Horizontal type, multistage, centrifugal pump of cast of iron body and bronze impeller with stainless steel shaft, mechanical seal as required.

(c) Suitable HP, 1500 RPM water cooled with radiator, diesel engine conforming to relevant IS standard complete with auto starting mechanism, 12 /24 volts electric starting equipment, diesel tank, exhaust pipe extended upto 10 m outside pump house duly insulated with 50 mm thick glass wool with 1.0 mm thick aluminium sheet cladding, residential silencer, instruments and protection as per standard specification, stop solenoid for auto stop in the event of fault with audio indications, painted with post office red colour etc. as required.

(d) M.S fabricated, common base plate, coupling, coupling guard, foundation bolts etc. as required.

(e) Suitable cement concrete foundation duly plastered and with anti vibration pads.

27.2.1	2850 lpm at 88 m Head	Each	718247
27.2.2	2850 lpm at 70 m Head	Each	647234
27.2.3	2850 lpm at 56 m Head	Each	643496
27.2.4	2280 lpm at 88 m Head	Each	667029
27.2.5	2280 lpm at 70 m Head	Each	642112
27.2.6	2280 lpm at 56 m Head	Each	639343
27.2.7	1620 lpm at 88 m Head	Each	660246
27.2.8	1620 lpm at 70 m Head	Each	621348
27.2.9	1620 lpm at 56 m Head	Each	538291

Note: * The head of the pump is selected in a manner so as to give a minimum 3.5kgf/cm² pressure at the highest/farthest point.

27.3 (a) Supplying, installation, testing and commissioning of electric driven pressurisation pump conforming to IS 12469: 2019 suitable for automatic operation and consisting of following, complete in all respects, as required : (Jockey Pump)

(b) Horizontal type, multistage, centrifugal pump of cast iron body and bronze impeller with stainless steel shaft, mechanical seal as required.

(c) Suitable HP squirrel cage induction motor TEFC type suitable for operation on 415 volts, 3 phase 50 Hz AC supply with IP 55 class of protection for enclosure, horizontal foot mounted type with Class-'F' insulation, conforming to IS-12615: 2018.

(d) M.S.fabricated Common base plate, coupling, coupling guard, foundation bolts etc. as required.

(e) Suitable cement concrete foundation duly plastered and with anti vibration pads.

27.3.1	180 lpm at 88 m Head	Each	134171
27.3.2	180 lpm at 70 m Head	Each	109254
27.3.3	180 lpm at 56 m Head	Each	87105
27.3.4	300 lpm at 88 m Head	Each	175699
27.3.5	300 lpm at 70 m Head	Each	134448
27.3.6	300 lpm at 56 m Head	Each	121712

27.4 (a) Supplying, installation, testing and commissioning of electric driven terrace pump conforming to IS 12469: 2019 suitable for automatic operation and consisting of following, complete in all respects, as required: (Terrace Pump)

(b) Horizontal type, multistage, centrifugal, split casing pump of cast iron body & bronze impeller with stainless steel shaft, mechanical seal as required.

(c) Suitable HP squirrel cage induction motor TEFC type suitable for operation on 415 volts, 3 phase, 50 Hz, AC supply with IP55 class of protection for enclosure, horizontal foot mounted type with Class-'F' insulation, conforming to IS-12615: 2018.

(d) M.S.fabricated common base plate, coupling, coupling guard, foundation bolts etc.as required.

(e) Suitable cement concrete foundation duly plastered and with anti vibration pads.

27.4.1	900 lpm at 35 m Head	Each	107744
27.4.2	450 lpm at 35 m Head	Each	91384

CONTROL PANELS

27.5 Fabrication, supply, Installation testing & commissioning of Electrical control panel of cubical construction, floor mounted type, fabricated out of 2mm thick CRCA sheet, compartmentalised with hinged lockable doors, dust and vermin proof, powder coated of approved shade after 7 tank treatment process, cable alley, inter-connection with suitable size copper

conductor cable/solid copper strip, having switchgears and accessories, mountings and internal wiring, earth terminals, numbering etc. complete in all respect, suitable for main fire pump, pressurisation pump & diesel pump set complete as per CPWD specification with following in coming and Outgoings, suitable for operation on 415V, 3 phase, 50Hz Ac Supply with enclosure protection class IP 42 as required :

27.5.1 INCOMING

250A, 50kA 4 Pole MCCB, Ics=100% Icu rating

Digital Voltmeter 0-500V with selector switch

Digital Ammeter (0-250 A) with selector switch & CTs etc.

LED type RYB phase indicating lamps, ON, OFF, trip indicating lamps

Set of Copper Bus Bar 300A

OUTGOING

(Note : All outgoing feeders for pumps should have digital Ammeter with selector switches, and LED type ON, OFF, trip indicating lamps)

Main Fire Pump

125 A, 50kA TPN MCCB, Ics=100% Icu, with fully automatic Star/Delta starter suitable for 60 HP pump with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation.

Jockey Pump

63 A, 50kA TPN MCCB, Ics=100% Icu, with suitable HP fully automatic Star/Delta starter with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation.

Diesel Engine Control

Control for diesel engine comprising -

Automatic/Manual selector switch & 3 attempts starting device, timers and relays as required, push buttons, start/stop in manual mode

Indicating lamp for high/ Low Lub. Oil pressure, High Water Temp and Engine on indication

Battery charger suitable for 12V/24 V DC with boost and trickle selector switch, 0-30 V DC volt meter, and 0-20 A DC Ammeter

All standard relays and accessories for automatic operation of diesel engine

System Controller

Designing, Supply, Installation, Testing and commissioning of system controller to control operation of main electric fire pump, diesel pump, Pressurization pump, Terrace pump in sequence as per specification consisting of relays, timers. Sensors, annunciation window for fault indication, complete as per specification

Each 256783

27.5.2 Incomings

400A, 50kA 4 Pole MCCB, Ics=100% Icu rating

Digital Voltmeter 0-500V with selector switch

Ammeter (0-400 A) with selector switch & CTs etc.

LED type RYB phase indicating lamps, ON, OFF, trip indicating lamps

Set of Copper Bus Bar 500A

Outgoings

(Note : All outgoing feeders for pumps should have digital Ammeter with selector switches, and LED type ON, OFF, trip indicating lamps)

Main Fire Pump

200 A, 50kA TPN MCCB, Ics=100% Icu, with fully automatic Star/Delta starter suitable for 75 hp pump with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation.

Jockey Pump

100 A, 50kA TPN MCCB, Ics=100% Icu, with suitable HP fully automatic Star/Delta starter with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation.

Diesel Engine Control

Control for diesel engine comprising -

Automatic/Manual selector switch & 3 attempt starting device, timers and relays as required, push buttons, start/stop in manual mode

Indicating lamp for high/ Low Lub. Oil pressure, High Water Temp and Engine on indication

Battery charger suitable for 12V/24 V DC with boost and trickle selector switch, 0-30 V DC volt meter, and 0-20 A DC Ammeter

All standard relays and accessories for automatic operation of diesel engine

System Controller

Designing, Supply, Installation, Testing and commissioning of system controller to control operation of main electric fire pump, diesel pump, Pressurization pump, Terrace pump in sequence as per specification consisting of relays, timers. Sensors, annunciation window for fault indication, complete as per specifications.

Each 300803

27.5.3 Incomings

630A, 50kA 4 Pole MCCB, Ics=100% Icu rating

Digital Voltmeter 0-500V with selector switch

Ammeter (0-630 A) with selector switch & CTs etc.

LED type RYB phase indicating lamps, ON, OFF, trip indicating lamps

Set of Copper Bus Bar 800Amps

Outgoings

(Note : All outgoing feeders for pumps should have digital Ammeter with selector switches, and LED type ON, OFF, trip indicating lamps)

Main Fire Pump

250 A, 50kA TPN MCCB, Ics=100% Icu, with fully automatic Star/Delta starter suitable for 100 hp pump with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation.

Jockey Pump

125 A, 50kA TPN MCCB, Ics=100% Icu, with Suitable HP fully automatic Star/Delta starter with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation.

Diesel Engine Control

Control for diesel engine comprising -

Automatic/Manual selector switch & 3 attempts starting device, timers and relays as required, push buttons, start/stop in manual mode

Indicating lamp for high/ Low Lub. Oil pressure, High Water Temp and Engine on indication

Battery charger suitable for 12V/24 V DC with boost and trickle selector switch, 0-30 V DC volt meter, and 0-20 A DC Ammeter

All standard relays and accessories for automatic operation of diesel engine

System Controller

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27.5.4	Designing, Supply, Installation, Testing and commissioning of system controller to control operation of main electric fire pump, diesel pump, Pressurization pump, Terrace pump in sequence as per specification consisting of relays, timers. Sensors, annunciation window for fault indication, complete as per specification Incomings 400A, 50kA 4 Pole MCCB, Ics=100% Icu Rating Digital Voltmeter 0-500V with selector switch Ammeter (0-400 A) with selector switch & CTs etc. LED type RYB phase indicating lamps, ON, OFF, trip indicating lamps Set of Copper Bus Bar 500Amps Outgoings (Note : All outgoing feeders for pumps should have digital Ammeter with selector switches, and LED type ON, OFF, trip indicating lamps) Main Fire Pumps 125 Amp, 50kA TPN MCCB, Ics=100% Icu, with fully automatic Star/Delta starter suitable for 60 hp pump with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation. - 2 sets Jockey Pump 63 Amp, 50kA TPN MCCB, Ics=100% Icu, with Suitable HP fully automatic Star/Delta starter with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation. - 2 sets Diesel Engine Control Control for diesel engine comprising - Automatic/Manual selector switch & 3 attempts starting device, timers and relays as required, push buttons, start/stop in manual mode Indicating lamp for high/ Low Lub. Oil pressure, High Water Temp and Engine on indication Battery charger suitable for 12V/24 V DC with boost and trickle selector switch, 0-30 V DC volt meter, and 0-20 A DC Ammeter All standard relays and accessories for automatic operation of diesel engine System Controller	Each	337486
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	Designing, Supply, Installation, Testing and commissioning of system controller to control operation of main electric fire pump, diesel pump, Pressurization pump, Terrace pump in sequence as per specification consisting of relays, timers. Sensors, annunciation window for fault indication, complete as per specification	Each	388843
27.5.5	Incomings 630A, 50kA 4 Pole MCCB, Ics=100% Icu Rating Digital Voltmeter 0-500V with selector switch Ammeter (0-630 A) with selector switch & CTs etc. LED type RYB phase indicating lamps, ON, OFF, trip indicating lamps Set of Copper Bus Bar 800Amps Outgoings (Note : All outgoing feeders for pumps should have digital Ammeter with selector switches, and LED type ON, OFF, trip indicating lamps) Main Fire Pump 200 Amp, 50kA TPN MCCB, Ics=100% Icu, with fully automatic Star/Delta starter suitable for 75 hp pump with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation. - 2 sets Jockey Pump 100 Amp, 50kA TPN MCCB, Ics=100% Icu, with Suitable HP fully automatic Star/Delta starter with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation. - 2 sets Diesel Engine Control Control for diesel engine comprising - Automatic/Manual selector switch & 3 attempts starting device, timers and relays as required, push buttons, start/stop in manual mode Indicating lamp for high/ Low Lub. Oil pressure, High Water Temp and Engine on indication Battery charger suitable for 12V/24 V DC with boost and trickle selector switch, 0-30 V DC volt meter, and 0-20 A DC Ammeter All standard relays and accessories for automatic operation of diesel engine System Controller		

Designing, Supply, Installation, Testing and commissioning of system controller to control operation of main electric fire pump, diesel pump, Pressurization pump, Terrace pump in sequence as per specification consisting of relays, timers. Sensors, annunciation window for fault indication, complete as per specification

Each 469546

27.5.6

Incomings

800A, 50kA 4 Pole MCCB, Ics=100% Icu Rating

Digital Voltmeter 0-500V with selector switch

Ammeter (0-800 A) with selector switch & CTs etc.

LED type RYB phase indicating lamps, ON, OFF, trip indicating lamps

Set of Copper Bus Bar 1000Amps

Outgoings

(Note : All outgoing feeders for pumps should have digital Ammeter with selector switches, and LED type ON, OFF, trip indicating lamps)

Main Fire Pump

250 Amp, 50kA TPN MCCB, Ics=100% Icu, with fully automatic Star/Delta starter suitable for 100 hp pump with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation. - 2 sets

Jockey Pump

125 Amp, 50kA TPN MCCB, Ics=100% Icu, with Suitable HP fully automatic Star/Delta starter with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation. - 2 sets

Diesel Engine Control

Control for diesel engine comprising -

Automatic/Manual selector switch & 3 attempts starting device, timers and relays as required, push buttons, start/stop in manual mode

Indicating lamp for high/ Low Lub. Oil pressure, High Water Temp and Engine on indication

Battery charger suitable for 12V/24 V DC with boost and trickle selector switch, 0-30 V DC volt meter, and 0-20 A DC Ammeter

Each 601606

PIPES & ACCESSORIES

27.6	Providing laying, testing & commissioning of 'C' class heavy duty MS Pipe conforming to IS 1239/3589 i/c fittings like elbows, tees, flanges, tapers, nuts bolts, gaskets etc. in ground including welding, excavation & providing cement concrete blocks as supports, anticorrosive treatment with coaltar/asphalt tape as per IS 10221, refilling the trench etc. of following sizes complete as required.		
27.6.1	200 mm. dia (wall thickness = 6.3 mm)	Meter	5940
27.6.2	150 mm. Dia	Meter	4104
27.7	Providing, laying, testing & commissioning of 'C' class heavy duty MS pipe conforming to IS 3589/IS 1239 including Welding, fittings like elbows, tees, flanges, tapers, nuts bolts, gaskets etc. and fixing the pipe on the wall/ceiling with suitable clamp/support frame and painting with two or more coats of synthetic enamel paint of required shade complete as required :		
27.7.1	25 mm dia	Meter	852
27.7.2	32 mm dia	Meter	965
27.7.3	40 mm dia	Meter	1177
27.7.4	50 mm dia	Meter	1436
27.7.5	65 mm dia	Meter	1805
27.7.6	80 mm dia	Meter	2088
27.7.7	100 mm dia	Meter	2800
27.7.8	150 mm dia	Meter	3886
27.7.9	200 mm dia (wall thickness 6.3 mm)	Meter	5882
27.7.10	250 mm dia (wall thickness 6.3 mm)	Meter	7131
27.7.11	300 mm dia (wall thickness 7.1 mm)	Meter	4674
27.8	Providing, laying, testing & commissioning of 'B' class medium duty G.I. pipe conforming to IS 1239 including welding, fittings like elbows, tees, flanges, tapers, nuts, bolts, gaskets etc. and fixing the pipe on the wall/ceiling with suitable clamp/support frame and painting with two or more coats of synthetic enamel paint of required shade complete as required :		

27.8.1	25 mm	Meter	711
27.8.2	32 mm	Meter	785
27.8.3	40 mm	Meter	966
27.8.4	50 mm	Meter	1066
27.8.5	65 mm	Meter	1330
27.8.6	80 mm	Meter	1514
27.8.7	100 mm	Meter	1925
27.8.8	150 mm	Meter	2633
27.9	Supplying and fixing single headed internal hydrant valve with instantaneous Gunmetal/Stainless Steel coupling of 63 mm dia with cast iron wheel ISI marked conforming to IS 5290 (Type -A) with blank Gunmetal/Stainless Steel cap and chain as required :		
27.9.1	Single headed Gunmetal	Each	8208
27.9.2	Single headed Stainless steel	Each	6632
27.10	Supplying and fixing Single headed external yard hydrant valve with 1 No. 63 mm dia instantaneous FM Gunmetal/Stainless Steel coupling and cast iron wheel, ISI marked, conforming to IS 5290 (type A) with blank Gunmetal/Stainless Steel cap and chain as required :		
27.10.1	Single headed Gunmetal	Each	8208
27.10.2	Single headed Stainless steel	Each	6632
27.11	Supplying, fixing, testing and commissioning of butterfly valve of PN 1.6 rating with bronze/gunmetal seat duly ISI marked complete with nuts, bolts, washers, gaskets conforming to IS 13095 of following sizes as required :		
27.11.1	40 mm dia	Each	3870
27.11.2	50mm dia	Each	4114
27.11.3	65 mm dia	Each	4712
27.11.4	80 mm dia	Each	5406

27.11.5	100 mm dia	Each	7271
27.11.6	150 mm dia	Each	9737
27.11.7	200 mm dia	Each	16248
27.11.8	250 mm dia	Each	21721
27.11.9	300 mm dia	Each	26141
27.12	Supplying, fixing, testing & commissioning of double flanged sluice valve of rating PN 1.6 with non rising spindle, bronze/gun metal seat, ISI marked complete with nuts, bolts, washers, gaskets and conforming to IS 780 of following sizes as required :		
27.12.1	40mm dia	Each	6073
27.12.2	50 mm dia	Each	7639
27.12.3	65mm dia	Each	9797
27.12.4	80mm dia	Each	11314
27.12.5	100mm dia	Each	15816
27.12.6	150mm dia	Each	23968
27.12.7	200mm dia	Each	41386
27.12.8	250mm dia	Each	48894
27.12.9	300mm dia	Each	68961
27.13	Supplying and fixing orifice plate made out of 6 mm thick stainless steel (Grade 304) with orifice of required size to be fitted between flange & landing valve of external and internal hydrants to reduce pressure at the outlet to the level of 3.5 kg/cm ² complete as required.		
		Each	1436
27.14	Providing, installation, testing and commissioning of non-return valve of following sizes confirming to IS: 5312 complete with rubber gasket, GI bolts, nuts, washers etc.as required :		
27.14.1	40mm dia	Each	6257
27.14.2	50mm dia	Each	6257

27.14.3	65mm dia	Each	6955
27.14.4	80mm dia	Each	8255
27.14.5	100mm dia	Each	11853
27.14.6	125mm dia	Each	16929
27.14.7	150mm dia	Each	18884
27.14.8	200mm dia	Each	30896
27.14.9	250mm dia	Each	47622
27.14.10	300mm dia	Each	63466
27.15	Providing, installation, testing and commissioning of stainless steel Y-strainer fabricated out of 1.6 mm thick stainless steel, Grade 304, sheet with 3 mm dia holes with stainless steel flange.		
27.15.1	80 mm dia	Each	5117
27.15.2	100mm dia	Each	7270
27.15.3	150mm dia	Each	11682
27.15.4	200mm dia	Each	23571
27.15.5	250mm dia	Each	28985
27.15.6	300mm dia	Each	38558
27.16	Supplying and fixing 63 mm dia, 15 m long RRL hose pipe with 63 mm dia male and female couplings duly bound with GI wire, rivets etc. conforming to IS 636 (type-A) as required :		
27.16.1	Gun Metal	Each	5455
27.16.2	Stainless Steel (Grade 304)	Each	4667
27.17	Supplying and fixing first-aid Hose Reel with MS construction spray painted in post office red, conforming to IS 884 complete with the following as required. (a) 20 mm nominal internal dia water hose thermoplastic (Textile reinforced) type -2 as per IS: 12585 (b) 20 mm nominal internal dia gun metal globe valve & nozzle.		

(c) Drum and brackets for fixing the equipments on wall.

(d) Connections from riser with 25 mm dia stop gun metal valve & M.S. Pipe and socket.

27.17.1	30 m	Each	9440
27.17.2	40 m	Each	12991
27.18	Supplying & fixing 63 mm dia gun metal short branch pipe with 20 mm nominal internal diameter size nozzle conforming to IS 903 suitable for instantaneous connection to interconnect hose pipe coupling as required:		
27.18.1	Gun metal	Each	4977
27.18.2	Stainless Steel (Grade 304)	Each	3402
27.19	Supplying and fixing of fire brigade connection of cast iron body with gun metal male instantaneous inlet couplings complete with cap and chain as reqd. for suitable dia MS pipe connection conforming to IS 904 as required:		
27.19.1	2 way-100 mm dia M.S. Pipe	Each	7262
27.19.2	4 way - 150 mm dia M.S. Pipe	Each	14820
27.19.3	3 way - 150 mm dia M.S. Pipe	Each	10097
27.20	Supplying and fixing air vessel made of 250 mm dia, 8 mm thick MS sheet, 1200 mm in height with air release valve on top and flanged connection to riser, drain arrangement with 25 mm dia gun metal wheel valve with required accessories, pressure gauge and painting with synthetic enamel paint of approved shade as required.		
27.21	Providing, fixing, testing & commissioning of 15mm dia quartzoid bulb type sprinklers of rating 68 degree centigrade with required accessories :		
27.21.1	Pendent Sprinkler	Each	599
27.21.2	Upright Sprinkler	Each	599
27.21.3	Horizontal side wall sprinkler	Each	699
27.21.4	Concealed sprinkler	Each	1553
27.22	Providing & fixing of pressure switch in M.S. pipe line including connection etc. as required.	Each	1670

27.23	Providing & fixing flow switch in following sizes M.S. pipe including connection etc as required.		
27.23.1	100mm dia	Each	8069
27.23.2	150mm dia	Each	9644
27.24	Providing, fixing, testing & commissioning of installation control valve of cast iron body, brass/bronze working parts comprising of water motor alarm, bronze seat clapper, clapper arm and hydraulically driven mechanical gong bell to sound continuous alarm when the wet riser/sprinkler system activates, pressure gauges, emergency releases, strainer, pressure switch, cock valve complete with drain valve and bypass, test control box, ball valves, MS pipe of required size, flanges, orifice plate, gasket etc of follwing sizes as required:		
27.24.1	80mm dia	Each	45052
27.24.2	100mm dia	Each	47460
27.24.3	150mm dia	Each	50805
27.25	Supplying, installation, testing & commissioning of sprinkler flexible pipe (UL Listed) of stainless steel complete with 15 NPT on reducer thread with maximum working pressure of 175 PSI test pressure of 875 PSI (Burst) with branch line (Inlet) 25mm NPT male thread to sprinkler head (Outlet) 15mm NPT female thread with reducer, nipple, 2 side brackets, center bracket, stockbar of following sizes complete as required.		
27.25.1	700mm	Each	1495
27.25.2	1000mm	Each	1703
27.25.3	1200mm	Each	1800
27.25.4	1500mm	Each	1969
27.26	Supplying, installation, testing & commissioning of deluge valve (to be used for water curtains) UL listed with wet pilot basic trim assembly for DVA with test and alarm, drip and drain valve with water motor gong complete as required.		
27.26.1	50 mm dia	Each	87237
27.26.2	80 mm dia	Each	87862
27.26.3	100mm dia	Each	98253

27.26.4	150mm dia	Each	121301
27.27	Providing, installation, testing & commissioning of water curtain nozzle 15mm dia nickel chrome plated brass, BSPT, K-factor as per design consideration complete as required.	Each	787
27.28	Providing, installation, testing & commissioning of electric release trim assembly for deluge wall with two way solenoid valve, weather proof operation on 24 volts DC supply complete as required.	Each	7755
27.29	Providing, installation, testing & commissioning of adjustable rosette plate for 15mm dia in white finish UL Listed or FM approved complete as required.	Each	264
Note:	For Cabling, earthing and cable trays please refer to DSR (E&M) For digging of hard & soft soil please refer to DSR (Civil)		

CHAPTER 28

FIRE DETECTION AND ALARM SYSTEM

Item No.	Description	Unit	Rate(₹)
28.1	FIRE DETECTION AND ALARM SYSTEM (CONVENTIONAL)		
28.1.1	Supplying, installation, testing & commissioning of heat detector operating at 54°C/57°C with rate of rise cum fixed temperature (dual thermistor) type with mounting base complete with all connection etc. as required.	Each	1136
28.1.2	Supplying, installation, testing & commissioning of smoke detector with builtin LED and mounting base complete with all connections etc. as required.	Each	1223
28.1.3	Supplying, installation, testing & commissioning of manual call boxes of MS construction in surface/recess with stainless steel chain & hammer assembly complete with glass and push button etc. as required.	Each	513
28.1.4	Supplying, installation, testing & commissioning of manual call box of ABS type in surface/recess with stainless steel chain & hammer assembly complete with glass and push button etc. as required.	Each	444
28.1.5	Supplying, installation, testing & commissioning response indicator on surface/recess MS box having two LEDs metallic cover complete with all connections etc. as required.	Each	222
28.1.6	Supplying, installation, testing & commissioning fire alarm sounder with facility to make announcement, mounted in M.S. box (16 SWG) with hinged cover plate & suitable for operation with amplifier i/c line matching transformer etc. complete as required.	Each	600
28.1.7	Supplying, installation, testing & commissioning fire alarm sounder with facility to make announcement, mounted in A.B.S. box with hinged cover plate & suitable for operation with amplifier i/c line matching transformer etc. complete as required.	Each	530
28.1.8	Supplying, installation, testing & commissioning talk back slave station in surface/recess suitable for operation on simplex mode complete with P.T.T. knob & speaker/microphone enclosed in a M.S.(16 SWG)/ ABS box with break glass in front etc. complete as required.	Each	963

28.1.9	Supplying, installation, testing & commissioning sector panel suitable for following zones, complete with visual indications for short circuit fault, open circuit fault, fire condition and all other standard facilities as per IS:2189 with mimic diagram for all area/zone covered, complete with all connections, interconnections as required.		
28.1.9.1	4 Zone	Each	9808
28.1.9.2	6 Zone	Each	13269
28.1.9.3	10 Zone	Each	17780
28.1.10	Supplying, installation, testing & commissioning of main control and indicating panel made out of 16 SWG MS sheet to accommodate the following items duly powder coated in approved colour with louvers for ventilation, locking arrangement, audio and visual indication for fire alarm and public address system, monitoring system including connections, interconnections etc complete as required. 10 Zone panel for fire alarm system 250 Watt amplifier racks suitable for operation on 230V AC/24V DC supply conforming to IEC-268-3 complete with all accessories as required - 2 Nos.(one to act as standby) Talk back master station with LED PTT (press to talk) push button for operation on 230V AC/24 V DC supply conforming to IEC-268 for simplex mode of operation/communication suitable for 20 Nos.talk back unit -1 set. Announcement control desk suitable for selection of different zones selectively and ON ALL CALL switch with visual indication etc. complete as required -1 set. Amplifier change over switch for inter changing amplifier -1 No. Monitor panel for loudspeaker complete with output selector, ON/OFF switch, fuse, visual indications etc. complete as required- 1 No. Gooseneck microphone with stand and ON/OFF switch- 1 No. Main ON/OFF switch, fuse indication lamps, DC and AC voltmeters & ammeters, terminal blocks etc. complete as required -1 set. Battery charger trickle cum boost to take complete load of fire alarm & PA system complete with all accessories including providing & fixing of 2 nos.12 volt, 60 AH each sealed maintenance free batteries -1 set.	Each	10033

28.1.11 Supplying, installation, testing & commissioning of main control and indicating panel made out of 16 SWG MS sheet to accommodate the following items duly powder coated in approved colour with louvers for ventilation, locking arrangement, audio and visual indication for fire alarm and public address system, monitoring system including connections, interconnections etc complete as required.

6 zone panel for fire alarm system

100 watt amplifier racks suitable for operation on 230V AC/24V DC supply conforming to IEC-268-3 complete with all accessories as required- 2 Nos.(one to act as standby)

Talk back master station with LED PTT (press to talk) push button for operation on 230V AC/24 V DC supply conforming to IEC-268 for simplex mode of operation/communication suitable for 20 Nos.talk back unit -1 set.

Announcement control desk suitable for selection of different zones selectively and ON ALL CALL switch with visual indication etc., complete as required- 1 set.

Amplifier change over switch for inter changing amplifier- 1 No.

Monitor panel for loudspeaker complete output selector ON/OFF switch, fuse visual indications etc. complete as required- 1 No.

Gooseneck microphone with stand and ON/OFF switch -1 No.

Mains ON/OFF switch, fuse indication lamps, DC and AC voltmeters & ammeters terminal blocks etc. complete as required -1 set.

Battery charger trickle cum boost to take complete load of fire alarm & PA system complete with all accessories including providing & fixing of 2 nos. 12 volt, 30 AH each sealed maintenance free batteries -1 set.

Each 52258

28.2 INTELLIGENT FIRE ALARM SYSTEM

28.2.1 Supplying, installation, testing and commissioning of micro processor based intelligent addressable main fire alarm panel, central processing unit with the following loop modules and capable of supporting not less than 240 devices (including detectors) and minimum 120 detectors per loop and loop length up to 2 km, network communication card, minimum 320 character graphics/ LCD display with touch screen or other keypad and minimum 4000 events history log in the non volatile memory (EPROM), power supply unit (230 $\pm$ 5 % V, 50 hz), 48 hrs back-up with 24 volt sealed maintenance free batteries with automatic charger. The panel shall have facility to connect printer to printout log and facility to have seamless integration with analog/digital voice evacuation system (which is part of the schedule of work under SH: PA System) and shall be complete with all accessories . The panel shall be compatible for IBMS

system with open protocol BACnet/ Modbus over IP complete as per specifications.

28.2.1.1	Ten Loop Panel.	Each	476893
28.2.1.2	Two Loop Panel.	Each	250420
28.2.2	Supplying, installation, testing & commissioning of central graphical fire alarm management system to centrally monitor and operate the fire alarm system complete as required.	Each	210264
28.2.3	Supplying, installation, testing & commissioning of repeater panel with 320 character/ Touch screen LCD display with inbuilt reset, acknowledge and silence switches complete as required.	Each	113734
28.2.4	Supplying, installation, testing & commissioning of intelligent analog addressable photothermal detector complete with mounting base complete as required.	Each	3004
28.2.5	Supplying, installation, testing & commissioning of response indicator on surface/recessed MS Box having two LED, metallic cover complete with all connections etc as required.	Each	306
28.2.6	Supplying, installation, testing & commissioning of intelligent addressable programmable sounder complete as required.	Each	2787
28.2.7	Supplying, installation, testing & commissioning of fault isolator complete with base as required.	Each	3434
28.2.8	Supplying, installation, testing & commissioning of intelligent aspiration detector for area coverage of minimum 5000 sq. ft. complete as required.	Each	309044
28.2.9	Supplying, installation, testing & commissioning of intelligent addressable thermal detector with rate of rise cum fixed temperature thermistor complete with base as required.	Each	2866
28.2.10	Supplying, installation, testing & commissioning of addressable fire control module complete as required.	Each	3156
28.2.11	Supplying, installation, testing & commissioning of addressable phone control module complete as required.	Each	3432
28.2.12	Supplying, installation, testing & commissioning of addressable beam detector with short circuit isolator (inbuilt or separate) complete with emitter and receiver including connections with remote test features etc complete as required.	Each	78525

28.2.13	Supplying, installation, testing & commissioning of intelligent addressable duct detector including suitable Photo detector complete with base as required.	Each	8183
28.2.14	Supplying, installation, testing & commissioning of addressable manual call point complete as required.	Each	4063
28.2.15	Supplying, installation, testing & commissioning of addressable horn cum strobe complete as required.	Each	3682
28.2.16	Supplying, installation, testing & commissioning of addressable strobe complete as required.	Each	3536
28.2.17	Supplying, installation, testing & commissioning of fire fighter telephone handset complete as required.	Each	6016
28.2.18	Supplying, installation, testing & commissioning of intelligent interface unit BACnet/ Modbus protocol i.e. supplying communication links between building management system and fire alarm control panel complete as required.	Each	196207
28.2.19	Supplying, installation, testing & commissioning of fire fighter phone jack complete as required.	Each	1689
28.3	PUBLIC ADDRESS SYSTEM		
28.3.1	Supplying, installation, testing & commissioning of 6 zone, voice alarm controller with USB, MP3 player (including 6 zone button paging station) with seamless integration facility with main fire alarm panel for voice evacuation complete as required.	Each	132115
28.3.2	Supplying, installation, testing & commissioning of 1.5/3/6W ceiling speaker complete as required.	Each	1025
28.3.3	Supplying, installation, testing & commissioning of 1.5/3/6W metal box ceiling/wall speakers complete as required.	Each	1891
28.3.4	Supplying, installation, testing & commissioning of ceiling/wall mounted loud speaker, 3/1.5 Watt in ABS enclosure complete as required.	Each	2555
28.3.5	Supplying, installation, testing & commissioning of 6 inches dia, 2 watts, 70/100 volts ceiling speaker complete as required.	Each	1939
28.3.6	Supplying, installation, testing & commissioning of digital audio amplifier 50 Watt, 25V rms operating at 240 volt AC supply complete as required.	Each	101150

28.3.7	Supplying, installation, testing & commissioning of digital audio amplifier 75 Watt, 25V rms operating at 240 Volt AC Supply complete as required.	Each	152368
28.3.8	Supplying, installation, testing & commissioning of exit point directional sound speaker with voice and integral audio amplifier with selectable sound pulse patterns complete as required.	Each	10618
28.3.9	Supplying, installation, testing & commissioning of Voice command keypad 6 zone, with microphone assembly complete as required.	Each	85450
28.4	ASPIRATION DETECTOR ACCESSORIES		
28.4.1	Supply, installation, testing and commissioning of 25 mm Outer dia CPVC PIPE with end caps including making air sampling opening of appropriate dia on appropriate interval and all accessories as required,(rate for area protected in sqm).	sq. m	65
28.5	CABLING & WIRING		
28.5.1	Supplying & laying of 2x1.5 sqmm fire survival armoured cable, 600/1000V rated with annealed copper conductor having glass mica fire barrier tape covered by an extruded layer of Cross Linkable Ethylene Propylene Rubber (EPR) insulation and LSZH inner bedding, steel wire armouring & LSZH outer sheath complete as required.	Meter	386
28.5.2	Supplying & laying of 2x1.5 sqmm fire alarm armoured cable, 600/1000V rated with annealed copper conductor having XLPE insulation, steel wire armouring & FRLS/HFFR outer sheath complete as required.	Meter	192
28.5.3	Supplying and drawing of cable Fire Retardant PVC insulated copper conductor cable in the existing surface / recessed steel conduit of following pairs, cores and size including connections and interconnections etc. as required.		
28.5.3.1	speaker cable Single pair, 2-core, 1.5 sqmm	Meter	61
28.5.3.2	speaker cable Two pair, 2-core, 1.5 sqmm	Meter	97
28.5.3.3	speaker cable Three pair, 2-core, 1.5 sqmm	Meter	134
28.5.3.4	speaker cable Four pair, 2-core, 1.5 sqmm	Meter	171
28.5.4	Supplying and fixing 25 mm dia MS flexible pipe with PVC coating along with all ancillaries and accessories like coupler etc. as required.	Meter	60
	CONDUITING AND WIRING		
	Please refer to CPWD Specification for internal electrical works		

CHAPTER 29

DIESEL GENERATOR SET

Item No.	Description	Unit	Rate(₹)
29.1	Supply, installation, Testing & Commissioning of 'Silent Type Diesel Generating set as per CPCB IV + or better norms along with having Prime Power Rating of KVA as below, 415 volts at 1500 RPM, 0.8 lagging power factor at 415 V suitable for 50 Hz, 3 phase system & for 0.85 Load Factor, including testing at factory and site with fuel, load for test and other necessary arrangements Complete as per CPWD specifications, should have QR code which should contain drawing, test report OEM manual, Geo- Tag of manufacturing location, rating plate as per relevant IS Code etc. and consisting of the followings: (A) Diesel Engine: Tourbocharged Diesel engine 4 stroke water cooled, multi cylinder, dynamically balanced fly wheel, electric start of suitable BHP at 1500 RPM suitable for above output of alternator at 40 Degree C, 50% RH & at 1000 Meter MSL , capable of taking 10% over loading for one hour after 12 hours of continuous operation. The engine will be with Electronic governor, Dry type Air filter with service indicator, first filling of engine fuel (after commissioning) lubricating Oil, Coolant and other consumables complete with all the required accessories, the Electronic governor shall be as per ISO 8528. The engine shall comply to the latest CPCB norms (CPCB IV + or better) and Conforming to BS 5514, BS 649, IS 10000, IS 10002, IS 13018 and as per CPWD specifications. (B) Engine mounted Instrument Panel fitted with and having digital display for following: (i) Start-stop switch with key (ii) Water temperature indication (iii) Lubrication oil pressure indication (iv) Lubrication oil temperature indication (v) Battery charging indication and Voltage indication (vi) RPM indication (vii) Over speed indication (viii) Low lubrication Oil trip indication (ix) Engine Running Hours indication (x) Fuel Level (C) Alternator: Synchronous alternator rated of appropriate KVA, 415 volts at 1500 RPM, 3 phase 50 Hz, AC supply with 0.8 lagging power factor at 40 Degree C, 50% RH & at 1000 Meter MSL. The alternator shall be having Screen Protected Drip Proof (SPDP) enclosure IP23, brushless, continuous duty, dynamically		

balanced rotor, capable of taking 10% over loading for one hour after 12 hours of continuous operation, self cooled, self-excited and self-regulated through AVR conforming to IS13364(Part 2)/IS: 4722/BS 2613 suitable for tropical conditions and with class- H insulation.

(D) Base Frame & Foundation:

Both the engine and alternator shall be mounted on suitable base frame made of MS channel with necessary reinforcement which shall be installed on suitable cement concrete foundation and vibration isolation arrangement as per recommendations of manufacturer.

(E) FUEL TANK:

Daily service fuel tank of suitable liters capacity as per CPWD Specifications, fabricated out of 3 mm thick M.S. sheet complete with all standard accessories and fuel piping between fuel tank and diesel engine with MS class 'C' pipes of suitable dia. Complete with valves, level indications & accessories as required as per specifications.

(F) Exhaust System:

Dry exhaust manifold with hospital type exhaust silencer and catalytic convertor.

(G) Starting System:

12V/24V DC starting system comprising of starter motors: voltage regulator and arrangement for initial excitation complete with suitable numbers of batteries (180 AH capacity lead acid SMF type) as required as per specifications. The battery shall be housed inside the acoustic enclosure of DG Set.

(H) Acoustic and weather proof enclosure with arrangement for fresh air intake for cooling of the engine & alternator, extraction, discharging hot air in to the atmosphere and the temperature rise inside the enclosure, noise level outside enclosure. The acoustic enclosure should be suitable for cable connection/connection through bus-trunking. Such arrangements on acoustic enclosure should be water proof & dust-proof conforming to IP-65 protection. The enclosure shall be as per CPCB IV + or better norms etc. and as per CPWD specifications.

(I) AMF Panel:

Free standing floor mounted IP 42 automatic mains failure control panel including auto by-pass, suitable for KVA as below for silent type DG set complete with relays, timers, set of CTs for metering & protection and energy analyser to indicate currents, phase and line voltages, frequency, power factor, KWH, Kilo Volt Ampere Reative Hour (KVARH), KVA (Phase & Total), KW & provision for overload, short circuit, restricted earth fault, under frequency, power (aluminium) and control (copper)

cabling of suitable size upto 15 meter between AMF panel, LT Panel and DG Set including connection interconnection etc. as required, all complete and inter locking and communication/ Ethernet /RS485/SNMP port open protocol for BMS integration including suitable software, the panel shall be of DG Set OEM make etc. as per approved by Engineering in charge and including the following:

1. Suitable numbers and appropriate capacity 4 pole motorised electrically operated draw out with cradle type 3 position ACB/ MCCB with electronic release for O/C & E/F and shunt trip.
2. Auto/Manual/Test/Off selector switch
3. Protection for under and over voltage phase reversal (2 nos Over voltage relay, 2 Nos. reverse power relay and 2 Nos. under voltage relay).
4. 3 Sets of current transformers 15 P 10 accuracy for protection and 15 VA class-I for metering
5. Energy analyser unit to indicate current, Voltage(L-N & L_L), kW, kVA (Phase & Total), Frequency, KWH, PF.
6. LED Indicating lamps for load on mains and load on set
7. Fuse/ MCB for instruments
8. Battery charger, complete with transformer/ rectifier, D.C. voltmeter and ammeter, selector switch for trickle, off and boost and current adjustment.
9. Main supply failure monitor
10. Supply failure timer
11. Restoration timer
12. Control unit with three impulse automatic engine start/stop and failure to start lockout.
13. Impulse counter with locking and reset facility.
14. ON/OFF/Control circuit switch with indicator
15. Audio/Video annunciation for
 - (i) High water temperature
 - (ii) Low lubricating oil pressure
 - (iii) Engine over speed
 - (iv) Engine fails to start
 - (v) Full load/maximum load warning
16. Protection for over/under Frequency, Loss of AC sensing, Over Current, Unbalancing load with suitable number of relays and accessories
17. Maintenance notification based on Engine Run Hour & due date.

18. Load Management through PLC to achieve auto opening and closing of incomer breakers, bus coupler switching of essential panel , interlocking providing signal to AMF Panel for load status and AMF shall give command to DG Set to auto start / auto stop depending upon load status and requirement etc. and necessary hardware and software required to perform the operation shall be provided by the contractor including all control wiring.

29.1.1	25KVA	Each	403101
29.1.2	35KVA	Each	446290
29.1.3	40KVA	Each	532669
29.1.4	50KVA	Each	565061
29.1.5	62.5KVA	Each	597453
29.1.6	82.5KVA	Each	741418
29.1.7	100KVA	Each	863788
29.1.8	125KVA	Each	885383
29.1.9	160KVA	Each	1249613
29.1.10	200KVA	Each	1547620
29.1.11	250KVA	Each	1871541
29.1.12	320KVA	Each	2447399
29.1.13	380KVA	Each	2843302
29.1.14	400 KVA	Each	2951276
29.1.15	500KVA	Each	3455152
29.1.16	625KVA	Each	5038764
29.1.17	750KVA	Each	6406428

CHAPTER 30

HVAC(ONLY PLUMBING,DUCTING & AHU)

Item No.	Description	Unit	Rate(₹)
	PLUMBING		
	INSULATED CHILLED WATER PIPING (resin bonded fiber glass glass INSULATION)		
30.1	Supplying, laying/ fixing, testing and commissioning of following nominal sizes of chilled water piping inside the building (with necessary clamps, vibration isolators and fittings but excluding valves, strainers, gauges etc.) duly insulated with 80 kg/cum density resin bonded fiber glass or 144 Kg/ cu m density mineral wool (non Combustible) pipe section insulation covered with a layer of 120 gm/sqm polythene sheet (vapour barrier) and finally applying 0.63mm aluminium sheet cladding complete with type3 , grade 1 roofing feltstrip(as per IS:1322 as amended up to date) at joints and repairing of damage to building etc. as per specifications and as required. Note: The Pipes of sizes 150mm & below shall be M.S. 'C' class as per IS : 1239 and pipes size above 150mm shall be welded black steel pipe heavy class as per IS: 3589, from minimum 6.35mm thick M.S. Sheet for pipes upto 350 mm dia. and from minimum 7mm thick MS sheet for pipes of 400 mm dia and above.		
30.1.1	400mm dia.(75 mm thick insulation)	Meter	22662
30.1.2	350 mm dia.(75 mm thick insulation)	Meter	13512
30.1.3	300 mm dia.(75 mm thick insulation)	Meter	11875
30.1.4	250mm dia.(75 mm thick insulation)	Meter	10027
30.1.5	200mm dia.(75 mm thick insulation)	Meter	8415
30.1.6	150mm dia. (50mm thick insulation)	Meter	5578
30.1.7	125mm dia.(50mm thick insulation)	Meter	4742
30.1.8	100mm dia. (50mm thick insulation)	Meter	3877
30.1.9	80mm dia. (50mm thick insulation)	Meter	3183
30.1.10	65mm dia.(50mm thick insulation)	Meter	2704
30.1.11	50mm dia.(50mm thick insulation)	Meter	2251

30.1.12	40mm dia. (50mm thick insulation)	Meter	1936
30.1.13	32 mm dia (50mm thick insulation)	Meter	1695
30.1.14	25 mm dia (50mm thick insulation)	Meter	1534
30.2	INSULATED CHILLED WATER PIPING (Nitrile rubber insulation) Supplying, laying/ fixing, testing and commissioning of following nominal sizes of chilled water piping inside the building (with necessary clamps, vibration isolators and fittings but excluding valves, strainers, gauges etc.) duly insulated with following closed cell elastometric nitrile rubber of minimum 45 Kg / cu m density, thermal conductivity 0.037 W/MK or better at 20 deg mean temperature class 'O' insulation applied by suitable adhesive complete including repairing of damage to building etc. as per specifications and as required complete in all respect. Note:-The Pipes of sizes 150mm & below shall be M.S. 'C' class as per IS : 1239 and pipes size above 150mm shall be welded black steel pipe heavy class as per IS: 3589, from minimum 6.35mm thick M.S. Sheet for pipes upto 350 mm dia. and from minimum 7mm thick MS sheet for pipes of 400 mm dia and above.		
30.2.1	400mm dia.(32 mm thick insulation)	Meter	19846
30.2.2	350mm dia.(32 mm thick insulation)	Meter	10921
30.2.3	300mm dia. (32 mm thick insulation)	Meter	9663
30.2.4	250mm dia.(32 mm thick insulation)	Meter	8276
30.2.5	200mm dia.(32 mm thick insulation)	Meter	6728
30.2.6	150mm dia. (32 mm thick insulation)	Meter	4676
30.2.7	125mm dia.(32 mm thick insulation)	Meter	4105
30.2.8	100mm dia. (32 mm thick insulation)	Meter	3279
30.2.9	80mm dia.(32 mm thick insulation)	Meter	2467
30.2.10	65mm dia(32 mm thick insulation)	Meter	1989
30.2.11	50mm dia(32 mm thick insulation)	Meter	1738

30.2.12	40mm dia. (32 mm thick insulation)	Meter	1428
30.2.13	32 mm dia(19 mm thick insulation)	Meter	1252
30.2.14	25 mm dia(19 mm thick insulation)	Meter	1042
30.3	INSULATED CHILLED WATER PIPING (expanded polystyrene insulation) Supplying, laying/ fixing, testing and commissioning of following nominal sizes of chilled water piping plumbing inside the building (with necessary clamps, vibration isolators and fittings but excluding valves, strainers, gauges etc.) duly insulated with fire retardant quality expanded polystyrene moulded pipe section of density 20 kg/cu.m after a thick coat of cold setting adhesive (CPRX compound) wrapping with 500g polythene faced hessain and finally applying 0.63mm aluminium sheet cladding complete with type3 , grade 1 roofing feltstrip(as per IS:1322 as amended up to date) at joints repairing of damage to building etc. as per specifications and as required complete in all respect. Note:-The Pipes of sizes 150mm & below shall be M.S. 'C' class as per IS : 1239 and pipes size above 150mm shall be welded black steel pipe heavy class as per IS: 3589, from minimum 6.35mm thick M.S. Sheet for pipes upto 350 mm dia. and from minimum 7mm thick MS sheet for pipes of 400 mm dia and above.		
30.3.1	400mm dia.(75 mm thick insulation)	Meter	21473
30.3.2	350 mm dia.(75 mm thick insulation)	Meter	12391
30.3.3	300 mm dia.(75 mm thick insulation)	Meter	11020
30.3.4	250mm dia.(75 mm thick insulation)	Meter	9291
30.3.5	200mm dia (75 mm thick insulation)	Meter	7810
30.3.6	150mm dia. (75 mm thick insulation)	Meter	5289
30.3.7	125mm dia.(50mm thick insulation)	Meter	4445
30.3.8	100mm dia.(50mm thick insulation)	Meter	3819
30.3.9	80mm dia. (50mm thick insulation)	Meter	3020
30.3.10	65mm dia.(50mm thick insulation)	Meter	2566

30.3.11	50mm dia.((50mm thick insulation)	Meter	2143
30.3.12	40mm dia. (50mm thick insulation)	Meter	1845
30.3.13	32 mm dia (50mm thick insulation)	Meter	1613
30.3.14	25 mm dia (50mm thick insulation)	Meter	1469

INSULATION ON EXISTING PIPE

30.4	Supplying, laying/ fixing, testing and commissioning of following thickness resin bonded fiber glass pipe section insulation having density 80 kg/cum or mineral wool (non combustible) having density of 144 Kg/ cu m duly covered with a layer of 120 gm/sqm polythene sheet (vapour barrier) on existing pipe and finally applying 0.63mm aluminium sheet cladding complete with type3 , grade 1 roofing feltstrip(as per IS:1322 as amended up to date) at joints etc. as per specifications and as required.		
30.4.1	75mm	per sqm	3240
30.4.2	65 mm	per sqm	2669
30.4.3	50mm	per sqm	2866
30.4.4	40mm	per sqm	2640
30.4.5	25mm	per sqm	2137
30.5	Supplying, laying,fixing, testing and commissioning of following thickness closed cell elastometric nitrile rubber of class 'O' applied by suitable adhesive , as per specifications and as required complete in all respect.		
30.5.1	32mm	per sqm	1203
30.5.2	19mm	per sqm	724
30.6	Supplying, laying/ fixing of following thickness insulation with fire retardant quality expanded polystrene moulded pipe section of density 20 kg/cu.m after a thick coat of cold setting adhesive (CPRX compound) wrapping with 500g polythene faced hessain and finally applying 0.63mm aluminium sheet cladding complete with type3 , grade 1 roofing feltstrip(as per IS:1322 as amended up to date) at joints repairing etc. as per specifications and as required complete in all respect..		

30.6.1	75mm	per sqm	1502
30.6.2	50mm	per sqm	1367
INSULATED VALVES			
30.7	Supplying, fixing, testing and commissioning of following valves, strainers, gauges in the chilled water plumbing duly insulated to the same specifications as the connected piping and adequately supported as per specifications.		
30.7.1	BUTTERFLY VALVE (MANUAL) with C I body SS Disc, Nitrile Rubber Seal & O- Ring PN 16 pressure rating for chilled water/hot water circulation as specified		
30.7.1.1	200 mm dia	Each	16887
30.7.1.2	150mm dia.	Each	9272
30.7.1.3	125mm dia.	Each	8505
30.7.1.4	100mm dia.	Each	7482
30.7.1.5	80mm dia.	Each	5419
30.7.1.6	65mm dia.	Each	5086
30.7.1.7	50mm dia.	Each	4627
30.7.1.8	40mm dia.	Each	3906
30.7.2	BALANCING VALVE WITH BUILT IN MEASURING FACILITY with C I body flanged construction with EPDM coated disc with long pitch with protected out pipe insulation & PN 16 pressure rating for chilled / hot water circulation as specified.		
30.7.2.1	200 mm dia	Each	73203
30.7.2.2	150mm dia.	Each	36463
30.7.2.3	125mm dia.	Each	28132
30.7.2.4	100mm dia.	Each	21241
30.7.2.5	80mm dia.	Each	14722
30.7.2.6	65 mm dia	Each	13214

30.7.2.7	50mm dia.	Each	11524
30.7.2.8	40 mm dia	Each	8200
30.7.3	NON - RETURN VALVE with dual plate of C I body SS plates vulcanized NBR seal flanged end & PN 16 pressure rating for chilled / hot water circulation including insulation as specified.		
30.7.3.1	200 mm dia	Each	15686
30.7.3.2	150mm dia.	Each	10155
30.7.3.3	125mm dia.	Each	8732
30.7.3.4	100 mm dia	Each	6945
30.7.3.5	80mm dia.	Each	4821
30.7.3.6	65 mm dia	Each	4307
30.7.4	Y - STRAINER of Ductile CI Body flanged ends with stainless steel strainer for chilled / hot water circulation including insulation as specified.		
30.7.4.1	200mm dia	Each	52447
30.7.4.2	150mm dia.	Each	29286
30.7.4.3	125mm dia.	Each	23397
30.7.4.4	100 mm dia	Each	15834
30.7.4.5	80 mm dia	Each	10666
30.7.4.6	65mm dia.	Each	8661
30.7.4.7	50mm dia.	Each	6461
30.7.4.8	40mm dia.	Each	4837
30.8	Providing and fixing in position the industrial type pressure gauges with gun metal / brass valves complete as required	Each	1251
30.9	Providing & fixing in position the mercury in glass industrial type thermometers.	Each	1134

CONDENSER WATER PIPE

- 30.10 Supplying, fixing, testing and commissioning of condenser water pipes of following sizes of MS 'C' class along with necessary clamps, vibration isolators and fittings such as bends, tees etc. but excluding valves, strainers, gauges etc. adequately supported on rigid supports duly painted/buried in ground excavation and refilling etc. as per specification and as required complete in all respect.

Note:-The Pipes size 150mm & below shall be M.S. 'C' class as per IS : 1239 and pipes size above 150mm shall be welded black steel pipe heavy class as per IS: 3589, from minimum 6.35mm thick M.S. Sheet for pipes upto 350 mm dia. And from minimum 7mm thick MS sheet for pipes of 400 mm dia and above.

30.10.1	300mm dia.	Meter	8445
30.10.2	250 mm dia	Meter	7124
30.10.3	200mm dia.	Meter	5962
30.10.4	150mm dia.	Meter	4045
30.10.5	125mm dia.	Meter	3506
30.10.6	100mm dia.	Meter	2973

VALVES WITHOUT INSULATION

- 30.11 Supplying, fixing, testing and commissioning of following valves, gauges and strainers for condenser water circulation as per specifications.

30.11.1 BUTTERFLY VALVE (MANUAL) with C I body SS disc nitrile sheet & O - ring & PN 16 pressure rating as specified.

30.11.1.1	200mm dia.	Each	15064
30.11.1.2	150mm dia.	Each	7648
30.11.1.3	125mm dia.	Each	6976
30.11.1.4	100mm dia.	Each	5962
30.11.1.5	80mm dia.	Each	4438
30.11.1.6	65mm dia.	Each	4210

30.11.1.7	50mm dia.	Each	3708
30.11.1.8	40mm dia.	Each	3113
30.11.2	NON - RETURN VALVE with dual plate of C I body SS plates vulcanized NBR seal flanged end & PN 16 pressure rating as specified.		
30.11.2.1	200mm dia.	Each	13833
30.11.2.2	150mm dia.	Each	8546
30.11.2.3	125mm dia.	Each	7334
30.11.2.4	100mm dia.	Each	5631
30.11.2.5	80mm dia.	Each	3837
30.11.2.6	65mm dia.	Each	3384
30.11.3	Supplying, fixing, testing and commissioning of following sizes Motorized Butter fly Valve with CI Body, SS Disc,O - ring and minimum PN-16 pressure rating , conforming to BS 5155, IS 13095, with IP-55 actuator, capable of accepting upto 10V DC , and upto 20mA electric signal and providing similar transduced feedback output to control system as required.		
30.11.3.1	350mm dia.	Each	117120
30.11.3.2	300mm dia.	Each	104953
30.11.3.3	250mm dia.	Each	75561
30.11.3.4	200mm dia.	Each	57795
30.11.3.5	150mm dia.	Each	44806
30.11.3.6	125mm dia.	Each	44420
30.11.3.7	100mm dia.	Each	43595
30.11.3.8	80mm dia.	Each	18641
30.11.3.9	65mm dia.	Each	18460
30.11.3.1	50mm dia.	Each	18083

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30.11.4	Supply, Installation, Testing and Commissioning of following sizes electronic, self-balancing, pressure independent type dynamic balancing valve with integrated 2 way modulating control valve in a single body. The actuator shall be capable of accepting upto 10V DC and upto 20mA electric signal and shall provide similar transduced feedback output to control system. Maximum close off pressure shall not be less than 6 Bar for upto 50 mm valves and 7 Bar for 65 mm & above. Valves should have pressure rating of 25 Bar minimum.		
30.11.4.1	150mm dia.	Each	168917
30.11.4.2	125mm dia.	Each	153790
30.11.4.3	100mm dia.	Each	121015
30.11.4.4	80mm dia.	Each	71582
30.11.4.5	65mm dia.	Each	56243
30.11.4.6	50mm dia.	Each	36431
30.11.4.7	40mm dia.	Each	33234
30.11.4.8	32mm dia.	Each	17895
30.11.4.9	25mm dia.	Each	16832
30.11.4.10	20mm dia.	Each	12271
30.12	DUCTING, GRILLS,DIFFUSER AND INSULATION		
30.12.1	Supply, installation, balancing and commissioning of factory fabricated GSS sheet metal rectangular/round ducting complete with neoprene rubber gaskets, elbows, splitter dampers, vanes, hangers, supports, bends, transitions, reducers, end caps, collars etc as required complete in all respect in confirmation to IS : 655 and approved drawings and specifications of following sheet thickness complete as required.		
30.12.1.1	Thickness 0.63 mm sheet	sq. m	1191
30.12.1.2	Thickness 0.80 mm sheet	sq. m	1382
30.12.1.3	Thickness 1.00 mm sheet	sq. m	1488

30.12.1.4	Thickness 1.25 mm sheet	sq. m	2013
30.12.2	Supply, installation, balancing and commissioning of fabricated at site GSS sheet metal rectangular/round ducting complete with neoprene rubber gaskets, elbows, splitter dampers, vanes, hangers, supports bends, transitions, reducers, end caps, collars etc as required complete in all respect in confirmation with IS : 655 and approved drawings and specifications of following sheet thickness complete as required.		
30.12.2.1	Thickness 0.63 mm sheet	sq. m	1213
30.12.2.2	Thickness 0.80 mm sheet	sq. m	1387
30.12.2.3	Thickness 1.00 mm sheet	sq. m	1851
30.12.2.4	Thickness 1.25 mm sheet	sq. m	2014
30.13	Supply, installation, testing and commissioning of GI volume control duct damper complete with neoprene rubber gaskets, nuts, bolts, screws linkages, flanges etc in confirmation to SMACNA/IS and as per specifications complete etc. as required.	sq. m	7580
30.14	Supply, installation, testing and commissioning of Motorized (ON-OFF Type) duct mounted GI volume control damper with enthalpy sensor and necessary control wire for integration within AHU room complete with neoprene rubber gaskets, nuts, bolts, screws linkages, flanges etc in confirmation to SMACNA/IS and as per specifications complete etc. as required.		
30.14.1	Damper	sq. m	9788
30.14.2	Actuator	Each	9798
30.15	Supply and Fixing installation, testing and commissioning of powder coated extruded aluminium Supply Air Grills with aluminium volume control dampers in confirmation to SMACNA/IS and as per specifications complete etc. as required.	sq. m	9623
30.16	Supplying & fixing of powder coated extruded aluminium Return Air Grills with louvers but without volume control dampers complete in confirmation to SMACNA/IS and as per specifications complete etc. as required.	sq. m	6285

30.17	Supply and Fixing installation, testing and commissioning of supply air diffusers of powder coated aluminium with aluminium volume control dampers with anti smudge ring & removable core in confirmation to SMACNA/IS and as per specifications complete etc. as required.	sq. m	12841
30.18	Supply and Fixing installation, testing and commissioning of Return air diffusers of powder coated aluminium without volume control dampers with anti smudge ring & removable core in confirmation to SMACNA/IS and as per specifications complete etc. as required.	sq. m	8595
30.19	Supply and Fixing installation, testing and commissioning of thermal insulated flexible duct of following sizes duly supported at regular interval as per site requirement etc. complete in confirmation to SMACNA/IS and as per specifications complete etc. as required.		
30.19.1	200 mm dia.	Meter	633
30.20	Supplying, Fixing,testing and commissioning of UL 555S certified fire dampers of minimum 90 minutes fire rated in supply air duct/main branch and return air path as and where required of required sizes i/c control wiring,the damper shall be motorized and spring return so as to close the damper in the event of power failure automatically and open the same in case of power being restored. The spring return action shall be inbuilt mechanism and not externally mounted. The damper shall also be closed in the event of fire signal complete as required and as per specifications.		
30.20.1	Fire damper	sq. m	11355
30.20.2	Actuator	Each	10627
30.21	Supply and fixing of acoustic lining of supply air duct and plenum with 25 mm thick resin bonded glass wool having density of 32 kg/m ³ , with 25 mm X 25 mm GI section of 1.25 mm thick, at 600 mm centre to centre covered with Reinforced Plastic tissue paper and 0.5 mm thick perforated aluminum sheet fixed to inside surface of ducts with cadmium plated nuts, bolts, stick pins, CPRX compound in confirmation to SMACNA/IS and as per specifications complete etc. as required.	sq.m	890

30.22	Supplying, fixing acoustic lining on wall and ceiling of AHU rooms with 50mm thick, density 32 kg/cu.m resin bonded glass fiber insulation friction fixed in 610mm x 610 mm frame work made of 25X50X50X50X25 mm made out of 0.6mm thick GI sheet U shaped channel and covered with reinforced fiber glass tissue and finished with 0.80 mm perforated aluminium sheet etc. complete as required and as per specifications.	sq.m	1294
30.23	Supplying and fixing of following thickness duly laminated aluminum foil of mat finish closed cell Nitrile rubber (Class "O") insulation on existing duct after applying suitable adhesive for Nitrile rubber. The joints shall be sealed with 50 mm wide and 3 mm thick self adhesive nitrile rubber tape insulation complete as per specifications and as required.		
30.23.1	19mm	sq.m	810
30.23.2	25mm	sq.m	1033
30.24	Supplying and fixing 50 mm thick aluminium foil faced resin bonded fibre glass insulation (on duct) of density 24 kg/cu.m or mineral wool insulation(non combustible) of density 44 kg/cu m after applying two coats of cold setting adhesive (CPRX compound) sealing all joints with self adhesive aluminium tape & covering with 0.63mmx19mm GI wire mesh netting butting all joints and laced with GI wire complete as per specifications and as required. (for indoor applications).	sq.m	548
30.25	Supplying and fixing Installation, Testing and commissioning of 50mm thick aluminium foil faced resin bonded fibre glass insulation (on duct) of density 24 kg/cu.m or mineral wool insulation(non combustible) of density 44 kg/cu m after applying two coats of cold setting adhesive (CPRX compound) sealing all joints with self adhesive aluminium tape & covering with 0.63mmx19mm GI wire mesh netting & butting all joints and laced with GI wire complete and finally covered with one layer of tar felt stuck with hot bitumen as per specifications and as required. (for outdoor applications)	sq.m	629
	AIR HANDLING UNITS		
30.26	Supply, Installation, Testing and commissioning of factory built floor mounted chilled water double skin type horizontal/vertical air handling units in confirmation to IS 18794 : 2024 made of 25mm thick panels consisting of pre plasticized G.I. casing of thickness 0.8mm outside layer and 0.8 mm inside layer with polyurethane foam (PUF) insulation factory injected between them by injection moulding machine, complete with blower		

section with blower suitable for static pressure as required, minimum 2 bend GSS/PVC eliminators, cooling coil section with aluminium finned copper tubes (tubes thickness not less than 0.5mm) cooling coil of 6 row deep, filter section with 50mm thick metal viscous/ washable synthetic type air prefilters, belt drive package with TEFC drive motor of efficiency class IE3 suitable for $415 \pm 10\%$ volts, 50Hz, 3 Phase AC supply suitably designed for variable frequency drive applications, drain connections, stainless steel (18G) drain pan with PUF insulation, 150 mm dia. dial type pressure gauges (2 nos.) and industrial type thermometers (2 nos.) at the inlet and outlet of coil, auto purge valve wherever required, necessary vibration isolation arrangement etc. complete as per specification and of following capacities.

30.26.1	4300 CMH	Set	110362
30.26.2	5100 CMH	Set	129837
30.26.3	6800 CMH	Set	145902
30.26.4	8500 CMH	Set	180747
30.26.5	10200 CMH	Set	208449
30.26.6	11900 CMH	Set	236902
30.26.7	13600 CMH	Set	254450
30.26.8	15300 CMH	Set	288419
30.26.9	17000 CMH	Set	305215
30.26.10	18700 CMH	Set	337303
30.26.11	20400 CMH	Set	362372
30.26.12	23800 CMH	Set	397358
30.26.13	28900 CMH	Set	419639
30.26.14	34000 CMH	Set	465104
30.26.15	39100 CMH	Set	644915

30.27	Supply, Installation, Testing and commissioning of factory built in confirmation to IS 18794 : 2024 floor mounted chilled water double skin type horizontal/vertical air handling units made of 25mm thick panels consisting of pre plasticized G.I. casing of thickness 0.8mm outside layer and 0.8 mm inside layer with polyurethane foam (PUF) insulation factory injected between them by injection moulding machine complete with blower section with blower suitable for static pressure as required, minimum 2 bend GSS/PVC eliminators, cooling coil section with aluminium finned copper tubes (tubes thickness not less than 0.5mm) cooling coil of 4 row deep, filter section having filters suitable for achieving air quality as per CPWD HVAC Specification 2024 belt drive package with TEFC drive motor of efficiency class IE3 suitable for $415 \pm 10\%$ volts, 50Hz, 3 Phase AC supply suitably designed for variable frequency drive applications, drain connections, stainless steel (18G) drain pan with PUF insulation, 150 mm dia. dial type pressure gauges (2 nos.) and industrial type thermometers (2 nos.) at the inlet and outlet of coil, auto purge valve wherever required, necessary vibration isolation arrangement etc. complete as per specification and of following capacities.		
30.27.1	4300 CMH*	Each	97252
30.27.2	5100 CMH*	Each	114150
30.27.3	6800 CMH*	Each	127371
30.27.4	8500 CMH*	Each	157605
30.27.5	10200 CMH*	Each	184382
30.27.6	11900 CMH*	Each	206403
30.27.7	13600 CMH*	Each	223114
30.27.8	15300 CMH*	Each	253573
30.27.9	17000 CMH*	Each	266859
30.27.10	18700 CMH*	Each	291987
30.27.11	20400 CMH*	Each	316621
30.27.12	23800 CMH*	Each	301047

30.27.13	28900 CMH*	Each	384197
30.27.14	34000 CMH*	Each	425772
30.27.15	39100 CMH*	Each	487835

*NOTE : THE ITEM OF 4 ROW COOLING COIL AHUs SHOULD BE USED ONLY FOR REPLACEMENT OF EXISTING AHUs WHERE HOT WATER GENERATOR AND HOT WATER LINES ARE NOT FEASIBLE TO BE PROVIDED BECAUSE OF THE SITE CONSTRAINTS.

CHAPTER 31

VRV/VRF

Item No.	Description	Unit	Rate(₹)
31.1	OUTDOOR UNIT Supply, Installation, Testing & Commissioning of Modular type Variable Refrigerant Flow/Variable Refrigerant Volume air cooled Outdoor units suitable for cooling/heating having 100% hermetically sealed inverter type twin Rotary/Scroll Compressor(s), minimum two compressors (with individual separate PCB) for above 14HP modules, microprocessor based Controller, top discharge type condensing unit(s), with R-410-A Refrigerant or equivalent, vibration Isolators with suitable foundation etc. complete as required. To have better efficiency condensor fan shall be capable to operate at different speed with respect to load. The unit shall deliver the rated capacity and in confirmation as per IS 18728:2024 and CPWD Specifications and work even at 50°C ambient temperature without tripping. The system shall be able to deliver 100% of the rated capacity upto 39 Degree Celcius. The unit shall be suitable to work on 400V +/- 10%, 3 Phase, 50Hz AC power supply and BMS compatible. The unit shall be filled with first charge of the refrigerant and ready for use as required. The condenser should be coated with a hydrophilic film to prevent water accumulation on the surface of the heat exchanger, enhance water dispersion, and reduce the risk of degradation, thereby improving overall performance and durability. The Indian Seasonal Energy Efficiency Ratio (ISEER) of the unit shall be as per Energy Conservation and Sustainable Building Code (ECSBC) 2024 as below and complete as per CPWD specification, connections, inter connections etc. as required. (For capacity <40 kW ISEER 5.4, Capacity ≥ 40 and <70 ISEER 5.5, Capacity ≥ 70 ISEER 5.6 for ECSBC Building) For Cooling or Heating or Both		
31.1.1	6 HP to 8 HP	Per HP	20947
31.1.2	10 HP to 12 HP	Per HP	20083
31.1.3	14 HP to 22 HP	Per HP	19111
31.2	Supply, Installation, Testing & Commissioning of Modular type Variable Refrigerant Flow/Variable Refrigerant Volume air cooled Outdoor units suitable for cooling/heating having 100% hermetically sealed inverter type twin Rotary/Scroll Compressor(s), minimum two compressors (with individual		

separate PCB) for above 14HP modules, microprocessor based Controller, top discharge type condensing unit(s), with R-410-A Refrigerant or equivalent, vibration Isolators with suitable foundation etc. complete as required. To have better efficiency condensor fan shall be capable to operate at different speed with respect to load. The unit shall deliver the rated capacity and in confirmation as per IS 18728:2024 and CPWD Specifications and work even at 50°C ambient temperature without tripping. The system shall be able to deliver 100% of the rated capacity upto 39 Degree Celcius. The unit shall be suitable to work on 400V +/- 10%, 3 Phase, 50Hz AC power supply and BMS compatible. The unit shall be filled with first charge of the refrigerant and ready for use as required. The condenser should be coated with a hydrophilic film to prevent water accumulation on the surface of the heat exchanger, enhance water dispersion, and reduce the risk of degradation, thereby improving overall performance and durability. The Indian Seasonal Energy Efficiency Ratio (ISEER) of the unit shall be as per Energy Conservation and Sustainable Building Code (ECSBC) 2024 as below and complete as per CPWD specification, connections, inter connections etc. as required. (For capacity <40 kW ISEER 6.4, Capacity ≥ 40 and <70 ISEER 6.5, Capacity ≥ 70 ISEER 6.6 for ECSBC+ Building)
For Cooling or Heating or Both

31.2.1	6 HP to 8 HP	Per HP	22343
31.2.2	10 HP to 12 HP	Per HP	21422
31.2.3	14 HP to 22 HP	Per HP	20385
31.3	Supply, Installation, Testing & Commissioning of Modular type Variable Refrigerant Flow/Variable Refrigerant Volume air cooled Outdoor units suitable for cooling/heating having 100% hermetically sealed inverter type twin Rotary/Scroll Compressor(s), minimum two compressors (with individual separate PCB) for above 14HP modules, microprocessor based Controller, top discharge type condensing unit(s), with R-410-A Refrigerant or equivalent, vibration Isolators with suitable foundation etc. complete as required. To have better efficiency condensor fan shall be capable to operate at different speed with respect to load. The unit shall deliver the rated capacity and in confirmation as per IS 18728:2024 and CPWD Specifications and work even at 50°C ambient temperature without tripping. The system shall be able to		

deliver 100% of the rated capacity upto 39 Degree Celcius. The unit shall be suitable to work on 400V +/- 10%, 3 Phase, 50Hz AC power supply and BMS compatible. The unit shall be filled with first charge of the refrigerant and ready for use as required. The condenser should be coated with a hydrophilic film to prevent water accumulation on the surface of the heat exchanger, enhance water dispersion, and reduce the risk of degradation, thereby improving overall performance and durability. The Indian Seasonal Energy Efficiency Ratio (ISEER) of the unit shall be as per Energy Conservation and Sustainable Building Code (ECSBC) 2024 as below and complete as per CPWD specification, connections, inter connections etc. as required. (For capacity <40 kW ISEER 7.4, Capacity ≥ 40 and <70 ISEER 7.5, Capacity ≥ 70 ISEER 7.6 for Super ECSBC Building)
For Cooling or Heating or Both

31.3.1	6 HP to 8 HP	Per HP	23740
31.3.2	10 HP to 12 HP	Per HP	22761
31.3.3	14 HP to 22 HP	Per HP	21659
31.4	INDOOR UNIT		
	Supply, Installation, Testing and Commissioning of following minimum capacity 4 way Cassette Type Indoor ceiling mounted unit equipped with synthetic washable media pre-filter, fan section with low noise fan/dynamically balanced blower, multispeed motor, coil section with DX Copper coil, electronic expansion valve, outer cabinet, drain pump, grill, necessary supports, vibration Isolation, Corded remote control etc., suitable for operation on single phase 230 V ± 10%, 50Hz AC supply, complete, as required. The Indoor units must shut down upon receiving a signal from the BMS System/Fire Signals. The system shall be capable to adjust air flow as per room requirement in auto mode. The cooling capacity of indoor unit will be at air inlet conditions of 27 Degree C DB and 19 Degree C WB temperature. (Make will be same as of Outdoor)		
31.4.1	0.8 TR	Each	22304
31.4.2	1.0 TR	Each	27880
31.4.3	1.2 TR	Each	38601

31.4.4	1.6 TR	Each	39464
31.4.5	2.0 TR	Each	39734
31.4.6	2.4 TR	Each	40814
31.4.7	2.6 TR	Each	40814
31.4.8	3.6 TR	Each	44269
31.4.9	4.1 TR	Each	45781
31.4.10	4.6TR	Each	52799
31.5	Supply, Installation, Testing and Commissioning of following minimum capacity 4-way compact VRV/VRF Cassette Type Indoor ceiling mounted unit equipped with synthetic washable media pre-filter, fan section with low noise fan/dynamically balanced blower, multispeed motor, coil section with DX Copper coil, electronic expansion valve, outer cabinet, drain pump, grill, necessary supports, vibration Isolation, Corded remote control etc., suitable for operation on single phase 230 V $\pm$ 10%, 50Hz AC supply, complete, as required. The Indoor units must shut down upon receiving a signal from the BMS System/Fire Signals. The system shall be capable to adjust air flow as per room requirement automatically. The cooling capacity of indoor unit will be at air inlet conditions of 27 Degree C DB and 19 Degree C WB temperature. (Make will be same as of Outdoor)		
31.5.1	0.6 TR	Each	22610
31.5.2	0.8 TR	Each	25122
31.5.3	1.0 TR	Each	39410
31.5.4	1.2 TR	Each	39950
31.5.5	1.6 TR	Each	39950

31.6	Supply, Installation, Testing and Commissioning of following minimum capacity Single way wall/corner VRV/VRF Cassette Type Indoor ceiling mounted unit equipped with synthetic washable media pre-filter, fan section with low noise fan/dynamically balanced blower, multispeed motor, coil section with DX Copper coil, electronic expansion valve, outer cabinet, drain pump, grill, necessary supports, vibration Isolation, Corded remote control etc., suitable for operation on single phase 230 V $\pm$ 10%, 50Hz AC supply, complete, as required. The Indoor units must shut down upon receiving a signal from the BMS System/Fire Signals. The system shall be capable to adjust air flow as per room requirement automatically. The cooling capacity of indoor unit will be at air inlet conditions of 27 Degree C DB and 19 Degree C WB temperature. (Make will be same as of Outdoor)		
31.6.1	0.6 TR	Each	30233
31.6.2	0.8 TR	Each	30880
31.6.3	1.0 TR	Each	32392
31.6.4	1.2 TR	Each	42326
31.6.5	1.6 TR	Each	42326
31.6.6	2.0 TR	Each	43621
31.7	Supply, Installation, Testing and Commissioning of following minimum capacity Double way VRV/VRF Cassette Type Indoor ceiling mounted unit equipped with synthetic washable media pre-filter, fan section with low noise fan/dynamically balanced blower, multispeed motor, coil section with DX Copper coil, electronic expansion valve, outer cabinet, drain pump, grill, necessary supports, vibration Isolation, Corded remote control etc., suitable for operation on single phase 230 V $\pm$ 10%, 50Hz AC supply, complete, as required. The Indoor units must shut down upon receiving a signal from the BMS System/Fire Signals. The cooling capacity of indoor unit will be at air inlet conditions of 27 Degree C DB and 19 Degree C WB temperature. (Make will be same as of Outdoor)		
31.7.1	0.6 TR	Each	44053
31.7.2	1.0 TR	Each	46861
31.7.3	2.0 TR	Each	57226

31.7.4	3.3 TR	Each	60541
31.7.5	4.2 TR	Each	64173
31.8	Supply, Installation, Testing and Commissioning of following minimum capacity High wall type Indoor unit equipped with and comfort washable synthetic media pre-filter, fan section with low noise fan/dynamically balanced blower, multispeed motor, coil section with DX copper coil, electronic expansion valve, outer cabinet, cord less remote control, drain pan, necessary accessories etc., suitable for operation on 230 V $\pm$ 10%, 50 Hz, single phase AC supply, complete as required. The Indoor units must shut down upon receiving a signal from the BMS System/Fire Signals. The system shall be capable to adjust air flow as per room requirement automatically. The cooling capacity of indoor unit will be at air inlet conditions of 27 Degree C DB and 19 Degree C WB temperature. (Make will be same as of Outdoor)		
31.8.1	0.6 TR	Each	16790
31.8.2	0.8 TR	Each	17276
31.8.3	1.0 TR	Each	17816
31.8.4	1.2 TR	Each	21595
31.8.5	1.6 TR	Each	22027
31.8.6	2.0 TR	Each	22674
31.9	Supply, Installation, Testing and Commissioning of following minimum capacity and external static pressure VRF/VRV ceiling mounted low static ductable type Indoor unit equipped with washable synthetic media pre-filter, fan section with low noise fan/dynamically balanced blower, multispeed motor, coil section with DX copper coil, electronic expansion valve, corded remote control, outer cabinet, vibration Isolators, drain pan, drain pump, other necessary supports etc., suitable for operation on single phase AC supply 230 V $\pm$ 10%, 50 Hz complete as required. The Indoor units must shut down upon receiving a signal from the BMS System/Fire Signals. The system shall be capable to adjust air flow as per room requirement automatically. The cooling capacity of indoor unit will be at air inlet conditions of 27 Degree C DB and 19 Degree C WB temperature. (Make will be same as of Outdoor)		

	Low static ductable units (minimum 19 to 29 pascal external static pressure)		
31.9.1	0.5 TR	Each	30233
31.9.2	0.6 TR	Each	30880
31.9.3	0.8 TR	Each	31204
31.9.4	1.03 TR	Each	31636
31.9.5	1.3 TR	Each	34444
31.9.6	1.6 TR	Each	34983
31.9.7	2.0 TR	Each	36063
31.10	Supply, Installation, Testing and Commissioning of following minimum capacity and external static pressure VRF/VRV ceiling mounted mid static ductable type Indoor unit equipped with washable synthetic media pre-filter, fan section with low noise fan/dynamically balanced blower, multispeed motor, coil section with DX copper coil, electronic expansion valve, corded remote control, outer cabinet, vibration Isolators, drain pan, drain pump, other necessary supports etc., suitable for operation on single phase AC supply 230 V $\pm$ 10%, 50 Hz complete as required. The Indoor units must shut down upon receiving a signal from the BMS System/Fire Signals. The cooling capacity of indoor unit will be at air inlet conditions of 27 Degree C DB and 19 Degree C WB temperature. (Make will be same as of Outdoor)		
	Mid static ductable units (minimum 30 to 48 pascal external static pressure)		
31.10.1	1.2 TR	Each	34336
31.10.2	1.6 TR	Each	34983
31.10.3	2.0 TR	Each	36063
31.10.4	2.4 TR	Each	37683
31.10.5	3.2 TR	Each	40166
31.11	Supply, Installation, Testing and Commissioning of following minimum capacity and external static pressure VRF/VRV ceiling mounted mid high static ductable type Indoor unit equipped with washable synthetic media pre-filter, fan section		

with low noise fan/dynamically balanced blower, multispeed motor, coil section with DX copper coil, electronic expansion valve, corded remote control, outer cabinet, vibration Isolators, drain pan, drain pump, other necessary supports etc., suitable for operation on single phase AC supply 230 V $\pm$ 10%, 50 Hz complete as required. The Indoor units must shut down upon receiving a signal from the BMS System/Fire Signals. The system shall be capable to adjust air flow as per room requirement automatically. The cooling capacity of indoor unit will be at air inlet conditions of 27 Degree C DB and 19 Degree C WB temperature. (Make will be same as of Outdoor)

High Static Ductable units (minimum 49 to 77 pascal external static pressure)

31.11.1	0.8 TR	Each	31510
31.11.2	1.03 TR	Each	32730
31.11.3	1.2 TR	Each	34514
31.11.4	1.6 TR	Each	34983
31.11.5	2.0 TR	Each	36063
31.11.6	2.4 TR	Each	37683
31.11.7	3.2 TR	Each	40166
31.11.8	4.0 TR	Each	45997
31.11.9	4.6 TR	Each	64568
31.12	Supply, Installation, Testing and Commissioning of following minimum capacity and external static pressure VRF/VRV ceiling mounted high ductable type Indoor unit equipped with washable synthetic media pre-filter, fan section with low noise fan/dynamically balanced blower, multispeed motor, coil section with DX copper coil, electronic expansion valve, corded remote control, outer cabinet, vibration Isolators, drain pan, drain pump, other necessary supports etc., suitable for operation on single phase AC supply 230 V $\pm$ 10%, 50 Hz complete as required. The Indoor units must shut down upon receiving a signal from the BMS System/Fire Signals. The cooling capacity of indoor unit will be at air inlet conditions of 27 Degree C DB and 19 Degree C WB temperature. (Make will be same as of Outdoor)		

31.12.1	High Static Ductable units (minimum 78 pascal external static pressure)		
31.12.1	5.5 TR	Each	74394
31.12.3	6.6 TR	Each	78065
31.12.4	8.0 TR	Each	85191
31.13	Supply, Installation, Testing and Commissioning of Y/T/Multi Joints. Joints shall be of same Original Equipment Manufacturer (OEM) make as of ODU's and IDUs		
31.13.1	Indoor Units	Each	4535
31.13.2	Outdoor Multi Joint	Each	8098
COPPER REFRIGERANT PIPING			
31.14	Supply, Installation, testing and commissioning including vacuumization and Nitrogen testing of following nominal sizes of soft/hard drawn copper refrigerant piping for VRV/VRF system, complete with fittings, with suitable adjustable ring type hanger supports, jointing/brazing including accessories, insulated with XPLE Class-O tubular insulation/with Class-O closed cell elastomeric nitrile rubber tubular sleeves sections of 19 mm thick insulation as given below for Suction and Liquid lines, all accessories as per specifications etc. as required :		
31.14.1	6.4 mm dia (OD) (Soft drawn) with tube thickness 1.2 mm	Meter	256
31.14.2	9.5 mm dia (OD) (Soft drawn) with tube thickness 1.2 mm	Meter	346
31.14.3	12.7 mm dia (OD) (Soft drawn) with tube thickness 1.2 mm	Meter	487
31.14.4	15.86 mm dia (OD) (Soft drawn) with tube thickness 1.2 mm	Meter	615
31.14.5	19 mm dia (OD) (Hard drawn) with tube thickness 1.2 mm	Meter	739
31.14.6	22.2 mm dia (OD) (Hard drawn) with tube thickness 1.2 mm	Meter	904
31.14.7	25.4 mm dia (OD) (Hard drawn) with tube thickness 1.2 mm	Meter	1068
31.14.8	28.58 mm dia (OD) (Hard drawn) with tube thickness 1.2 mm	Meter	1157
31.14.9	31.8 mm dia (OD) (Hard drawn) with tube thickness 1.62 mm	Meter	1222
31.14.10	34.9 mm dia (OD) (Hard drawn) with tube thickness 1.62 mm	Meter	1286

31.14.11	38.1 mm dia (OD) (Hard drawn) with tube thickness 1.62 mm	Meter	1322
31.14.12	41.27 mm dia (OD) (Hard drawn) with tube thickness 1.62 mm	Meter	1368

CHAPTER 32 UNITARY SYSTEM

Item No.	Description	Unit	Rate(₹)
	WINDOW AC UNITS		
32.1	Supply , Installation, Testing and Comissioning of Window type Air conditioners complete with copper power cable upto 3 Mtr, wireless Remote, suitable for working between 180- 260V with low & high voltage cutoff and 50 hz ,1 phase AC supply capable of performing, cooling, dehumidification, air circulation, R-32/R-410A /R-407B Green Refrigerant with Scroll / rotary compressor with min 5 year Original Equipment Manufacturer (OEM) warranty both compressor and Printed Circuit Board (PCB), antifreeze thermostat on the coil as a safety feature, complete with fixing including T&P & labour etc as required complete in all respect as specified of following capacity.Sound level of up to 50dB inside the room is acceptable. The unit shall be in confirmation with IS 1391 Part-I 2023 and CPWD Specification. The system shall be able to deliver 100% of the rated capacity upto 42 Degree Celcius. The system shall be able to operate up to 50 °C (out door ambient temperature). Non Inverter Type		
32.1.1	1.0 TR with fixed speed 5 Star BEE rating	Each	32848
32.1.2	1.5 TR with fixed speed 5 Star BEE rating	Each	36532
	Inverter Type		
32.1.3	1.0 TR with Inverter 5 Star BEE rating	Each	34407
32.1.4	1.5 TR with Inverter 5 Star BEE rating	Each	40074
	HI - WALL SPLIT SYSTEMS		
32.2	Supply , Installation, Testing and Comissioning of Air Cooled Hi Wall split type Air conditioners complete with Indoor unit(IDU), Out door unit (ODU), surface / concealed copper Refrigerant piping with insulation (closed cell elastomeric nitrile rubber tubular pipe section) upto 3 Mtr (IDU to ODU), copper power cable upto 3.5 Mtr (IDU to ODU) i/c drain pipe R-32/R-410/ R-407 Green Refrigerant, wireless Remote control, suitable for working between 180-260V with low & high voltage cutoff and 50 hz ,1 phase AC supply capable of performing cooling, dehumidification, air circulation of following capacity with Scroll / rotary compressor. The system shall be able to deliver 100% of the rated capacity upto 42 Degree Celcius. Min 5 year Original Equipment Manufacturer (OEM) warranty both compressor and		

Printed Circuit Board (PCB). Must comply : Electrical cable IS 694 or IS 9968 temperature sensing control IS /International Electrotechnical Commission (IEC) 60730, hermetic compressor IS 10617, heat exchanger IS 11329, capacitor IS 2993 and motor IS 12615. Complete as per CPWD specification and IS : 1391 Part II 2023. The system shall be able to operate up to 50 °C (out door ambient temperature).

Inverter Type - Cooling only

32.2.1	0.75 TR with 5 Star BEE Rating	Each	39095
32.2.2	1.0 TR with 5 Star BEE Rating	Each	40580
32.2.3	1.5 TR with 5 Star BEE Rating	Each	43539
32.2.4	2.0 TR with 5 Star BEE Rating	Each	57396

Inverter Type - Hot & Cold

32.2.5	1.0 TR with 3 Star BEE Rating	Each	38408
32.2.6	1.5 TR with 3 Star BEE Rating	Each	41768

DUCTABLE TYPE SPLIT UNITS

- 32.3 Supply, Installtion, Testing and Commissioning of air cooled ducted split type air conditioning machine with each having a capacity and details as mentioned below suitable for operation on R 32/R-410A /R-407 Green refrigerant comprising of Scroll type compressor hermetically sealed complete with automatic capacity, safety switches, lubrication system with min 5 year (OEM) warranty for both compressor and Printed Circuit Board (PCB) , Suitable capacity squirrel cage induction motor having class 'B' insulation suitable for operation on $415 \pm 10\%$ volts, 50 Hz, A.C. supply for Blower motor, Necessary drive arrangement for blower motor, Matching Air cooled condenser with necessary fittings for refrigerant piping connections, necessary structural support for mounting condensers, Microprocessor based control panel complete with accessories, machine Isolation / disconnect switch, valves and accessories to inter connect compressor and condenser including pressure testing, vacuum. Necessary starters suitable for Indoor & outdoor unit complete with O/L ,U/V, phase reversal protection, single phase preventors i/c copper conductor control and power cable and drain pipe of suitable size and length etc complete as required. The total cooling capacity/heating capacity of tested unit shall have a capacity as per relevant IS code. The lab testing reports

as per IS: 8148 shall be submitted from National Accreditation Board for Testing and Calibration Laboratories (NABL) accredited as per ISO/International Electrotechnical Commission (IEC) 17025 standards / Central Power Research Institute (CPRI)/Electrical Research and Development Association (ERDA)with Original Equipment Manufacturer (OEM) etc. complete as per CPWD specification as required.

	Inverter		
32.3.1	1.5 TR (BEE 4 Star Rated)	Each	57010
32.3.2	2.2 TR (BEE 4 Star Rated)	Each	64676
32.3.3	3.0 TR (BEE 4 Star Rated)	Each	87998
32.3.4	3.5 TR (3.2 EER)	Each	127031
32.3.5	4.0 TR (3.2 EER)	Each	143821
	Non Inverter		
32.3.6	1.0 TR (BEE 4 Star Rated)	Each	41678
32.3.7	1.5 TR (BEE 4 Star Rated)	Each	43621
32.3.8	2.0 TR (BEE 4 Star Rated)	Each	56837
32.3.9	2.5 TR (BEE 4 Star Rated)	Each	74070
32.3.10	3.0 TR (BEE 4 Star Rated)	Each	80116
32.3.11	3.5 TR (3.2 EER)	Each	88214
32.3.12	4.0 TR (3.2 EER)	Each	97932
32.3.13	4.5 TR (3.2 EER)	Each	113897
32.3.14	5.5 TR (3.2 EER)	Each	124062
32.3.15	8.5 TR (3.2 EER)	Each	164444
32.3.16	11.0 TR (3.2 EER)	Each	211520
32.3.17	16.7 TR (3.2 EER)	Each	327160

CASSETTE TYPE SPLIT UNITS

- 32.4 Supply, Installation, Testing and Commissioning of Air Cooled Cassette type Air conditioners complete with Indoor unit(IDU), Out door unit (ODU), R-32/R410A/R-407 Green Refrigerant, wireless Remote, inbuilt drain pump, suitable for 400/230V, 50 Hz ,1 /3 phase AC supply, including surface / concealed copper Refrigerant piping with insulation (closed cell elastomeric nitrile rubber tubular pipe section) upto 5.5 Mtr (IDU to ODU), copper power and control cable upto 5.5 Mtr (IDU to ODU) including drain pipe, the system shall be capable of performing cooling, dehumidification, Air circulation, filtration & ventilation of following capacity with Scroll/rotary compressor with min 5 year Original Equipment Manufacturer (OEM) warranty both compressor and Printed Circuit Board (PCB) as specified. The system shall be able to deliver 100% of the rated capacity as per relevant IS Code. The lab testing reports as per IS: 1391 shall be submitted from National Accreditation Board for Testing and Calibration Laboratories (NABL) accredited as per International Electrotechnical Commission (IEC) 17025 standards / Central Power Research Institute (CPRI)/Electrical Research and Development Association (ERDA) with Original Equipment Manufacturer (OEM) etc. complete as per CPWD specification and as per IS: 1391 as required.

Inverter Type- Cooling only

32.4.1	1.5 TR with 5 Star BEE Rating	Each	77417
32.4.2	2.0 TR with 5 Star BEE Rating	Each	85731
32.4.3	2.5 TR with 5 Star BEE Rating	Each	118123
32.4.4	3.0 TR with 5 Star BEE Rating	Each	123684
32.4.5	3.5 TR with 5 Star BEE Rating	Each	139610
32.4.6	4.0 TR with 5 Star BEE Rating	Each	142201

Heating & Cooling

32.4.7	1.5 TR with 3 Star BEE Rating	Each	60141
32.4.8	2.0 TR with 3 Star BEE Rating	Each	67052
32.4.9	2.5 TR with 3 Star BEE Rating	Each	80764

TOWER TYPE SPLIT UNITS

- 32.5 Supply, Installation, Testing and Commissioning of Air Cooled Floor standing Tower type split Air conditioners complete with Indoor unit(IDU), Out door unit (ODU), surface / concealed copper Refrigerant piping with insulation (closed cell elastomeric nitrile rubber tubular pipe section) upto 5 Mtr (IDU to ODU), copper power cable upto 5.5 Mtr (IDU to ODU), i/c drain pipe of suitable length and size. R-32/R-410/R-407C Green Refrigerant, wireless Remote control, suitable for working between 180-260V with low & high voltage cutoff and 50 hz ,1 phase AC supply capable of performing cooling, dehumidification, air circulation of following capacity with Scroll / rotary with min 5 year Original Equipment Manufacturer (OEM) warranty both compressor and Printed Circuit Board (PCB). as specified. The system shall be able to deliver 100% of the rated capacity as per relevant IS Code. The lab testing reports as per IS: 1391 shall be submitted from National Accreditation Board for Testing and Calibration Laboratories (NABL) accredited as per International Electrotechnical Commission (IEC) 17025 standards / Central Power Research Institute (CPRI)/Electrical Research and Development Association (ERDA) with Original Equipment Manufacturer (OEM) etc. complete as per CPWD specification and as per IS: 1391 as required.

Heating & Cooling

32.5.1	3.3 TR BEE 4 Star Rating	Each	126142
32.5.2	3.8 TR BEE 4 Star Rating	Each	133410
32.5.3	4.6 TR BEE 4 Star Rating	Each	140677

Cooling Only

32.5.4	2.4 TR BEE 5 Star Rating	Each	93613
32.5.5	3.3 TR BEE 5 Star Rating	Each	104086
32.5.6	3.8 TR BEE 5 Star Rating	Each	110025
32.5.7	4.6 TR BEE 5 Star Rating	Each	116287

AIR COOLED PACKAGE UNITS

- 32.6 Supplying, installation , testing and commissioning of Air cooled ductable type Packaged air-conditioning units complete with Hermetically sealed Scroll compressors fitted inside the indoor unit & first charge of refrigerant R410A or equilant permitted green refrigerant & oil ,air cooled condenser, fan section with statically/dynamically balanced centrifugal blower driven by a Totally Enclosed Fan Cooled (TEFC) squirrel cage three speed motor, Multi rows cooling coil of copper with aluminium fins etc. The enclosures shall be fabricated of M.S. The Package unit shall be equipped with synthetic fiber filter, insulated drain pan, controls all encased in a unit. The casing shall be factory powder coated. Electrical panel board for Package units shall comprise of control and power panel with including all associries i/c Voltage scanner, overload, low voltage, high voltage & phase imbalance protection, along with VI Pads complete with all ancillaries including MS painted stand for Outdoor units of suitable size, foundation and allied minor civil works as per instructions of Engineer-in-charge of following ratings including electric control panel & fitting as per CPWD specifications and as per IS: 8148 complete as required.

Inverter

32.6.1	5.0 TR (2.8 EER)	Each	131382
32.6.2	8.0 TR (2.8 EER)	Each	177609
32.6.3	11.0 TR (2.8 EER)	Each	214946
32.6.4	16.5 TR (2.8 EER)	Each	301037
32.6.5	22.0 TR (2.8 EER)	Each	393585

CHAPTER 33 CHILLERS

Item No.	Description	Unit	Rate(₹)
	AIR COOLED CHILLERS		
	AIR COOLED SCREW CHILLERS		
33.1	Supply, installation, testing and commissioning of following capacity Air-Cooled (suitable for out door installation) Screw Chiller package complete with VFD (Variable Frequency Drive), hermetic/semi hermetic, screw type compressor each with step less capacity control of 25 % to 100 % of the rated capacity, with microprocessor based control panel compatible for BMS operation, motor, starter panel (VFD), machine mounted, air- cooled condensers with Copper tube and Aluminium fins, factory fitted chiller insulation, water flow switch, vibration spring Isolators, victaulic couplings, integral refrigerant piping and wiring with single/ two circuits, automatic and safety controls mounted in central console panel and all mounted on a steel frame (complete as per specifications) i/c suitable foundation/mounting structure made of RCC/MS Structure i/c anticorrosive paint, anti vibration pad, power control cable and connection inter connection etc. as per design approved by engineer-in-charge. Motor shall be suitable for 415±10% 50 cycles. 3 phase AC supply. Refrigerant gas used shall be R-134A. The chiller shall be Building Management System (BMS) compatible The system shall be in confirmation to IS : 16590, BEE Star labeling and CPWD Specification as amended upto date and tested as per IS 16590. Chilled water Leaving Temp. (6.67 deg. C) Chilled water Entering Temp. (12.2 deg. C) Evaporator fouling factor = 0.018 m ² . °C/kW Condenser air entering temp. : As per Site Dry Bulb Temperature Suitable for Seismic Zone and Altitude as per location/site. Microprocessor based controller.		
33.1.1	Upto 74 TR BEE 3 Star rated	Per TR	28073
33.1.2	75 TR - 140 TR BEE 3 Star rated	Per TR	27137
33.1.3	141 TR - 200 TR BEE 3 Star rated	Per TR	25734
33.2	Supply, installation, testing and commissioning of following capacity Air-Cooled Screw Chiller package complete with VFD (Variable Frequency Drive), hermetic/semi hermetic, multiple screw type compressor each with step less capacity control of 25 % to 100 % of the rated capacity, with		

microprocessor based control panel, motor, starter panel (VFD), machine mounted, air- cooled condensers with Copper tube and Aluminium fins, factory fitted chiller insulation, water flow switch , vibration spring Isolators, victaulic couplings, integral refrigerant piping and wiring with multiple circuits, automatic and safety controls mounted in central console panel and all mounted on a steel frame (complete as per specifications) i/c suitable foundation/mounting structure made of RCC/MS Structure i/c anticorrosive paint, anti vibration pad, power control cable and connection inter connection etc. as per design approved by engineer-in-charge. Motor shall be suitable for 415±10% 50 cycles. 3 phase AC supply. Refrigerant gas used shall be R-134a. The chiller shall be Building Management System (BMS) compatible The system shall be in confirmation to IS : 16590 and CPWD Specification as amended upto date. BEE Star labeling and CPWD Specification as amended upto date and tested as per IS 16590.

Chilled water Leaving Temp. (6.67 deg. C)

Chilled water Entering Temp. (12.2 deg. C)

Evaporator fouling factor = 0.018 m². 0C/kW

Condenser air entering temp. : As per Site Dry Bulb Temperature

Suitable for Seismic Zone and Altitude as per location/site.

Microprocessor based controller.

33.2.1	upt to 200 TR - BEE 4 Star Rated	Per TR	29009
33.2.2	201 TR to 250 TR BEE 4 Star Rated	Per TR	26669
33.2.3	251 TR - 300 TR BEE 4 Star Rated	Per TR	25266
33.2.4	301 TR - 350 TR BEE 4 Star Rated	Per TR	24330
33.2.5	351 TR - 400 TR BEE 4 Star Rated	Per TR	23394

AIR COOLED SCROLL CHILLERS

- 33.3 Supply, installation, testing and commissioning of following capacity Air-Cooled , Scroll Chiller package complete with VFD (Variable Frequency Drive), hermetic/semi hermetic,multiple scroll type compressors each with step less capacity control of 25 % to 100 % of the rated capacity, with microprocessor based control panel, motor, starter panel (VFD), machine mounted, air- cooled condensers, factory fitted chiller insulation, water flow switch , vibration spring Isolators,

victaulic couplings, integral refrigerant piping and wiring with single/ two circuits, automatic and safety controls mounted in central console panel and all mounted on a steel frame (complete as per specifications) i/c suitable foundation/mounting structure made of RCC/MS Structure i/c anticorrosive paint, anti vibration pad, power control cable and connection inter connection etc. as per design approved by engineer-in-charge. Motor shall be suitable for 415±10% 50 cycles. 3 phase AC supply. Refrigerant gas used shall be R-410A. The chiller shall be Building Management System (BMS) compatible. The system shall be in confirmation to IS : 16590 and CPWD Specification as amended upto date. BEE Star labeling and CPWD Specification as amended upto date and tested as per IS 16590.

Chilled water Leaving Temp. (6.67 deg. C)

Chilled water Entering Temp. (12.2 deg. C)

Evaporator fouling factor = 0.018 m². 0C/kW

Condenser air entering temp. : As per Site Dry Bulb Temperature

Suitable for Seismic Zone and Altitude as per location/site.

Microprocessor based controller.

33.3.1	Upto 50 TR BEE 4 Star Rated	Per TR	22926
33.3.2	51 TR - 70 TR BEE 4 Star Rated	Per TR	22458
33.3.3	71 TR - 100 TR BEE 4 Star Rated	Per TR	21991
33.4	Supply, installation, testing and commissioning of following capacity Air-Cooled Scroll Chiller package complete with VFD (Variable Frequency Drive), hermetic/semi hermetic, multiple scroll type compressors each with step less capacity control of 25 % to 100 % of the rated capacity, with microprocessor based control panel, motor, starter panel (VFD), machine mounted, air- cooled condensers, factory fitted chiller insulation, water flow switch , vibration spring Isolators, victaulic couplings, integral refrigerant piping and wiring with single/ two circuits, automatic and safety controls mounted in central console panel and i/c suitable foundation/mounting structure made of RCC/MS Structure i/c anticorrosive paint, anti vibration pad, power control cable and connection inter connection etc. as per design approved by engineer-in-charge. Motor shall be suitable for 415±10% 50 cycles. 3 phase AC supply. Refrigerant gas used shall be R-410A. The chiller shall be Building Management System (BMS) compatible The system shall be in confirmation to IS : 16590 and CPWD		

Specification as amended upto date. BEE Star labeling and CPWD Specification as amended upto date and tested as per IS 16590.

Chilled water Leaving Temp. (6.67 deg. C)
 Chilled water Entering Temp. (12.2 deg. C)
 Evaporator fouling factor = 0.018 m². 0C/kW
 Condenser air entering temp. : As per Site Dry Bulb Temperature
 Suitable for Seismic Zone and Altitude as per location/site.
 Microprocessor based controller.

33.4.1	upto 50 TR BEE 3 Star Rated	Per TR	21991
33.4.2	51 TR - 70 TR BEE 3 Star Rated	Per TR	20587
33.4.3	71 TR - 100 TR BEE 3 Star Rated	Per TR	20119

WATER COOLED CHILLERS
 WATER COOLED SCREW CHILLERS

33.5 Supply, installation, testing and commissioning of following capacity floor-mounted AHRI (Air-Conditioning, Heating, and Refrigeration Institute) Certified water cooled screw-type chiller machine complete with VFD (Variable Frequency Drive), single/multi semi-hermetic twin screw type compressor, water-cooled Shell & Tube type condenser, Shell & Tube horizontal flooded type evaporator with carbon steel shell and seamless copper tubes with 19 mm nitrile rubber insulation i/c suitable foundation/mounting structure made of RCC/MS Structure i/c anticorrosive paint, anti vibration pad, power control cable and connection inter connection etc. as per design approved by engineer-in-charge, interconnected copper refrigerant piping and wiring, vibration Isolators, gauge panel, automatic safety controls, flow switch at evaporator and condenser and ozone friendly Chlorofluorocarbons (CFC)-free refrigerant gas R-134A. The refrigerant flow control shall use an electronic expansion valve. The chiller shall be designed for a Water Side working pressure of 150 PSI and hydraulically tested at 1.3 times of design pressure . A number of properly spaced baffles shall be provided for maintaining optimum water velocity and heat transfer and the tubes shall be adequately supported. The chiller shall be Building Management System (BMS) compatible The system shall be in confirmation to IS : 16590 and CPWD Specification as amended upto date. BEE Star labeling and CPWD Specification as amended upto date and tested as per IS 16590.

Chilled water Leaving Temp. (6.67 deg. C)

Chilled water Entering Temp. (12.2 deg. C)

Evaporator fouling factor = 0.018 m². 0C/kW

Condenser water Entering Temp. (32.2 deg. C)

Condenser water Leaving Temp. (36.4 deg. C)

Suitable for Seismic Zone and Altitude as per location/site.

Microprocessor based controller.

33.5.1	Upto 70 TR BEE 3 Star Rated	Per TR	19183
33.5.2	71 TR - 110 TR BEE 3 Star Rated	Per TR	16844
33.5.3	111 TR - 150 TR BEE 3 Star Rated	Per TR	16844
33.5.4	151 TR - 210 TR BEE 3 Star Rated	Per TR	16844
33.5.5	211 TR - 260 TR BEE 3 Star Rated	Per TR	16376

33.5.6	261 TR - 300 TR BEE 3 Star Rated	Per TR	15908
33.5.7	301 TR - 450 TR BEE 3 Star Rated	Per TR	14972
33.5.8	451 TR -600 TR BEE 3 Star Rated	Per TR	14504
33.6	Supply, Installation, Testing and Commissioning of following capacity floor-mounted AHRI (Air-Conditioning, Heating, and Refrigeration Institute) Certified VFD (Variable Frequency Drive) Operated water cooled screw-type chiller machine complete with single/multi semi-hermetic twin screw type compressor, with independent circuits, water-cooled Shell & Tube type condenser, Shell & Tube horizontal flooded type evaporator with carbon steel shell and seamless copper tubes with 19 mm nitrile rubber insulation i/c suitable foundation/mounting structure made of RCC/MS Structure i/c anticorrosive paint, anti vibration pad, power control cable and connection inter connection etc. as per design approved by engineer-in-charge, common base frame, interconnected copper refrigerant piping and wiring, vibration Isolators, gauge panel, automatic safety controls, flow switch at evaporator and condenser and ozone friendly Chlorofluorocarbons (CFC)-free refrigerant gas R-134A. The refrigerant flow control shall use an electronic expansion valve. The chiller shall be designed for a Water Side working pressure of 150 psig and hydraulically tested at 1.5 times of design pressure . A number of properly spaced baffles shall be provided for maintaining optimum water velocity and heat transfer and the tubes shall be adequately supported. The chiller shall be Building Management System (BMS) compatible The system shall be in confirmation to IS : 16590 and CPWD Specification as amended upto date. BEE Star labeling and CPWD Specification as amended upto date and tested as per IS 16590. Chilled water Leaving Temp. (6.67 deg. C) Chilled water Entering Temp. (12.2 deg. C) Evaporator fouling factor = 0.018 m². 0C/kW Condenser water Entering Temp. (32.2 deg. C) Condenser water Leaving Temp. (36.4 deg. C) Suitable for Seismic Zone and Altitude as per location/site. Microprocessor based controller.		
33.6.1	Upto 74 TR BEE 4 Star Rated	Per TR	20587
33.6.2	75 TR - 150 TR BEE 4 Star Rated	Per TR	19651

33.6.3	151TR - 300 TR BEE 4 Star Rated	Per TR	17780
33.6.4	301TR - 450 TR BEE 4 Star Rated	Per TR	16844
33.6.5	451 TR - 525 TR BEE 4 Star Rated	Per TR	14972
33.6.6	526 TR - 600 TR BEE 4 Star Rated	Per TR	14037
33.7	Supply, installation, testing and commissioning of following capacity floor-mounted Variable Frequency Drive (VFD) water cooled scroll-type chiller machine complete with hermetic scroll type single/ multi compressors with independent circuits, water-cooled carbon steel shell, seamless copper tubes condenser and evaporator with 19 mm nitrile rubber insulation, i/c suitable foundation/mounting structure made of RCC/MS Structure with anti-corrosive paint, anti vibration pad, power control cable and connection inter connection etc. as per design approved by engineer-in-charge, gauge panel, automatic safety controls, flow switch at evaporator and condenser and ozone friendly Chlorofluorocarbons (CFC)-free refrigerant gas R-410A. The refrigerant flow control shall use an electronic expansion valve. The chiller shall be designed for a Water Side working pressure of 150 psig and hydraulically tested at 1.5 times of design pressure. A number of properly spaced baffles shall be provided for maintaining optimum water velocity and heat transfer and the tubes shall be adequately supported. The chiller shall be Building Management System (BMS) compatible The system shall be in confirmation to IS : 16590 and CPWD Specification as amended upto date. BEE Star labeling and CPWD Specification as amended upto date and tested as per IS 16590. Chilled water Leaving Temp. (6.67 deg. C) Chilled water Entering Temp. (12.2 deg. C) Evaporator fouling factor = 0.018 m². 0C/kW Condenser water Entering Temp. (32.2 deg. C) Condenser water Leaving Temp. (36.4 deg. C) Suitable for Seismic Zone and Altitude as per location/site. Microprocessor based controller.		
33.7.1	Upto 40 TR BEE 3 Star Rated	Per TR	22926
33.7.2	41 TR - 75 TR BEE 3 Star Rated	Per TR	22458
33.7.3	76 TR - 150 TR BEE 3 Star Rated	Per TR	21991

WATER COOLED CENTRIFUGAL CHILLERS

- 33.8 Supplying, Installation, Testing & Commissioning of following capacity Centrifugal Water Cooled Chilling Machine complete with factory fitted (unit mounted/ free standing) (Variable Frequency Drive (VFD)) with active harmonic filter with IP54 protection having actual capacity as below. The refrigerant shall be ozone friendly Chlorofluorocarbons (CFC)-free refrigerant gas R-134A. The chiller machine shall be test and performance as per IS : 16590 and BEE star labeling. The scope of work shall include Lifting, shifting & positioning of the equipment at location shown on the drawing. Chiller given hereunder, comprising of following and complete as per specification/drawings and as directed by Engineer-in-charge.
- Chilled water inlet temperature of 12.2°C (54°F) & Chilled water outlet temperature 6.7°C (44°F) with chilled water circulation, Evaporator side fouling factor 0.018 m².°C /kW - Condenser water inlet temperature Inlet - 32.2°C (90°F) & condenser water outlet temperature of 36.4°C (97.5°F) with water circulation, Condensor side fouling factor 0.044 m². 0C/Kw.
 - a) Open/ Semi-Hermetic/ Fully hermetic Centrifugal Compressor complete with automatic capacity control system, safety switches, speed increasing mechanism, forced feed lubrication system etc. as per detailed specifications and compressor extended warranty of 1 year for refrigerent leakage & mechanical seal.
 - b) Suitable capacity TEFC/SPDP Squirrel Cage Induction Motor with enclosure IP 23/ as per Original Equipment Manufacturer (OEM) standard & class 'F' insulation suitable for operation on 415±10% Volt, 3 Phase, 50 HZ, AC Supply. Vendor must provide Junction box along with each set of unit including cable works from junction box to chiller.
 - c) Unit Mounted/ Free standing IP-54 protection (UL /EN certified) Variable Frequency Drive (VFD) Starter panel with air Cooled/ Refrigerant Cooled or as per Original Equipment Manufacturer (OEM) standard, suitable for compressor motor, complete having over-load protection, under-voltage protection, protection against phase reversal, current sensing independent single phasing protection etc. including multi-function meter and CTs, complete as per detailed specifications. Variable Frequency Drive (VFD)s shall comply with International Electrotechnical Commission (IEC) 61800-3 & have THD less than 5% at all Loads Active / passive filters must be use to achieve desired THD levels and other parameters as per IEEE - 519. Variable Frequency Drive (VFD)s shall be compatible for Modbus/BACnet Protocols. The

power factor shall be > 0.95 at all loads. Original Equipment Manufacturer (OEM) shall ensure quality for each set of chiller & Variable Frequency Drive (VFD) before dispatch.

d) Lubrication Device consisting of automatic electric oil pump, oil cooler, head tank, oil strainer, automatic pressure regulating valve, oil heater, thermal switch etc, as per detailed specifications and as required.

e) Matching Shell and Tube Water Cooled Condenser of M.S. Shell and integrally finned Copper Tubes, 2 pass heat exchanger. The Condenser shall have U- stamping / PED Certification.

f) Matching Shell and Tube Flooded type Chiller for centrifugal unit consisting of MS Shell and Copper Tubes, 2 pass Heat Exchanger, duly insulated at factory complete as per specifications and as required. The Evaporator shall have U-stamping.

h) Refrigerant Line Accessories comprising of safety valves, angle valve, liquid line indications, liquid level control, liquid line Isolation valve, etc. OR as per Original Equipment Manufacturer (OEM) design standard complete as per specifications.

i) DP/ Water Flow Switches at inlet and outlet of the condenser & chiller, water drain & air purge valves wherever required, complete as per specifications.

g) Suction Line and Chiller Insulation with minimum 19mm thick elastomeric nitrile rubber insulation complete as required from factory.

h) Foundation Frame Work for mounting the above condenser, chiller, compressor and motor with base plate, panel complete with anti-vibration pads (set of spring type), vibration isolators with Isolation efficiency more than 90%. Numbers shall be as per Original Equipment Manufacturer (OEM) standards.), complete as per specifications.

i) Initial/ First Charge of Refrigerant Gas and Compressor Oil.

j) Chiller shall be factory tested at 25%, 50%, 75% and 100% load at Constant Condenser Water inlet and as per relevant IS.

k) Each chiller shall be provided with set of grooved coupling along with the chiller for cooler and condenser inlet / outlet connection.

l) Chiller Original Equipment Manufacturer (OEM) shall provide undertaking in the name of end user for providing support for maintenance & spare availability for next 7 yrs from the date of Handover.

m) Software Selection Sheet to be as per relevant IS based on latest version.

n) Sound performance shall be as per relevant IS for all loads. This data shall be provided as apart of chiller technical submittal.

o) Suitable for Seismic Zone and Altitude as per location/site.

p) The chiller shall be Building Management System (BMS) compatible and communication/ Ethernet /RS485/SNMP port open protocol for BMS integration including suitable software Microprocessor based controller.

33.8.1	300 TR - 450 TR BEE 3 Star Rated	Per TR	25266
33.8.2	451 TR - 600 TR BEE 3 Star Rated	Per TR	24798
33.8.3	601 TR - 1000 TR BEE 3 Star Rated	Per TR	24330
33.8.4	1001 TR - 1600 TR BEE 3 Star Rated	Per TR	23862
33.8.5	1601 TR - 2000 TR BEE 3 Star Rated	Per TR	23862

WATER COOLED MAGNETIC CENTRIFUGAL CHILLERS

- 33.9 Supply, installation, testing & commissioning of following capacity water cooled Magnetic centrifugal type chiller machine complete with hermetic single/ multi compressors with independent circuits. The refrigerant shall be ozone friendly Chlorofluorocarbons (CFC)-free refrigerant gas R-134A. The chiller machine shall be test and performance as per IS : 16590 and BEE star labeling. The chiller shall have single Semi/hermetically sealed refrigerant cooled motor of working on 415 + 10% volts, 3 Phase, 50 Hz AC supply. Shell & tube flooded chiller & condenser with descaling & drain valves, victaulic /Flange coupling on condenser & evaporator, microprocessor panel for multiple start ups, i/c suitable foundation/mounting structure made of RCC/MS Structure with anti-corrosive paint, anti vibration pad, power control cable and connection inter connection etc. as per design approved by engineer-in-charge, electrical termination suitable for aluminium conductors along with thermal insulations anti vibration pads, flow switch and required accessories etc, Movable diffuser, Sight Glass at evaporator, Liquid line Isolation valves, Liquid Crystal Display (LCD) Human Machine Interface (HMI) . complete as per specifications and drawings. Complete with first charge of Refrigerant (Preferably at factory charge) . Starter shall be Variable Frequency Drive (VFD) type and shall be Unit Mounted/ Floor Mounted ≥IP42 (UL Listed / CE Marked). Each Compressor shall be equipped with Suitable

capacity Permanent Magnet Motor with class 'F' Insulation suitable for operation on 415 +/- 10% volts, 50 HZ, A.C. Supply. Chillers shall be factory AHRI tested at design conditions at 100%, 75%, 50% and 25% load respectively; test certificates shall be produced for all chillers. The chiller shall be Building Management System (BMS) compatible and shall have RS485/RS232 serial communication protocol; the motor shall be suitable for 3- Phase, 415 V $\pm$ 10%, 50 Hz AC electric supply. The system shall be in confirmation to IS : 16590 and CPWD Specification as amended upto date. BEE Star labeling and CPWD Specification as amended upto date and tested as per IS 16590.

Chilled water Leaving Temp. (6.67deg. C)

Chilled water Entering Temp. (12.2 deg. C)

Evaporator fouling factor = 0.018 m². 0C/kW

Condenser water Entering Temp. (32.2 deg. C)

Condenser water Leaving Temp. (36.4 deg. C)

Condenser fouling factor = 0.044 m². °C/kW

Suitable for Seismic Zone and Altitude as per location/site.

Microprocessor based controller.

33.9.1	121 TR - 150 TR BEE 3 Star Rated	Per TR	27605
33.9.2	151 TR - 300 TR BEE 3 Star Rated	Per TR	22458
33.9.3	301 TR - 450 TR BEE 3 Star Rated	Per TR	21991
33.9.4	451 TR - 600 TR BEE 3 Star Rated	Per TR	20213
33.10	Supply, installation, testing & commissioning of following capacity water cooled Magnetic centrifugal type chiller machine complete with hermetic single/ multi compressors with independent circuits, The refrigerant shall be ozone friendly Chlorofluorocarbons (CFC)-free refrigerant gas R-134A. The chiller machine shall be test and performance as per IS : 16590 and BEE star labeling. The chiller shall have dual Semi/hermetically sealed refrigerant cooled motor of working on 415 + 10% volts, 3 Phase, 50 Hz AC supply. Shell & tube flooded chiller & condenser with descaling & drain valves, victaulic /Flange coupling on condenser & evaporator, microprocessor panel for multiple start ups, i/c suitable foundation/mounting structure made of RCC/MS Structure with anti-corrosive paint, anti vibration pad, power control cable and connection inter connection etc. as per design approved by engineer-in-charge, electrical termination suitable for		

aluminium conductors along with thermal insulations anti vibration pads, flow switch and required accessories etc, Movable diffuser, Sight Glass at evaporator, Liquid line Isolation valves, Liquid Crystal Display (LCD) Human Machine Interface (HMI). complete as per specifications and drawings. Complete with first charge of Refrigerant (Preferably at factory charge) .

Starter shall be Variable Frequency Drive (VFD) type and shall be Unit Mounted/ Floor Mounted $\geq$ IP42 (UL Listed / CE Marked). Each Compressor shall be equipped with Suitable capacity Permanent Magnet Motor with class 'F' Insulation suitable for operation on 415 +/- 10% volts, 50 HZ, A.C. Supply. Chillers shall be factory AHRI tested at design conditions at 100%, 75%, 50% and 25% load respectively; test certificates shall be produced for all chillers. The chiller shall be Building Management System (BMS) compatible and shall have RS485/RS232 serial communication protocol; the motor shall be suitable for 3- Phase, 415 V $\pm$ 10%, 50 Hz AC electric supply. The system shall be in confirmation to IS : 16590 and CPWD Specification as amended upto date.

Chilled water Leaving Temp. (6.67deg. C)

Chilled water Entering Temp. (12.2 deg. C)

Evaporator fouling factor = 0.018 m². °C/kW

Condenser water Entering Temp. (32.2 deg. C)

Condenser water Leaving Temp. (36.4 deg. C)

Condenser fouling factor = 0.044 m². °C/kW

Suitable for Seismic Zone and Altitude as per location/site.

Microprocessor based controller.

33.10.1	120 TR - 210 TR BEE 4 Star Rated	Per TR	29945
33.10.2	211 TR - 300 TR BEE 4 Star Rated	Per TR	27137
33.10.3	301 TR - 355 TR BEE 4 Star Rated	Per TR	23862
33.10.4	356 TR - 450 TR BEE 4 Star Rated	Per TR	21991
33.10.5	451 TR - 600 TR BEE 4 Star Rated	Per TR	21897

CHAPTER 34 COOLING TOWERS

Item No.	Description	Unit	Rate(₹)
34.1	COOLING TOWER Supply, Installation, testing and commissioning of Induced Draft counterflow cooling Towers(CTI approved). The Cooling Tower shall be of Fiber Reinforced Plastic (FRP) Construction. The casing, basin/sump, fan deck and fan cylinder shall be of FRP, with direct driven fans, Galvanised hardware complete with sump and drain connection with suitable valve, PVC Honey comb fill, louvers, drift eliminator complete with spray nozzle having self rotating sprinklers, steel ladder, Isolating switch and other accessories to make it fully operational and maintainance National Accreditation Board for Testing and Calibration Laboratories (NABL) & positioning of cooling tower at Terrace of Building. Propeller Type Fan, weather proof IP 55 and Direct driven.The fan motor shall be premium efficiency IE3 class , as per IS 12615 The Cooling tower shall be capable to communicate effectively with Building Management System (BMS). Range of CT: 6 deg C. Designed Duty Conditions :- EWT,LWT,D/WBT Complete as per CPWD specification/ drawings and as directed by Engineer-in-charge. (Note - Cooling tower size depends on the ambient temperature conditions , contractor must check the required design temperature).		
34.1.1	300 GPM	Each	320033
34.1.2	450 GPM	Each	427148
34.1.3	600 GPM	Each	597021
34.1.4	750 GPM	Each	800884
34.1.5	900 GPM	Each	931092
34.1.6	1050 GPM	Each	1021370
34.1.7	1200 GPM	Each	1179071
34.1.8	1350 GPM	Each	1246446
34.1.9	1500 GPM	Each	1373711
34.1.10	1800 GPM	Each	1691873

34.1.11	2100 GPM	Each	1916501
34.1.12	2400 GPM	Each	2276196
34.1.13	2700 GPM	Each	2383197
34.1.14	3000 GPM	Each	2549102

CHAPTER 35

AHU & FCU

Item No.	Description	Unit	Rate(₹)
35.1	CEILING SUSPENDED AHU Supply, Installation, testing and commissioning of Factory built ceiling suspended chilled water double skin type horizontal/vertical air handling units of following capacity, made of 25mm thick panels consisting of pre plasticized G.I. casing of thickness 0.8mm outside layer and 0.8 mm inside layer with Polyurethane Foam (PUF) insulation factory injected between them by injection moulding machine, complete with blower section with blower suitable for static pressure as required, minimum 2 bend PVC eliminators, cooling coil section with aluminium finned copper tubes (tubes thickness not less than 0.5mm) cooling coil of 4 row deep, filter section with 50mm thick metal viscous/ washable synthetic type air prefilters, belt drive package with Totally Enclosed Fan Cooled (TEFC) drive motor of efficiency class IE3 suitable for 415 ± 10% volts, 50Hz, 3 Phase AC supply suitably designed for Variable Frequency Drive (VFD) applications, drain connections, stainless steel (18G) drain pan with PUF insulation, 150 mm dia. dial type pressure gauges (2 nos.) and industrial type thermometres (2 nos.) and industrial type thermometres (2 nos.) at the inlet and outlet of coil, auto purge valve wherever required, necessary vibration Isolation arrangement, noise level shall not exceed 70 dBA. AHU shall be AHRI/Eurovent certified, fan shall be AMCA certified etc. Complete as per CPWD specification/drawings and as directed by Engineer-in-Charge. (Total static pressure considered is max. 50 mm WC).		
35.1.1	1000 CFM	Each	64662
35.1.2	1600 CFM	Each	75330
35.1.3	2000 CFM	Each	80476
35.1.4	2500 CFM	Each	91705
35.1.5	3000 CFM	Each	101063
35.1.6	4000 CFM	Each	116036
35.1.7	5000 CFM	Each	142237
35.1.8	6000 CFM	Each	160017
35.1.9	8000 CFM	Each	198383

35.1.10	10000 CFM	Each	254530
35.1.11	12000 CFM	Each	305997
35.2	FCU (FAN COIL UNIT) DUCTABLE FAN COIL UNIT Supply, installation ,testing and commisioning of Ceiling Concealed Fan Coil Unit comprising of 3 rows deep chilled water cooling coil, centrifugal blowers, fractional horse power (FHP) motor, synthetic fibre filters, insulated & extended condensate drain pan along with L-type auxillary tray, casing, coil piping connections, condensate drain piping connections & wiring. Fan coil units shall be suitable for operation on 220 +/- 6% Volts, 50Hz, single phase power supply of following sizes & capacities.Complete as per CPWD specification and as directed by Engineer-in-charge.		
35.2.1	3.0 TR nominal capacity with 1200 CFM air quantity.	Each	23114
35.2.2	2.5 TR nominal capacity with 1000 CFM air quantity.	Each	21336
35.2.3	2.0 TR nominal capacity with 800 CFM air quantity.	Each	18341
35.2.4	1.5 TR nominal capacity with 600 CFM air quantity.	Each	16002
35.2.5	1.0 TR nominal capacity with 400 CFM air quantity.	Each	14130
35.3	CASSETTE FAN COIL UNIT Supply, installation, testing and commisioning of Chilled Water Ceiling Suspended Hydronic Cassette type fan coil unit, four(4) way directional flow, low noise, each complete with two(2) rows of deep chilled water cooling coil, multi-blade centrifugal fan, test reports from National Accreditation Board for Testing and Calibration Laboratories (NABL)/AHRI accerdited lab, electronic air cleaning system, required set of ball valves with & without strainers & 2 way Motorized valve, insulated condensate drain pans with drain pump assembly & drain pump failure alarm, pipe connections through copper pipes, Infra-red remote control, Liquid Crystal Display (LCD), four (4) speed motor , fan four(4) direction air flow, auto swing louver, decorative panel etc., condensation drain connections, All units shall be suitable for 220 +/- 10% Volts, 50 Hz, single phase power supply etc. complete as per specification. The wireless Remote temperature control / thermostat shall have memory back up for set point restore in case of power failure and re-start. Four(4)hanger rods with required anchoring fasteners, hooks, washers etc.complete		

as per CPWD specification and as directed by Engineer-in-charge.

35.3.1	4.0 TR nominal capacity with 1600 CFM air quantity.	Each	46817
35.3.2	3.5TR nominal capacity with 1400 CFM air quantity.	Each	43512
35.3.3	3.0 TR nominal capacity with 1200 CFM air quantity.	Each	24049
35.3.4	2.5 TR nominal capacity with 1000 CFM air quantity.	Each	21897
35.3.5	2.0 TR nominal capacity with 800 CFMm air quantity.	Each	16844
35.3.6	1.5 TR nominal capacity with 600 CFM air quantity.	Each	14411
35.3.7	1.0TR nominal capacity with 400 CFM air quantity.	Each	13569
35.4	HIGH WALL FAN COIL UNIT Supply, installation ,testing and commisioning of High wall Fan Coil Unit comprising of two 2 rows deep chilled water cooling coil, centrifugal blowers, fractional horsepower (FHP) motor , synthetic fibre filters, insulated & extended condensate drain pan, casing, coil piping connections, condensate drain piping connections & wiring. Fan coil units shall be suitable for operation on 220 +/- 6% Volts, 50Hz, single phase power supply of following sizes & capacities.Complete as per CPWD specification and as directed by Engineer-in-charge.		
35.4.1	2.0 TR nominal capacity with 800 Cfm air quantity.	Each	24933
35.4.2	1.5 TR nominal capacity with 600 Cfm air quantity.	Each	21922
35.4.3	1.0 TR nominal capacity with 400 Cfm air quantity.	Each	17889

CHAPTER 36

EVAPORATIVE COOLING

Item No.	Description	Unit	Rate(₹)
36.1	EVAPORATIVE COOLING Supply, Installation, Testing and Commissioning of factory assembled double skin central evaporative cooling plant having specifications as per A, B, C, D, E, F,G A. Air washer section comprising 50 mm thick pre-air filter made out from washable Aluminium wire mesh filter with 90 % down to 10 microns. B. Humidification section comprising of Wet pads 200 mm thick impregnated cellulose paper media (Celdec pads) of imported origin with two (2) bend PVC eliminator, internal casing with blank off's of wet section in SS-304 construction. C. Fan Section comprising of belt driven, Double Inlet Double Width (DIDW) backward curved fan with outlet velocity less than or equal to 10 m/s and minimum efficiency of 70% Air Movement & Control Association International (AMCA) certified centrifugal fan suitable for required cfm at 50 mm WC static pressure. D. Totally Enclosed Fan Cooled (TEFC) motor of IE-3 class as required with pulley, belt. E. The unit shall be fabricated with frame work hollow extruded aluminium profile with 0.80 mm precoated GSS on outside and on inside complete with 25 mm thick Chloroflouro Carbon (CFC) free Polyurthane Frame (PUF) insulation of minimum 40 kg/cum density sandwiched in between inner and outer skins, SS -304 (18 g) Sump tank, 25mm C-PVC piping, make up, drain & quick fill and drain connection, Butterfly/Gate valves for pumps, make up, drain & quick fill and drain connections of sump, 2 no. Pumps of suitable capacity and necessary fittings, stand, anti vibration pads etc.as required. F. Starter panel DOL/Star-Delta suitable for operation of Blower motor & pump made out of 1.6 mm thick sheet steel powder coated enclosure comprising of over load protection relay, short circuit & single phasing protection, ON / OFF push buttons, ammeter, voltmeter, indicating lamps, MCB, contactor etc. (As per Specification of CPWD & direction of Engineer- in-charge) complete in all respect. G. All as per pre approved by Engineer-in-charge.		
36.1.1	5000 CFM	Each	99192
36.1.2	8000 CFM	Each	131944
36.1.3	10000 CFM	Each	174053

36.1.4	12000 CFM	Each	206337
36.1.5	15000 CFM	Each	246108
36.1.6	20000 CFM	Each	345299
36.1.7	25000 CFM	Each	396767
36.1.8	30000 CFM	Each	483793

CHAPTER 37

AIR COOLED HEAT PUMP FOR HOT WATER

Item No.	Description	Unit	Rate(₹)
	AIR COOLED HEAT PUMP (FOR HOT WATER)		
37.1	Supply, installation, testing & commissioning of Heat pumps system for hot water using heat energy source from ambient air to Hot water, of High efficiency and energy saving operation, capable of heating water at 55° to 60° C with silent operation (the sound level should not exceed 65 dB). The Heat Pump shall have LCD display control panel with built in diagnostic and troubleshooting information and an inbuilt cycle for defrosting in case icing occurs on evaporator including all other mounting, fitting and controls, all interconnecting wiring/cabling between heat pump and electric panel etc complete in all respect with but not limited to following specifications. Power Supply V/Ph/Hz : 400~440V/3 PH/50Hz. Suggested Maximum output water temperature in Deg C : 55 ° to 60° C, ambient temperature range in Deg C : -5 °C~45 °C , Type of Fan : Low Noise axial fan , Suggested Noise level: DBA <63 , COP: 3.0 to 4.0. Hot Water Storage Tank consisting of GI/MS/SS cylindrical shape clarifier tank. (inlet temperature of hot water storage tank 60- 65 deg.C) suitable for minimum 4 Kg /Sqm working pressure. Tank shall be provided with water flow meter, inlet / outlet, overflow, drain connection with MH cover, 6 mm thick tank, pressure relief valves, pressure gauge at inlet / outlet with isolation cock, thermometer at inlet / outlet, ball Valve, safety valve,check valve etc and capacity of the tank shall be for 03 Hr rated output. The complete system to be tested to a pressure of 10 Kg/cm2 complete in all respects including temperature indications, thermostat and other required accessories. Tank shall be insulated with 100 mm thick crown 150 grade & 50 mm rock wool pads of approved quality and cladded with 24 SWG aluminium sheet cladding.		
37.1.1	200 LPH	Each	112292
37.1.2	300 LPH	Each	149723
37.1.3	500 LPH	Each	196512

CHAPTER-38

IPABX/EPABX System

Item No.	Description	Unit	Rate(₹)
36.1	Supply, installation, testing and Commissioning of IP based voice communication system with 04 Port Voicemail, 04 Port FXS, 04 Port FXO, 1 PRI Trunk lines (30 Channels) Circuit with CLIP Facility, min 100 IP users License with provision for further additions, 100 analog users, 01 Number IP based Operator Console, 100 Party Conference, Speed Dial, Music on Hold, Internal/ External ring difference, Call Barring, Call Pickup, Reducdant server in active -Active mode and as per technical specification including supporting accessories etc complete as required.	Each	1532182
36.2	Supply, installation, testing and Commissioning of IP at its core with IP switching technology and 100% non blocking communication system with 04 Port Voicemail, 04 Port FXS, 04 Port FXO, 1 PRI Trunk lines (30 Channels) Circuit with CLIP Facility, min 100 IP users License with provision for further additions, 100 analog users, 01 Number IP based Operator Console, 45 Party Conference, Speed Dial, Music on Hold, Internal/ External ring difference, Call Barring, Call Pickup. The system should be reducdant which offer 100% duplication of power supply and CPU card and to match the hybrid IP-PBX as per technical specification including supporting accessories etc complete as required.	Each	1210038
36.3	Supply, Installation, Testing & Commissioning of Type 1 IP Phone having 2 VoIP account, 120 x40 DOT matrix screen with backlight, Full HD duplex speaker phone, IPV6, Gigabit Etherent, PoE etc. complete as required.	Each	11582.00
36.4	Supply, Installation, Testing & Commissioning of Analog phones having standard feautres like 16 digit LCD display, 2 way speaker, caller ID and number storage facility etc. complete as required.	Each	1529.00
36.5	Supplying and drawing 20 Pair 0.5 mm dia FRLS PVC insulated annealed copper conductor, unarmored telephone cable in the existing surface/ recessed steel/ PVC Conduit etc. complete as required.	Meter	340.00

CHAPTER -39

LAN

Item No.	Description	Unit	Rate(₹)
39.1	Supplying, installation, Testing and commissioning of following capacity 8 port Layer 2 indoor Network Switch having features and specifications etc. as mentioned here under: Minimum 8 X RJ-45 Gigabit Ethernet Ports and additional 2 X 1G SFP Ports with non-blocking architecture by having Switching capacity of minimum 20Gbps and packet forwarding rate of 14Mpps or higher, 8K MAC table. Internal dual AC Power supply with operating temperature of -5 degree Celsius to +50 degree Celsius. Jumbo frame: 10KB or higher, IGMP Snooping, IGMP snooping querier and support for IGMP Snooping Fast Leave, Should create 250 or more MLD groups, MLD Snooping, Per VLAN MLD Snooping C MLD Snooping Fast Leave, STP, RSTP, MSTP, Loopback detection, Multicast filtering, VLAN Tagging, QOS: 8 Que per port, WRR, IPv4/IPv6 Interface : minimum 100 IPv4 static route entries and minimum 50 IPv6 static route entries, 700 Access Control entries, SSH CSSL for IPv4 and IPv6, 802.1x, ARP Spoofing, DHCP Snooping and server screening, IP+MAC+Port Binding, Per Port Bandwidth Control, 802.1X Authentication (Supports local/RADIUS database, Port-based Access Control and EAP, OTP, TLS, TTLS, PEAP Support) Cable Diagnostic feature, LLDP, LLDP-MED, SNTP/NTP, RMON, SNMP v1,v2c, v3 and SNMP Traps. Certifications: UL, CE, FCC, RoHS, MTCTE, OEM/Product must be on Trusted Telecom Portal of Department of Telecommunication, Government of India,		
39.1.1	8 port PoE Layer 2 Network Switch with PoE Support of 130W or higher with each copper port supporting 802.3at PoE+.	Each	35091
39.1.2	8 port Layer 2 Non-PoE Network Switch	Each	28073
39.2	Supplying, installation, Testing and commissioning of following capacity 8 port PoE Layer 2 Industrial Grade outdoor Network Switch having features and specifications etc. as mentioned here under: At least 8 X RJ-45 PoE/PoE+ Gigabit Ethernet Ports and additional 4 X 1G SFP Ports with non-blocking architecture by having Switching capacity of min. 24Gbps and packet forwarding rate of 17Mpps or higher, console port, 16K MAC table, IPv4 and IPv6 ready, IGMP snooping v1, v2, v3, DHCP, DHCP Relay, NTP/SNTP, SNMPv1, v2 C v3, TELNET/SSH, LACP, STP, RSTP, MSTP, ERPS. VLAN: Port based, Mac based, IP subnet based, tag based, protocol based. Q-in Q, L2/L3 ACL, 1588v2 Precision Time Protocol HW-Based E2E Transparent clock, SNTP, TFTP, SNMP, Radius, Syslog. Dual DC Power input, IP30 rated, DIN Rail, -5° to 70° C operating		

	temperature. Certificate: UL, CE, FCC, NEMA-TS2, IEC/MIL-STD for Shock , Vibration & Drop Certificate. OEM Should have Certificate for Industrial Cyber Security Capability IEC-62443-4-1:2018, ISO 45001, ISO 27001 and ISO 14001 Certified or Equivelent Indian Certificate. OEM/Product must be on Trusted Telecom Portal of Department of Telecommunication, Government of India,	Each	106678
39.3	Supplying, installation, Testing and commissioning of following capacity 24 port Layer 2 indoor Network Switch having features and specifications etc. as mentioned here under: At least 24 X RJ-45 Gigabit Ethernet Ports and additional 2 X 10G Base -T with 4 X SFP Ports with non-blocking architecture by having Switching capacity of min. 168Gbps and packet forwarding rate of 125Mpps or higher, 16K MAC table. Console Port, USB port, Stacking support of min. 8 units per stack. Internal dual AC Power supply, STP, RSTP, MSTP, BPDU Filter, BPDU Restriction, Min. 9K Jumbo Frame, LBD, IGMP Snooping V1/V2/V3, MLD Snopping V1/V2. IGMP /MLD Groups 1K or more, IPv4/IPv6 Loopback Interface, 16 L3 IP Interface, Ipv6 ND, VRRPv3, UDP Helper, ECMP. VLAN: 802.1Q, Port based, Q-in-Q, Multicast VLAN, Protocol VLAN, VLAN Trunking, DHCP Snopping, Server, server Screening. RADIUS , TACACS+ Authentication, QoS: 802.1P, 8 queues per port, QoS : WRR, Strict+WRR, WRED, 802.1p. ACL: MAC based, IPv4C IPv6, TCP/UDP Port number, time based ACL, TFTP Client, SNMP V1, v2c, v3, SNMP traps, RMON, DHCP server, relay, client, LLDP, LLDP-MED, OAM, Dying Gasp, 802.3ah, sflow, RIP, OSPF v2/v3, policy based route, SSL, SSH. 6 kV surge protection on all Gigabit Ethernet ports and on all GE RJ-45 access ports. Certifications: UL, CE, FCC, RoHS, MTCTE, OEM/Product must be on Trusted Telecom Portal of Department of Telecommunication, Government of India,		
39.3.1	24 port PoE Layer 2 Network Switch with PoE Support of 370W or higher with each cooper port supporting 802.3at PoE+ min.	Each	125393
39.3.2	24 port Layer 2 Non-PoE Network Switch	Each	86163
39.4	Supplying, installation, Testing and commissioning of following capacity 24 port Layer 3 Network Switch having features and specifications etc. as mentioned here under: At least 20 X RJ-45 Gigabit Ethernet Ports and additional 4 combo 10/100/1000 base-t/SFP Ports with additional 4 SFP+ ports non-blocking architecture by having Switching capacity of min. 128Gbps and packet forwarding rate of 95Mpps or higher, 16K MAC table. Console Port, USB port, Internal AC Power supply with		

39.5	operating temperature of 0 to +50 degree Celsius. Stacking port with stacking support of minimum 8 units per stack. 48K Mac address, 9K or more Jumbo frame, ERPS with enabling ring to converge in less than 50 ms from node or link failure, Static routing, Static, 6to4, ISATAP and GRE, IPv6 Neighbor Discovery (ND), BGP, ISIS, MPLS, VRRP v2, OSPFv2/v3, PBR, Route Redistribution, RIPv1/v2/ng, IP Helper all L3 Protocols should support from day 1 for both IPv6 and IPv4, 802.1x, Port security , Radius C TACACS+ authentication C accounting, DAD, DAI, DoS attack prevention, SSH, Guest VLAN, IP Access List, MAC Access List. Multicast: VLAN, PIM-SM, PIM-DM, PIM- SDM, PIM-SMv6, DVMRP v3 and MSDP, IGMP C MLD Snooping, Flow mirroring, IGMP C MLD Snooping Querier, Per-VLAN IGMP C MLD Snooping. SNMP, Dual configurations, Multiple images, RMON, LLDP/LLDP-MED, sFlow, DHCP/BOOTP Client, IPv4/IPv6, Syslog server, Debug command, 802.3ah Ethernet Link OAM, 802.1ag Connectivity Fault, Y.1731 OAM, 802.1Qbb Priority- based Flow Control (PFC), Dying Gasp, Cable Diagnostics , 6 kV surge protection on all Gigabit Ethernet ports and on all GE RJ-45 access ports. Certifications: UL, CE, FCC, RoHS, MTCTE, OEM/Product must be on Trusted Telecom Portal of Department of Telecommunication, Government of India. Supplying, installation, Testing and commissioning of 10 Giga Bandwidth Data Centre Network Switch of 48 port Layer 3 having features and specifications etc. as mentioned here under: - ToR/EOR/Core Network Switch 48x 10GbE SFP+, 6x 100GbE QSFP28, RPS, with Console Port, Management Port and 1 x USB 2.0 Type A port, The switch must provide a switching capacity of no less than 2.16 Tbps and a forwarding rate of at least 1600 Mpps, ensuring non-blocking performance for high-bandwidth applications. It should support hot-swappable dual power supply modules (AC or DC) for 1+1 redundancy and load sharing, as well as hot-swappable, front-to-back airflow fan trays with N+1 redundancy to ensure continuous availability. IEEE 802.1X support for port-based and MAC-based authentication should be available, along with integration for RADIUS and TACACS+ authentication services. Guest VLANs, DoS prevention, and management access restrictions are essential. SNMP v1/v2c/v3, syslog, sFlow, and support for auto-configuration via DHCP. Support for OpenFlow v1.4, Netconf/YANG, and LLDP is required for modern network automation. Support ONIE (Open Network Install Environment), IEEE 802.1Q bb Priority-based Flow Control (PFC), OpenFlow v1.4, ERPS, IPv6 Tunneling, IP interfaces and supported, VXLAN, Private VLAN, Multicast	Each	193791
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	VLAN, Guest VLAN Double VLAN (Q-in-Q), RoCEv2, PIM-SSM/SM, MSDP. Support High Availability with MLAG/VSS and Layer 2/3 VPNs Critical for Multi-tenant, Virtualized, and Containerized Environments. BGPv4/v6, OSPFv2/v3, IS-IS, RIPng. VRRP, ECMP, BFD, and route redistribution for high-availability routing. Support flexible routing domains and secure inter-segment traffic flow. Certifications: CE, FCC, MTCTE, OEM/Product must be on Trusted Telecom Portal of Department of Telecommunication, Government of India.	Each	1055981
39.6	Supplying, Installation, Testing and commissioning of small form-Pluggable (SFP) for Network Switches. Must be hot Pluggable, RoHS Complaint, etc. complete as required.		
39.6.1	1G SFP Single Mode/Multi Mode		4211
39.6.2	1G Copper SFP		7018
39.6.3	10 G SFP + Single Mode/Multi Mode		14972
39.6.4	40G SFP		42110
39.6.5	100G SFP		126329
39.7	Supplying, Installation, Testing and commissioning of indoor Wireless Access Point for Low Density use having features and specifications etc. as mentioned here under - Dual-band Wi-Fi6 (802.11ax) 574Mbps (2.4GHz) + 1200Mbps (5GHz), WI-FI 6 Certified, 1 x RJ45 console port, 1G LAN POE Port, factory reset, WPA/WPA2/WPA3™ Personal/Enterprise, WEP 64/128-bit, SSID broadcast disable, MAC address access control, Internal RADIUS server, SNMP, Customizable Captive Portal, Auto Channel, Works as Access Point, WDS, WDS with AP, Wireless Client, Web (HTTP), Secure Socket Layer (SSL), Traffic control, Support ATF/Fast Roaming/Band Steering, Supports enhanced security – WPA-PSK/WPA2-PSK/WPA3-PSK and RADIUS client, and Cipher negotiation, and MAC/ IP ACL for networks, Certifications: MTCTE, CE, FCC. Should also work as Standalone and must be Compatible with S/W & H/W Controller with additional features like - Real-Time Monitoring and Alerts, Rich Advanced Features, Centralized Device Management, Role-Based Administration, Real-Time Monitoring, VLAN and Access Control, Auto RF Management and Bandwidth Optimization, Multi-SSID and Captive Portal Authentication.OEM/Product must be on Trusted Telecom Portal of Department of Telecommunication, Government of India.	Each	22739

- 39.8 Supplying, Installation, Testing and commissioning of indoor Wireless Access Point for High density use having features and specifications etc. as mentioned here under- WiFi6 802.11ax 574Mbps (2.4GHz) + 2402Mbps (5GHz), WIFI 6 Certified, 1 x RJ45 console port, MG 2.5G LAN POE Port, factory reset, WPA/WPA2/WPA3™ Personal/Enterprise, WEP 64/128-bit, SSID broadcast disable, MAC address access control, Internal RADIUS server, SNMP, Customizable Captive Portal, Auto Channel, Works as Access Point, WDS, WDS with AP, Wireless Client, Web (HTTP), Secure Socket Layer (SSL), Traffic control, Support ATF/Fast Roaming/Band Steering, Supports enhanced security – WPA-PSK/WPA2-PSK/WPA3-PSK and RADIUS client, and Cipher negotiation, and MAC/ IP ACL for networks, Certifications: MTCTE, CE, FCC.
Works as Standalone, Compatible with S/W & H/W Controller with additional features like - Real-Time Monitoring and Alerts, Rich Advanced Features, Centralized Device Management, Role-Based Administration, Real-Time Monitoring, VLAN and Access Control, Auto RF Management and Bandwidth Optimization, Multi-SSID and Captive Portal Authentication.
- Each 35559
- 39.9 Supplying, Installation, Testing and commissioning of Outdoor Wireless Access Point having features and specifications etc. as mentioned here under - WIFI 6 Certified, 1 x RJ45 console port, MG 2.5G LAN POE Port, factory reset, 4 x internal antennas, the transmission coverage of shall be minimum 80 meters in radius, AP should have transmission power minimum of 22 dBm, The AP shall provide a minimum of 29 dBm EIRP for both 2.4 GHz and 5 GHz frequencies. Field deployment shall be with EIRP as per regulatory guidelines, AP shall have dual-band Omni-directional Antenna, either internal or external. Field deployment shall be with EIRP as per the WPC guideline, WPA/WPA2/WPA3™ Personal/Enterprise, WEP 64/128-bit, SSID broadcast disable, MAC address access control, Internal RADIUS server, SNMP, Customizable Captive Portal, Auto Channel, Works as Access Point, WDS, WDS with AP, Wireless Client, Web (HTTP), Secure Socket Layer (SSL), Traffic control, Support ATF/Fast Roaming/Band Steering, Supports enhanced security – WPA-PSK/WPA2-PSK/WPA3-PSK and RADIUS client, and Cipher negotiation, and MAC/ IP ACL for networks, IP68 water/dustproof standard, 6KV Surge Protection; 4KV ESD Protection (contact), 6KV (air), Operating ambient temperature (-5~50 °C), Grounding point support, IEC 60601-1-2 EMC Medical Electrical Equipment Testing and Certification, EN50121-1 & EN50121-4 for Wireless

	Application. Certifications: MTCTE, CE, FCC. Works as Standalone, Compatible with S/W & H/W Controller with additional features like - Real-Time Monitoring and Alerts, Rich Advanced Features, Centralized Device Management, Role-Based Administration, Real-Time Monitoring, VLAN and Access Control, Auto RF Management and Bandwidth Optimization, Multi-SSID and Captive Portal Authentication. OEM/Product must be on Trusted Telecom Portal of Department of Telecommunication, Government of India.	Each	46171
39.10	Supplying, Installation, Testing and commissioning of indoor Wireless Controller having features and specifications etc. as mentioned here under- The network controller shall be a dedicated hardware appliance designed for centralized management of wireless access points, It must support the management of up to 500 networking devices, The system should be on-premise, ensuring data privacy and complete control over network configurations. It must include an intuitive user interface for real-time monitoring, device discovery, firmware upgrades, scheduling, and mass configuration deployment. Features such as role-based administration, multi-site management. AP to controller communication shall be use of industry standards-based (IEEE or IETF) CAPWAP or any tunneling protocols as per guidelines of DoT The controller should support Layer 2 and Layer 3 device discovery, VLAN configuration, MAC ACL groups, IGMP snooping, and bandwidth optimization. Automatic radio frequency (RF) management and access policy enforcement are required for reliable wireless performance. The solution must offer a comprehensive dashboard with network topology, floor plan views, client and device lists, and alert notifications. For authentication, it should support a range of methods including local database, external RADIUS, LDAP, and passcode or MAC-based access. It should provide detailed traffic analytics, logs, and customizable reporting for network visibility and planning. Firmware and configuration backup and restoration functions must be available, with USB-based log storage supported. Physical specifications should include a compact form factor with a minimum of one 2.5G Ethernet port and USB 3.0 port, and it must be MTCTE, CE, FCC, and RoHS certified. The controller should operate reliably within a 0°C to 40°C temperature range and 10–90% non-condensing humidity. OEM/Product must be on Trusted Telecom Portal of Department of Telecommunication, Government of India.	Each	113228

39.11	Supply of 24 port Cat6 Patch Panel loaded. Must be of 1U height with clear label holders and white label with the panel. 24 Ports Cat-6 Patch Panel should have ETL/UL verification program certificate for compliance with ANSI/TIA-568.2-D	Each	5278
39.12	Supplying, Installation, Testing and commissioning of following CAT6 Patch Cord should have ETL/UL verification program certificate for compliance with ANSI/TIA-568.2-D		
39.12.1	Copper Patch Cords of length 1m (3ft)	Each	151
39.12.2	Copper Patch Cords of length 3m (10ft)	Each	227
39.13	Supplying, Installation, Testing and commissioning of following CAT6A Patch Cord should have ETL/UL verification program certificate for compliance with ANSI/TIA-568.2-D		
39.13.1	Copper Patch Cords of length 1m (3ft)	Each	173
39.13.2	Copper Patch Cords of length 3m (10ft)	Each	405
39.14	Supplying, Installation, Testing and commissioning of CAT6 Copper Information Outlet (IO) with face plate of color as per site requirement, should have ETL/UL verification program certificate for compliance with ANSI/TIA-568.2-D. All copper Cable and Components should be from same OEM to maintain compatibility and interoperability.	Each	212
39.15	Supplying, Installation, Testing and commissioning of CAT6A Copper Information Outlet (IO) with face plate of color as per site requirement, should have ETL/UL verification program certificate for compliance with ANSI/TIA-568.2-D. All copper Cable and Components should be from same OEM to maintain compatibility and interoperability.	Each	363
39.16	Supplying, Installation, Testing and commissioning of RJ45 Connector for CAT 6/6A Cables. RJ45 modular plug supports 4 twisted pairs, 8 positions, 8 connectors. Housing: PC, UL94V-2, transparent color. Use for 24- 26 AWG stranded wires.	Each	9
39.17	Supplying, drawing, Installation, Testing and commissioning of CAT6 UTP LSZH 23AWG Twisted Pair Cable in existing conduit/ on surface, Category 6 Unshielded Twisted Pair, 4 pair should be complied as per UL/ETL verification program for compliance with ANSI/TIA-568.2-D standard. Outer diameter should be in the range of 6.1mm nominal with Operating Temperature Range : -5° to +60°C. Bending Radius : < 4 X Cable Diameter at -5°C ± 1°C and Pulling Force : 11.5Kg		
39.17.1	1 Run of cable	Meter	60
39.17.2	2 Run of cable	Meter	96

39.17.3	3 Run of cable	Meter	131
39.17.4	4 Run of cable	Meter	167
39.18	Supplying, drawing, Installation, Testing and commissioning of Cat6A UTP 4 pair, 23 AWG solid copper cable in existing conduit/ on surface, U/FTP, LSZH, Non-Plenum, Horizontal (solid) Cable suitable for high speed data networking application supporting upto 10Gbps over a 100 meter channel. The 4 Unshielded Twisted Pairs (UTP) cable with color coded insulation for easy identification should have FLAME PROPERTIES i.e. Flammability Test - IEC 60332-1, Smoke Density - IEC 61034, LSZH standards compliance: ANSI/TIA-568 C.2, ISO/IEC 11801, IEEE 802.3an, RoHS. Delay Skew should be < 45NS. The outer Cable Diameter should be 7.5 + 2 mm. Cable should have been tested and verified by UL/ ETL.		
39.18.1	1 Run of cable	Meter	74
39.18.2	2 Run of cable	Meter	124
39.18.3	3 Run of cable	Meter	174
39.18.4	4 Run of cable	Meter	224
39.19	Supplying and drawing following core Fiber Optic Cable having corrugated steel armoring. The Fiber should be SM Fiber Central -loose tube filled with Thixotropic jelly, duly following Standards: ISO 11801, IEC 60793-1/60794-1-2, ITU-T-REC G.652D and Telecordia GR-20-core, High quality Electro Chromium Coated Corrugated Steel tape (ECCS) and HDPE-Sheath, Operating ambient Temperature should be – 5 deg C to +50 deg C and Storage Temperature shall be – 5 deg C to +50 deg C, Max Attenuation ± 0.36 (db / km) at Operational Wavelength 1310 nm and ± 0.22 db / km at Operational Wavelength 1550 nm. type of fiber should be 9/125 / G.652D & Refractive Index should be 1.4670/1.4675. The value for Mode-field, Cladding Diameter $9.2 \pm 0.4 \mu\text{m}$ and $125 \pm 0.7 \mu\text{m}$ correspondingly. The Dispersion value $\leq 3.5 \leq 18$ ps/nm-km and PMD value ≤ 0.2 ps/km and Cable Cut-off wavelength ≤ 1260 nm etc complete as required.		
39.19.1	Single Mode 6 Core Optical fiber cable	Meter	64
39.19.2	Single Mode Optical 12 Core fiber cable	Meter	89

39.20	Supplying and drawing following core Multi mode Fiber Optic Cable having corrugated steel armoring, lesser optimize 50/125 µm with HDPE jacket, LSZH (Low smoke Zero Halogen), Tight-buffered or loose tube, maximum attenuation: ≤ 3.5 dB/km at 850 nm, ≤ 1.5 dB/km at 1300 nm, bandwidth ≥ 2000 MHz-km at 850 nm, ≥ 500 MHz-km at 1300 nm compliance and certifications: ISO/IEC 11801 Ed. 2.2, ANSI/TIA-568-C.3, ITU-T G.651.1 RoHS compliant. Multi Mode Fiber shall be Central-loose tube filled with Thixotropic jelly, duly following Standards: ISO 11801, IEC 60793-1/60794-1-2, ITU-T-REC G.652D and Telecordia GR-20-core, High quality Electro Chromium Coated Corrugated Steel tape (ECCS) and HDPE-Sheath, Operating ambient Temperature should be -5 deg C to $+50$ deg C and Storage Temperature should be -5 deg C to $+50$ deg C, Maximum Attenuation ± 3.5 db / km at Operational Wavelength 850 nm and ± 1.5 db / km at Operational Wavelength 1300 nm . type of fiber should be 50/125 / G.652D & Refractive Index should be 1.483/1.478, The value for Mode-field,Cladding Diameter 50 ± 2.5 µm and 125 ± 1 µm corrspondingly etc complete as required.		
39.20.1	6 Core Multi Mode Optical Fiber Cable	Meter	87
39.20.2	12 Core Multi Mode Optical Fiber Cable	Meter	109
39.21	Supply and fixing following port Rack Mount loaded LIU with pigtail, Front-mounted cable saddles for jumper management, suitable to manage both splices and terminations, Pre-assembled shelves in multiple configurations, Rubber fiber slotted bracket built-in, metal splice shelf to protect the fibers, 2 fiber spools built-in for 900µm tight buffered fiber storing, Capable of storing up to 3 meters of 900µm tight buffered fiber per adapter, Removable front and rear covers for better access to interior of LIU, Should be Single / Multi mode LC Type fully loaded. Accessory kit consists of cable ties, mounting ear screws, and spiral wrap tube etc complete as required.		
39.21.1	6 Port LIU	Each	5471
39.21.2	12Port LIU	Each	6970
39.21.3	24 Port LIU	Each	9949

39.22	Supply and fixing Patch Cord of Optical Fiber Multi Mode or Single Mode LC to LC Fiber Duplex having high precision ceramic ferrule with good concentricity, Fiber corning single mode G652D, Cable Type 2mm with Mechanical specification Apex Offset $\leq 50 \mu\text{m}$, Fiber Height $\pm 100 \text{ nm}$, End-face radius of Curvature $7\text{mm} < R < 25\text{mm}$, Repeatability $\leq 0.2 \text{ db}$, Working ambient Temperature – 5 deg C to +50 deg C, Storage Temperature – 10 deg C to +50 deg C, Length -- 2meter, Type -- LC-LC Duplex etc complete as required.	Each	698
39.23	Supply and fixing of following size Wall Mount Rack having Fixed Structure with 0.8mm CRCA Sheet, completely knocked-down condition (CKD) Shape, Vertical Mounting Rail 1.6 mm with 'U'Marking, Minimum 2 cable entry/exit provision at Top and bottom with rubber protection, Front 5mm Toughened Glass with lock, wall mounting kit, Powder coated Color- RAL- 7035, complied with UL & RoHS, 2 x 6 sockets 16A power distribution units, 3 numbers closed cable organizer, Hardware mounting screws packet of 20 x 1 number, etc. complete as required.		
39.23.1	6U Rack	Each	6870
39.23.2	9U Rack	Each	8872
39.23.3	15U Rack	Each	13198
39.24	Supply and fixing 24U Rack with 4 inch Castor wheels and front brake Floor Standing Rack with Main Frame Pillar of 1.25mm CRCA Sheet with removable side panels, L- Shape adjustable Vertical Mounting Rail of 2mm with 'U'Marking completely knocked-down condition (CKD) Shape, fitted with 4 number cooling fans, minimum 5nos two line cable entry/exit provision at top and bottom with rubber protection, Front door 5mm Toughened Glass with MS frame or MS perforated with lock, Rear MS vented or perforated door, Powder coated Color- RAL- 7035 or black, complied with UL & RoHS, 2 x 6 sockets 16A power distribution units, 3 numbers closed cable organizer, Hardware mounting screws packet of 20 x 1 number, etc. complete as required.	Each	20991
39.25	Supply and fixing 42U Rack with 6 inch Castor wheels and front brake Floor Standing Rack with Main Frame Pillar of 1.6mm CRCA Sheet with removable side panels, L- Shape adjustable Vertical Mounting Rail of 2mm with 'U'Marking completely knocked-down condition (CKD) Shape, fitted with 4 number cooling fans, minimum 8nos two line cable entry/exit provision at top and bottom with rubber protection, Front		

	door 5mm Toughened Glass with MS frame or MS perforated with lock, Rear MS vented or perforated door, Powder coated Color- RAL- 7035 or black, complied with UL & RoHS, 2 x 6 sockets 16A power distribution units, 3 numbers closed cable organizer, Hardware mounting screws packet of 20 x 1 number, etc. complete as required.	Each	32628
39.26	Supply and fixing Outdoor Pole/ Wall mount 6U Rack, IP55, Front Door with Filter, Louver & Unique key Lock. 3 Socket PDU 5Amp - 1 No. Self adhesive thermal foam from inside. Hood For Air Inlet at front side.1U Cantilever Tray 250mmDepth, Gasket. etc complete as required.	Each	22008

CHAPTER-40 CCTV CAMERA

Item No.	Description	Unit	Rate(₹)
	2 MP IP IR Dome Camera		
40.1	Supply Installation Testing and Commissioning of 2 MP IP IR Dome Camera having following specifications and features etc :- 1) Signal System: PAL/NTSC, Signal to Noise Ratio: >50 dB, Camera should display Camera title, Date & Time in live & recorded video 2) Image Sensor: 1/2.8" or better progressive Scan CMOS to get color image even at night condition (Minimum Illumination: 0.006 Lux@ F1.4, AGC ON, 0 lux with IR or better) True Day & Night High Performance Mechanical IR cut filter with auto switch, Integrated IR Source (Auto, Manual)- Inbuilt IR LED's with effective distance upto 50 meter or better and 30 meter for colour view in night, Imaging: 1/3s to 1/30000s electronic shutter support, Auto Gain Control, White Balance- Auto, Back Light Compensation, Multi zone Privacy Masking, HLC, Digital Watermarking. 3) Compression (Minimum):- Video:- H.265 or better, H.264H, H.264, Audio:- G.711U/A, G.711Mu, G.726, AAC, G.723 4) Wide Dynamic Range:- WDR (120db or more) 5) Digital Noise Reduction:- DNR (3D) On/Off 6) Video Streaming & Frame Rates :- Triple streaming, configurable (Main stream: 2MP (1920 × 1080)@25/30 fps, Sub streams minimum: 720P@25/30 fps). 7) Image Setting: Rotate Mode, saturation, brightness, contrast, sharpness adjustable through client software or web browser, Edge Analytics: Tripwire, Intrusion, Motion Detection 8) Cyber Security: AES 256-bit Encryption, Configuration encryption, trusted execution, Digest, security logs, account lockout, syslog, video encryption, IP/MAC filtering, HTTPS, trusted upgrade, trusted boot. 9) Onboard Storage: Camera should support built-in Micro SD/SDHC/SDXC Card slot upto 512 GB. It should be supplied with minimum 128GB memory Card. 10) Recording Management: Format SD, overwrite, storage management, video to NAS device 11) Alarm Trigger : Motion/tampering detection; audio detection; network disconnection detection; IP conflict detection; memory card state detection; memory space detection		

12) Network Protocol: SFTP, IPv6, IPv4, DNS, RTCP, NTP, RTP, HTTP, HTTPS, SNMP, TCP/IP, PPPoE, NFS, UDP, ICMP, SSL, DHCP, SMTP, RTSPS, unicast

13) System Capability: ONVIF, Camera shall support open source VMS

14) Ethernet: 1 RJ 45 10/100 Ethernet port

15) Audio : It should support 1 x Built-In Mic and 1/1 Alarm In/ Out for External Mic. and Speakers as per site requirement.

16) Power Input: The camera should support simultaneous dual power input—12 VDC (via power adapter) and PoE (802.3af)—to ensure continuous operation in the event of a failure in one power source.

17) Power Requirement: 12VDC/24 VAC/PoE (802.3af)/ePoE

18) Housing/ Enclosure:- IP67 weather proof, IK10, Metallic body

19) Operating Condition:- Ambient Temperature:- (-)05°C to 50°C, humidity 95% (max) (non-condensing)

20) IR life: 40000 hours or higher

21) Video Bit rate: 32 KBPS - 8 MBPS or better

22) Standards: BIS with ER, STQC Certified, CE, FCC and RoHS

40.1.1 2MP (@ 25/30fps@1080P (1920×1080)) IP IR Dome Camera with 2.8/ 3.6mm fixed lens

Each 8048

40.1.2 2MP @ 25/30fps@1080P (1920×1080) Dome Camera with 2.7mm ~13.5mm Motorized lens, Vari Focal (view Angle for H: 104°–29°; V: 54°–16°; D: 125°–34°)

Each 19796

2 MP IP IR Outdoor Bullet Camera

40.2 Supplying Installation Testing and Commissioning of 2MP Fixed Lens IP IR Outdoor Bullet Camera having following specifications and features etc :-

1) Signal System: PAL/NTSC, Signal to Noise Ratio: > 50 dB, Camera should display Camera title, Date & Time in live & recorded video

2) Image Sensor: 1/2.8" or better progressive Scan CMOS to get color image even at night condition (Minimum Illumination: 0.006 Lux@ F1.4, AGC ON, 0 lux with IR or better) True Day & Night High Performance Mechanical IR cut filter with auto switch, Integrated IR Source (Auto, Manual)- Inbuilt Smart IR LED's with effective distance upto 50 meter or better and 30 meter for colour view in night, Imaging: 1/3s to 1/30000s electronic shutter support, Auto Gain Control, White Balance-Auto, Back Light Compensation, Multi zone Privacy Masking, HLC, Digital Watermarking.

3) Compression (Minimum):- Video:- H.265 or better, H.264H, H.264, Audio:- G.711U/A, G.711Mu, G.726, AAC, G.723

- 4) Wide Dynamic Range:- WDR (120db or more)
- 5) Digital Noise Reduction:- DNR (3D) On/Off
- 6) Video Streaming & Frame Rates :- Triple streaming , configurable
(Main stream: 2MP (1920 × 1080)@25/30 fps Sub streams minimum: 720P@25/30 fps).
- 7) Image Setting: Rotate Mode, saturation, brightness, contrast, sharpness adjustable through client software or web browser, Edge Analytics: Tripwire, Intrusion, -Motion Detection
- 8) Cyber Security: AES 256-bit Encryption, Configuration encryption, trusted execution, Digest, security logs, account lockout, syslog, video encryption, IP/MAC filtering, HTTPS, trusted upgrade, trusted boot.
- 9) Onboard Storage: Camera should support built-in Micro SD/SDHC/SDXC Card slot upto 512 GB. It should be supplied with minimum 128GB memory Card.
- 10) Recording Management: Format SD, overwrite, storage management, video to NAS device,
- 11) Alarm Trigger : Motion/tampering detection; audio detection; network disconnection detection; IP conflict detection; memory card state detection; memory space detection
- 12) Network Protocol: SFTP, IPv6, IPv4, DNS, RTCP, NTP, RTP, HTTP, HTTPS, SNMP— TCP/IP, PPPoE, NFS, UDP, ICMP, SSL, DHCP, SMTP, RTSPS, unicast,
- 13) System Capability: ONVIF, Camera shall support open source VMS
- 14) Ethernet: 1 RJ 45 10/100 Ethernet port
- 15) Audio : It should support 1 x Built-In Mic and 1/1 Alarm In/ Out for External Mic. and Speakers as per site requirement.
- 16) Power Input: The camera should support simultaneous dual power input—12 VDC (via power adapter) and PoE (802.3af)—to ensure continuous operation in the event of a failure in one power source.
- 17) Power Requirement: 12VDC/24 VAC/PoE (802.3af)/ePoE
- 18) Housing/ Enclosure:- IP67 weather proof, IK10, Metallic body
- 19) Operating Condition:- Ambient Temperature:- (-)05°C to 50°C, humidity 95% (max) (non-condensing)
- 20) IR life: 40000 hours or higher
- 21) Video Bit rate: 32 KBPS - 8 MBPS or better
- 22) Standards: BIS with ER, STQC Certified, CE, FCC and RoHS

40.2.1 2MP @ 25/30fps@1080P (1920×1080) Outdoor Bullet Camera with 2.8/ 3.6mm fixed lens

Each 7760

40.2.2	2MP @ 25/30fps@1080P (1920×1080) Outdoor Bullet Camera with 2.7mm ~13.5mm Motorized lens, Vari Focal (view Angle for H: 97°–28°; V: 52°–16°; D: 117°–32°)	Each	19796
	4MP IP IR Dome Camera		
40.3	Supplying Installation Testing and Commissioning of 4MP IP IR Dome Camera having following specifications and features etc :- 1) Signal System: PAL/NTSC, Signal to Noise Ratio: > 50 dB, Camera should display Camera title, Date & Time in live & recorded video 2) Image Sensor: 1/2.8" or better progressive Scan CMOS to get color image even at night condition (Minimum Illumination: 0.006 Lux@ F1.4, AGC ON, 0 lux with IR or better) True Day & Night High Performance Mechanical IR cut filter with auto switch, Integrated IR Source (Auto, Manual)- Inbuilt IR LED's with effective distance upto 50 meter or better and 30 meter for colour view in night, Imaging: 1/3s to 1/30000s electronic shutter support, Auto Gain Control, White Balance- Auto, Back Light Compensation, Multi zone Privacy Masking, HLC, Digital Watermarking. 3) Compression (Minimum): Video:- H.265 or better, H.264H, H.264, Audio:- G.711U/A, G.711Mu, G.726, AAC, G.723 4) Wide Dynamic Range:- WDR (120db or more) 5) Digital Noise Reduction:- DNR (3D) On/Off 6) Video Streaming & Frame Rates:- Triple streaming, configurable Main stream: 4MP (2560×1440)25/30fps, Sub streams minimum: 720P@25/30 fps 7) Image Setting: Rotate Mode, saturation, brightness, contrast, sharpness adjustable through client software or web browser, Edge Analytics: Tripwire, Intrusion, -Motion Detection 8) Cyber Security: AES 256-bit Encryption, Configuration encryption, trusted execution, Digest, security logs, account lockout, video encryption, IP/MAC filtering, HTTPS, trusted upgrade 9) Onboard Storage: Camera should support built-in Micro SD/SDHC/ SDXC Card slot upto 512 GB. It should be supplied with minimum 128GB memory Card. 10) Recording Management: Format SD, overwrite, storage management, video to NAS device, 11) Alarm Trigger : Motion/tampering detection; network disconnection detection; IP conflict detection; memory card state detection; memory space detection		

12) Network Protocol: SFTP, IPv6, IPv4, DNS, RTCP, NTP, RTP, HTTP, HTTPS, SNMP TCP/IP, PPPoE, NFS, UDP, ICMP, SSL, DHCP, SMTP, RTSPS, unicast,

13) System Capability: ONVIF, Camera shall support open source VMS

14) Ethernet: 1 RJ 45 10/100 Ethernet port

15) Audio : It should support 1 x Built-In Mic and 1/1 Alarm In/ Out for External Mic. and Speakers as per site requirement.

16) Power Input: The camera should support simultaneous dual power input—12 VDC (via power adapter) and PoE (802.3af)—to ensure continuous operation in the event of a failure in one power source.

17) Power Requirement: 12VDC/24 VAC/PoE (802.3af)/ePoE

18) Housing/ Enclosure:- IP67 weather proof, IK10, Metallic body

19) Operating Condition:- Ambient Temperature:- (-)05°C to 50°C, humidity 95% (max) (non-condensing)

20) IR life: 40000 hours or higher

21) Video Bit rate: 32 KBPS - 8 MBPS or better

22) Standards: BIS with ER, STQC Certified, CE, FCC and RoHS

40.3.1	4MP @ 25/30fps@1440P (2560 x 1440) Dome Camera with 2.8/ 3.6/ 6/8/12 mm fixed lens (as per site requirement).	Each	10365
40.3.2	4MP @ 25/30fps@1440P (2560x1440) Dome Camera with 2.7mm ~13.5mm Motorized lens, Vari Focal (view Angle for H: 104°–29°; V: 54°–16°; D: 125°–34°)	Each	23754

4MP IP IR Outdoor Bullet Camera

40.4 Supplying Installation Testing and Commissioning of 4MP IP IR Outdoor Bullet Camera having following specifications and features etc :-

1) Signal System: PAL/NTSC, Signal to Noise Ratio: > 50 dB, Camera should display Camera title, Date & Time in live & recorded video

2) Image Sensor: 1/2.8" or better progressive Scan CMOS get color image even at night condition (Minimum Illumination: 0.006 Lux@ F1.4, AGC ON, 0 lux with IR or better) True Day & Night High Performance Mechanical IR cut filter with auto switch, Integrated IR Source (Auto, Manual)- Inbuilt IR LED's with effective distance upto 50 meter or better and 30 meter for colour view in night, Imaging: 1/3s to 1/30000s electronic shutter support, Auto Gain Control, White Balance- Auto, Back Light Compensation, Multi zone Privacy Masking, HLC, Digital Watermarking.

- 3) Compression (Minimum): Video:- H.265–or better, H.265, H.264H, H.264, Audio:- G.711U/A, G.711Mu, G.726, AAC, G.723
- 4) Wide Dynamic Range:- WDR (120db or more)
- 5) Digital Noise Reduction:- DNR (3D) On/Off
- 6) Video Streaming & Frame Rates :-
Triple streaming , configurable
Main stream: 4MP (2560×1440)25/30fps
Sub streams minimum: 720P@25/30 fps
- 7) Image Setting: Rotate Mode, saturation, brightness, contrast, sharpness adjustable through client software or web browser, Edge Analytics: Tripwire, Intrusion, Motion Detection
- 8) Cyber Security: AES 256-bit Encryption, Configuration encryption, trusted execution, Digest, security logs, account lockout, video encryption, IP/MAC filtering, HTTPS, trusted upgrade
- 9) Onboard Storage: Camera should support built-in Micro SD/ SDHC/ SDXC Card slot upto 512 GB. It should be supplied with minimum 128GB memory Card.
- 10) Recording Management: Format SD, overwrite, storage management, video to NAS device,
- 11) Alarm Trigger : Motion/tampering detection; network disconnection detection; IP conflict detection; memory card state detection; memory space detection
- 12) Network Protocol: SFTP, IPv6, IPv4, DNS, RTCP, NTP, RTP, HTTP, HTTPS, SNMP –TCP/IP, PPPoE, NFS, UDP, ICMP, SSL, DHCP, SMTP, RTSPS, unicast,
- 13) System Capability: ONVIF, Camera shall support open source VMS
- 14) Ethernet: 1 RJ 45 10/100 Ethernet port
- 15) Audio : It should support 1 x Built-In Mic and 1/1 Alarm In/ Out for External Mic. and Speakers as per site requirement.
- 16) Power Input: The camera should support simultaneous dual power input—12 VDC (via power adapter) and PoE (802.3af)—to ensure continuous operation in the event of a failure in one power source.
- 17) Power Requirement: 12VDC/24 VAC/PoE (802.3af)/ePoE
- 18) Housing/ Enclosure:- IP67 weather proof, IK10, Metallic body
- 19) Operating Condition:- Ambient Temperature:- (-)05°C to 50°C, humidity 95% (max) (non-condensing)
- 20) IR life: 40000 hours or higher
- 21) Video Bit rate: 32 KBPS - 8 MBPS or better
- 22) Standards: BIS with ER, STQC Certified, CE, FCC and

	RoHS		
40.4.1	4MP @ 25/30fps@1440P (2560 x 1440) Outdoor Bullet Camera with 2.8/ 3.6/6/8/12 mm fixed lens (as per site requirement)	Each	10063
40.4.2	4MP @ 25/30fps@1440P (2560x1440) Outdoor Bullet Camera with 2.7mm ~13.5mm Motorized lens, Vari Focal (view Angle for H: 97°–28°; V: 52°–16°; D: 117°–32°)	Each	23754
	5/6MP IP IR Dome Camera		
40.5	Supplying Installation Testing and Commissioning of 5/6MP IP IR Dome Camera having following specifications and features etc :- 1) Type of Camera: Dome Camera 2) Image Sensor: 1/2.8" or better progressive Scan CMOS get color image even at night condition 3) Signal System: PAL/NTSC 4) Minimum Illumination: 0.008Lux@ F1.4, AGC ON, 0 lux with IR or better 5) Imaging: 1/3s to 1/30000s Shutter Support, Auto Gain Control , White Balance- Auto, Back Light Compensation, Multi zone Privacy Masking, HLC, Digital Watermarking. 6) On Screen Display: Camera should display Camera title, Date & Time in live & recorded video both. 7) Signal to Noise Ratio: > 50 dB 8) Day & Night: True Day & Night High Performance Mechanical IR cut filter with auto switch, IR Source- Inbuilt IR LED's with effective distance upto 50 Mtrs integrated IR 30 Mtrs for colour view in night. 9) Video Compression (Minimum): H.265 or better, H.265, H.264H, H.264, Audio:- G.711U/A, G.711Mu, G.726, AAC, G.723 10) Wide Dynamic Range: WDR (120db or more) 11) Digital Noise Reduction: DNR (2D/3D) On/Off 12) Streaming: Triple streaming, configurable 13) Video Streaming & Frame Rates: Triple streaming , configurable Main stream: 5/6MP @25/30 fps, Sub streams: D1@25/30 fps or better 14) Image Setting: Rotate Mode, saturation, brightness, contrast, sharpness adjustable through client software or web browser 15) Profile Management: User configuration import, export 16) Security: User Authentication, Water Marking		

- 17) Onboard Storage: Camera should support built-in Micro SD/SDHC/SDXC Card slot upto 512 GB. It should be supplied with minimum 128GB memory Card.
- 18) Recording Management: Format SD, overwrite, storage management, video to NAS device,
- 19) Edge Analytics : Tripwire, Intrusion, Motion Detection
- 20) Alarm Trigger : Motion/tampering detection; network disconnection detection; IP conflict detection; memory card state detection; memory space detection
- 21) Alarm Support: It should have 1/1 Alarm In/ Out Port
- 22) Audio Support: It should have 1x Built-In Mic and 1/1 Audio In/ Out Port for external Mic. and Speaker (As per site requirement) with G.711U/A/ G.711Mu/ AAC/ G.726 audio compresion
- 23) Network Protocol: SFTP, IPv6, IPv4, DNS,RTCP, NTP, RTP, HTTP, HTTPS, SNMP TCP/IP, PPPoE, NFS, UDP, ICMP, SSL, DHCP, SMTP, RTSPS, unicast,
- 24) System Capability: ONVIF
- 25) VMS: Camera shall support open source VMS
- 26) Cyber Security: trusted boot, AES 256-bit Encryption, Configuration encryption, trusted execution, Digest, security logs, , account lockout, video encryption, IP/MAC filtering, HTTPS, trusted upgrade.
- 27) Ethernet: 1 RJ 45 10/100 Ethernet port
- 28) Power Input: The camera should support simultaneous dual power input—12 VDC (via power adapter) and PoE—to ensure continuous operation in the event of a failure in one power source.
- 29) Power Requirement: 12VDC/24 VAC/PoE (802.3af)/ePoE
- 30) Enclosure: IP67 weather proof, IK10
- 31) Operating Condition: Ambient Temperature:- (-)05°C to 50°C, humidity 95% (max) (non-condensing)
- 32) IR life: 40000 hours or higher
- 33) Video Bit rate: 32 KBPS - 8 MBPS or better
- 34) Standards: BIS with ER, STQC Certified, CE, FCC and RoHS

40.5.1	5/6MP ((3200 × 1800)@20 fps or better) IP IR 2.8/ 3.6/6/8/12 mm fixed lens (as per site requirement) Dome Camera.	Each	12800
40.5.2	5/6MP ((3200 × 1800)@20 fps or better) IP IR 2.7mm ~13.5mm Motorized lens, Vari Focal Dome Camera,Angle of View: H: 104°–32°, V: 54°–18°, D: 125°–36° or better 5/6MP IP IR Bullet Camera	Each	26993
40.6	Supplying Installation Testing and Commissioning of 5/6 MP IP IR Bullet Camera having following spcifications and features		

etc :-

- 1) Type of Camera: Bullet Camera
- 2) Image Sensor: 1/2.8" or better progressive Scan CMOS get color image even at night condition
- 3) Signal System: PAL/NTSC
- 4) Minimum Illumination: 0.006Lux@ F1.4, AGC ON, 0 lux with IR or better
- 5) Imaging: 1/3s to 1/30000s, Auto Gain Control , White Balance- Auto, Back Light Compensation, Multi zone Privacy Masking, HLC, Digital Watermarking.
- 6) On Screen Display: Camera should display Camera title, Date & Time in live & recorded video both.
- 7) Signal to Noise Ratio: > 50 dB
- 8) Lens Type & Focus: 3.6mm/6mm fixed lens (optional as per site requirement)
- 9) Day & Night: True Day & Night High Performance Mechanical IR cut filter with auto switch, IR Source- Inbuilt IR LED's with effective distance upto 50 Mtrs integrated IR and 30 Mtrs for colour view in night.
- 10) Video Compression (Minimum): H.265 or better, H.265, H.264H, H.264, Audio:- G.711U/A, G.711Mu, G.726, AAC, G.723
- 11) Wide Dynamic Range: WDR (120db or more)
- 12) Digital Noise Reduction: DNR (2D/3D) On/Off
- 13) Streaming: Triple streaming, configurable
- 14) Video Streaming & Frame Rates: Triple streaming , configurable
Main stream: 5/6MP @25/30 fps, Sub streams: D1@25/30 fps or better
- 15) Image Setting: Rotate Mode, saturation, brightness, contrast, sharpness adjustable through client software or web browser
- 16) Profile Management: User configuration import, export
- 17) Security: User Authentication, Digital Water Marking
- 18) Onboard Storage: Camera should support built-in Micro SD/SDHC/SDXC Card slot upto 512 GB. It should be supplied with minimum 128GB memory Card.
- 19) Recording Management: Format SD, overwrite, storage management, video to NAS device,
- 20) Edge Analytics: Tripwire, Intrusion, Motion Detection
- 21) Alarm Trigger : Motion/tampering detection; network disconnection detection; IP conflict detection; memory card state detection; memory space detection
- 22) Alarm Support: It should have 1/1 Alarm In/ Out Port

23) Audio Support: It should have 1x Built-In Mic and 1/1 Audio In/ Out Port for external Mic. and Speaker (As per site requirement) with G.711U/A / G.711Mu/ AAC/ G.726 audio compression

24) Network Protocol: SFTP, IPv6, IPv4, DNS, RTCP, NTP, RTP, HTTP, HTTPS, SNMP TCP/IP, PPPoE, NFS, UDP, ICMP, SSL, DHCP, SMTP, RTSPS, unicast,

25) System Capability: ONVIF

26) VMS: Camera shall support open source VMS

27) Cyber Security: trusted boot, AES 256-bit Encryption, Configuration encryption, trusted execution, Digest, security logs, account lockout, , video encryption, IP/MAC filtering, HTTPS, trusted upgrade.

28) Ethernet: 1 RJ 45 10/100 Ethernet port

29) Power Input: The camera should support simultaneous dual power input—12 VDC (via power adapter) and PoE—to ensure continuous operation in the event of a failure in one power source.

30) Power Requirement: 12VDC/24 VAC/PoE (802.3af)/ePoE

31) Enclosure: IP67 weather proof, IK10

32) Operating Condition: Ambient Temperature:- (-)05°C to 50°C, humidity 95% (max) (non-condensing)

33) IR life: 40000 hours or higher

34) Video Bit rate: 32 KBPS - 8 MBPS or better

35) Standards: BIS with ER, STQC Certified, CE, FCC and RoHS

40.6.1	5/6MP ((3200 × 1800)@20 fps or better) IP IR 2.8/ 3.6/6/8/12 mm fixed lens (as per site requirement) outdoor bullet Camera	Each	12525
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40.6.2	5/6MP ((3200 × 1800)@20 fps or better) IP IR 2.7mm ~13.5mm Motorized lens, Vari Focal outdoor Bullet Camera, Angle of View: H: 104°–32°, V: 54°–18°, D: 125°–36° or better	Each	26993
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8 MP IP IR Dome Camera

40.7 Supplying Installation Testing and Commissioning of 8 MP IP IR Dome Camera having following specifications and features etc :-

1) Type of Camera: Dome Camera

2) Image Sensor: 1/2.8" or better progressive Scan CMOS get color image even at night condition

3) Signal System: PAL/NTSC

4) Minimum Illumination: 0.006 Lux@ F1.4, AGC ON, 0 lux with IR or better

- 5) Imaging: 1/3s to 1/30000s, Auto Gain Control , White Balance- Auto, Back Light Compensation, Multi zone Privacy Masking, HLC, Digital Watermarking.
- 6) On Screen Display: Camera should display Camera title, Date & Time in live & recorded video both.
- 7) Signal to Noise Ratio: > 50 dB
- 8) Day & Night: True Day & Night High Performance Mechanical IR cut filter with auto switch, IR Source- Inbuilt -IR LED's with effective distance upto 50 Mtrs integrated IR and 30 meter for colour view in night.
- 9) Video Compression (Minimum): H.265 or better, H.265, H.264H, H.264, Audio:- G.711U/A, G.711Mu, G.726, AAC, G.723
- 10) Wide Dynamic Range: WDR (120db or more)
- 11) Digital Noise Reduction: DNR (2D/3D) On/Off
- 12) Streaming: Triple streaming, configurable
- 13) Video Streaming & Frame Rates: Triple streaming , configurable
Main stream: 8MP (3840 × 2160)@20 fps, Sub streams: 4MP (2688×1520)@25/30 fps & D1@25/30 fps;
- 14) Image Setting: Rotate Mode, saturation, brightness, contrast, sharpness adjustable through client software or web browser
- 15) Profile Management: User configuration import, export
- 16) Security: User Authentication, Water Marking
- 17) Onboard Storage: Camera should support built-in Micro SD/SDHC/SDXC Card slot upto 512 GB. It should be supplied with minimum 128GB memory Card.
- 18) Recording Management: Format SD, overwrite, storage management, video to NAS device,
- 19) Edge Analytics: Tripwire, Intrusion, Motion Detection
- 20) Alarm Trigger: Motion/tampering detection; network disconnection detection; IP conflict detection; memory card state detection; memory space detection
- 21) Alarm Support: It should have 1/1 Alarm In/ Out Port
- 22) Audio Support: It should have 1x Built-In Mic and 1/1 Audio In/ Out Port for external Mic. and Speaker (As per site requirement) with G.711U/A/ G.711Mu/ AAC/ G.726 audio compresion
- 23) Network Protocol: SFTP, IPv6, IPv4, DNS,RTCP, NTP, RTP, HTTP, HTTPS, SNMP TCP/IP, PPPoE, NFS, UDP, ICMP, SSL, DHCP, SMTP, RTSPS, unicast,
- 24) System Capability: ONVIF
- 25) VMS: Camera shall support open source VMS

26) Cyber Security: trusted boot, AES 256-bit Encryption, Configuration encryption, trusted execution, Digest, security logs, account lockout, video encryption, IP/MAC filtering, HTTPS, trusted upgrade.

27) Ethernet: 1 RJ 45 10/100 Ethernet port

28) Power Input: The camera should support simultaneous dual power input—12 VDC (via power adapter) and PoE—to ensure continuous operation in the event of a failure in one power source.

29) Power Requirement: 12VDC/24 VAC/PoE (802.3af)/ePoE

30) Enclosure: IP67 weather proof, IK10

31) Operating Condition: Ambient Temperature:- (-)05°C to 50°C, humidity 95% (max) (non-condensing)

32) IR life: 40000 hours or higher

33) Video Bit rate: 32 KBPS - 8 MBPS or better

34) Standards: BIS with ER, STQC Certified, CE, FCC and RoHS

40.7.1	8 MP (3840 × 2160)@20 fps or better IP IR 2.8/ 3.6/6/8/12 mm fixed lens (as per site requirement) Dome Camera,	Each	19435
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40.7.2	8 MP (3840 × 2160)@20 fps or better IP IR 2.7mm ~13.5mm Motorized lens, Vari Focal Dome Camera, Angle of View: H: 104°–32°, V: 54°–18°, D: 125°–36° or better	Each	28073
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40.8 8 MP IP IR Outdoor Bullet Camera

Supplying Installation Testing and Commissioning of 8 MP IP IR Outdoor Bullet Camera having following specifications and features etc :-

1) Type of Camera: Outdoor Bullet Camera

2) Image Sensor: 1/2.8" or better progressive Scan CMOS get color image even at night condition

3) Signal System: PAL/NTSC

4) Minimum Illumination: 0.006 Lux@ F1.4, AGC ON, 0 lux with IR or better

5) Imaging: 1/3s to 1/30000s, Auto Gain Control , White Balance- Auto, Back Light Compensation, Multi zone Privacy Masking, HLC, Digital Watermarking.

6) On Screen Display: Camera should display Camera title, Date & Time in live & recorded video both.

7) Signal to Noise Ratio: > 50 dB

8) Day & Night: True Day & Night High Performance Mechanical IR cut filter with auto switch, IR Source- Inbuilt Smart IR LED's with effective distance upto 50 Mtrs integrated IR and 30 meter for colour view in night

- 9) Video Compression (Minimum): H.265 or better, H.265, H.264H, H.264, Audio:- G.711U/A, G.711Mu, G.726, AAC, G.723
- 10) Wide Dynamic Range: WDR (120db or more)
- 11) Digital Noise Reduction: DNR (2D/3D) On/Off
- 12) Streaming: Triple streaming, configurable
- 13) Video Streaming & Frame Rates: Triple streaming , configurable
- Main stream: 8MP (3840 × 2160)@20 fps, Sub streams: 4MP (2688 × 1520)@25/30 fps & D1@25/30 fps;
- 14) Image Setting: Rotate Mode, saturation, brightness, contrast, sharpness adjustable through client software or web browser
- 15) Profile Management: User configuration import, export
- 16) Security: User Authentication, Digital Water Marking
- 17) Onboard Storage; Camera should support built-in Micro SD/SDHC/SDXC Card slot upto 512 GB. It should be supplied with minimum 128GB memory Card.
- 18) Recording Management: Format SD, overwrite, storage management, video to NAS device,
- 19) Edge Analytics: Tripwire, Intrusion, Motion Detection
- 20) Alarm Trigger: Motion/tampering detection; network disconnection detection; IP conflict detection; memory card state detection; memory space detection
- 21) Alarm Support: It should have 1/1 Alarm In/ Out Port
- 22) Audio Support: It should have 1x Built-In Mic and 1/1 Audio In/ Out Port for external Mic. and Speaker (As per site requirement) with G.711U/A/ G.711Mu/ AAC/ G.726 audio compression
- 23) Network Protocol: SFTP, IPv6, IPv4, DNS,RTCP, NTP, RTP, HTTP, HTTPS, SNMP TCP/IP, PPPoE, NFS, UDP, ICMP, SSL, DHCP, SMTP, RTSPS, unicast,
- 24) User Access: Minimum 08 Users
- 25) System Capability: ONVIF Profile S, G & T. CCTV Camera OEM should be fulltime member of ONVIF and quoted models should be listed on ONVIF official website (All the certifications & Credentials should be valid on the date of technical evaluation).
- 26) VMS: Camera shall support open source VMS
- 27) Cyber Security: trusted boot, AES 256-bit Encryption, Configuration encryption, trusted execution, Digest, security logs, account lockout, –video encryption, IP/MAC filtering, HTTPS, trusted upgrade.
- 28) Ethernet: 1 RJ 45 10/100 Ethernet port

	29) Power Input: The camera should support simultaneous dual power input—12 VDC (via power adapter) and PoE—to ensure continuous operation in the event of a failure in one power source.		
	30) Power Requirement: 12VDC/24 VAC/PoE (802.3af)/ePoE		
	31) Enclosure: IP67 weather proof, IK10		
	32) Operating Condition: Ambient Temperature:- (-)05°C to 50°C, humidity 95% (max) (non-condensing)		
	33) IR life: 40000 hours or higher		
	34) Video Bit rate: 32 KBPS - 8 MBPS or better		
	35) Standards: BIS with ER, STQC Certified, CE, FCC and RoHS		
40.8.1	8 MP (3840 × 2160)@20 fps or better IP IR 2.8/ 3.6/6/8/12 mm fixed lens	Each	19435
40.8.2	8 MP (3840 × 2160)@20 fps or better IP IR 2.7mm ~13.5mm Motorized lens, Vari Focal outdoor Bullet Camera, Angle of View: H: 104°–32°, V: 54°–18°, D: 125°–36° or better PTZ (Pan. Tilt and Zoom) IP IR Camera	Each	28073
40.9	Supplying Installation Testing and Commissioning of PTZ (Pan,Tilt and Zoom) IP IR Camera having following specifications and features etc :- PTZ (Pan,Tilt and Zoom) IP IR Camera having following specifications and features etc :- 1) IP IR PTZ Camera: Motorized PAN TILT ZOOM with IR 2) Image Sensor: 1/2.8" or better progressive Scan CMOS 3)Signal System: PAL/NTSC 4) Minimum Illumination: 0.008 Lux@ F1.6, AGC ON,, 0 lux with IR, or better 5) Imaging: 1/1s to 1/30000s, Auto Gain Control , White Balance- Auto, Back Light Compensation, Image Rotation, Multi zone Privacy Masking (Upto 24 Area), HLC, Defog, Region Of Interest and EIS features. 6) On Screen Display: Camera should display Camera title, Date & Time in live & recorded video both. 7) Signal to Noise Ratio: > 50 dB 8) Event Notification: Through Relays, E-Mails or FTP 9) Day & Night: True Day & Night High Performance Mechanical IR cut filter with auto switch, IR Source- Inbuilt IR LED's with effective distance. The camera should have IR LED's and cover distance up to 300 meter or above		

- 10) Auto Tracking: The camera should be equipped with Deep-learning-based auto tracking function using simultaneously all of the panning, tilting and zooming should be available. When a motion is detected in a registered monitoring area, the camera should track the motion (object) and capture it.
- 11) Pre/Post Event Buffering: The camera should support atleast of 5 seconds of pre & post event buffering.
- 12) Presets: 300 Presets
- 13) Video Compression (Minimum): H.265 or better, H.265, H.264H, H.264, Audio:- G.711U/A, G.711Mu, G.726, AAC, G.723
- 14) Wide Dynamic Range: WDR (120db or more), HLC & BLC
- 15) Digital Noise Reduction: DNR (2D+3D) On/Off
- 16) Image Setting: Rotate Mode, ROI ,EIS, Defog, saturation, brightness, contrast, sharpness adjustable through client software or web browser
- 17) Profile Management: User configuration import, export
- 18) Cyber Security: Configuration encryption, AES 256-bit Encryption, Digest, account lockout, video encryption, IP/MAC filtering, trusted boot, trusted upgrade, trusted execution
- 19) Onboard Storage: Camera should support built in Micro SD/SDHC/SDXC Card slot upto 512 GB . It should be supplied with minimum 128GB Memory.
- 20) Recording Management: Format SD, overwrite, storage management, video to NAS device, remote archive access via FTP login
- 21) Edge based Video Analytics & Alarm Trigger: Motion/tampering detection; network disconnection detection; IP conflict detection; memory card state detection; memory space detection, Tripwire, Intrusion, Object Abandon/ Missing, SMD, Face Detection
- 22) Network Protocol: SFTP, IPv6, IPv4, DNS,NTP, HTTP, HTTPS, SNMP, TCP/IP, PPPoE, NFS, ICMP, DHCP, SMTP, RTSPS.
- 23) System Capability: ONVIF Profile S, G & T. CCTV Camera OEM should be fulltime member of ONVIF and quoted models should be listed on ONVIF official website (All the certifications & Credentials should be valid on the date of technical evaluation).
- 24) VMS: Camera shall support open source VMS
- 25) Connectivity: 1x LAN RJ-45 (10/100Base-T)
- 26) Audio Support: Audio Interface : The camera should have 1/1 Audio In/Out to connect External Mic and Speaker Audio Compression : PCM, G.711U/A, G.711MU, G.726, MPEG2-Layer2, G.722.1

- 27) Alarm In/Out : Alarm In/out- 7/2 Ch In/Out
- 28) Power Input: The camera should support simultaneous dual power input—DC/ AC (via power adapter) and PoE (802.3af)—to ensure continuous operation in the event of a failure in one power source.
- 29) Power Requirement: 24 VDC, 2.5 A ($\pm 25\%$), PoE+ (802.3at)
- 30) Enclosure & Weather Proof Standard: IP67 weather proof and IK10, TVS 8000V lightning proof, surge protection, voltage transient protection
- 31) Operating Condition:- Ambient Temperature:- (-)05°C to 50°C, humidity 95% (max) (non-condensing)
- 32) IR life: 40000 hours or higher
- 33) Video Bit rate: 32 KBPS - 8 MBPS or better
- 34) Standards: BIS with ER, STQC Certified, CE, FCC and RoHS

40.9.1	2 MP IP IR motorized PTZ (1920 × 1080 @ 25/30fps or better) camera, Triple streaming , configurable on resolution Main stream: 1080p@25/30 fps Sub streams :1080p@25/30 fps & D1@25/30 fps or better, varifocal lens 4.3-137 mm or better with Automatic & manual Focus Adjustment provisions, 32x Optical zoom and 12x Digital zoom with angle of view 57°–2.4° , PAN Travel: Pan: 0° ~ 360° endless,Manual Pan: 260° /s, Preset : 300° /s, Tilt, Travel: Tilt: -20° ~ 90°, auto flip 180°, Manual Tilt: 120° /s, Preset : 200° /s	Each	70111
40.9.2	4 MP IP IR motorized PTZ (4MP @25/30fps or better) camera, Triple streaming , configurable on resolution Main stream: 4MP@25/30 fps Sub streams :1080p@25/30 fps & D1@25/30 fps or better, varifocal lens 4.5-135mm or better with Automatic & manual Focus Adjustment provisions, 30x Optical zoom and 12x Digital zoom with angle of viewH : 57°–2.4°, m, PAN Travel: Pan: 0° ~ 360° endless;Manual Pan: 300° /s, Preset : 300° /s, Tilt Travel: Tilt: 0-90°, auto flip 180°, Manual Tilt: 200° /s, Preset : 300° /s	Each	78389
40.9.3	5MP IP IR motorized PTZ (2960 x 1668@25/30fps or better) camera, Triple streaming , configurable on resolution:- Main stream: 5MP@25/30 fps Sub streams:1080p@25/30 fps, varifocal lens 3.95mm (± 5 mm) ~ 177.75mm (± 5 mm) or better with Automatic & manual Focus Adjustment provisions, 45x Optical zoom and 16x Digital zoom with angle of view H: 65.7°–1.9°V: 39.4°–1.1°D: 73.1°–2.1, PAN Travel: Pan: 0° ~ 360° endless,Manual Pan: 260° /s, Preset : 300° /s, Tilt Travel: Tilt: -20° ~ 90°, auto flip 180°, Manual Tilt: 120° /s, Preset : 200° /s	Each	107974

Network Video Recorder (NVR)

- 40.10 Supplying Installation Testing and Commissioning of following Channel Network Video Recorder (NVR) with camera licenses to record for all channels having specifications and features etc as mentioned below :
- 1) Network Video Recorder Embedded/ Installed OS (Linux) along with Camera Licenses for number of channels of NVR to record per NVR and to provide a live view, storage and simultaneously Multi-channel playback of all IP, IR camera or more and must be ONVIF with minimum support of 256Mbps incoming Bandwidth.
 - 2) NVR should support H.265 or better, H.264, MJPEG, MPEG4 support
 - 3) Must support 1 channel Input, 1 channel Output, RCA for Two-way Talk
 - 4) Intelligent auto power on when power resumes after power outage.
 - 5) Storage: It should support minimum 2 SATA Slots with 20TB capacity/ Slot.
 - 6) Connectivity Interface : 1 Nos. x 10/100/1000 Mbps Ethernet Ports, 1x RS485, 1x RS232
 - 7) Backup Interface : Its should have 2 Nosx USB port (1x USB3.0, 1xUSB2.0)
 - 8) Video Output Ports: 1x HDMI and 1 VGA
 - 9) Alarm Ports: It should have 4/1 Ch In/ Out ports to connect various type of external sensors and output devices like hooter/ Siren etc.
 - 10) Email & SMS Alert options: Option for SMS/ Email Alerts to minimum 5 designated mobile number for power failure, HDD failure, vandalism, tampering, network disconnection and panic
 - 11) Web & Mobile Application: Web, Mobile app (For iPhone, iPad, Android Phone) for alerts and viewing.
 - 12) Protocols: HTTP, HTTPS, TCP/IP, IPv4, UDP, NTP, DHCP, DNS, SMTP, UPnP, DDNS, Alarm Server, IP Search, Multicast, Auto Registration, ONVIF 21.12(Profile T, Profile S, Profile G), CGI, SDK and OEM Cloud for remote monitoring without any public IP need.
 - 13) Standards: CE, FCC, RoHS and BIS Certified
 - 14) Power Supply : Should support 12VDC, 4Amp or AC100-240V, 50/60Hz Power supply.
 - 15) Operating Condition : Ambient Temperature (-5°C to 50°C), humidity 90% (max) (non-condensing)
 - 16) The VMS, NVR application shall support all the features & functionalities of the offered cameras.

- 17) VMS should consist of Base license and Channel Licenses. VMS should be provided with camera Licenses , with no dependency of VMS licenses by binding with the MAC address of the cameras to achieve the functionality.
- 18) The NVR OEM shall be responsible for providing a mobile application compatible with both Android and iOS devices, enabling remote monitoring and playback of cameras/NVR footage.
- 19) The OEM must provide its own DDNS server hosted in India, eliminating the need for a public IP address for remote monitoring over the Internet.
- 20) Must support Resolution: 7680x4320 (8K), 3840*2160, 1920×1080, 1280×1024,1280×720, 1024×768
- 21) Must support recording resolution upto 32MP
- 22) Must support Continuous, Alarm, Motion, Instant, Panic Recording Mode
- 23) When alarm recording is enabled and an event occurs, you can click the alarm icon on monitoring page to view the alert details. The snapshot function is supported on monitoring and playback page
- 24) The Network Video Recorder (NVR) shall be configured to send email whenever a system message is created or an alarm event occurs. The email server shall be a valid SMTP server. Each recipient email address shall be configured to receive any combination of critical, warning, or informational messages or alarm notifications. When an alarm occurs, the email message includes the NVR name, time of alarm and a list of camera that is configured to record upon alarm
- 25) It should support Network Support: HTTP, TCP/IP, SMTP, DHCP, DNS, DDNS, FTP, NTP, UPnP, Multi IP Setting. Convert multiple recording files to one avi/MP4 file.
- 26) General AI Based Search: Search Pictures by channel, time, event type, target classification (Fall Detection, People Approach Detection, People No. Exception Detection, People Staying Detection, Violence Detection.
- 27) Alarm Notifications based on: Motion detection, video tampering, video loss, scene changing, PIR alarm, Camera external alarm, Face detection, face recognition, perimeter protection (intrusion and tripwire), ANPR, people counting, stereo analysis, crowd distribution, heat map, Disk Full, Storage Error, IP Conflict and abnormal behavior of fan
- 28) Alarm Notification should be linked with Recording, snapshots, Camera external alarm output, buzzer, logs, presets and email.

- 29) Built-In Artificial Intelligence: NVR should have built-in AI :-
 - 2 Channel face detection and recognition, - Minimum 4 Channel perimeter protection, - Minimum 8 Channel Smart Motion Detection
- 30) Face Recognition Database Capacity: It should support total Blacklist and Whitelist capacity of Minimum 20,000 Faces or more with Face Detection speed of 12 face images/sec.
- 31) Face & Human Attributes Search: Search Pictures/ Video by Gender, age group, glasses, expressions, face mask, beard, Top color, top type, hat, bag, age, gender and umbrella.
- 32) ANPR Capability: It should support ANPR Camera with License plate, plate color, vehicle body, vehicle model, vehicle logo, calling, seatbelt, vehicle registration location etc vehicle attributes.
- 33) Alarm Notifications based on: Motion detection, video tampering, video loss, scene changing, PIR alarm, Camera external alarm, Face detection, face recognition, perimeter protection (intrusion and tripwire), ANPR, people counting, stereo analysis, crowd distribution, heat map, Disk Full, Storage Error, IP Conflict and abnormal behavior of fan, cybersecurity exception
- 34) Alarm Notification should be linked with Recording, snapshots, Camera external alarm output, buzzer, logs, presets and email.
- 35) General AI Based Search: Search Pictures by channel, time, event type, target classification (Fall Detection, People Approach Detection, People No. Exception Detection, People Staying Detection, Violence Detection).
- 36) Smart playback function: Should support smart search for the selected area in the video and smart playback to improve the playback efficiency
- 37) VCA (Video Content Analytic): Should support multiple video contented analytics based on camera analytics

40.10.1	08 Channel Network Video Recorder (NVR) having display split :- Main screen: 1/4/8/9, 2nd screen: 1/4/8/9	Each	10078
40.10.2	16 Channel Network Video Recorder (NVR) having display split :- Main screen: 1/4/8/9/16, 2nd screen: 1/4/8/9/16	Each	12799
40.11	Supplying Installation Testing and Commissioning of following Channel Network Video Recorder (NVR) with camera licenses to record for all channels having specifications and features etc as mentioned below :		

- 1) Network Video Recorder Embedded/ Installed OS (Linux) along with Camera Licenses to record per NVR and to provide a live view, storage and simultaneously Multi-channel playback of all IP camera or more and must be ONVIF with minimum support of 384 Mbps incoming Bandwidth.
- 2) NVR should support video compressions : H.265 or better, H.264, MJPEG.
- 3) Must support 1 channel RCA Input, 2 channel RCA Output for Two-way Talk with G.711U/A, G.711u, PCM, G726 audio compressions.
- 4) Intelligent auto power on when power resumes after power outage.
- 5) Storage: It should support minimum 8 SATA Slots with 20TB capacity/ Slot and RAID support of RAID 0/1/5/6/10.
- 6) Connectivity Interface : 2 Nos. x 10/100/1000 Mbps Ethernet Ports, 1x RS485, 1x RS232, 1x eSATA Port
- 7) Backup Interface : Its should have 4 Nosx USB port (2x USB3.0, 2xUSB2.0)
- 8) Video Output Ports: 2x HDMI and 2 VGA
- 9) Alarm Ports: It should have 16/8 Ch In/ Out ports to connect various type of external sensors and output devices like hooter/ Siren etc.
- 10) Email & SMS Alert options: Option for SMS/ Email Alerts to minimum 5 designated mobile number for power failure, HDD failure, vandalism, tampering, network disconnection and panic
- 11) Web & Mobile Application: Web, Mobile app (For iPhone, iPad, Android Phone) for alerts and viewing.
- 12) Protocols: HTTP, HTTPS, TCP/IP, IPv4/IPv6, UDP, DHCP, DNS, SMTP, UPnP, IP Filter, PPPoE, FTP, DDNS, Alarm Server, IP Search, Multicast, Auto Registration, ONVIF 21.12 (Profile T, Profile S, Profile G), CGI, SDK and OEM Cloud for remote monitoring without any public IP need.
- 13) Standards: CE, FCC, RoHS, BIS Certified
- 14) Power Supply : Should support AC100-240V, 50/60Hz Power supply.
- 15) Operating Condition : -5°C to 50°C, humidity 90% (max) (non-condensing)
- 16) The VMS application shall support all the features & functionalities of the offered cameras.
- 17) VMS should consist Licenses for all channels to record Cameras with General, motion detection, intelligent, alarm and POS recording modes. VMS should be provided with Camera Licenses , with no dependency of VMS licenses by binding with the MAC address of the cameras to achieve the functionality.

- 18) The NVR OEM shall be responsible for providing a mobile application compatible with both Android and iOS devices, enabling remote monitoring and playback of cameras/NVR footage.
- 19) The OEM must provide its own DDNS server hosted in India, eliminating the need for a public IP address for remote monitoring over the Internet.
- 20) Must support Continuous, Alarm, Motion, Instant, Panic Recording Mode
- 21) It should support Resolution: 32MP; 24MP; 16MP; 12MP; 8MP; 5MP; 4MP; 1080p; 720p; D1; CIF; QCIF
- 22) When alarm recording is enabled and an event occurs, you can click the alarm icon on monitoring page to view the alert details. The snapshot function is supported on monitoring and playback page
- 23) The Network Video Recorder (NVR) shall be configured to send email whenever a system message is created or an alarm event occurs. The email server shall be a valid SMTP server. Each recipient email address shall be configured to receive any combination of critical, warning, or informational messages or alarm notifications. When an alarm occurs, the email message includes the NVR name, time of alarm and a list of camera that is configured to record upon alarm
- 24) It should have Web and GUI interface.
- 25) Built-In Artificial Intelligence: NVR should have built-in AI :-
- 2 Channel face detection and recognition, - Minimum 4 Channel perimeter protection, - Minimum 8 Channel Smart Motion Detection
- 26) Face Recognition Database Capacity: It should support total Blacklist and Whitelist capacity of Minimum 20,000 Faces or more with Face Detection speed of 12 face images/sec and facility to add Name, gender, birthday, address, credential type, credential No., countries & regions and state to each face image.
- 27) Face & Human Attributes Search: Search Pictures/ Video by Gender, age group, glasses, expressions, face mask, beard, Top color, top type, hat, bag, age, gender and umbrella.
- 28) ANPR Capability: It should support ANPR Camera with License plate, plate color, vehicle body, vehicle model, vehicle logo, calling, seatbelt, vehicle registration location etc vehicle attributes.

29) Alarm Notifications based on: Motion detection, video tampering, video loss, scene changing, PIR alarm, Camera external alarm, Face detection, face recognition, perimeter protection (intrusion and tripwire), ANPR, people counting, stereo analysis, crowd distribution, heat map, Disk Full, Storage Error, IP Conflict and abnormal behavior of fan, cybersecurity exception

30) Alarm Notification should be linked with Recording, snapshots, Camera external alarm output, buzzer, logs, presets and email.

31) General AI Based Search: Search Pictures by channel, time, event type, target classification (Fall Detection, People Approach Detection, People No. Exception Detection, People Staying Detection, Violence Detection.

32) Smart playback function: Should support smart search for the selected area in the video and smart playback to improve the playback efficiency

34) VCA (Video Content Analytic): Should support multiple video contented analytics based on camera analytics

35) Analytics by NVR: Perimeter protection and face recognition

40.11.1	32 Channel Network Video Recorder (NVR) having display split :- Main screen: 1/4/8/9/16/25/32(36), 2nd screen: 1/4/8/9/16	Each	64784
40.11.2	64 Channel Network Video Recorder (NVR) having display split :- Main screen: 1/4/8/9/16/25/36/64, 2nd screen: 1/4/8/9/16	Each	82600
	Hard Disk Drive		
40.12	Supplying, Installation, Testing and Commissioning following capacity Surveillance grade Hard Disk with upto 256MB/s Transfer Rate, 256 MB Cache, 7200 RPM Disk Speed, 3.5 inch form factor, SATA Interface, BSMI, ICES-003/NNB-003, CE, FCC, KC, Maghreb, RCM, UKCA, VCCI, CB-Scheme, TUV, UL Certifications.		
40.12.1	4TB (Terabytes)	Each	16088
40.12.2	6TB (Terabytes)	Each	18895
40.12.3	8TB (Terabytes)	Each	23214
40.12.4	10TB (Terabytes)	Each	28613

CHAPTER – 41

ACCESS CONTROL SYSTEM

Item No.	Description	Unit	Rate(₹)
41.1	Access Control System with Face recognition		
	Supplying, Installation, Testing and Commissioning of Access Control System comprising of Face and fingerprint Recognition, Smart Card, QR Code, NFC & PIN Code based Reader, following sizes of 5" IPS TFT Touch Screens, 1GB RAM, 8 GB ROM, Live Face Detection with anti-spoofing against 2D photograph, mobile selfie & video, Fake Fingerprint and face Detection Technology, 30,000 Face storage (1:N) /, 3,000 fingerprint (1:N), QR code 30,000, 30,000 card storage (1:N), 2 lakh transaction storage, 20,000 image logs, onboard dual camera (colour & IR) Primary camera min 2MP, Support Face authentication & verification from up to 1 meters, Communication protocols WiFi, TCP/IP, Wiegand Out, Wiegand IN, USB, MQTT protocol, Blacklist Employee Management, User wise authentication modes; Built in Lock Control, Mask Detection & Alert (configurable), 40 to 30,000 lux illumination, IP65 for Ingress Protection, Certification: CE, FCC, BIS, ROHS, etc. Metal Enclosure with battery backup, suitable for mounting on metal surface/metal frames/ wooden frames/wall/flap barriers including all mounting bracket & accessories complete etc. as required.		
41.1.1	5" IPS TFT Touch Screen	Each	58,306
41.2	Access Control System (without Finger Print)		
	Supplying, Installation, Testing and Commissioning of Access Control System comprising of Face Recognition, Smart Card, QR Code, NFC & PIN Code based Reader, following sizes of 5" IPS TFT Touch Screens, 1GB RAM, 8 GB ROM, AES 128 bit encrypted, Live Face Detection with anti-spoofing against 2D photograph, mobile selfie & video, Fake Fingerprint Detection Technology, 10,000 Face storage (1:N) /, 10,000 card storage (1:N), QR code storage 10,000 (1:N), 2 lakh transaction storage, 20,000 image logs, onboard dual camera (colour & IR) Primary camera min 2MP, Support Face authentication & verification from up to 1 meters, Communication protocols WiFi, TCP/IP, Wiegand Out, USB, MQTT protocol, Blacklist Employee Management, User wise authentication modes; Built in Lock Control, Mask Detection & Alert (configurable), 40 to 30,000 lux illumination, IP54 for Ingress Protection, Certification: CE, FCC, BIS, ROHS, WPC etc. Metal enclosure with battery backup, suitable for		

	mounting on metal surface/metal frames/ wooden frames/wall/flap barriers including all mounting bracket & accessories complete etc. as required.	Each	38,870
41.3	Access Control System with 2.8" display (without Face Recognition Readers) Supplying, Installation, Testing and Commissioning of This system is a compact solution with fingerprint, card, and password authentication: ☐ Authentication Methods: Fingerprint, Card & Password with Capacitive Touch screen. ☐ Fingerprint Detection: Fake Fingerprint Detection Technology. ☐ Display: 2.8" TFT touch Screen display. ☐ Capacity: 10,000 Fingerprint storage (1:N). ☐ Logs: 200,000 transaction logs. ☐ Power: 12V DC. (UL listed) ☐ Communication Protocols: TCP/IP, WiFi, USB, Wiegand Out / IN, RS 485. ☐ Additional Features: User-wise authentication modes, Built-in Lock Control. ☐ Certifications: CE, FCC, RoHS, BIS. ☐ Environmental: IP 54 Ingress Protection. ☐ Mounting: Suitable for metal surfaces/frames, wooden frames, walls, and flap barriers.	Each	32,392
41.4	Access Control System with 1.77" display (without Face Recognition Readers) Supplying, Installation, Testing and Commissioning of Access Control System comprising of Fingerprint & Card based Reader with 1.77" LCD Display, Fake Fingerprint Detection Technology, supports ANSI 378 encryption, AES 128 bit encrypted algorithm to protect important info such as personal info transmitted, 5,000 Fingerprint storage (1 :N) / 10,000 Fingerprint storage (1 :1), 50,000 transaction logs, Onboard Camera to capture live images against transactions, Communication protocols like TCP/IP, Wiegand Out / IN, RS 485, OSDP; Supports Mobile Key through Bluetooth, User wise authentication modes; Built in Lock Control; UL Listed Power Supply, Certification: CE, FCC, RoHS, BIS, WPC, Ingress Protection IP 65 etc. suitable for mounting on metal surface/metal frames/ wooden frames / wall / flap barriers including all mounting bracket & accessories complete etc. as required.	Each	28,073
41.5	Software for Web based Access Control System Supplying, Installation, Testing and Commissioning of Web based Access Control Enrolment and Device Configuration Software, Should be independent of OS (Windows), Unified Management of Biometric, Card, and Controllers, Import Functionality for bulk user details, Biometric Template Retrieval		

	for Data Recovery, User Management, Logs and Terminal Management, Support for IP-based and USB Biometric Devices, User-specific Access Policies, Different User Policies on the Same Terminal, Assignment of Specific Access Schedules, Pushing Users to Terminals or Groups, Admin Authority Management to define rights of each user to perform, Blacklist User Management, Access Schedule Creation, Real-Time Log Retrieval, Real- Time User Identification, Secure Storage of Biometric/Users Data, Capable to integrate with 3rd party applications over Web API's, MQTT protocol support	Each	97,176
41.6	Electromagnetic locks Supplying, Installation, Testing and Commissioning of Electromagnetic locks (fail safe type) for following type of Doors suitable for operation on 12 V DC having holding force 600 lbs for Single Leaf Door and 1200lbs for Double leaf door with suitable feedback i/c connection complete etc. as required.		
41.6.1	Single leaf door	Each	4831
41.6.2	Double leaf door (Each leaf 600lbs & Total 1200 lbs)	Each	7,593
41.7	Exit Push Button Supplying, Installation, Testing and Commissioning of Heavy Duty SS Finish Exit Push Button for releasing access doors with all necessary installation accessories i/c connection complete etc. as required.	Each	1,173
41.8	Door Position Sensor Supplying, Installation Insta, Testing and Commissioning of Door Position Sensor i/e connection complete etc. as required.	Each	622

CHAPTER - 42

BUILDING MANAGEMENT SYSTEM (BMS)

Item No.	Description	Unit	Rate(₹)
42.1	CENTRAL CONTROL SERVER Supply Installation Testing Commissioning of Central Control Server System comprising of Intel® Xeon® CPU E5-2640 x64 (or better) compatible with dual and quad core processors, Windows 11 (Professional, Enterprise, Ultimate, 64 bit), Windows Server 2013 R2 (Standard, Enterprise, 64 bit) 16 GB or more, Video card and monitor capable of displaying at 1024 x 768 resolution or better, Latest Microsoft Licence Version Operating System Loaded with MS OFFICE 2019 Professional, Word, EXCEL, Power Point, Outlook Express, Microsoft Team official version lifetime licences suitable for running BMS software complete etc. as required.	Each	186434
42.2	PC System Supply Installation Testing Commissioning of Client PC System comprising of PC Hardware with minimum 11th Gen Intel® i7 processor CPU E5-2640 x64 (or better), compatible with dual and quad core processors minimum 3.1 Ghz and 8GB or higher RAM, Hard Drive Space of 256 GB SSD or better, Video card and monitor capable of 1920 x 1080 pixel resolution or better with Ethernet adapter (10/100/1000 Mbps with RJ-45 connector) with 2 GB or higher Graphics card suitable for BMS System complete etc. as required.	Each	90698
42.3	LaserJet printer Supply Installation Testing Commissioning of LaserJet printer having printing resolution 1200 x 1200 dpi, (dot/inch) or better, 21 to 30 ppm printing speed (LaserJet) complete etc. as required	Each	22674
42.4	LED Display Supplying, Installation, Testing and Commissioning of following size LED display (LED monitor) industrial grade with 3840x2160 resolution or better, USB playback, bluetooth and miracast connectivity, 4X HDMI 2.0, DP 1.2, HDR 10/10+, brightness: 500-nits or better, Video wall mode should be available, contrast ratio: 1200:1, OPS slot, viewing angle (H/V): 170°/ 170°, response time less than 12ms, Display control shall be on monitor screen and programmable with remote (remote shall be supplied with system), Key Board, Optical Mouse, etc. as required.		
42.4.1	32 inch or larger	Each	39590

42.4.2	42 inch or larger	Each	51459
42.4.3	55 inch or larger	Each	82057
42.5	Software for Building Management System : Supply Installation Testing Commissioning of Necessary Software (UL listed) Packages containing Building Management with Capability of Collection, Storage, Processing and Controlling all the Data. Software shall support Open Standard Protocols like XML, BACnet, Modbus, Lon works etc. and without the use of Proprietary Protocols, with Enterprise Level Database Like MS SQL complete Lifetime Licenced & capable to collect, store & access all points from Fire Alarm Systems, IT Systems, Automation Systems, HVAC System, Lift, Energy management, Water Management System, etc. to create & monitor historical & event data, event management data, alarm management data & reporting data, Email Alerts for Critical alarms for all the systems. The scope includes additional software required by BMS software (to enable a standard Web browser) to be resident on the DDC/client machine (manufacture-specific browsers shall not be acceptable). User log-on identification and password shall be required & a minimum three-layered security system (User interface Level, Application Level and Data Base Level) to be deployed for safety. The supplied computer software shall employ object-oriented technology (OOT) for representation of all data and control devices within the system. This data shall reside on a server for all database access (Systems requiring proprietary database and user interface programs shall not be acceptable). All components and controllers supplied under this contract shall be true "peer-to-peer" communicating devices (Components or controllers requiring "polling" by a Master / Global / Host to pass data shall NOT be acceptable). Physical connection of BACnet/MOD BUS devices shall be via Ethernet at all levels. The License must be provided by default. The BMS software shall be complete in all respect to cater the following different numbers of points managed in the installation.		
42.5.1	For Upto 500 point included	Each	97377
42.5.2	For Upto 2500 point included	Each	167892
42.5.3	For Upto 5000 point included	Each	246899
42.6	Touchscreen Color portable operator terminal		

Supply Installation Testing Commissioning of following sizes Touchscreen Color portable operator terminal capable of connecting to one of the controllers on the communication bus and view parameters of the complete system. The terminal should be capable of viewing /changing parameters and trends, the specified parameters as per requirement etc. as required.

42.6.1	4" Touch Screen	Each	62985
42.6.2	7" Touch screen	Each	74468

42.7 DDC CONTROLLERS

Supply Installation Testing Commissioning of IP Based DDC controller capable of fully “stand- alone” operation i.e. In the event of loss of communication with other DDC’s or Control Station, they shall be able to function on their own. The controllers shall consist of dual/quad core 32 bit microprocessors with EPROM based operating system on BACNET. The network Topology shall not allow single point failure. Each DDC on field level shall have embedded TCP/IP (10/100Mbps) connectivity so that it can be hooked into the Local Area Network (LAN) provided by the client or on dedicated network . Each DDC can be accessed from the Graphical User Interface (GUI) or from a standard Web browser (WBI) by connecting the server alongwith inbuilt web browser. All controllers shall accept 230V+/- 10%, 50Hz Uninterrupted power supply provided by end user. Controller shall support DHCP addressing over Local Area Network (LAN) so that the static IP requirements get reduced however a single static IP shall be provided for system so that it can be hosted on to internet in consultation with end user. The microprocessor based DDC’s shall be provided with power supply, analoge to Digital and Digital to analoge converters with memory and capacity to accommodate input/output (I/O) hardware points (with or without an expansion board). Each DDC should have own board and shall accept 0-10 V / 4-20mA input signal having Minimum 20% spare capacity considered for expansion. Each DDC on field level should have minimum 128MB RAM & 64MB Flash memory. Following type of All controllers should be mandatorily BTL approved with B-BC profile complete etc. as required.

42.7.1	DDC Controller-Type-I (Support upto 32 Points)	Each	46861
42.7.2	DDC Controller-Type-II (Support upto 64 Points)	Each	66008

42.7.3	DDC Controller-Type-III (Support upto 96 Points)	Each	71047
42.7.4	DDC Controller-Type-IV (Support upto 128 Points)	Each	74070
42.7.5	DDC Controller-Type-V (Support upto 160 Points)	Each	86264
42.8	EXPNSION IO MODULE		
	Supply Installation Testing Commissioning of following type of input output (IO) expansion module complete etc. as required.		
42.8.1	I/O Module with 4 UIO channels	Each	17636
42.8.2	I/O Module with 8 UIO channels	Each	18492
42.8.3	I/O Module with 8 UI channels	Each	19601
42.8.4	I/O Module with 8 DI channels	Each	15580
42.8.5	I/O Module with 16 DI channels	Each	18029
42.8.6	I/O Module with 8 DO channels	Each	16910
42.8.7	I/O Module with 4 DO channels	Each	15721
42.9	Enclosure for DDC		
	Supply Installation Testing Commissioning of following sizes Powder coated wall mounted lockable and secure enclosure for DDC panel made of 16 SWG sheet steel IP -55 rated with proper internal mounting accessories for DDC such as DIN Rails, terminal block for cable terminations, well dressed for all internal cabling, ferruling, tagging etc. The DDC panels shall have power supply and connections for other peripheral equipment, switchgear protections, isolation transformer, earthing, relays, fuses, auxiliary power output socket & lamp etc. as required.		
42.9.1	500X500X200 mm	Each	10581
42.9.2	600mmX600mmX200mm	Each	12657
42.9.3	800mmX1000mmX200mm	Each	18643
42.10	Integration Unit		
	Supply Installation Testing Commissioning following type 3rd party Integration unit to provide the interface between Ethernet		

LAN and the 3rd party field control devices or any other devices which need to be integrated. These shall also provide supervisory capability of functions over the devices connected to it. The purpose of using these units should be limited to integrate the devices only. The Unit must have two onboard Ethernet port, minimum 2/3 on board RS485 ports. The Integration unit should have inbuilt 4GB memory for program storage. The Unit must communicate over TCP/IP with communication speed of 10/100MBPS. The Integration unit should be capable of handling multiple protocol simultaneously such as Bacnet, Modbus, SNMP restricted to single protocol. All integrators should be mandatorily BTL listed. The integration unit shall be housed in suitable size sheet steel enclosure made of 1.6mm thick CRCA sheet, lockable front door with glass, arrangement for cable entry and duly powder coated complete etc as required.

42.10.1	Integrator for 500 Points	Each	146124
42.10.2	Integrator for 1250 Points	Each	176357
42.10.3	Integrator for 5000 Points	Each	226744
42.11	Field Devices		
	Supply Installation Testing Commissioning of following Field Devices including connection etc. as required		
42.11.1	Water Differential Pressure (DP) Switch for Pump Status.	Each	21299
42.11.2	High/Low Level Switch for Tank Level Monitoring.	Each	5782
42.11.3	Air DP Switch for Fan Run Status.	Each	2141
42.11.4	Duct CO2 Sensor, 0-2000ppm, 0-10V/4-20mA Output.	Each	2570
42.11.5	Outside Temp & Humidity Sensor with radiation shield. Measuring Range: Temp:- -15 to 50 Deg C & RH 0-100%, Accuracy: +/- 1 Degc +/- 3%	Each	9876
42.11.6	Immersion Temp Sensor for CHW line. Measuring range: -30 to 110 Deg C, Accuracy: +/- 1.3 Deg C	Each	9322
42.11.7	Water Pressure Sensor . Accuracy: +/-0.3% FSL	Each	4585
42.11.8	Duct type temperature sensor Measuring range: -30 to 110 Deg C, Accuracy: +/- 1.3 Deg C	Each	4656

42.11.9	Current Sensing Relay	Each	3174
42.11.10	Wall Mount Temp. Sensor	Each	4283
42.11.11	Wall/Ceiling Mount CO Sensor	Each	6550
42.11.12	Differential pressure switch for filters & Blowers	Each	3411
42.11.13	Bi-Level Switches (Hi/Low)	Each	5543
42.11.14	Water Level Sensor	Each	6621
42.11.15	Explosion proof level Switch	Each	7619
42.11.16	Water Flow Switch	Each	3779
42.11.17	Indoor Air Quality Sensor (CO2,T,H,PM,TVOC)	Each	32752
42.11.18	Ultrasonic Water BTU Meter including digital display (Clamp-On) suitable for Pipe sizes dia upto 250mm	Each	357752

APPENDIX-I

BASIC RATE OF LABOUR & HIRE CHARGES

Code No.	DESCRIPTION	Unit	Rate
1001	Wireman	day	981
1002	Cable jointer	day	981
1003	Lineman	day	981
1004	Fitter, Grade 1	day	981
1005	Fitter, Grade 2	day	893
1006	Painter	day	893
1007	Khallasi	day	805
1008	Carpenter, Grade 1	day	981
1009	Blacksmith, Grade 2	day	893
1010	Mason, Grade 2	day	893
1011	Stone Chiseler	day	893
1012	Beldar/ coolie	day	805
1013	Bhisti	day	805
1014	Excavator	day	805
1015	Stone Breaker	day	893
1016	Mate	day	805
1017	Engineer	day	1500
1018	Highly Skilled Worker	day	876
1019	Electrician	day	981
1020	Electronic Technician	day	981
1020	Electronic Technician	day	981
1022	Software Engineer	day	2100
1081	Hire charges for 5 ton truck	day	4000
1082	Hire charges for compressor and spray gun	day	350

1083	Drilling of 46 Nos 12 mm dia holes on G.I. pipe	L.S.	300
1084	Drilling holes	Each	6
1085	Solder jointing	Each	12
1086	Welding charges	Meter	1
1087	Welder	day	981

APPENDIX-II

BASIC RATE OF MATERIAL

Note:- These rates are exclusive of Contractor's profit, Overheads and Carriage and any taxes etc.

Code No.	Description	Unit	Rate
1101	1.5 sq. mm ISI marked, FRLS/HFFR PVC insulated, single core copper conductor cable	Meter	18
1102	2.5 sq. mm ISI marked, FRLS/HFFR PVC insulated, single core copper conductor cable	Meter	28
1103	4.0 sq. mm ISI marked, FRLS/HFFR PVC insulated, single core copper conductor cable	Meter	42
1104	6.0 sq. mm ISI marked, FRLS/HFFR PVC insulated, single core copper conductor cable	Meter	63
1105	10 sq. mm ISI marked, FRLS/HFFR PVC insulated, single core copper conductor cable	Meter	106
1106	16 sq. mm ISI marked, FRLS/HFFR PVC insulated, single core copper conductor cable	Meter	164
1108	16/0.20 mm (0.5 sqmm) twin core FRLS/HFFR PVC sheathed, flat flexible copper cable	Meter	19
1109	16/0.20 mm (0.5 sqmm) twin circular, FRLS/HFFRPVC sheathed, workshop flexible copper cable	Meter	18
1110	32/0.20 mm (1.0 sqmm) twin circular, FRLS/HFFRPVC sheathed, workshop flexible copper cable	Meter	13
1111	1 pair, 0.5 mm dia annealed copper conductor, FRLS/HFFRPVC insulated, unarmoured, telephone cable	Meter	6
1112	2 pair, 0.5 mm dia annealed copper conductor, FRLS/HFFRPVC insulated, unarmoured, telephone cable	Meter	10
1113	4 pair, 0.5 mm dia annealed copper conductor, FRLS/HFFRPVC insulated, unarmoured, telephone cable	Meter	19
1114	Co-axial TV cable RG-6 grade, 0.7 mm solid copper conductor PE insulated, shielded with fine tinned copper braid protected with PVC sheath	Meter	20

CONDUIT (STEEL & PVC) AND ACCESSORIES

1201	20 mm diA, ISI marked, steel conduit	Meter	73
1202	25 mm diA, ISI marked, steel conduit	Meter	93
1203	32 mm diA, ISI marked, steel conduit	Meter	130
1204	40 mm diA, ISI marked, steel conduit	Meter	218
1205	50 mm diA, ISI marked, steel conduit	Meter	285
1206	20 mm inspection/ solid bends	Each	24
1207	25 mm inspection/ solid bends	Each	29
1208	32 mm inspection/ solid bends	Each	48
1209	40 mm inspection/ solid bends	Each	85
1210	50 mm inspection/ solid bends	Each	160
1211	20 mm sockets	Each	8
1212	25 mm sockets	Each	9
1213	32 mm sockets	Each	13
1214	40 mm sockets	Each	20
1215	50 mm sockets	Each	36
1216	20 mm junction box, one way	Each	34
1217	20 mm junction box, two way	Each	34
1218	20 mm iron staples/ saddles/ screws	Each	8
1219	25 mm iron staples/ saddles/ screws	Each	10
1220	32 mm iron staples/ saddles/ screws	Each	16
1221	40 mm iron staples/ saddles/ screws	Each	19
1222	50 mm iron staples/ saddles/ screws	Each	22

1224	20 mm diA, ISI marked, PVC conduit	Meter	21
1225	25 mm diA, ISI marked, PVC conduit	Meter	29
1226	32 mm diA, ISI marked, PVC conduit	Meter	46
1227	40 mm diA, ISI marked, PVC conduit	Meter	64
1228	50 mm diA, ISI marked, PVC conduit	Meter	97
1229	20 mm PVC bends	Each	8
1230	25 mm PVC bends	Each	11
1231	32 mm PVC bends	Each	18
1232	40 mm PVC bends	Each	28
1233	50 mm PVC bends	Each	47
1234	20 mm PVC couplers	Each	6
1235	25 mm PVC couplers	Each	7
1236	32 mm PVC couplers	Each	11
1237	40 mm PVC couplers	Each	14
1238	50 mm PVC couplers	Each	18
1239	20 mm PVC junction box, one way	Each	18
1240	20 mm PVC junction box, two way	Each	19
1241	75 mm X 75 mm X 50 mm PVC box	Each	22
	Metal Boxes		
1300	Modular GI box for 2 module	Each	33
1301	Modular GI box for 3 module	Each	41
1302	Modular GI box for 4 module	Each	48
1303	Modular GI box for 6 module	Each	74

1304	Modular GI box for 8 module	Each	90
1305	Modular GI box for 12 module	Each	114
1306	75 mm X 75 mm X 60 mm deep metal box	Each	23
1307	100 mm X 100 mm X 60 mm deep metal box	Each	35
1308	150 mm X 75 mm X 60 mm deep metal box	Each	37
1309	150 mm X 150 mm X 60 mm deep metal box	Each	53
1310	180 mm X 100 mm X 60 mm deep metal box	Each	45
1311	200 mm X 125 mm X 60 mm deep metal box	Each	56
1312	200 mm X 150 mm X 60 mm deep metal box	Each	70
1313	200 mm X 250 mm X 60 mm deep metal box	Each	90
1314	200 mm X 300 mm X 60 mm deep metal box	Each	112
1315	250 mm X 300 mm X 60 mm deep metal box	Each	139
1316	200 mm X 150 mm X 75 mm deep metal box	Each	76
1317	200 mm X 250 mm X 75 mm deep metal box	Each	105
1318	200 mm X 150 mm X 100 mm deep metal box	Each	92
1319	200 mm X 250 mm X 100 mm deep metal box	Each	119
1320	200 mm X 300 mm X 100 mm deep metal box	Each	139
1321	250 mm X 300 mm X 100 mm deep metal box	Each	174

WIRING SWITCHES AND ACCESSORIES

1322	3 mm thick phenolic laminated sheet	Sq.cm	0.20
1401	Ceiling rose, 3 pin, 5 A ISI marked	Each	20
1402	S.P. 5/6 A, one way modular switch, ISI marked	Each	34
1403	S.P. 5/6 A, two way modular switch, ISI marked	Each	68

1404	S.P. 15/16 A, one way modular switch, ISI marked	Each	74
1405	3 pin 5/6 A modular socket outlet, ISI marked	Each	63
1406	6 pin 15/16 A modular socket outlet, ISI marked	Each	105
1407	Modular bell push, ISI marked	Each	62
1408	Stepped type Modular Fan regulator (2 module)	Each	220
1409	Telephone Socket outlet modular type	Each	68
1410	T.V. Socket outlet modular type	Each	68
1411	Modular blanking plate	Each	16
1412	6 pin 25 amp modular type socket	Each	125
1420	Modular base & cover plate for 1 module	Each	42
1421	Modular base & cover plate for 2 module	Each	42
1422	Modular base & cover plate for 3 module	Each	56
1423	Modular base & cover plate for 4 module	Each	61
1424	Modular base & cover plate for 6 module	Each	79
1425	Modular base & cover plate for 8 module	Each	101
1426	Modular base & cover plate for 12 module	Each	147
1431	S.P. 5/6 A, one way switch, piano type ISI marked	Each	10
1432	S.P. 5/6 A, two way switch, piano type ISI marked	Each	19
1433	S.P. 15/16 A, one way switch, piano type ISI marked	Each	48
1434	3 pin 5/6 A socket outlet, piano type ISI marked	Each	23
1435	6 pin 15/16 & 5/6 A socket outlet, piano type ISI marked	Each	60
1436	Bell push, piano type	Each	19
1437	Telephone Socket outlet piano type	Each	39

1438	T.V. Socket outlet piano type	Each	28
1441	Brass pendant holder	Each	34
1442	Brass batten/ angle holder	Each	38
1443	Brass bracket holder 16 mm	Each	47
1444	Call bell/ buzzer, single phase	Each	44
1445	PIR Occupancy sensor	Each	3183
1446	PIR Occupancy sensor With Day light dimming	Each	6632
1447	Microwave occupancy Sensor	Each	8687
1448	Astronomical time switch suitable for 1 output per phase and suitable for single phase supply	Each	3022
1449	Astronomical time switch suitable for 2 output per phase and suitable for single phase supply	Each	6169
1450	Astronomical time switch suitable for 3 output (1 output per phase) and suitable for three phase supply	Each	7709
1451	Deleted		
1452	PVC Batten/ Angle Holder	Each	22
RISING MAINS BUS TRUNKING AND OVER HEAD BUS-BAR			
1501	200 A, TPN, rising mains in metal enclosure with aluminium busbar including fixing clamp	Meter	5056
1502	300 A, TPN, rising mains in metal enclosure with aluminium busbar including fixing clamp	Meter	5286
1503	400 A, TPN, rising mains in metal enclosure with aluminium busbar including fixing clamp	Meter	6550
1504	600 A, TPN, rising mains in metal enclosure with aluminium busbar including fixing clamp	Meter	7355
1505	800 A, TPN, rising mains in metal enclosure with aluminium busbar including fixing clamp	Meter	8906
1511	16 A, TPN, 2 way, distribution tap off box with ISI marked HRC fuses for rising mains	Each	9029

1512	16 A, TPN, 4 way, distribution tap off box with ISI marked HRC fuses for rising mains	Each	20357
1513	16 A, TPN, 6 way, distribution tap off box with ISI marked HRC fuses for rising mains	Each	28894
1514	16 A, TPN, 8 way, distribution tap off box with ISI marked HRC fuses for rising mains	Each	40057
1515	32 A, TPN, 2 way, distribution tap off box with ISI marked HRC fuses for rising mains	Each	11164
1516	32 A, TPN, 4 way, distribution tap off box with ISI marked HRC fuses for rising mains	Each	22984
1517	32 A, TPN, 6 way, distribution tap off box with ISI marked HRC fuses for rising mains	Each	30206
1518	32 A, TPN, 8 way, distribution tap off box with ISI marked HRC fuses for rising mains	Each	42026
1519	63 A, TPN, 2 way, distribution tap off box with ISI marked HRC fuses for rising mains	Each	12739
1520	63 A, TPN, 4 way, distribution tap off box with ISI marked HRC fuses for rising mains	Each	21670
1521	63 A, TPN, 6 way, distribution tap off box with ISI marked HRC fuses for rising mains	Each	23377
1522	63 A, TPN, 8 way, distribution tap off box with ISI marked HRC fuses for rising mains	Each	40320
1526	200 A, TPN, adaptor box with cable end box, with TPN disconnecter FSU, ISI marked HRC fuses and brass compression gland for rising mains	Each	18124
1527	300 A, TPN, adaptor box with cable end box, with TPN disconnecter FSU, ISI marked HRC fuses and brass compression gland for rising mains	Each	19700
1528	400 A, TPN, adaptor box with cable end box, with TPN disconnecter FSU, ISI marked HRC fuses and brass compression gland for rising mains	Each	21670
1529	600 A, TPN, adaptor box with cable end box, with TPN disconnecter FSU, ISI marked HRC fuses and brass compression gland for rising mains	Each	23377
1530	800 A, TPN, adaptor box with cable end box, with TPN disconnecter FSU, ISI marked HRC fuses and brass compression gland for rising mains	Each	26004
1531	800 A, TPN, bus trunking with aluminium busbars	Meter	10178

1532	1000 A, TPN, bus trunking with aluminium busbars	Meter	12083
1533	1250 A, TPN, bus trunking with aluminium busbars	Meter	15498
1534	1400 A, TPN, bus trunking with aluminium busbars	Meter	19438
1535	1600 A, TPN, bus trunking with aluminium busbars	Meter	19438
1536	200 A, TPN, overhead busbars with aluminium busbars	Meter	5779
1537	400 A, TPN, overhead busbars with aluminium busbars	Meter	7486
1538	32 A, TPN, plug-in-box with TPN disconnecter FSU and ISI marked HRC fuses for overhead busbars	Each	7561
1539	63 A, TPN, plug-in-box with TPN disconnecter FSU and ISI marked HRC fuses for overhead busbars	Each	8523
1540	100 A, TPN, plug-in-box with TPN disconnecter FSU and ISI marked HRC fuses for overhead busbars	Each	10535
1541	16 A, TPN, one way, tap off box with TPN disconnecter FSU and ISI marked HRC fuses for rising mains	Each	6000
1542	32 A, TPN, one way, tap off box with TPN disconnecter FSU and ISI marked HRC fuses for rising mains	Each	7561
1543	63 A, TPN, one way, tap off box with TPN disconnecter FSU and ISI marked HRC fuses for rising mains	Each	8523
1544	100 A, TPN, one way, tap off box with TPN disconnecter FSU and ISI marked HRC fuses for rising mains	Each	10535
1545	200 A, TPN, one way, tap off box with TPN disconnecter FSU and ISI marked HRC fuses for rising mains	Each	14006
1546	315 A, TPN, one way, tap off box with TPN disconnecter FSU and ISI marked HRC fuses for rising mains	Each	19124
1547	400 A, TPN, one way, tap off box with TPN disconnecter FSU and ISI marked HRC fuses for rising mains	Each	21800
1551	200 A, TPN, compact type rising mains with aluminium busbar including all accessories	Meter	5610
1552	315 A, TPN, compact type rising mains with aluminium busbar including all accessories	Meter	6624
1553	400 A, TPN, compact type rising mains with aluminium busbar including all accessories	Meter	7932
1554	500 A, TPN, compact type rising mains with aluminium busbar including all accessories	Meter	7932

1555	630 A, TPN, compact type rising mains with aluminium busbar including all accessories	Meter	9192
1556	800 A, TPN, compact type rising mains with aluminium busbar including all accessories	Meter	13620
1557	1000 A, TPN, compact type rising mains with aluminium busbar including all accessories	Meter	14124
1558	1250 A, TPN, compact type rising mains with aluminium busbar including all accessories	Meter	16842
1559	200 A, TPN, compact type bus trunking with aluminium busbar including elbows and other all accessories	Meter	5666
1560	315 A, TPN, compact type bus trunking with aluminium busbar including elbows and other all accessories	Meter	6690
1561	400 A, TPN, compact type bus trunking with aluminium busbar including elbows and other all accessories	Meter	8011
1562	500 A, TPN, compact type bus trunking with aluminium busbar including elbows and other all accessories	Meter	8011
1563	630 A, TPN, compact type bus trunking with aluminium busbar including elbows and other all accessories	Meter	9284
1564	800 A, TPN, compact type bus trunking with aluminium busbar including elbows and other all accessories	Meter	13756
1565	1000 A, TPN, compact type bus trunking with aluminium busbar including elbows and other all accessories	Meter	13756
1566	1250 A, TPN, compact type bus trunking with aluminium busbar including elbows and other all accessories	Meter	16065
1567	200 A, TPN, End Feed Unit for compact rising mains including all accessories	Each	6252
1568	315 A, TPN, End Feed Unit for compact rising mains including all accessories	Each	7566
1569	400 A, TPN, End Feed Unit for compact rising mains including all accessories	Each	8358
1570	500 A, TPN, End Feed Unit for compact rising mains including all accessories	Each	8358
1571	630 A, TPN, End Feed Unit for compact rising mains including all accessories	Each	9900

1572	800 A, TPN, End Feed Unit for compact rising mains including all accessories	Each	13260
1573	1000 A, TPN, End Feed Unit for compact rising mains including all accessories	Each	13260
1574	1250 A, TPN, End Feed Unit for compact rising mains including all accessories	Each	14778
1575	125 A, TPN, TAP OFF Box for compact rising mains including all accessories	Each	6400
1576	200 A, TPN, TAP OFF Box for compact rising mains including all accessories	Each	6400
1577	315 A, TPN, TAP OFF Box for compact rising mains including all accessories	Each	7840
1578	400 A, TPN, TAP OFF Box for compact rising mains including all accessories	Each	7840
1579	500 A, TPN, TAP OFF Box for compact rising mains including all accessories	Each	11840
1580	630 A, TPN, TAP OFF Box for compact rising mains including all accessories	Each	11840
	Sandwich Rising Mains with Aluminium bus bar		
1581	400 A, Isc=25 kA for 1 sec	per m	11040
1582	500 A, Isc=30 kA for 1 sec	per m	11730
1583	630 A, Isc=50 kA for 1 sec	per m	13950
1584	800 A, Isc=50 kA for 1 sec	per m	15840
1585	1000 A, Isc=50 kA for 1 sec	per m	17700
1586	1250 A, Isc=50 kA for 1 sec	per m	19374
1587	1600 A, Isc=50 kA for 1 sec	per m	23160
1588	2000 A, Isc=50 kA for 1 sec	per m	29760
1589	2500 A, Isc=50 kA for 1 sec	per m	33108
1590	3200 A, Isc=50 kA for 1 sec	per m	41508
1591	4000 A, Isc=50 kA for 1 sec	per m	70839

SWITCH DISCONNECTOR SFU AND MCCB

1601	32 A, TPN, switch disconnecter fuse unit (Panel mounted type) with ISI marked HRC fuses	Each	1443
1602	63 A, TPN, switch disconnecter fuse unit (Panel mounted type) with ISI marked HRC fuses	Each	2127
1603	100 A, TPN, switch disconnecter fuse unit (Panel mounted type) with ISI marked HRC fuses	Each	4346
1604	125 A, TPN, switch disconnecter fuse unit (Panel mounted type) with ISI marked HRC fuses	Each	5151
1605	160 A, TPN, switch disconnecter fuse unit (Panel mounted type) with ISI marked HRC fuses	Each	5851
1606	200 A, TPN switch fuse unit with ISI marked HRC fuses	Each	7009
1607	315 A, TPN switch fuse unit with ISI marked HRC fuses	Each	10647
1608	400 A, TPN switch fuse unit with ISI marked HRC fuses	Each	13025
1610	3 pole MCCB, 100A, 16kA Ics=100% Icu and Operational Voltage 690V	Each	2644
1611	3 pole MCCB, 125A, 16kA Ics=100% Icu and Operational Voltage 690V	Each	3403
1612	3 pole MCCB, 150A, 16kA Ics=100% Icu and Operational Voltage 690V	Each	4287
1613	3 pole MCCB, 200A, 16kA Ics=100% Icu and Operational Voltage 690V	Each	5541
1614	3 pole MCCB, 200A, 25kA Ics=100% Icu and Operational Voltage 690V	Each	8051
1615	3 pole MCCB, 250A, 25kA Ics=100% Icu and Operational Voltage 690V	Each	9522
1616	3 pole MCCB, 250A, 35kA Ics=100% Icu and Operational Voltage 690V	Each	10199
1617	3 pole MCCB, 315A, 35kA Ics=100% Icu and Operational Voltage 690V	Each	15986
1618	3 pole MCCB, 400A, 35kA Ics=100% Icu and Operational Voltage 690V	Each	15986
1619	3 pole MCCB, 500A, 35kA Ics=100% Icu and Operational Voltage 690V	Each	18909
1620	3 pole MCCB, 630A, 50kA Ics=100% Icu and Operational Voltage 690V	Each	20650

1621	3 pole MCCB, 800A, 50kA Ics=100% Icu and Operational Voltage 690V	Each	26915
1622	4 pole MCCB,100A,30kA Ics=100% Icu and Operational Voltage 690V	Each	5623
1623	4 pole MCCB, 125A, 36kA Ics=100% Icu and Operational Voltage 690V	Each	5961
1624	4 pole MCCB, 200A, 36kA Ics=100% Icu and Operational Voltage 690V	Each	11243
1625	4 pole MCCB,250A, 36kA Ics=100% Icu and Operational Voltage 690V	Each	12953
1626	4 pole MCCB,250A, 50kA Ics=100% Icu and Operational Voltage 690V	Each	13633
1627	4 pole MCCB,400A, 50kA Ics=100% Icu and Operational Voltage 690V	Each	31928
1628	4 pole MCCB,630A, 50kA Ics=100% Icu and Operational Voltage 690V MCB'S,ISOLATORS,RCCB'S AND MCB DB'S	Each	32250
1705	25 A modular SPMCB, 'C' curve	Each	161
1706	6 A, to 32 A, ratings , SP MCB, "C" curve, 10 KA breaking capacity	Each	135
1707	6 A, to 32 A, ratings , SPN MCB, "C" curve, 10 KA breaking capacity	Each	394
1708	6 A, to 32 A, ratings , DP MCB, "C" curve, 10 KA breaking capacity	Each	432
1709	6 A, to 32 A, ratings , TP MCB, "C" curve, 10 KA breaking capacity	Each	702
1710	6 A, to 32 A, ratings , TPN MCB, "C" curve, 10 KA breaking capacity	Each	869
1711	Single pole, blanking plate	Each	8
1712	40 A, 2 pole isolator	Each	270
1713	63 A, 2 pole isolator	Each	340
1714	40 A, 4 pole isolator	Each	616

1715	63 A, 4 pole isolator	Each	664
1716	100 A, 4 pole isolator	Each	810
1717	25 A, rating, 2 pole RCCB, 30mA	Each	1458
1718	40 A, rating, 2 pole RCCB, 30mA	Each	1726
1719	63 A, rating, 2 pole RCCB, 30mA	Each	1982
1720	25 A, rating, 4 pole RCCB, 30mA	Each	1836
1721	40 A, rating, 4 pole RCCB, 30mA	Each	2290
1722	63 A, rating, 4 pole RCCB, 30mA	Each	2052
1723	16/25 A, rating, 2 pole RCBO 100mA/ 300mA sensitivity	Each	0
1724	32 A, rating, 2 pole RCBO 100mA/ 300mA sensitivity	Each	0
1725	40 A, rating, 2 pole RCBO 100mA/ 300mA sensitivity	Each	0
1726	16/25 A, rating, 4 pole RCBO 100mA/ 300mA sensitivity	Each	0
1727	32 A, rating, 4 pole RCBO 100mA/ 300mA sensitivity	Each	0
1728	40 A, rating, 4 pole RCBO 100mA/ 300mA sensitivity	Each	0
1730	20 A, SPN, industrial type socket outlet, with plug top and metal chained cover in sheet steel enclosure	Each	680
1731	20 A, TPN, industrial type socket outlet, with plug top and metal chained cover in sheet steel enclosure	Each	997
1732	30 A, TPN, industrial type socket outlet, with plug top and metal chained cover in sheet steel enclosure	Each	2142

1734	2+2 way, SPN, single door, MCB DB	Each	820
1735	2+4 way, SPN, single door, MCB DB	Each	786
1736	2+6 way, SPN, single door, MCB DB	Each	1007
1737	2+10 way, SPN, single door, MCB DB	Each	1189
1738	6 way, SPN, double door, MCB DB	Each	1347
1739	8 way, SPN, double door, MCB DB	Each	1624
1740	12 way, SPN, double door, MCB DB	Each	1429
1741	14 way, SPN, double door, MCB DB	Each	2053
1742	Sheet steel DP MCB enclosure	Each	300
1743	Sheet steel TP MCB enclosure	Each	354
1751	4 way (4+12), TPN, MCB DB, single door, horizontal type	Each	0
1752	6 way (4+18), TPN, MCB DB, single door, horizontal type	Each	0
1753	8 way (4+24), TPN, MCB DB, single door, horizontal type	Each	0
1754	4 way (4+12), TPN, MCB DB, double door, horizontal type	Each	2712
1755	6 way (4+18), TPN, MCB DB, double door, horizontal type	Each	3378
1756	8 way (4+24), TPN, MCB DB, double door, horizontal type	Each	4128
1757	4 way (4+12), TPN, vertical type, MCB DB, single door	Each	0

1758	8 way (4+24), TPN, vertical type, MCB DB, single door	Each	0
1759	12 way (4+36), TPN, vertical type, MCB DB, single door	Each	0
1760	4 way (4+12), TPN, vertical type, MCB DB, double door	Each	5294
1761	8 way (4+24), TPN, vertical type, MCB DB, double door	Each	7297
1762	12 way (4+36), TPN, vertical type, MCB DB, double door	Each	9311
1763	4 way, TPN, vertical type, single door, MCB DB with provision to mount MCCB as incomer	Each	0
1764	8 way, TPN, vertical type, single door, MCB DB with with provision to mount MCCB as incomer	Each	0
1765	12 way, TPN, vertical type, single door, MCB DB with with provision to mount MCCB as incomer	Each	0
1771	2 + 4 way, SPN, prewired MCB DB with extended loose wire box, single door	Each	2494
1772	2 + 8 way, SPN, prewired MCB DB with extended loose wire box, single door	Each	2903
1773	2 + 12 way, SPN, prewired MCB DB with extended loose wire box, single door	Each	4299
1774	2 + 4 way, SPN, prewired MCB DB with extended loose wire box, double door	Each	2809
1775	2 + 8 way, SPN, prewired MCB DB with extended loose wire box, double door	Each	3277
1776	2 + 12 way, SPN, prewired MCB DB with extended loose wire box, double door	Each	4865
1779	4 way, (4+12) TPN, prewired MCB DB with extended loose wire box, single door	Each	0
1780	6 way, (4+18) TPN, prewired MCB DB with extended loose wire box, single door	Each	0
1781	8 way, (4+24) TPN, prewired MCB DB with extended loose wire box, single door	Each	0

1782	12 way, (4+36) TPN, prewired MCB DB with extended loose wire box, single door	Each	0
1783	4 way, (4+12) TPN, prewired MCB DB with extended loose wire box, double door	Each	6244
1784	6 way, (4+18) TPN, prewired MCB DB with extended loose wire box, double door	Each	7521
1785	8 way, (4+24) TPN, prewired MCB DB with extended loose wire box, double door	Each	8653
1786	12 way, (4+36) TPN, prewired MCB DB with extended loose wire box, double door	Each	10075
1787	4 way (4+4), TPN, vertical type, prewired, MCB DB with extended loose wire box, single door	Each	4490
1788	6 way (4+6), TPN, vertical type, prewired, MCB DB with extended loose wire box, single door	Each	0
1789	8 way (4+8), TPN, vertical type, prewired, MCB DB with extended loose wire box, single door	Each	0
1790	12 way (4+12), TPN, vertical type, prewired, MCB DB with extended loose wire box, single door	Each	0
1791	4 way (4+4), TPN, vertical type, prewired, MCB DB with extended loose wire box, double door	Each	5925
1792	6 way (4+6), TPN, vertical type, prewired, MCB DB with extended loose wire box, double door	Each	0
1793	8 way (4+8), TPN, vertical type, prewired, MCB DB with extended loose wire box, double door	Each	0
1794	12 way (4+12), TPN, vertical type, prewired, MCB DB with extended loose wire box, double door	Each	0
	Cable End Box		
1801	Cable End Boxes for 6 Way SPN DD DB	Each	372
1802	Cable End Boxes for 8 Way SPN DD DB	Each	432
1803	Cable End Boxes for 10 Way SPN DD DB	Each	416
1804	Cable End Boxes for 14 Way SPN DD DB	Each	485
1805	Cable End Boxes for 4 Way TPN DD DB	Each	620
1806	Cable End Boxes for 6 Way TPN DD DB	Each	653

1807	Cable End Boxes for 8 Way TPN DD DB	Each	815
1808	Cable End Boxes for Vertical TPN DD DB	Each	688
BRASS COMPRESSION GLANDS			
2101	Brass compression gland (19 mm) for 2 X 6 sq. mm 1.1 kV grade cable	Set	32
2102	Brass compression gland (19 mm) for 2 X 10 sq. mm 1.1 kV grade cable	Set	32
2103	Brass compression gland for (22 mm) 2 X 16 sq. mm 1.1 kV grade cable	Set	42
2104	Brass compression gland for (22 mm) 2 X 25 sq. mm 1.1 kV grade cable	Set	42
2105	Brass compression gland for (25 mm) 2 X 35 sq. mm 1.1 kV grade cable	Set	50
2106	Brass compression gland for (28 mm) 2 X 50 sq. mm 1.1 kV grade cable	Set	76
2107	Brass compression gland for (22 mm) 3 X 10 sq. mm 1.1 kV grade cable	Set	42
2108	Brass compression gland for (25 mm) 3 X 16 sq. mm 1.1 kV grade cable	Set	50
2109	Brass compression gland for (25 mm) 3 X 25 sq. mm 1.1 kV grade cable	Set	50
2110	Brass compression gland for (28 mm) 3 X 35 sq. mm 1.1 kV grade cable	Set	76
2111	Brass compression gland for (32 mm) 3 X 50 sq. mm 1.1 kV grade cable	Set	85
2112	Brass compression gland for (35 mm) 3 X 70 sq. mm 1.1 kV grade cable	Set	111
2113	Brass compression gland for (38 mm) 3 X 95 sq. mm 1.1 kV grade cable	Set	140
2114	Brass compression gland for (45 mm) 3 X 120 sq. mm 1.1 kV grade cable	Set	167
2115	Brass compression gland for (50 mm) 3 X 150 sq. mm 1.1 kV grade cable	Set	214

2116	Brass compression gland for (57 mm) 3 X 185 sq. mm 1.1 kV grade cable	Set	274
2117	Brass compression gland for (62 mm) 3 X 225 sq. mm 1.1 kV grade cable	Set	323
2118	Brass compression gland for (62 mm) 3 X 240 sq. mm 1.1 kV grade cable	Set	323
2119	Brass compression gland for (70 mm) 3 X 300 sq. mm 1.1 kV grade cable	Set	386
2120	Brass compression gland for (28 mm) 3½ X 25 sq. mm 1.1 kV grade cable	Set	76
2121	Brass compression gland for (32 mm) 3½ X 35 sq. mm 1.1 kV grade cable	Set	85
2122	Brass compression gland for (35 mm) 3½ X 50 sq. mm 1.1 kV grade cable	Set	111
2123	Brass compression gland for (38 mm) 3½ X 70 sq. mm 1.1 kV grade cable	Set	140
2124	Brass compression gland for (45 mm) 3½ X 95 sq. mm 1.1 kV grade cable	Set	167
2125	Brass compression gland for (45 mm) 3½ X 120 sq. mm 1.1 kV grade cable	Set	167
2126	Brass compression gland for (50 mm) 3½ X 150 sq. mm 1.1 kV grade cable	Set	214
2127	Brass compression gland for (57 mm) 3½ X 185 sq. mm 1.1 kV grade cable	Set	274
2128	Brass compression gland for (62 mm) 3½ X 225 sq. mm 1.1 kV grade cable	Set	323
2129	Brass compression gland for (62 mm) 3½ X 240 sq. mm 1.1 kV grade cable	Set	323
2130	Brass compression gland for (70 mm) 3½ X 300 sq. mm 1.1 kV grade cable	Set	386
2131	Brass compression gland for (82 mm) 3½ X 400 sq. mm 1.1 kV grade cable	Set	551
2132	Brass compression gland for (25 mm) 4 X 10 sq. mm 1.1 kV grade cable	Set	50

2133	Brass compression gland for (28 mm) 4 X 16 sq. mm 1.1 kV grade cable	Set	76
2134	Brass compression gland for (28 mm) 4 X 25 sq. mm 1.1 kV grade cable	Set	76
2135	Brass compression gland for (32 mm) 4 X 35 sq. mm 1.1 kV grade cable	Set	85
2136	Brass compression gland for (35 mm) 4 X 50 sq. mm 1.1 kV grade cable	Set	111

ALUMINIUM LUGS

2201	Aluminium lugs for 6 sq. mm cable	Each	2
2202	Aluminium lugs for 10 sq. mm cable	Each	2
2203	Aluminium lugs for 16 sq. mm cable	Each	3
2204	Aluminium lugs for 25 sq. mm cable	Each	4
2205	Aluminium lugs for 35 sq. mm cable	Each	5
2206	Aluminium lugs for 50 sq. mm cable	Each	7
2207	Aluminium lugs for 70 sq. mm cable	Each	12
2208	Aluminium lugs for 95 sq. mm cable	Each	12
2209	Aluminium lugs for 120 sq. mm cable	Each	17
2210	Aluminium lugs for 150 sq. mm cable	Each	23
2211	Aluminium lugs for 185 sq. mm cable	Each	28
2212	Aluminium lugs for 225 sq. mm cable	Each	40
2213	Aluminium lugs for 240 sq. mm cable	Each	48
2214	Aluminium lugs for 300 sq. mm cable	Each	68
2215	Aluminium lugs for 400 sq. mm cable	Each	98

MV CABLE JOINTING KITS

2300	Outdoor cable jointing kit with cast resin compound with lugs for 2 X 16 sq. mm 1.1 kV grade cable	Set	625
2301	Outdoor cable jointing kit with cast resin compound with lugs for 2 X 25 sq. mm 1.1 kV grade cable	Set	701
2302	Outdoor cable jointing kit with cast resin compound with lugs for 2 X 35 sq. mm 1.1 kV grade cable	Set	701

2303	Outdoor cable jointing kit with cast resin compound with lugs for 2 X 50 sq. mm 1.1 kV grade cable	Set	701
2304	Outdoor cable jointing kit with cast resin compound with lugs for 3 X 16 sq. mm 1.1 kV grade cable	Set	701
2305	Outdoor cable jointing kit with cast resin compound with lugs for 3 X 25 sq. mm 1.1 kV grade cable	Set	701
2306	Outdoor cable jointing kit with cast resin compound with lugs for 3 X 35 sq. mm 1.1 kV grade cable	Set	701
2307	Outdoor cable jointing kit with cast resin compound with lugs for 3 X 50 sq. mm 1.1 kV grade cable	Set	771
2308	Outdoor cable jointing kit with cast resin compound with lugs for 3 X 70 sq. mm 1.1 kV grade cable	Set	771
2309	Outdoor cable jointing kit with cast resin compound with lugs for 3 X 95 sq. mm 1.1 kV grade cable	Set	878
2310	Outdoor cable jointing kit with cast resin compound with lugs for 3 X 120 sq. mm 1.1 kV grade cable	Set	878
2311	Outdoor cable jointing kit with cast resin compound with lugs for 3 X 150 sq. mm 1.1 kV grade cable	Set	878
2312	Outdoor cable jointing kit with cast resin compound with lugs for 3 X 185 sq. mm 1.1 kV grade cable	Set	1326
2313	Outdoor cable jointing kit with cast resin compound with lugs for 3 X 225 sq. mm 1.1 kV grade cable	Set	1326
2314	Outdoor cable jointing kit with cast resin compound with lugs for 3 X 240 sq. mm 1.1 kV grade cable	Set	1604
2315	Outdoor cable jointing kit with cast resin compound with lugs for 3 X 300 sq. mm 1.1 kV grade cable	Set	1604
2316	Outdoor cable jointing kit with cast resin compound with lugs for 3½ X 25 sq. mm 1.1 kV grade cable	Set	701
2317	Outdoor cable jointing kit with cast resin compound with lugs for 3½ X 35 sq. mm 1.1 kV grade cable	Set	771
2318	Outdoor cable jointing kit with cast resin compound with lugs for 3½ X 50 sq. mm 1.1 kV grade cable	Set	771
2319	Outdoor cable jointing kit with cast resin compound with lugs for 3½ X 70 sq. mm 1.1 kV grade cable	Set	878

2320	Outdoor cable jointing kit with cast resin compound with lugs for 3½ X 95 sq. mm 1.1 kV grade cable	Set	878
2321	Outdoor cable jointing kit with cast resin compound with lugs for 3½ X 120 sq. mm 1.1 kV grade cable	Set	878
2322	Outdoor cable jointing kit with cast resin compound with lugs for 3½ X 150 sq. mm 1.1 kV grade cable	Set	1326
2323	Outdoor cable jointing kit with cast resin compound with lugs for 3½ X 185 sq. mm 1.1 kV grade cable	Set	946
2324	Outdoor cable jointing kit with cast resin compound with lugs for 3½ X 225 sq. mm 1.1 kV grade cable	Set	1604
2325	Outdoor cable jointing kit with cast resin compound with lugs for 3½ X 240 sq. mm 1.1 kV grade cable	Set	1604
2326	Outdoor cable jointing kit with cast resin compound with lugs for 3½ X 300 sq. mm 1.1 kV grade cable	Set	1604
2327	Outdoor cable jointing kit with cast resin compound with lugs for 3½ X 400 sq. mm 1.1 kV grade cable	Set	1931
2328	Outdoor cable jointing kit with cast resin compound with lugs for 4 X 16 sq. mm 1.1 kV grade cable	Set	701
2329	Outdoor cable jointing kit with cast resin compound with lugs for 4 X 25 sq. mm 1.1 kV grade cable	Set	701
2330	Outdoor cable jointing kit with cast resin compound with lugs for 4 X 35 sq. mm 1.1 kV grade cable	Set	771
2331	Outdoor cable jointing kit with cast resin compound with lugs for 4 X 50 sq. mm 1.1 kV grade cable	Set	771
2332	Straight through cable jointing kit with cast resin compound with ferrules for 2 X 16 sq. mm 1.1 kV grade cable	Set	1654
2333	Straight through cable jointing kit with cast resin compound with ferrules for 2 X 25 sq. mm 1.1 kV grade cable	Set	1654
2334	Straight through cable jointing kit with cast resin compound with ferrules for 2 X 35 sq. mm 1.1 kV grade cable	Set	1654
2335	Straight through cable jointing kit with cast resin compound with ferrules for 2 X 50 sq. mm 1.1 kV grade cable	Set	1931
2336	Straight through cable jointing kit with cast resin compound with ferrules for 3 X 16 sq. mm 1.1 kV grade cable	Set	1931

2337	Straight through cable jointing kit with cast resin compound with ferrules for 3 X 25 sq. mm 1.1 kV grade cable	Set	1654
2338	Straight through cable jointing kit with cast resin compound with ferrules for 3 X 35 sq. mm 1.1 kV grade cable	Set	1654
2339	Straight through cable jointing kit with cast resin compound with ferrules for 3 X 50 sq. mm 1.1 kV grade cable	Set	1931
2340	Straight through cable jointing kit with cast resin compound with ferrules for 3 X 70 sq. mm 1.1 kV grade cable	Set	2310
2341	Straight through cable jointing kit with cast resin compound with ferrules for 3 X 95 sq. mm 1.1 kV grade cable	Set	2373
2342	Straight through cable jointing kit with cast resin compound with ferrules for 3 X 120 sq. mm 1.1 kV grade cable	Set	2714
2343	Straight through cable jointing kit with cast resin compound with ferrules for 3 X 150 sq. mm 1.1 kV grade cable	Set	2714
2344	Straight through cable jointing kit with cast resin compound with ferrules for 3 X 185 sq. mm 1.1 kV grade cable	Set	3307
2345	Straight through cable jointing kit with cast resin compound with ferrules for 3 X 225 sq. mm 1.1 kV grade cable	Set	3799
2346	Straight through cable jointing kit with cast resin compound with ferrules for 3 X 240 sq. mm 1.1 kV grade cable	Set	3799
2347	Straight through cable jointing kit with cast resin compound with ferrules for 3 X 300 sq. mm 1.1 kV grade cable	Set	4783
2348	Straight through cable jointing kit with cast resin compound with ferrules for 3½ X 25 sq. mm 1.1 kV grade cable	Set	1931
2349	Straight through cable jointing kit with cast resin compound with ferrules for 3½ X 35 sq. mm 1.1 kV grade cable	Set	1931
2350	Straight through cable jointing kit with cast resin compound with ferrules for 3½ X 50 sq. mm 1.1 kV grade cable	Set	2310
2351	Straight through cable jointing kit with cast resin compound with ferrules for 3½ X 70 sq. mm 1.1 kV grade cable	Set	2333
2352	Straight through cable jointing kit with cast resin compound with ferrules for 3½ X 95 sq. mm 1.1 kV grade cable	Set	2714
2353	Straight through cable jointing kit with cast resin compound with ferrules for 3½ X 120 sq. mm 1.1 kV grade cable	Set	3307

2354	Straight through cable jointing kit with cast resin compound with ferrules for 3½ X 150 sq. mm 1.1 kV grade cable	Set	3307
2355	Straight through cable jointing kit with cast resin compound with ferrules for 3½ X 185 sq. mm 1.1 kV grade cable	Set	3799
2356	Straight through cable jointing kit with cast resin compound with ferrules for 3½ X 225 sq. mm 1.1 kV grade cable	Set	3875
2357	Straight through cable jointing kit with cast resin compound with ferrules for 3½ X 240 sq. mm 1.1 kV grade cable	Set	4783
2358	Straight through cable jointing kit with cast resin compound with ferrules for 3½ X 300 sq. mm 1.1 kV grade cable	Set	5730
2359	Straight through cable jointing kit with cast resin compound with ferrules for 3½ X 400 sq. mm 1.1 kV grade cable	Set	3076
2360	Straight through cable jointing kit with cast resin compound with ferrules for 4 X 16 sq. mm 1.1 kV grade cable	Set	1654
2361	Straight through cable jointing kit with cast resin compound with ferrules for 4 X 25 sq. mm 1.1 kV grade cable	Set	1931
2362	Straight through cable jointing kit with cast resin compound with ferrules for 4 X 35 sq. mm 1.1 kV grade cable	Set	1931
2363	Straight through cable jointing kit with cast resin compound with ferrules for 4 X 50 sq. mm 1.1 kV grade cable	Set	2310
2364	Straight through cable jointing kit with heat shrinkable kit with ferrules for 2 X 16 sq. mm 1.1 kV grade cable	Set	1273
2365	Straight through cable jointing kit with heat shrinkable kit with ferrules for 2 X 25 sq. mm 1.1 kV grade cable	Set	1507
2366	Straight through cable jointing kit with heat shrinkable kit with ferrules for 2 X 35 sq. mm 1.1 kV grade cable	Set	1507
2367	Straight through cable jointing kit with heat shrinkable kit with ferrules for 2 X 50 sq. mm 1.1 kV grade cable	Set	1507
2368	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3 X 16 sq. mm 1.1 kV grade cable	Set	1273
2369	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3 X 25 sq. mm 1.1 kV grade cable	Set	1507
2370	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3 X 35 sq. mm 1.1 kV grade cable	Set	1507

2371	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3 X 50 sq. mm 1.1 kV grade cable	Set	1507
2372	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3 X 70 sq. mm 1.1 kV grade cable	Set	2175
2373	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3 X 95 sq. mm 1.1 kV grade cable	Set	2175
2374	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3 X 120 sq. mm 1.1 kV grade cable	Set	2946
2375	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3 X 150 sq. mm 1.1 kV grade cable	Set	2946
2376	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3 X 185 sq. mm 1.1 kV grade cable	Set	2946
2377	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3 X 225 sq. mm 1.1 kV grade cable	Set	3233
2378	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3 X 240 sq. mm 1.1 kV grade cable	Set	3233
2379	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3 X 300 sq. mm 1.1 kV grade cable	Set	3883
2380	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3½ X 25 sq. mm 1.1 kV grade cable	Set	1507
2381	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3½ X 35 sq. mm 1.1 kV grade cable	Set	1507
2382	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3½ X 50 sq. mm 1.1 kV grade cable	Set	1507
2383	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3½ X 70 sq. mm 1.1 kV grade cable	Set	2175
2384	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3½ X 95 sq. mm 1.1 kV grade cable	Set	2175
2385	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3½ X 120 sq. mm 1.1 kV grade cable	Set	2946
2386	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3½ X 150 sq. mm 1.1 kV grade cable	Set	2946
2387	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3½ X 185 sq. mm 1.1 kV grade cable	Set	2946

2388	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3½ X 225 sq. mm 1.1 kV grade cable	Set	3233
2389	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3½ X 240 sq. mm 1.1 kV grade cable	Set	3233
2390	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3½ X 300 sq. mm 1.1 kV grade cable	Set	3883
2391	Straight through cable jointing kit with heat shrinkable kit with ferrules for 3½ X 400 sq. mm 1.1 kV grade cable	Set	5286
2392	Straight through cable jointing kit with heat shrinkable kit with ferrules for 4 X 16 sq. mm 1.1 kV grade cable	Set	1273
2393	Straight through cable jointing kit with heat shrinkable kit with ferrules for 4 X 25 sq. mm 1.1 kV grade cable	Set	1507
2394	Straight through cable jointing kit with heat shrinkable kit with ferrules for 4 X 35 sq. mm 1.1 kV grade cable	Set	2175
2395	Straight through cable jointing kit with heat shrinkable kit with ferrules for 4 X 50 sq. mm 1.1 kV grade cable	Set	2175
11 kV & 33 kV CABLE JOINTING KITS			
2413	Indoor cable jointing kit with cast resin compound with lugs for 11 kV grade XLPE cable for 3 core 70 sq. mm.	Set	839
2414	Indoor cable jointing kit with cast resin compound with lugs for 11 kV grade XLPE cable for 3 core 120 sq. mm.	Set	839
2415	Indoor cable jointing kit with cast resin compound with lugs for 11 kV grade XLPE cable for 3 core 240 sq. mm.	Set	1225
2416	Indoor cable jointing kit with cast resin compound with lugs for 11 kV grade XLPE cable for 3 core 300 sq. mm.	Set	1270
2417	Outdoor cable jointing kit with cast resin compound with lugs for 11 kV grade XLPE cable for 3 core 70 sq. mm.	Set	2873
2418	Outdoor cable jointing kit with cast resin compound with lugs for 11 kV grade XLPE cable for 3 core 120 sq. mm.	Set	2911
2419	Outdoor cable jointing kit with cast resin compound with lugs for 11 kV grade XLPE cable for 3 core 240 sq. mm.	Set	3296
2420	Outdoor cable jointing kit with cast resin compound with lugs for 11 kV grade XLPE cable for 3 core 300 sq. mm.	Set	3296

2421	Straight through cable jointing kit with cast resin compound with ferrules for 11 kV grade XLPE cable for 3 core 70 sq. mm.	Set	2253
2422	Straight through cable jointing kit with cast resin compound with ferrules for 11 kV grade XLPE cable for 3 core 120 sq. mm.	Set	2714
2423	Straight through cable jointing kit with cast resin compound with ferrules for 11 kV grade XLPE cable for 3 core 240 sq. mm.	Set	4294
2424	Straight through cable jointing kit with cast resin compound with ferrules for 11 kV grade XLPE cable for 3 core 300 sq. mm.	Set	5088
2437	Indoor heat shrinkable cable jointing kit with lugs for 11 kV grade XLPE cable for 3 core 70 sq. mm.	Set	8468
2438	Indoor heat shrinkable cable jointing kit with lugs for 11 kV grade XLPE cable for 3 core 120 sq. mm.	Set	10135
2439	Indoor heat shrinkable cable jointing kit with lugs for 11 kV grade XLPE cable for 3 core 240 sq. mm.	Set	10811
2440	Indoor heat shrinkable cable jointing kit with lugs for 11 kV grade XLPE cable for 3 core 300 sq. mm.	Set	10811
2441	Outdoor heat shrinkable cable jointing kit with lugs for 11 kV grade XLPE cable for 3 core 70 sq. mm.	Set	13048
2442	Outdoor heat shrinkable cable jointing kit with lugs for 11 kV grade XLPE cable for 3 core 120 sq. mm.	Set	14009
2443	Outdoor heat shrinkable cable jointing kit with lugs for 11 kV grade XLPE cable for 3 core 240 sq. mm.	Set	15753
2444	Outdoor heat shrinkable cable jointing kit with lugs for 11 kV grade XLPE cable for 3 core 300 sq. mm.	Set	15753
2445	Straight through heat shrinkable cable jointing kit with ferrules for 11 kV grade XLPE cable for 3 core 70 sq. mm.	Set	20834
2446	Straight through heat shrinkable cable jointing kit with ferrules for 11 kV grade XLPE cable for 3 core 120 sq. mm.	Set	27585
2447	Straight through heat shrinkable cable jointing kit with ferrules for 11 kV grade XLPE cable for 3 core 240 sq. mm.	Set	30170
2448	Straight through heat shrinkable cable jointing kit with ferrules for 11 kV grade XLPE cable for 3 core 300 sq. mm.	Set	30170

2528	Indoor heat shrinkable cable jointing kit with lugs for 33 kV grade XLPE cable for 3 core 70 sq. mm.	Set	14250
2529	Indoor heat shrinkable cable jointing kit with lugs for 33 kV grade XLPE cable for 3 core 120 sq. mm.	Set	21548
2530	Indoor heat shrinkable cable jointing kit with lugs for 33 kV grade XLPE cable for 3 core 240 sq. mm.	Set	24375
2531	Outdoor heat shrinkable cable jointing kit with lugs for 33 kV grade XLPE cable for 3 core 70 sq. mm.	Set	23100
2532	Outdoor heat shrinkable cable jointing kit with lugs for 33 kV grade XLPE cable for 3 core 120 sq. mm.	Set	29941
2533	Outdoor heat shrinkable cable jointing kit with lugs for 33 kV grade XLPE cable for 3 core 240 sq. mm.	Set	52629
2534	Straight through heat shrinkable cable jointing kit with ferrules for 33 kV grade XLPE cable for 3 core 70 sq. mm.	Set	40672
2535	Straight through heat shrinkable cable jointing kit with ferrules for 33 kV grade XLPE cable for 3 core 120 sq. mm.	Set	51398
2536	Straight through heat shrinkable cable jointing kit with ferrules for 33 kV grade XLPE cable for 3 core 240 sq. mm.	Set	72995
OVER HEAD LINE ACCESSORIES			
2601	Stay rod (1.8 M long, 19/20 mm diA,) with anchor plate 45 cm X 45 cm X 7.5 mm complete with thimble etc.	Set	432
2602	Stay wire (7/4.00 mm diA,)	kg	37
2603	Stay wire (7/3.15 mm diA,)	kg	37
2604	Turn buckle (20 mm X 60 cm)	Each	158
2605	Strain insulator	Each	25
2606	Bow tightner	Each	16
2607	Shackle insulator (75 mm X 90 mm) with G.I. bolts and nuts	Set	29
2608	Shackle insulator (100 mm X 110 mm) with G.I. bolts and nuts	Set	50
2609	Pin insulator (100 mm X 65 mm) with G.I. spindle and nuts	Set	50
2610	Pin insulator (100 mm X 80 mm) with G.I. spindle and nuts	Set	50

2611	MV horn gap lightning arrestor with pin insulator (100 mm X 65 mm), spindle and brass metal parts etc.	Set	79
2615	15 A aerial fuse complete with porcelain tube as required	Each	13
2616	30 A aerial fuse complete with porcelain tube as required	Each	20
2618	Stay clamp	Set	50
2619	Brace (50 mm X 50 mm X 6 mm angle iron)	Meter	169
2620	Angle iron bracket (50 mm X 50 mm X 6 mm) 65 cm long	Set	112
2623	Eye hook	Set	24
2624	Guy clamp	Set	54
2635	D' iron clamp (with coach screws)	Set	61
2643	Strain insulator 11 kV	Set	18
2644	Strain insulator 33 kV	Set	27
2645	11 kV pin insulator with pin washers and nuts	Set	90
2646	11 kV disc insulator	Set	405
2647	Galvanised insulator hardware fitting ball and socket type with strain clamp, bolts, nuts and washers	Set	119
2648	33 kV pin insulator with pin washers and nuts	Set	405
2649	Galvanised insulator hardware fitting ball and socket type with strain clamp, bolts, nuts and washers for 3 Nos 11 kV disk insulator	Set	347
2650	3 piece lightning arrestor set for 11 kV O.H. lines complete with G.I. clamp bolts and nuts with washers	Set	1620
2651	Single piece lightning arrestor set for 33 kV O.H. lines complete with G.I. clamp bolts and nuts with washers	Set	12600
2665	G.I. strap for shackle insulator	Set	43
	CABLE TRAYS		0
2701	MS perforated cable tray painted with powder coating 100 X 50 X 1.6 mm	Meter	158

2702	MS perforated cable tray painted with powder coating 150 X 50 X 1.6 mm	Meter	197
2703	MS perforated cable tray painted with powder coating 225 X 50 X 1.6 mm	Meter	256
2704	MS perforated cable tray painted with powder coating 300 X 50 X 1.6 mm	Meter	315
2705	MS perforated cable tray painted with powder coating 375 X 50 X 2 mm	Meter	467
2706	MS perforated cable tray painted with powder coating 450 X 50 X 2 mm	Meter	541
2707	MS perforated cable tray painted with powder coating 600 X 50 X 2 mm	Meter	688
2708	MS perforated cable tray painted with powder coating 300 X 62.5 X 2 mm	Meter	418
2709	MS perforated cable tray painted with powder coating 375 X 62.5 X 2 mm	Meter	492
2710	MS perforated cable tray painted with powder coating 450 X 62.5 X 2 mm	Meter	565
2711	MS perforated cable tray painted with powder coating 600 X 62.5 X 2 mm	Meter	713
2712	MS perforated cable tray painted with powder coating 750 X 62.5 X 2 mm	Meter	860
2713	MS perforated cable tray painted with powder coating 900 X 62.5 X 2 mm	Meter	1007
2714	MS perforated cable tray painted with powder coating 600 X 75 X 2 mm	Meter	737
2715	MS perforated cable tray painted with powder coating 750 X 75 X 2 mm	Meter	870
2716	MS perforated cable tray painted with powder coating 900 X 75 X 2 mm	Meter	1032
2717	MS perforated cable tray Connector 100 X 50 X 1.6 mm	Each	30
2718	MS perforated cable tray Connector 150 X 50 X 1.6 mm	Each	39

2719	MS perforated cable tray Connector 225 X 50 X 1.6 mm	Each	40
2720	MS perforated cable tray Connector 300 X 50 X 1.6 mm	Each	40
2721	MS perforated cable tray Connector 375 X 50 X 2 mm	Each	44
2722	MS perforated cable tray Connector 450 X 50 X 2 mm	Each	51
2723	MS perforated cable tray Connector 600 X 50 X 2 mm	Each	56
2724	MS perforated cable tray Connector 300 X 62.5 X 2 mm	Each	44
2725	MS perforated cable tray Connector 375 X 62.5 X 2 mm	Each	50
2726	MS perforated cable tray Connector 450 X 62.5 X 2 mm	Each	55
2727	MS perforated cable tray Connector 600 X 62.5 X 2 mm	Each	61
2728	MS perforated cable tray Connector 750 X 62.5 X 2 mm	Each	67
2729	MS perforated cable tray Connector 900 X 62.5 X 2 mm	Each	71
2733	MS perforated cable tray Bend 100 X 50 X 1.6 mm	Each	410
2734	MS perforated cable tray Bend 150 X 50 X 1.6 mm	Each	511
2735	MS perforated cable tray Bend 225 X 50 X 1.6 mm	Each	664
2736	MS perforated cable tray Bend 300 X 50 X 1.6 mm	Each	818
2737	MS perforated cable tray Bend 375 X 50 X 2 mm	Each	1214
2738	MS perforated cable tray Bend 450 X 50 X 2 mm	Each	1405

2739	MS perforated cable tray Bend 600 X 50 X 2 mm	Each	1789
2740	MS perforated cable tray Bend 300 X 62.5 X 2 mm	Each	1086
2741	MS perforated cable tray Bend 375 X 62.5 X 2 mm	Each	1278
2742	MS perforated cable tray Bend 450 X 62.5 X 2 mm	Each	1469
2743	MS perforated cable tray Bend 600 X 62.5 X 2 mm	Each	1853
2744	MS perforated cable tray Bend 750 X 62.5 X 2 mm	Each	2235
2745	MS perforated cable tray Bend 900 X 62.5 X 2 mm	Each	2618
2746	MS perforated cable tray Bend 600 X 75 X 2 mm	Each	1916
2747	MS perforated cable tray Bend 750 X 75 X 2 mm	Each	2300
2748	MS perforated cable tray Bend 900 X 75 X 2 mm	Each	2683
2749	MS perforated cable tray Reducer 100 X 50 X 1.6 mm	Each	512
2750	MS perforated cable tray Reducer 150 X 50 X 1.6 mm	Each	638
2751	MS perforated cable tray Reducer 225 X 50 X 1.6 mm	Each	830
2752	MS perforated cable tray Reducer 300 X 50 X 1.6 mm	Each	1022
2753	MS perforated cable tray Reducer 375 X 50 X 2 mm	Each	1518
2754	MS perforated cable tray Reducer 450 X 50 X 2 mm	Each	1756
2755	MS perforated cable tray Reducer 600 X 50 X 2 mm	Each	2236

2756	MS perforated cable tray Reducer 300 X 62.5 X 2 mm	Each	1357
2757	MS perforated cable tray Reducer 375 X 62.5 X 2 mm	Each	1597
2758	MS perforated cable tray Reducer 450 X 62.5 X 2 mm	Each	1836
2759	MS perforated cable tray Reducer 600 X 62.5 X 2 mm	Each	2315
2760	MS perforated cable tray Reducer 750 X 62.5 X 2 mm	Each	2793
2761	MS perforated cable tray Reducer 900 X 62.5 X 2 mm	Each	3273
2762	MS perforated cable tray Reducer 600 X 75 X 2 mm	Each	2395
2763	MS perforated cable tray Reducer 750 X 75 X 2 mm	Each	2874
2764	MS perforated cable tray Reducer 900 X 75 X 2 mm	Each	3354
2765	MS perforated cable tray Tee 100 X 50 X 1.6 mm	Each	512
2766	MS perforated cable tray Tee 150 X 50 X 1.6 mm	Each	638
2767	MS perforated cable tray Tee 225 X 50 X 1.6 mm	Each	830
2768	MS perforated cable tray Tee 300 X 50 X 1.6 mm	Each	1022
2769	MS perforated cable tray Tee 375 X 50 X 2 mm	Each	1518
2770	MS perforated cable tray Tee 450 X 50 X 2 mm	Each	1756
2771	MS perforated cable tray Tee 600 X 50 X 2 mm	Each	2236
2772	MS perforated cable tray Tee 300 X 62.5 X 2 mm	Each	1357

2773	MS perforated cable tray Tee 375 X 62.5 X 2 mm	Each	1597
2774	MS perforated cable tray Tee 450 X 62.5 X 2 mm	Each	1836
2775	MS perforated cable tray Tee 600 X 62.5 X 2 mm	Each	2315
2776	MS perforated cable tray Tee 750 X 62.5 X 2 mm	Each	2793
2777	MS perforated cable tray Tee 900 X 62.5 X 2 mm	Each	3273
2778	MS perforated cable tray Tee 600 X 75 X 2 mm	Each	2395
2779	MS perforated cable tray Tee 750 X 75 X 2 mm	Each	2874
2780	MS perforated cable tray Tee 900 X 75 X 2 mm	Each	3354
2781	MS perforated cable tray Cross member 100 X 50 X 1.6 mm	Each	614
2782	MS perforated cable tray Cross member 150 X 50 X 1.6 mm	Each	766
2783	MS perforated cable tray Cross member 225 X 50 X 1.6 mm	Each	996
2784	MS perforated cable tray Cross member 300 X 50 X 1.6 mm	Each	1227
2785	MS perforated cable tray Cross member 375 X 50 X 2 mm	Each	1821
2786	MS perforated cable tray Cross member 450 X 50 X 2 mm	Each	2108
2787	MS perforated cable tray Cross member 600 X 50 X 2 mm	Each	2683
2788	MS perforated cable tray Cross member 300 X 62.5 X 2 mm	Each	1628
2789	MS perforated cable tray Cross member 375 X 62.5 X 2 mm	Each	1917

2790	MS perforated cable tray Cross member 450 X 62.5 X 2 mm	Each	2204
2791	MS perforated cable tray Cross member 600 X 62.5 X 2 mm	Each	2779
2792	MS perforated cable tray Cross member 750 X 62.5 X 2 mm	Each	3352
2793	MS perforated cable tray Cross member 900 X 62.5 X 2 mm	Each	3927
2794	MS perforated cable tray Cross member 600 X 75 X 2 mm	Each	2874
2795	MS perforated cable tray Cross member 750 X 75 X 2 mm	Each	3450
2796	MS perforated cable tray Cross member 900 X 75 X 2 mm	Each	4025
ANGLE/FLAT IRON AND STEEL SHEET			
2801	MS Suspender 6 mm dia 0.75m long	Each	23
2802	MS Suspender 8 mm dia 0.75m long	Each	28
2803	MS Suspender 10 mm dia 0.75m long	Each	40
2804	25 mm X 25 mm X 3 mm angle iron	kg	56
2805	35 mm X 35 mm X 4 mm angle iron	kg	56
2806	35 mm X 35 mm X 5 mm angle iron	kg	56
2807	50 mm X 50 mm X 6 mm angle iron	kg	56
2808	65 mm X 65 mm X 6 mm angle iron	kg	56
2809	20/25 mm X 3 mm flat iron	kg	56

2810	25 mm X 4 mm flat iron	kg	56
2811	25 mm X 6 mm flat iron	kg	56
2812	40 mm X 3 mm flat iron	kg	56
2813	50 mm X 6 mm flat iron	kg	56
2814	50 mm X 8 mm flat iron	kg	56
2815	100 mm X 50 mm X 6 mm channel iron (9.56 kg/ mtr)	kg	56
2816	75 mm X 40 mm X 6 mm channel iron (7.14 kg/ mtr)	kg	56
2817	1.6 mm thick M.S. sheet	kg	56
2818	3 mm thick M.S. sheet	kg	56

GI AND RCC PIPE

2819	20 mm dia G.I. Pipe (light class)	Meter	172
2820	32 mm dia G.I. Pipe (light class)	Meter	283
2821	50 mm dia G.I. Pipe (light class)	Meter	459
2822	80 mm dia G.I. Pipe (light class)	Meter	716
2823	100 mm dia G.I. Pipe (light class)	Meter	1037
2824	15 mm dia, G.I. pipe (medium class)	Meter	151
2825	20 mm dia, G.I. pipe (medium class)	Meter	190

2826	40 mm dia, G.I. pipe (medium class)	Meter	383
2827	15 mm dia, G.I. pipe (heavy class)	Meter	169
2828	20 mm dia, G.I. pipe (heavy class)	Meter	214
2829	Check nut 20mm	Each	6
2831	32 mm dia, G.I. pipe (medium class)	Meter	335
2832	50 mm dia, G.I. pipe (medium class)	Meter	538
2833	80 mm dia, G.I. pipe (medium class)	Meter	871
2834	100 mm dia, G.I. pipe (medium class)	Meter	1260
2835	150 mm dia, G.I. pipe (medium class)	Meter	1965
2836	40 mm to 20 mm reducer	Each	36
2837	Nipple 50 mm dia	Each	68
2838	40 mm dia, G.I. bend (medium class)	Each	61
2839	50 mm dia, G.I. bend (medium class)	Each	94
2841	100 mm dia RCC pipe NP2 class	Meter	210
2842	150 mm dia RCC pipe NP2 class	Meter	220
2843	250 mm dia RCC pipe NP2 class	Meter	365
2844	300 mm dia RCC pipe NP2 class	Meter	450

2845	100 mm dia RCC collar NP2 class	Each	32
2846	150 mm dia RCC collar NP2 class	Each	33
2847	250 mm dia RCC collar NP2 class	Each	50
2848	300 mm dia RCC collar NP2 class	Each	63
SCREWS, NUT BOTS AND OTHER ACCESSORIES			
2851	Al. Alloy/ cadmium plated iron screws, 20 mm	Each	1
2852	Iron screws, 35 mm X 6 mm	Each	2
2853	Iron screws, 40 mm X 6 mm	Each	2
2854	Iron screws, 45 mm X 6 mm	Each	2
2855	Steel fastener 6 mm X 75 mm	Each	7
2856	Steel fastener 8 mm X 75 mm	Each	11
2857	PVC fastener 40mm long	Each	1
2858	PVC clip for fixing cable	Each	0
2859	Rubber/ PVC bushes	Each	1
2860	6 mm dia rivet/ stud/ bolts and nuts	Each	5
2861	25mm X 3mm bolts & nuts	Each	4
2862	32mm X 8mm bolts & nuts	Each	5

2863	38mm X 10mm bolts & nuts	Each	9
2864	38 mm X 7 mm, bolts and nuts	Each	7
2865	15 mm long X 6 mm dia G.I. bolts and nuts	Each	5
2866	10 mm X 25 mm long G.I. bolt with nut etc	Each	11
2867	16 mm X 50 mm bolts and nuts with washers	Set	9
2868	16 mm X 40 mm bolts and nuts with washers	Set	9
2869	16 mm X 125 mm bolts and nuts with washers	Set	22
2870	16 mm X 150 mm bolts and nuts with washers	Set	25
2871	10 mm X 25 mm long tinned brass bolt with nut etc.	Each	32
2881	GI saddles 19mm x 0.55mm for conduit	Each	2
2882	GI saddles 25mm x 0.90mm for conduit above 25mm	Each	2
HDPE Pipe			
2891	63 mm dia	Meter	78
2892	90 mm dia	Meter	108
2893	120 mm dia	Meter	156
2894	160 mm dia	Meter	228
2895	200 mm dia	Meter	360

HDPE Pipe Coupler

2896	63 mm dia	Each	16
2897	90 mm dia	Each	72
2898	120 mm dia	Each	84
2899	160 mm dia	Each	168
2900	200 mm dia	Each	234
2901	20 mm X 3 mm copper tape (0.533 kg/mtr)	kg	612
2902	25 mm X 5 mm copper tape (1.15 kg/mtr)	kg	612
2903	32 mm X 6 mm copper tape (1.705 kg/mtr)	kg	612
2904	8 SWG copper wire (4.0 mm dia)	kg	630
2905	600 mm X 600 mm X 3 mm thick copper plate (10.5 kg)	Each	6426
2906	Copper saddle	Each	36
2907	50 mm X 5mm copper strip (2.30 kg/mtr)	kg	612
2908	Brass nipple	Each	14
2909	Lightning finial, 25 mm dia X 300 mm long, copper	Each	585
2910	Lightning finial, 25 mm dia X 300 mm long, G.I.	Each	135
2911	20 mm X 3 mm G.I. Tape (0.461 kg/mtr)	kg	72

2912	25 mm X 5 mm G.I. strip (1.0 kg/mtr)	kg	63
2913	25 mm X 6 mm G.I. strip (1.2 kg/mtr)	kg	59
2914	32 mm X 6 mm G.I. Tape (1.475 kg/mtr)	kg	59
2915	600 mm X 600 mm X 6 mm thick G.I. plate	Each	1620
2916	6 SWG G.I. wire	kg	63
2917	GI saddle 20mm x 3mm	Each	4
2918	Funnel	E	23
2919	G.I. nuts and through bolts with washer	Each	32
2920	Washers	Each	0
2921	G.I. Hooks made of 8 SWG GI wire/ GI clip	Each	4
2922	CI/MS cover plate hinged to FRLSame with Locking arrangement	Each	315
2923	Pulley of 50 mm dia	Each	90
2924	Clamp, bolts, nuts etc.	Each	117
2925	Flat iron clamp (50 mm X 6 mm) for G.I. pipe	Each	45
2926	G.I. plate (10 cm X 10 cm X 5 mm)	kg	68
2927	Ball and socket	Each	23
2928	Back plate	Each	9

2929	Check nut 20mm	Each	5
2930	Rubber reel, nut & bolts with washers and safety pin	Each	68
2931	PVC sleeve	meter	7
2932	250 mm X 200 mm H.T. danger notice plate	Each	72
2933	200 mm X 150 mm M.V. danger notice plate	Each	54
2934	Earthing thimbles and solder	Each	18
2935	Cement, paint, sand etc.	Each	7
2936	Painting with primer and finish paint	Each	7
2937	Diesel	Litre	81
2938	Cotton waste, cleaning cloth etc.	kg	50
2939	Cleaning materials like soap/ detergent	kg	54
2940	Tin solder etc.	Each	7
2941	Charcoal	kg	9
2942	Coke	kg	14
2943	Salt	kg	11
2944	Paint	Litre	180
2945	Common burnt clay F.P.S. (non modular) bricks class designation 7.5	Each	5

2946	Fine sand	cum	900
2947	Bricks ballast	cum	675
2948	Cement	ton	7200
2949	Stone aggregate (single size) 40mm nominal size	cum	1350
2950	Stone aggregate (single size) 20mm nominal size	cum	1620
2951	Stone aggregate (single size) 10mm nominal size	cum	1620
2952	Coarse sand (Zone III)	cum	1710
9999	Sundries/bending charges	Each	3

HVAC Items

The Pipes of sizes 150mm & below shall be M.S. 'C' class as per IS : 1239 and pipes size above 150mm shall be welded black steel pipe heavy class as per IS: 3589, from minimum 6.35mm thick M.S. Sheet for pipes upto 350 mm diA, and from minimum 7mm thick MS sheet for pipes of 400 mm dia and above.

3001	M.S. Pipe 400 mm dia	Meter	10270
3002	M.S. Pipe 350 mm dia	Meter	5026
3003	M.S. Pipe 300 mm dia	Meter	4369
3004	M.S. Pipe 250mm dia	Meter	3684
3005	M.S. Pipe 200mm dia	Meter	2937
3006	M.S. Pipe 150mm dia	Meter	1816
3007	M.S. Pipe 125mm dia	Meter	1526

3008	M.S. Pipe 100mm dia	Meter	1238
3009	M.S. Pipe 80mm dia	Meter	848
3010	M.S. Pipe 65mm dia	Meter	679
3011	M.S. Pipe 50 mm dia	Meter	531
3012	M.S. Pipe 40 mm dia	Meter	377
3013	M.S. Pipe 32 mm dia	Meter	326
3014	M.S. Pipe 25 mm dia	Meter	256
3023	Pipe aluminium Cladding 0.63mm	Sq.m	360
	BUTTERFLY VALVE (MANUAL) with C I body SS Disc, Nitrile Rubber Seal & O- Ring PN 16 pressure rating for chilled water/hot water circulation as specified		
3024	200 mm dia	Each	9777
3025	150mm dia	Each	4307
3026	125mm dia	Each	3814
3027	100mm dia	Each	3061
3028	80mm dia	Each	2387
3029	65mm dia	Each	2179
3030	50mm dia	Each	2020
3031	40mm dia	Each	1764
	BALANCING VALVE WITH BUILT IN MEASURING FACILITY with C I body flanged construction with EPDM coated disc with long pitch with protected out pipe insulation & PN 16 pressure rating for chilled / hot water circulation as specified.		
3032	200 mm dia	Each	50680
3033	150mm dia	Each	24320

3034	125mm dia	Each	18280
3035	100mm dia	Each	13320
3036	80mm dia	Each	9200
3037	65 mm dia	Each	8120
3038	50mm dia	Each	7040
3039	40 mm dia	Each	4880
	NON - RETURN VALVE with duel plate of C I body SS plates vulcanized NBR seal flanged end & PN 16 pressure rating for chilled / hot water circulation including insulation as specified.		
3040	200 mm dia	Each	8834
3041	150mm dia	Each	4989
3042	125mm dia	Each	4074
3043	100 mm dia	Each	2818
3044	80mm dia	Each	1942
3045	65 mm dia	Each	1599
	Y - STRAINER of Ductile CI Body flanged ends with stainless steel strainer for chilled / hot water circulation including insulation as specified.		
3046	200mm dia	Each	35614
3047	150mm dia	Each	18992
3048	125mm dia	Each	14769
3049	100 mm dia	Each	9325
3050	80 mm dia	Each	6237
3051	65mm dia	Each	4804
3052	50mm dia	Each	3375
3053	40mm dia	Each	2471
3054	Industrial type pressure gauges with gun metal / brass valves	Each	812

3055	Industrial type thermometers.	Each	641
	Factory fabricated GSS sheet metal rectangular/round ducting of following sheet thickness		
3056	Thickness 0.63 mm sheet	sq. m	500
3057	Thickness 0.80 mm sheet	sq. m	615
3058	Thickness 1.00 mm sheet	sq. m	740
3059	Thickness 1.25 mm sheet	sq. m	886
	GSS sheet metal of following sheet thickness		0
3060	Thickness 0.63 mm sheet	sq. m	398
3061	Thickness 0.80 mm sheet	sq. m	531
3062	Thickness 1.00 mm sheet	sq. m	664
3063	Thickness 1.25 mm sheet	sq. m	803
3064	GI volume control duct damper	sq. m	4592
3015	Motorized (ON-OFF Type) duct mounted GI volume control damper with enthalpy sensor and necessary control wire (minimum 1.5 sqmm) for integration within AHU room	sq. m	10800
3065	Actuator	sq. m	4599
3066	powder coated extruded aluminium Supply Air Grills with aluminium volume control dampers	sq. m	6484
3067	Powder coated extruded aluminium Return Air Grills with louvers but without volume control dampers	sq. m	4072
3068	Supply air diffusers of powder coated aluminium with aluminium volume control dampers with anti smudge ring & removable core.	sq. m	8809
3069	Return air diffusers of powder coated aluminium without volume control dampers with anti smudge ring & removable core.	sq. m	5742
3070	Thermal insulated flexible duct- 200 mm dia	sq. m	352
	Fire dampers in supply air duct/main branch and return air path with inbuilt mechanism of spring return action		0
3071	Fire damper	sq. m	7319

3072	Actuator	sq. m	6735
3073	25 mm thick resin bonded glass wool having density of 32 kg/m ³ ,	sq. m	117
3074	50mm thick, density 32 kg/cu.m resin bonded glass fiber insulation.	sq. m	234
	Duly laminated aluminum foil of mat finish closed cell Nitrile rubber (class "O") insulation for use on existing duct		
3076	19mmthickness	sq. m	452
3077	25mmthickness	sq. m	598

AIR HANDLING UNITS

Factory built floor mounted chilled water double skin type horizontal/vertical air handling units details and specifications as in the item with cooling coil of 6 row deep etc. complete as per specification and of following capacities.

3078	4300 CMH	Each	75213
3079	5100 CMH	Each	88485
3080	6800 CMH	Each	99433
3081	8500 CMH	Each	123181
3082	10200 CMH	Each	142060
3083	11900 CMH	Each	161451
3084	13600 CMH	Each	173410
3085	15300 CMH	Each	196560
3086	17000 CMH	Each	208007
3087	18700 CMH	Each	229875
3088	20400 CMH	Each	246960

Factory built floor mounted chilled water double skin type horizontal/vertical air handling units details and specifications as in the item with cooling coil of 4 row deep etc. complete as per specification and of following capacities.

3089	4300 CMH	Each	66278
3090	5100 CMH	Each	77794
3091	6800 CMH	Each	86804
3092	8500 CMH	Each	107409
3093	10200 CMH	Each	125658
3094	11900 CMH	Each	140665
3095	13600 CMH	Each	152054
3096	15300 CMH	Each	172812
3097	17000 CMH	Each	181867
3098	18700 CMH	Each	198992
3099	20400 CMH	Each	215780
	Motorized Butter fly Valves with CI Body, SS Disc,O - ring and minimum PN-16 pressure rating , conforming to BS 5155, IS 13095, with IP-55 actuator as specified.		
3101	350 mm	Each	79863
3102	300 mm	Each	71567
3103	250 mm	Each	51524
3104	200 mm	Each	38865
3105	150 mm	Each	30130
3106	125 mm	Each	29871
3107	100 mm	Each	29316
3108	80 mm	Each	12535
3109	65 mm	Each	12414
3110	50 mm	Each	12160
	Electronic, self-balancing, pressure independent type dynamic balancing valve with integrated 2 way modulating control valves in a single body, alongwith actuator as specified		
3111	150 mm	Each	115183

3112	125 mm	Each	104868
3113	100 mm	Each	82519
3114	80 mm	Each	48136
3115	65 mm	Each	37821
3116	50 mm	Each	24498
3117	40 mm	Each	22349
3118	32 mm	Each	12034
3119	25 mm	Each	11319
3120	20 mm	Each	8252

AIR HANDLING UNIT

Factory built floor mounted chilled water double skin type horizontal/vertical air handling units details and specifications as in the item with cooling coil of 6 row deep etc. complete as per specification and of following capacities.

3142	23800 CMH	Each	270803
3143	28900 CMH	Each	285988
3144	34000 CMH	Each	316972
3145	39100 CMH	Each	439515
3162	23800 CMH	Each	205166
3163	28900 CMH	Each	261834
3164	34000 CMH	Each	290168
3165	39100 CMH	Each	332464
3201	Resin bonded fiber glass insulation of density 80 kg/cu m of thickness 65 mm on existing pipe	Sq.m	991

Hot Dipped Galvanized CABLE TRAYS

3701	MS perforated cable tray Hot Dipped Galvanized 100 X 50 X 1.6 mm	Meter	256
3702	MS perforated cable tray Hot Dipped Galvanized 150 X 50 X 1.6 mm	Meter	286
3703	MS perforated cable tray Hot Dipped Galvanized 225 X 50	Meter	413

	X 1.6 mm				
3704	MS perforated cable tray Hot Dipped Galvanized 300 X 50 X 1.6 mm	Meter	458		
3705	MS perforated cable tray Hot Dipped Galvanized 375 X 50 X 2 mm	Meter	624		
3706	MS perforated cable tray Hot Dipped Galvanized 450 X 50 X 2 mm	Meter	693		
3707	MS perforated cable tray Hot Dipped Galvanized 600 X 50 X 2 mm	Meter	1014		
3708	MS perforated cable tray Hot Dipped Galvanized 300 X 62.5 X 2 mm	Meter	617		
3709	MS perforated cable tray Hot Dipped Galvanized 375 X 62.5 X 2 mm	Meter	726		
3710	MS perforated cable tray Hot Dipped Galvanized 450 X 62.5 X 2 mm	Meter	835		
3711	MS perforated cable tray Hot Dipped Galvanized 600 X 62.5 X 2 mm	Meter	1056		
3712	MS perforated cable tray Hot Dipped Galvanized 750 X 62.5 X 2 mm	Meter	1298		
3713	MS perforated cable tray Hot Dipped Galvanized 900 X 62.5 X 2 mm	Meter	1521		
3714	MS perforated cable tray Hot Dipped Galvanized 600 X 75 X 2 mm	Meter	1089		
3715	MS perforated cable tray Hot Dipped Galvanized 750 X 75 X 2 mm	Meter	1307		
3716	MS perforated cable tray Hot Dipped Galvanized 900 X 75 X 2 mm	Meter	1525		
3717	MS perforated Hot Dipped Galvanized cable tray Connector 100 X 50 X 1.6 mm	Each	32		
3718	MS perforated Hot Dipped Galvanized cable tray Connector 150 X 50 X 1.6 mm	Each	38		
3719	MS perforated Hot Dipped Galvanized cable tray Connector 225 X 50 X 1.6 mm	Each	40		
3720	MS perforated Hot Dipped Galvanized cable tray Connector 300 X 50 X 1.6 mm	Each	40		
3721	MS perforated Hot Dipped Galvanized cable tray Connector 375 X 50 X 2 mm	Each	40		
3722	MS perforated Hot Dipped Galvanized cable tray Connector 450 X 50 X 2 mm	Each	40		
3723	MS perforated Hot Dipped Galvanized cable tray Connector 600 X 50 X 2 mm	Each	40		
3724	MS perforated Hot Dipped Galvanized cable tray Connector 300 X 62.5 X 2 mm	Each	40		

3725	MS perforated Hot Dipped Galvanized cable tray Connector 375 X 62.5 X 2 mm	Each	40
3726	MS perforated Hot Dipped Galvanized cable tray Connector 450 X 62.5 X 2 mm	Each	40
3727	MS perforated Hot Dipped Galvanized cable tray Connector 600 X 62.5 X 2 mm	Each	40
3728	MS perforated Hot Dipped Galvanized cable tray Connector 750 X 62.5 X 2 mm	Each	40
3729	MS perforated Hot Dipped Galvanized cable tray Connector 900 X 62.5 X 2 mm	Each	40
3733	MS perforated Hot Dipped Galvanized cable tray Bend 100 X 50 X 1.6 mm	Each	640
3734	MS perforated Hot Dipped Galvanized cable tray Bend 150 X 50 X 1.6 mm	Each	715
3735	MS perforated Hot Dipped Galvanized cable tray Bend 225 X 50 X 1.6 mm	Each	1030
3736	MS perforated Hot Dipped Galvanized cable tray Bend 300 X 50 X 1.6 mm	Each	1146
3737	MS perforated Hot Dipped Galvanized cable tray Bend 375 X 50 X 2 mm	Each	1559
3738	MS perforated Hot Dipped Galvanized cable tray Bend 450 X 50 X 2 mm	Each	1733
3739	MS perforated Hot Dipped Galvanized cable tray Bend 600 X 50 X 2 mm	Each	2536
3740	MS perforated Hot Dipped Galvanized cable tray Bend 300 X 62.5 X 2 mm	Each	1543
3741	MS perforated Hot Dipped Galvanized cable tray Bend 375 X 62.5 X 2 mm	Each	1816
3742	MS perforated Hot Dipped Galvanized cable tray Bend 450 X 62.5 X 2 mm	Each	2088
3743	MS perforated Hot Dipped Galvanized cable tray Bend 600 X 62.5 X 2 mm	Each	2422
3744	MS perforated Hot Dipped Galvanized cable tray Bend 750 X 62.5 X 2 mm	Each	3243
3745	MS perforated Hot Dipped Galvanized cable tray Bend 900 X 62.5 X 2 mm	Each	3794
3746	MS perforated Hot Dipped Galvanized cable tray Bend 600 X 75 X 2 mm	Each	2723
3747	MS perforated Hot Dipped Galvanized cable tray Bend 750 X 75 X 2 mm	Each	3267
3748	MS perforated Hot Dipped Galvanized cable tray Bend 900 X 75 X 2 mm	Each	3812
3749	MS perforated Hot Dipped Galvanized cable tray Reducer 100 X 50 X 1.6 mm	Each	1014

3750	MS perforated Hot Dipped Galvanized cable tray Reducer 150 X 50 X 1.6 mm	Each	1207
3751	MS perforated Hot Dipped Galvanized cable tray Reducer 225 X 50 X 1.6 mm	Each	1573
3752	MS perforated Hot Dipped Galvanized cable tray Reducer 300 X 50 X 1.6 mm	Each	1871
3753	MS perforated Hot Dipped Galvanized cable tray Reducer 375 X 50 X 2 mm	Each	2306
3754	MS perforated Hot Dipped Galvanized cable tray Reducer 450 X 50 X 2 mm	Each	2509
3755	MS perforated Hot Dipped Galvanized cable tray Reducer 600 X 50 X 2 mm	Each	3571
3756	MS perforated Hot Dipped Galvanized cable tray Reducer 300 X 62.5 X 2 mm	Each	2299
3757	MS perforated Hot Dipped Galvanized cable tray Reducer 375 X 62.5 X 2 mm	Each	2706
3758	MS perforated Hot Dipped Galvanized cable tray Reducer 450 X 62.5 X 2 mm	Each	3200
3759	MS perforated Hot Dipped Galvanized cable tray Reducer 600 X 62.5 X 2 mm	Each	4101
3760	MS perforated Hot Dipped Galvanized cable tray Reducer 750 X 62.5 X 2 mm	Each	4425
3761	MS perforated Hot Dipped Galvanized cable tray Reducer 900 X 62.5 X 2 mm	Each	5162
3762	MS perforated Hot Dipped Galvanized cable tray Reducer 600 X 75 X 2 mm	Each	3578
3763	MS perforated Hot Dipped Galvanized cable tray Reducer 750 X 75 X 2 mm	Each	4486
3764	MS perforated Hot Dipped Galvanized cable tray Reducer 900 X 75 X 2 mm	Each	5225
3765	MS perforated Hot Dipped Galvanized cable tray Tee 100 X 50 X 1.6 mm	Each	767
3766	MS perforated Hot Dipped Galvanized cable tray Tee 150 X 50 X 1.6 mm	Each	801
3767	MS perforated Hot Dipped Galvanized cable tray Tee 225 X 50 X 1.6 mm	Each	1237
3768	MS perforated Hot Dipped Galvanized cable tray Tee 300 X 50 X 1.6 mm	Each	1374
3769	MS perforated Hot Dipped Galvanized cable tray Tee 375 X 50 X 2 mm	Each	1871
3770	MS perforated Hot Dipped Galvanized cable tray Tee 450 X 50 X 2 mm	Each	2079

3771	MS perforated Hot Dipped Galvanized cable tray Tee 600 X 50 X 2 mm	Each	3043
3772	MS perforated Hot Dipped Galvanized cable tray Tee 300 X 62.5 X 2 mm	Each	1852
3773	MS perforated Hot Dipped Galvanized cable tray Tee 375 X 62.5 X 2 mm	Each	2220
3774	MS perforated Hot Dipped Galvanized cable tray Tee 450 X 62.5 X 2 mm	Each	2505
3775	MS perforated Hot Dipped Galvanized cable tray Tee 600 X 62.5 X 2 mm	Each	3229
3776	MS perforated Hot Dipped Galvanized cable tray Tee 750 X 62.5 X 2 mm	Each	3888
3777	MS perforated Hot Dipped Galvanized cable tray Tee 900 X 62.5 X 2 mm	Each	4554
3778	MS perforated Hot Dipped Galvanized cable tray Tee 600 X 75 X 2 mm	Each	3267
3779	MS perforated Hot Dipped Galvanized cable tray Tee 750 X 75 X 2 mm	Each	3921
3780	MS perforated Hot Dipped Galvanized cable tray Tee 900 X 75 X 2 mm	Each	4575
3781	MS perforated Hot Dipped Galvanized cable tray Cross member 100 X 50 X 1.6 mm	Each	767
3782	MS perforated Hot Dipped Galvanized cable tray Cross member 150 X 50 X 1.6 mm	Each	801
3783	MS perforated Hot Dipped Galvanized cable tray Cross member 225 X 50 X 1.6 mm	Each	1252
3784	MS perforated Hot Dipped Galvanized cable tray Cross member 300 X 50 X 1.6 mm	Each	1533
3785	MS perforated Hot Dipped Galvanized cable tray Cross member 375 X 50 X 2 mm	Each	2072
3786	MS perforated Hot Dipped Galvanized cable tray Cross member 450 X 50 X 2 mm	Each	2402
3787	MS perforated Hot Dipped Galvanized cable tray Cross member 600 X 50 X 2 mm	Each	3054
3788	MS perforated Hot Dipped Galvanized cable tray Cross member 300 X 62.5 X 2 mm	Each	1857
3789	MS perforated Hot Dipped Galvanized cable tray Cross member 375 X 62.5 X 2 mm	Each	2179
3790	MS perforated Hot Dipped Galvanized cable tray Cross member 450 X 62.5 X 2 mm	Each	2509
3791	MS perforated Hot Dipped Galvanized cable tray Cross member 600 X 62.5 X 2 mm	Each	3168

3792	MS perforated Hot Dipped Galvanized cable tray Cross member 750 X 62.5 X 2 mm	Each	3888
3793	MS perforated Hot Dipped Galvanized cable tray Cross member 900 X 62.5 X 2 mm	Each	4554
3794	MS perforated Hot Dipped Galvanized cable tray Cross member 600 X 75 X 2 mm	Each	3270
3795	MS perforated Hot Dipped Galvanized cable tray Cross member 750 X 75 X 2 mm	Each	2798
3796	MS perforated Hot Dipped Galvanized cable tray Cross member 900 X 75 X 2 mm	Each	3263
3801	GI Suspender 6 mm dia 0.75m long	Each	21
3802	GI Suspender 8 mm dia 0.75m long	Each	27
3803	GI Suspender 10 mm dia 0.75m long	Each	32
3804	25 mm X 25 mm X 3 mm Galvannised slotted/angle iron/channel	kg	43
3805	35 mm X 55 mm X 4 mm Galvannised slotted/angle iron/channel	kg	43
3806	35mm X 35 mm X 5 mm Galvannised slotted/angle iron/channel	kg	43
4001	Resin bonded fiber glass pipe section insulation having density 80 kg/cum-75 mm thickness for 400 mm dia pipe	Meter	1850
4002	Resin bonded fiber glass pipe section insulation having density 80 kg/cum-75 mm thickness for 350 mm dia pipe	Meter	1654
4003	Resin bonded fiber glass pipe section insulation having density 80 kg/cum-75 mm thickness for 300 mm dia pipe	Meter	1356
4004	Resin bonded fiber glass pipe section 250mm insulation having density 80 kg/cum-75 mm thickness for 250 mm dia pipe	Meter	1184
4005	Resin bonded fiber glass pipe section 200 mm insulation having density 80 kg/cum-75 mm thickness for 200 mm dia pipe	Meter	1003
4006	Resin bonded fiber glass pipe section 150 mm insulation having density 80 kg/cum-50 mm thickness for 150 mm dia pipe	Meter	507
4007	Resin bonded fiber glass pipe section insulation having density 80 kg/cum-50 mm thickness for 125mm dia pipe	Meter	448
4008	Resin bonded fiber glass pipe section insulation having density 80 kg/cum-50 mm thickness for 100 mm dia pipe	Meter	201

4009	Resin bonded fiber glass pipe section insulation having density 80 kg/cum-50 mm thickness for 80 mm dia pipe	Meter	331
4010	Resin bonded fiber glass pipe section insulation having density 80 kg/cum-50 mm thickness for 65 mm dia pipe	Meter	300
4011	Resin bonded fiber glass pipe section insulation having density 80 kg/cum-50 mm thickness for 50 mm dia pipe	Meter	262
4012	Resin bonded fiber glass pipe section insulation having density 80 kg/cum-50mm thickness for 40 mm dia pipe	Meter	236
4013	Resin bonded fiber glass pipe section insulation having density 80 kg/cum-50 mm thickness for 32 mm dia pipe	Meter	222
4014	Resin bonded fiber glass pipe section insulation having density 80 kg/cum-50 mm thickness for 25 mm dia pipe	Meter	202
4015	Nitrile rubber (closed cell elastometric nitrile rubber of class 'O') Insulation -32 mm thickness	Per sq meter	670
4016	Nitrile rubber (closed cell elastometric nitrile rubber of class 'O') Insulation -19 mm thickness	Per sq meter	371
4017	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation -75mm thickness for 400 mm dia pipe	Meter	1047
4018	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation -75mm thickness for 350 mm dia pipe	Meter	895
4019	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation -75mm thickness for 300mm dia pipe	Meter	772
4020	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation -75mm thickness for 250 mm dia pipe	Meter	644
4021	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation -75 mm thickness for 200 mm dia pipe	Meter	549
4022	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation -50 mm thickness for 150 mm dia pipe	Meter	261
4023	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation - 50 mm thickness for 125 mm dia pipe	Meter	230
4024	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation -50 mm thickness for 100 mm dia pipe	Meter	202
4025	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation -50 mm thickness for 80 mm dia pipe	Meter	167
4026	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation -50 mm thickness for 65 mm dia pipe	Meter	153

4027	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation -50 mm thickness for 50 mm dia pipe	Meter	134
4028	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation -50 mm thickness for 40 mm dia pipe	Meter	120
4029	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation -50 mm thickness for 32 mm dia pipe	Meter	112
4030	Expanded polystyrene moulded pipe section of density 20kg/cum Insulation -50 mm thickness for 25 mm dia pipe	Meter	103
4031	Resin bonded fiber glass insulation of thickness 75 mm on existing pipe	Per sq meter	1366
4032	Resin bonded fiber glass insulation of density 80 kg/cu.mt. of thickness 50 mm on existing pipe	Per sq meter	1120
4033	Resin bonded fiber glass insulation of density 80 kg/cu.mt. Of thickness 40 mm on existing pipe	Per sq meter	972
4034	Resin bonded fiber glass insulation of thickness 25 mm on existing pipe	Per sq meter	641
4037	Expanded polystyrene insulation of thickness 75 mm on existing pipe	Per sq meter	224
4038	Expanded polystyrene insulation of thickness 50 mm on existing pipe	Per sq meter	149
4069	Acoustic lining of supply air duct with 25mm thick resin bonded glass wool having density of 32kg/m3	Per sq meter	117
4070	Reinforced fiber glass tissue and 0.08mm perforated aluminium sheet	Per sq meter	320
4071	50 mm thick resin bonded glass wool having density of 32kg/m3	Per sq meter	234
4072	50 mm thick resin bonded glass wool having density of 24 kg/m3 with factory laminated aluminium foil	Per sq meter	173
4073	Resin bonded fiber glass insulation of thickness 65 mm on existing pipe	Per sq meter	1239

FIRE DETECTION AND ALARM SYSTEM

4101	Heat Detectors of electronic Rate of Rise cum fixed temperature (Dual Thermistor) type with mounting base complete with all connection etc.	Each	750
4102	Smoke Detector with built in LED, mounting base complete with all connection etc.	Each	813

4103	Manual call boxes of MS Type in surface/recess with stainless steel chain & hammer assembly complete with glass and push button etc.	Each	300
4104	Manual call boxes of ABS Type in surface/recess with stainless steel chain & hammer assembly complete with glass and push button etc.	Each	250
4105	Response indicator on surface/recessed MS Box having two LED's metallic covers complete with all connections etc.	Each	90
4106	Fire alarm sounders (electronic) with facility to make announcement, mounted in M.S. Box (16 SWG) with hinged cover plate & suitable for operation with amplifier i/c line matching transformer etc.	Each	363
4107	Fire alarm sounders (electronic) with facility to make announcement, mounted in A.B.S. Box with hinged cover plate & suitable for operation with amplifier i/c line matching transformer etc.	Each	313
4108	Talk back slave station in surface/recess suitable for operation on simplex mode complete with P.T.T. Knob & speaker/microphone enclosed in a M.S (16 SWG)/ABS.Box, break glass in front etc. Sector panels suitable for following zones, complete with visual indications for short circuit fault, open circuit fault fire conditions and all other standard facilities as per IS:2189, mimic diagram for all area/zone covered complete with all connections, interconnections	Each	625
4109	4 Zone fire alarm control panel as per IS: 2189 with backlit LCD Display	Each	4875
4110	6 Zone fire alarm control panel as per IS: 2189 with backlit LCD Display	Each	7375
4111	10 Zone fire alarm control panel as per IS: 2189 with backlit LCD Display Main control and indication panel made out of 16 SGW MS sheet to accommodate the following items duly stove anameled painted in approved colour with louvers for ventilation's locking arrangement including audio and visual indication for fire alarm and public address system, monitoring system including providing and fixing the following connections, interconnections etc.	Each	9750
4112	10 zone control panel for fire alarm system with all accessories and specification mentioned in the item	Each	47177
4113	6 zone control panel for fire alarm system with following:-	Each	32272

Micro processor based. Intelligent Addressable Main Fire alarm panel , Central Processing Unit ,with the following loop module and with each loop capacity of 120 detector per loop with total of 240 detectors and devices with loop length up to 2 kmtr, network communication card, minimum 320 character graphics, LCD display with touch screen or other keypad and 1000 event history logs in the non volatile memory (EEPROM) power supply unit at 230+ 5 % V, 48 Hrs back-up with 24 volt sealed maintenance free batteries with automatic charger. The panel shall have facility to connect printer to printout logs and facility to have integration with analog voice evacuation system(which is part of the schedule of work under SH: PA System) and shall be complete with all accessories . The panel should be compatible for integration with IBMS system with open protocol Bacnet/IP, complete as per specs and as required for following loops.

4114	10 Loop Control Panel having CPU with 640 character display including backlite LCD display qwerty programming and operating manuals, supporting 1 to 10 signaling line circuits (upto 5 extendable cards & supporting 103 network modes) complete as required.	Each	338000
4115	Two Loop Control Panel having CPU with 640 character display including backlite LCD display qwerty programming and operating manuals, supporting 1 to 10 signaling line circuits (upto 5 extendable cards & supporting 103 network modes) complete as required.	Each	177650
4116	Central graphical fire alarm management system i/c software	Each	147250
4117	320 character LCD display repeater panel.	Each	81000
4118	Intelligent photothermal detector	Each	2100
4119	Response indicator on surface/recessed MS Box having two LED's metallic covers complete with all connections etc.	Each	150
4120	Intelligent addressable programmable sounder.	Each	1824
4121	Cost of fault isolator complete with base	Each	2291
4122	Cost of intelligent aspiration detector	Each	220000
4123	Cost of intelligent addressable thermal detector	Each	1881
4124	Cost of control module	Each	2090

4125	Cost of addressable fire phone control module	Each	2290
4126	Cost of addressable beam detector with remote test features	Each	55100
4127	Cost of intelligent duct detector unit with photothermal detector complete with base	Each	5722
4128	Cost of manual call point	Each	2746
4129	Cost of Horn cum strobe	Each	2470
4130	Cost of Strobe.	Each	2365
4131	Cost of fire fighter telephone handset	Each	4275
4132	Intelligent interface unit BACnet protocol for communication link between building management system and fire alarm control panel.	Each	141550
4133	Cost of fire fighter phone jack	Each	1150
4134	6 Zone, voice alarm controller with USB, MP3, AM/FM player (including zone button paging station).	Each	95250
4135	Cost of 1.5/3/6W Ceiling Speaker	Each	670
4136	Cost of 1.5/3/6W Metal Ceiling Speakers.	Each	1295
4137	Cost of wall mounted loud speaker, 3/1.5 Watt in ABS enclosure	Each	1775
4138	Cost of 2 watt, 70 Volts ceiling speaker.	Each	1330
4139	Cost of Digital audio amplifier 50 Watt 25VRMS operating of 240 Volt AC Supply.	Each	73000
4140	Digital audio amplifier 75 Watt 25VRMS operating of 240 Volt AC Supply.	Each	110000
4141	Cost of Exit point directional sound speaker with voice and integral audio amplifier with selectable sound pulse patterns	Each	7600
4142	Cost of Digital voice command keypad	Each	37083
4143 (A)	Micro phone assembly	Each	24576
4143 (B)	Cost of 25 mm Outer dia PVC/ABS Pipe with end caps including making air sampling opening of appropriate dia on appropriate interval.	meter	78
4144	Cost of Fire Survival armoured cable,Fibre glass tape wrapped, ZH inner, outer sheath 2x1.5 Sq. mm cable.	meter	196
4145	Cost of Fire alarm armoured FRLS/HFFR cable 2 x 1.5 sqmm	meter	58

	Speaker cable of following pairs ,cores and size of PVC insulated FRLS/HFFRcopper conductor cable with LHC in the existing surface / recessed steel conduit / PVC casing and capping including connections etc.		
4146	Cost of single pair 2-core 48/0.2 mm, speaker cable	meter	25
4147	Cost of 25 mm dia MS flexible pipe with PVC coating along with all ancillaries & accessories like coupler etc.	meter	25
4148	Two pair 0.5 Sq. mm copper twisted cable for talk back system.	meter	8
4149	Addressable beam detector	Each	55100
4150	Cost of two pair 2-core 48/0.2 mm for PA System.(wastage 5%)	meter	49
4151	Cost of three pair 2-core 48/0.2 mm Speaker Cable for PA System.(wastage 5%)	meter	74
4152	Cost of four pair 2-core 48/0.2 mm Speaker Cable for PA System. (wastage 5%)	meter	99
WET RISER & SPRINKLER SYSTEM			
4201	Fire Hydrant single outlet gunmetal NB inlet brass spindle 63mm female instantaneous/outlet with PVC blank cap and GI chain as per IS:5290 (ISI Marked) Single head gunmetal hydrant valve	Each	4685
4202	Short stainless steel branch pipe 63mm male instantaneous inlet male threaded outlet fitted with 15mm bore nozzle as per IS:903 (ISI Marked) Single Head Stainless Steel Hydrant Valve	Each	3547
4203	Short gunmetal pipe 63mm male instantaneous inlet male threaded outlet fitted with 15mm bore nozzle as per IS:903 (ISI Marked) Gunmetal short branch pipe	Each	1758
4204	Short stainless steel branch pipe	Each	1189
4205	FBC CI Body having two nos. gunmetal 63mm male instantaneous inlet coupling conforming to IS:903 fitted with non return valve flange outlet of 100mm dia complete with 25mm size gunmetal drain valve, two nos. rubber blank caps and chain Two way gunmetal FBC connection	Each	4002

4206	Three way gunmetal FBC connection	Each	6050
4207	Four way gunmetal FBC connection	Each	9462
	Sythetic fibre circular woven jackated rubber lined fire hose delivery having 38Kg. Per Cm. Sq. burst pressure, 28Kg. Per Cm. proof pressure, 14Kg. Per Cm. working pressure as per IS:636 (ISI Marked) Type A in 15 Meters length, 63mm dia duly copper via bounded two coupling pair of male female parts any duly intantanous type, 63mm size as per IS:903 (ISI Marked)		
4208	RRL Hose with male female gunmetal coupling	Each	3861
4209	RRL Hose with male female stainless steel coupling	Each	3292
	19mm swinging hose reel complete reinforced braided thermoplastic red/black hose		
4210	Swinging type hose reel complete with 19mm x 30 Mtrs. Thermolplastic hoseType-II	Each	2043
4211	Swinging type hose reel complete with 19mm x 40 Mtrs. Thermolplastic hose Type-II	Each	3232
4212	Hose reel drum	Each	2350
4213	25mm dia gun metal globe valve & nozzle.	Each	691
	Sprinkler operating at 68 ⁰ centigrade standard response, standard coverage 15mm BPST, K56 UL listed and FM approved		
4214	68 ⁰ centigrade Pendant Sprinkler	Each	168
4215	68 ⁰ centigrade Upright Sprinkler	Each	168
4216	68 ⁰ centigrade Sidewall Sprinkler	Each	240
4217	68 ⁰ centigrade Concealed Sprinkler	Each	857
	Sprinkler flexible drops stainless steel complete with 15 NPT on reducer thread with maximum working pressure of 175 PSI test pressure of 875 PSI (Burst) with branch line (Inlet) 25mm NPT male thread to sprinkler head (Outlet) 15mm NPT female thread 1 No. reducer + 1 No. Nipple + 2 Side Brackets + 1 Center Bracket + 625 mm stockbar		
4218	700mm flexible drop	Each	795
4219	1000mm flexible drop	Each	945
4220	1200mm flexible drop	Each	1015

4221	1500mm flexible drop	Each	1138
4222	Water flow switch for 100mm dia pipe	Each	4945
4223	Water flow switch for 150mm dia pipe	Each	6083
	hydraulic alarm valve with water motor gong bell and constant pressure trim assembly complete as required		
4224	150mm dia hydraulic alarm valve with water motor gong	Each	34933
4225	100mm dia hydraulic alarm valve with water motor gong	Each	32517
4226	80mm dia hydraulic alarm valve with water motor gong	Each	30777
	Deluge valve with wet pilot basic trim assembly with test and alarm, drip and drain valve with water motor gong		
4227	150mm dia deluge valve	Each	84888
4228	100mm dia deluge valve	Each	68238
4229	80mm dia deluge valve	Each	60732
4230	50mm dia deluge valve	Each	60280
	Water curtain nozzle, 15mm dia BPST K-Factor as per design consideration		
4231	Water curtain nozzle, nickel chrome plated brass	Each	313
4232	Electric release trim assembly for deluge valve with two way solenoid valve, weather proof operation on 24 volts DC	Each	5382
4233	Adjustable rosette plate for 15mm dia (Powder Coated)	Each	80
	BUTTERFLY VALVE PN 1.6 rating		
4234	40mm dia butterfly valve	Each	1948
4235	50mm dia butterfly valve	Each	1948
4236	65mm dia butterfly valve	Each	2150
4237	80mm dia butterfly valve	Each	2382
4238	100mm dia butterfly valve	Each	3305
4239	125mm dia butterfly valve	Each	4175
4240	150mm dia butterfly valve	Each	4803
4241	200mm dia butterfly valve	Each	9188

4242	250mm dia butterfly valve	Each	12912
4243	300mm dia butterfly valve	Each	16105
	MS pipe as per IS/1239 ISI MARKED C Class (heavy)		
4244	25mm dia MS Pipe	m	118
4245	32mm dia MS Pipe	m	155
4246	40mm dia MS Pipe	m	180
4247	50mm dia MS Pipe	m	265
4248	65mm dia MS Pipe	m	350
4249	80mm dia MS Pipe	m	424
4250	100mm dia MS Pipe	m	634
4251	125mm dia MS Pipe	m	819
4252	150mm dia MS Pipe	m	938
4253	200mm dia MS Pipe (6.3 mm thick)	m	1426
4254	250mm dia MS Pipe (6.3 mm thick)	m	1798
4255	300mm dia MS Pipe (7.1 mm thick)	m	2106
	GI pipe as per IS/1239 ISI MARKED C Class (heavy)		
4256	25mm dia G.I. Pipe	m	171
4257	32mm dia G.I. Pipe	m	215
4258	40mm dia G.I. Pipe	m	250
4259	50mm dia G.I. Pipe	m	310
4260	65mm dia G.I. Pipe	m	395
4261	80mm dia G.I. Pipe	m	505
4262	100mm dia G.I. Pipe	m	714
4263	150mm dia G.I. Pipe	m	1066

NON RETURN VALVE cast Iron double flange as per IS/5312 PN 1.6 rating complete with Bronze body & wedge Stainless Steel spindle, Gland packing, Rubber gasket

4264	40mm dia Non Return Valve	Each	3311
4265	50mm dia Non Return Valve	Each	3311
4266	65mm dia Non Return Valve	Each	3785
4267	80mm dia Non Return Valve	Each	4366
4268	100mm dia Non Return Valve	Each	6607
4269	125mm dia Non Return Valve	Each	10290
4270	150mm dia Non Return Valve	Each	11718
4271	200mm dia Non Return Valve	Each	20013
4272	250mm dia Non Return Valve	Each	31946
4273	300mm dia Non Return Valve	Each	43392

NON RISING cast Iron double flange SLUICE Valve as per ISI/14846 PN 1.6 rating complete with Bronze body & wedge Stainless Steel spindle, Gland packing, Rubber gasket

4274	40mm DIA Sluice Valve	Each	3540
4275	50mm DIA Sluice Valve	Each	4532
4276	65mm DIA Sluice Valve	Each	5875
4277	80mm DIA Sluice Valve	Each	6663
4278	100mm DIA Sluice Valve	Each	9324
4279	125mm DIA Sluice Valve	Each	12511
4280	150mm DIA Sluice Valve	Each	15461
4281	200mm DIA Sluice Valve	Each	27639
4282	250mm DIA Sluice Valve	Each	32913
4283	300mm DIA Sluice Valve	Each	47409

Cast iron double flange strainer with stainless steel mesh, PN 1.6 rating complete with accessories as required.

4284	80mm dia CIDF SS mesh Strainer	Each	2531
4285	100mm dia CIDF SS mesh Strainer	Each	3636
4286	150mm dia CIDF SS mesh Strainer	Each	6791
4287	200mm dia CIDF SS mesh Strainer	Each	14690
4288	250mm dia CIDF SS mesh Strainer	Each	18451
4289	300mm dia CIDF SS mesh Strainer	Each	25367
4290	25mm Air release valve as per IS:14845 ISI Marked	Each	1350
	electric driven main fire pump with motor, base plate coupling, coupling guard, foundation bolts and anti vibration pads		
4291	2850 lpm at 88 m Head	Each	331000
4292	2850 lpm at 70 m Head	Each	295000
4293	2850 lpm at 56 m Head	Each	294000
4294	2280 lpm at 88 m Head	Each	294000
4295	2280 lpm at 70 m Head	Each	268000
4296	2280 lpm at 56 m Head	Each	240000
4297	1620 lpm at 88 m Head	Each	268000
4298	1620 lpm at 70 m Head	Each	240000
4299	1620 lpm at 56 m Head	Each	204000
	diesel engine driven main fire pump, tank, exhaust, foundation bolts, anti vibration pads etc.		
4300	2850 lpm at 88 m Head	Each	505000
4301	2850 lpm at 70 m Head	Each	453700
4302	2850 lpm at 56 m Head	Each	451000
4303	2280 lpm at 88 m Head	Each	468000
4304	2280 lpm at 70 m Head	Each	450000
4305	2280 lpm at 56 m Head	Each	448000
4306	1620 lpm at 88 m Head	Each	463100
4307	1620 lpm at 70 m Head	Each	435000
4308	1620 lpm at 56 m Head	Each	375000
	electric driven pressurisation pump (Jockey pump)		

4309	180 lpm at 88 m Head	Each	90000
4310	180 lpm at 70 m Head	Each	72000
4311	180 lpm at 56 m Head	Each	56000
4312	300 lpm at 88 m Head	Each	120000
4313	300 lpm at 70 m Head	Each	90200
4314	300 lpm at 56 m Head	Each	81000
	electric driven pressurisation pump (Terrace pump)		
4315	900 lpm at 35 m Head	Each	70909
4316	450 lpm at 35 m Head	Each	59091
4317	Air Vessel 250 mm	Each	12646
4318	Pressure switch	Each	986
4319	Orifice plate made of 6 mm. thick stainless steel with orifice of required size in between flange & landing valve of external and internal hydrant to reduce pressure to working pressure of 3.5 kg/cm ²	Each	802
4320	Cost of Cement concrete foundation with 35x35x5 mm angle frame on edges duly plastered Cum with inertia blocks.	Cum	4500
4321	Control Panel having Incomer of 250A, 50kA 4 Pole MCCB, Ics=100% Icu rating, Ammeter, Voltmeter, phase indicating lamps, ON, OFF, trip indicating lamps, Copper Bus Bar 300A. OUTGOING for main fire pump (125A, 50 KA TPN MCCB with Starter etc. Jockey Pump (63 A, 50kA TPN MCCB, Ics=100% Icu, with suitable HP fully automatic Star/Delta starter). DIESEL ENGINE CONTROL etc. as per details in items. i/c System Controller to control operation of all pumps, relays, timers Sensors etc. as detail in item.	Each	175000

4322	Control Panel having Incomer of 400A, 50kA 4 Pole MCCB, lcs=100% lcu rating, Ammeter, Voltmeter, phase indicating lamps, ON, OFF, trip indicating lamps, Copper Bus Bar 500A. OUTGOING for main fire pump (200A, 50 KA TPN MCCB with Starter etc. Jockey Pump (100 A, 50kA TPN MCCB, lcs=100% lcu, with suitable HP fully automatic Star/Delta starter). DIESEL ENGINE CONTROL etc. as per details in items. Designing, Supply, Installation, Testing and commissioning of system controller to control operation of main electric fire pump, diesel pump, Pressurization pump, Terrace pump in sequence as per specification consisting of relays, timers. Sensors, annunciation window for fault indication, complete as per specifications.	Each	205000
4323	Control Panel having Incomer of 630A, 50kA 4 Pole MCCB, lcs=100% lcu rating, Ammeter, Voltmeter, phase indicating lamps, ON, OFF, trip indicating lamps, Copper Bus Bar 800A. OUTGOING for main fire pump (250A, 50 KA TPN MCCB with Starter etc. Jockey Pump (125 A, 50kA TPN MCCB, lcs=100% lcu, with suitable HP fully automatic Star/Delta starter). DIESEL ENGINE CONTROL etc. as per details in items i/c System Controller to control operation of all pumps, relays, timers Sensors etc. as detail in item.	Each	230000
4324	Control Panel having Incomer of 400A, 50kA 4 Pole MCCB, lcs=100% lcu rating, Ammeter, Voltmeter, phase indicating lamps, ON, OFF, trip indicating lamps, Copper Bus Bar 500A. OUTGOING 2 nos. for main fire pump (250A, 50 KA TPN MCCB with Starter etc. and 2 nos. for Jockey Pump (125 A, 50kA TPN MCCB, lcs=100% lcu, with suitable HP fully automatic Star/Delta starter). DIESEL ENGINE CONTROL etc. as per details in items i/c System Controller to control operation of all pumps, relays, timers Sensors etc. as detail in item.	Each	265000
4325	Control Panel having Incomer of 630A, 50kA 4 Pole MCCB, lcs=100% lcu rating, Ammeter, Voltmeter, phase indicating lamps, ON, OFF, trip indicating lamps, Copper Bus Bar 800A. OUTGOING 2 nos. for main fire pump (200A, 50 KA TPN MCCB with Starter etc. and 2 nos. for Jockey Pump (100 A, 50kA TPN MCCB, lcs=100% lcu, with suitable HP fully automatic Star/Delta starter). DIESEL ENGINE CONTROL etc. as per details in items i/c System Controller to control operation of all pumps, relays, timers Sensors etc. as detail in item.	Each	320000

4326	Control Panel having Incomer of 800A, 50kA 4 Pole MCCB, Ics=100% Icu rating, Ammeter, Voltmeter, phase indicating lamps, ON, OFF, trip indicating lamps, Copper Bus Bar 1000A. OUTGOING 2 nos. for main fire pump (250A, 50 KA TPN MCCB with Starter etc. and 2 nos. for Jockey Pump (125 A, 50kA TPN MCCB, Ics=100% Icu, with suitable HP fully automatic Star/Delta starter). DIESEL ENGINE CONTROL etc. as per details in items i/c System Controller to control operation of all pumps, relays, timers Sensors etc. as detail in item.	Each	410000
4327	Basic cost of 4 mm thick Anti Corrosive Pipe Protection Bituminous Tape,	sq m	210
4328	Basic cost of anticorrosive Black paint	Litre	110
4329	Cost of 4 Nos. Nuts bolts and washers	Set	50
4330	Cost of 1 No. 3 mm thick rubber gasket	Each	70
4331	Cost of 1 No. MS flange as per IS drilling 16 mm thick LS	Set	200
4332	Cost of Welding rod	Job	40
4333	Cost of 40 mm dia Flange	Each	110
4334	Cost of 8 Nos. Nut bolt fastner and packing gaskets	Each	90
4335	Cost of Welding rod LS	Each	60
4336	Cost of 50 mm dia butterfly valve	Each	110
4337	Cost of 50mm dia Flange	Each	110
4338	Cost of 65 mm dia Flange	Each	110
4339	Cost of 65 mm dia butterfly valve	Each	110
4340	Cost of 80 mm dia Flange	Each	220
4341	Cost of 8 Nos. Nut bolt fastner and packing gaskets	Each	180
4342	Cost of Welding rod LS	Each	90
4343	Cost of 100 mm dia Flange	Each	220
4344	Cost of 150 mm dia Flange	Each	220
4345	Cost of 200 mm dia Flange	Each	340
4346	Cost of 8 Nos. Nut bolt fastner and packing gaskets	Each	300

4347	Cost of Welding rod LS	Each	120
4348	Cost of 250 mm dia Flange	Each	340
4349	Cost of 300 mm dia Flange	Each	340
4350	Cost of packing gaskets	LS	50
4351	Cost of 125 mm dia Flange	Each	220
4352	Cost of 80 mm dia Flange	Each	110
4353	Cost of 100 mm dia Flange	Each	110
4354	Cost of 150 mm dia Flange	Each	110
4355	Cost of 25mm dia socket, union, nipples (Lumpsum)	Each	210
4356	Cost of 25mm dia gunmetal gate valve	Each	770
4357	Cost of 250mm dia air vessel wit air release valve and 25mm dia gunmetal wheel valve for drainage	Each	1400
4358	Cost of 15mm dia MS Socket & teflon tape (Lumpsum)	Each	45
4359	Cot of teflon tape	Each	20
4360	25mm dia MS Socket	Each	45
4361	Cost of MS Socket 15mm dia	Each	35
4362	Cost of water flow switch	Each	6500
4363	Cost of Packing gaskets	LS	50
4364	Cost of 4 Nos. Nut Bolt fastner & packing gaskets	LS	90
4365	Cost of 12 Nos. Nut bolt fastner and packing gaskets	Each	450

LED LIGHTS

LED Down lighter (SMD Type) (System lumen efficacy $\geq 105 < 120$ lm/Watt) of following body materials.

Powder coated die cast /Extruded aluminium Body including trim with Aluminium Reflector

4401	5 - 7 watt	Each	204
4402	8 - 10 watt	Each	263
4403	12 -15 watt	Each	329

4404	18 watt	Each	478
4405	22 watt	Each	494
4406	30 watt	Each	853

LED Down lighter (SMD Type) (System lumen efficacy ≥ 120 <135 lm/Watt) of following body materials.

Powder coated die cast /Extruded aluminium Body including trim with Aluminium Reflector

4407	5 - 7 watt	Each	204
4408	8 - 10 watt	Each	263
4409	12 -15 watt	Each	329
4410	18 watt	Each	494
4411	22 watt	Each	512
4412	30 watt	Each	925

LED Down lighter (SMD Type) (System lumen efficacy >135 lm/Watt) of following body materials.

Powder coated die cast /Extruded aluminium Body including trim with Aluminium Reflector

4413	5 - 7 watt	Each	223
4414	8 - 10 watt	Each	287
4415	12 -15 watt	Each	359
4416	18 watt	Each	538
4417	22 watt	Each	556
4418	30 watt	Each	953

LED Down lighter (COB Type) (System lumen efficacy ≥ 105 < 120 lm/Watt) of following body materials.

Powder coated die cast /Extruded aluminium Body including trim with Aluminium Reflector

4419	5 - 7 watt	Each	410
4420	8 - 10 watt	Each	447
4421	12 -15 watt	Each	566
4422	18 watt	Each	666
4423	22 watt	Each	774
4424	30 watt	Each	1008

LED Down lighter (COB Type) (System lumen efficacy ≥ 120 lm/Watt) of following body materials.

Powder coated die cast /Extruded aluminium Body including trim with Aluminium Reflector

4425	5 - 7 watt	Each	482
4426	8 - 10 watt	Each	495
4427	12 -15 watt	Each	670
4428	18 watt	Each	875
4429	22 watt	Each	901
4430	30 watt	Each	1295

LED Panel light 2x2 ft. (System lumen efficacy $\geq 105 < 120$ lm/Watt) of following body materials.

Powder coated die cast /Extruded aluminium Body (Thickness > 1.20 mm)

4431	15 watt	Each	903
4432	18 watt	Each	1301
4433	22 watt	Each	1525
4434	36 watt	Each	1625
4435	40 watt	Each	1750
4436	45 watt	Each	2100

CRCA Sheet Body (Thickness > 0.50 mm)

4437	15 watt	Each	816
4438	18 watt	Each	1200
4439	22 watt	Each	1220
4440	36 watt	Each	1300
4441	40 watt	Each	1400
4442	45 watt	Each	1638

Panel Down light 2x2 ft.

LED Panel light 2x2 ft. (System lumen efficacy $\geq 120 < 135$ lm/Watt) of following body materials.

Powder coated die cast /Extruded aluminium Body (Thickness > 1.20 mm)

4443	15 watt	Each	993
4444	18 watt	Each	1431

4445	22 watt	Each	1760
4446	36 watt	Each	1953
4447	40 watt	Each	2475
4448	45 watt	Each	2640
	CRCA Sheet Body (Thickness > 0.50 mm)		
4449	15 watt	Each	932
4450	18 watt	Each	1200
4451	22 watt	Each	1280
4452	36 watt	Each	1420
4453	40 watt	Each	1638
4454	45 watt	Each	1679
	LED Panel light 2x2 ft., (System lumen efficacy >135 lm/Watt) of following body materials.		
	Powder coated die cast /Extruded aluminium Body (Thickness > 1.20 mm)		
4455	15 watt	Each	1083
4456	18 watt	Each	1561
4457	22 watt	Each	2400
4458	36 watt	Each	2700
4459	40 watt	Each	2880
4460	45 watt	Each	3300
	CRCA Sheet Body (Thickness > 0.50 mm)		
4461	15 watt	Each	1049
4462	18 watt	Each	1800
4463	22 watt	Each	1831
4464	36 watt	Each	1869
4465	40 watt	Each	1884
4466	45 watt	Each	1925
	LED Batten light		
	LED Batten light (System lumen efficacy ≥105 <120 lm/Watt) of following body materials.		
	Powder coated die cast /Extruded aluminium Body (Thickness > 1.20 mm)		
4467	18- 22 Watt	Each	305

4468	24 -26watt	Each	313
4469	36 watt	Each	322
4470	40 watt	Each	334
	CRCA Sheet Body (Thickness > 0.50 mm)		
4471	18- 22 Watt	Each	267
4472	24 -26watt	Each	275
4473	36 watt	Each	282
4474	40 watt	Each	292
	LED Batten light (System lumen efficacy $\geq 120 < 135$ lm/Watt) of following body materials.		
	Powder coated die cast /Extruded aluminium Body (Thickness > 1.20 mm)		
4475	18- 22 Watt	Each	343
4476	24 -26watt	Each	351
4477	36 watt	Each	362
4478	40 watt	Each	376
	CRCA Sheet Body (Thickness > 0.50 mm)		
4479	18- 22 Watt	Each	305
4480	24 -26watt	Each	313
4481	36 watt	Each	322
4482	40 watt	Each	334
	LED Batten light (System lumen efficacy >135 lm/Watt) of following body materials.		
	Powder coated die cast /Extruded aluminium Body (Thickness > 1.20 mm)		
4483	18- 22 Watt	Each	381
4484	24 -26watt	Each	389
4485	36 watt	Each	403
4486	40 watt	Each	417
	CRCA Sheet Body (Thickness > 0.50 mm)		
4487	18- 22 Watt	Each	343
4488	24 -26watt	Each	351
4489	36 watt	Each	362
4490	40 watt	Each	376

LED Street light fixture, powder coated pressure die cast aluminium body (System lumen efficacy $\geq 105 < 120$ lm/Watt) of following body materials.

4491	10 watt	Each	375
4492	14 watt	Each	475
4493	15 watt	Each	485
4494	18 watt	Each	516
4495	20 watt	Each	523
4496	24 watt	Each	525
4497	25 watt	Each	536
4498	30 watt	Each	550
4499	36 watt	Each	700
4500	40 watt	Each	750
4501	45 watt	Each	800
4502	50 watt	Each	1100
4503	72 watt	Each	1200
4504	90 watt	Each	1375
4505	100 watt	Each	1650
4506	120 watt	Each	1700
4507	150 watt	Each	2250
4508	180 watt	Each	3100
4509	200 watt	Each	3200

LED Street light fixture, powder coated pressure die cast aluminium body (System lumen efficacy $\geq 120 < 135$ lm/Watt) of following body materials.

4510	10 watt	Each	413
4511	14 watt	Each	523
4512	15 watt	Each	543
4513	18 watt	Each	660
4514	20 watt	Each	750
4515	24 watt	Each	880
4516	25 watt	Each	915
4517	30 watt	Each	935
4518	36 watt	Each	990

4519	40 watt	Each	1000
4520	45 watt	Each	1073
4521	50 watt	Each	1210
4522	72 watt	Each	1350
4523	90 watt	Each	1500
4524	100 watt	Each	1750
4525	120 watt	Each	1950
4526	150 watt	Each	2300
4527	180 watt	Each	3300
4528	200 watt	Each	3450

LED Street light fixture, powder coated pressure die cast aluminium body (System lumen efficacy >135 lm/Watt) of following body materials.

4529	10 watt	Each	474
4530	14 watt	Each	575
4531	15 watt	Each	598
4532	18 watt	Each	675
4533	20 watt	Each	700
4534	24 watt	Each	925
4535	25 watt	Each	962
4536	30 watt	Each	1025
4537	36 watt	Each	1075
4538	40 watt	Each	1175
4539	45 watt	Each	1233
4540	50 watt	Each	1392
4541	72 watt	Each	1675
4542	90 watt	Each	1850
4543	100 watt	Each	2250
4544	120 watt	Each	2750
4545	150 watt	Each	3490
4546	180 watt	Each	3750
4547	200 watt	Each	4250

LED Flood Light, powder coated pressure die cast aluminium (System lumen efficacy 105 <120 lm/Watt) of following body materials.

4548	50 watt	Each	850
4549	70 watt	Each	1340
4550	100 watt	Each	1450
4551	150 watt	Each	2100
4552	200 watt	Each	3100
4553	250 watt	Each	3750

LED Flood Light, powder coated pressure die cast aluminium (System lumen efficacy ≥ 120 and <135 lm/Watt) of following body materials.

4554	50 watt	Each	1265
4555	70 watt	Each	1474
4556	100 watt	Each	1595
4557	150 watt	Each	2310
4558	200 watt	Each	3200
4559	250 watt	Each	4125

LED Flood Light, powder coated pressure die cast aluminium (System lumen efficacy >135 lm/Watt) of following body materials.

4560	50 watt	Each	1322
4561	70 watt	Each	1541
4562	100 watt	Each	1668
4563	150 watt	Each	2415
4564	200 watt	Each	3789
4565	250 watt	Each	4313

LED Smart Street light fixture, powder coated pressure die cast aluminium (System. System lumen efficacy ≥ 105 and <120 lm/Watt) as details in items.

4566	45 watt	Each	3483
4567	50 watt	Each	3642
4568	72 watt	Each	4021

4569	90 watt	Each	4211
4570	100 watt	Each	5033
4571	120 watt	Each	5286
4572	150 watt	Each	6665
4573	180 watt	Each	6791
4574	200 watt	Each	8006
LED Smart Street light fixture, powder coated pressure die cast aluminium (System. System lumen efficacy ≥ 120 and < 135 lm/Watt) as details in items.			
4575	45 watt	Each	3832
4576	50 watt	Each	4006
4577	72 watt	Each	4423
4578	90 watt	Each	4632
4579	100 watt	Each	5536
4580	120 watt	Each	5815
4581	150 watt	Each	7331
4582	180 watt	Each	7470
4583	200 watt	Each	8806
LED Smart Street light fixture, powder coated pressure die cast aluminium (System. System lumen efficacy > 135 lm/Watt) as details in items.			
4584	45 watt	Each	4006
4585	50 watt	Each	4188
4586	72 watt	Each	4624
4587	90 watt	Each	4842
4588	100 watt	Each	5788
4589	120 watt	Each	6079
4590	150 watt	Each	7665
4591	180 watt	Each	7810
4592	200 watt	Each	9207
Solar outdoor light			

4593	Integrated type solar PV lighting system on the existing pole structure, comprising of 20 watt, 6V Mono Passivated Emitter and Rear Contact as detail in item.	Each	13250
4594	Supplying of the integrated type solar PV lighting system on the existing pole structure, comprising of 30 watt, 6V Mono Passivated Emitter and Rear Contact (PERC) as details in items.	Each	16500
4595	Supplying of the integrated type solar PV lighting system on the existing pole structure, comprising of 35 watt, 6V Mono Passivated Emitter and Rear Contact (PERC) as details in items.	Each	21000

BLDC FAN

	ceiling fan with Brush Less Direct Current (BLDC) Motor, etc. complete as req		
4596	900mm, BEE 5 star rating, service value ≥ 5.1 CM/Min/Watt, air delivery 130 CM/Min (Minimum)	Each	1785
4597	1050mm, BEE 5 star rating, service value ≥ 5.1 CM/Min/Watt, air delivery 150 CM/Min (Minimum)	Each	1806
4598	1200mm, BEE 5 star rating, service value ≥ 6.0 CM/Min/Watt, air delivery 210 CM/Min (Minimum)	Each	1890
4599	1400mm, BEE 5 star rating, service value ≥ 6.0 CM/Min/Watt, air delivery 245 CM/Min (Minimum)	Each	1913
	ceiling fan with Brush Less Direct Current (BLDC) Motor, etc. complete as req		
4600	900mm, service value ≥ 5.1 CM/Min/Watt, air delivery 130 CM/Min (Minimum)	Each	1658
4601	1050mm, service value ≥ 5.1 CM/Min/Watt, air delivery 150 CM/Min (Minimum)	Each	1677
4602	1200mm, service value ≥ 6.0 CM/Min/Watt, air delivery 210 CM/Min (Minimum)	Each	1755
4603	1400mm, service value ≥ 6.0 CM/Min/Watt, air delivery 245 CM/Min (Minimum)	Each	1785

TRANSFORMER

	Oil Type		
	33/0.433 KV, 3 Phase, 50 Hz outdoor/ indoor mounting(Mineral oil filled) ONAN copper wound transformer BEE 3 Star rated		
4604	500 KVA	Each	922793
4605	630KVA	Each	1162719
4606	1000KVA	Each	1395000
4607	1250 KVA	Each	1743750
4608	1600KVA	Each	2232000

4609	2000KVA	Each	2790000
4610	2500KVA	Each	3487500
	BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 33/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer.		
4611	500 KVA	Each	1107351
4612	630KVA	Each	1395262
4613	1000KVA	Each	1674000
4614	1250 KVA	Each	2092500
4615	1600KVA	Each	2678400
4616	2000KVA	Each	3348000
4617	2500KVA	Each	4185000
	BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 33/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer.		
4618	500KVA	Each	1291910
4619	630KVA	Each	1627806
4620	1000KVA	Each	1953000
4621	1250 KVA	Each	2441250
4622	1600KVA	Each	3124800
4623	2000KVA	Each	3906000
4624	2500 KVA	Each	4882500
	11/0.433 KV, 3 Phase, 50 Hz outdoor/ indoor mounting(Mineral oil filled) BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) outdoor/ indoor mounting(Mineral oil filled) copper wound transformer		
4625	1000KVA	Each	1215000
4626	1250 KVA	Each	1518750
4627	1600KVA	Each	1944000
4628	2000KVA	Each	2430000

	BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) outdoor/ indoor mounting(Mineral oil filled) copper wound transformer.		
4629	1000KVA	Each	1458000
4630	1250 KVA	Each	1822500
4631	1600KVA	Each	2332800
4632	2000KVA	Each	2916000
	BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) outdoor/ indoor mounting(Mineral oil filled) copper wound transformer.		
4633	1000KVA	Each	1701000
4634	1250 KVA	Each	2126250
4635	1600KVA	Each	2721600
4636	2000KVA	Each	3402000
	BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) outdoor/ indoor mounting(Mineral oil filled) copper wound transformer.		
4637	200KVA	Each	234000
4638	250KVA	Each	292500
4639	315KVA	Each	368550
4640	400KVA	Each	468000
4641	500KVA	Each	585000
4642	630KVA	Each	737100
	BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) outdoor/ indoor mounting(Mineral oil filled) copper wound transformer.		
4643	200KVA	Each	262080
4644	250KVA	Each	327600
4645	315KVA	Each	412776
4646	400KVA	Each	524160
4647	500KVA	Each	655200

4648	630KVA BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) outdoor/ indoor mounting(Mineral oil filled) copper wound transformer.	Each	825552
4649	200KVA	Each	345800
4650	250KVA	Each	432250
4651	315KVA	Each	544635
4652	400KVA	Each	691600
4653	500KVA	Each	864500
4654	630KVA BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer.	Each	1089270
4655	63KVA	Each	77805
4656	100KVA	Each	123500
4657	160KVA BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer.	Each	197600
4658	63KVA	Each	87142
4659	100KVA	Each	138320
4660	160KVA BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, ONAN (Oil Natural Air Natural) copper wound transformer.	Each	221312
4661	63KVA	Each	108927
4662	100KVA	Each	172900
4663	160KVA 33/0.433 KV, 3 Phase, 50 Hz outdoor/ indoor mounting (Synthetic Organic Ester Oil Filled) BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 33/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer.	Each	276640

4664	500 KVA	Each	1025325
4665	630KVA	Each	1291910
4666	1000KVA	Each	1550000
4667	1250 KVA	Each	1937500
4668	1600KVA	Each	2480000
4669	2000KVA	Each	3100000
4670	2500KVA	Each	3875000
	BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 33/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer.		
4671	500 KVA	Each	1230390
4672	630KVA	Each	1550291
4673	1000KVA	Each	1860000
4674	1250 KVA	Each	2325000
4675	1600KVA	Each	2976000
4676	2000KVA	Each	3720000
4677	2500KVA	Each	4650000
	BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 33/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer.		
4678	500KVA	Each	1435455
4679	630KVA	Each	1808673
4680	1000KVA	Each	2170000
4681	1250 KVA	Each	2712500
4682	1600KVA	Each	3472000
4683	2000KVA	Each	4340000
4684	2500 KVA	Each	5425000
	11/0.433 KV, 3 Phase, 50 Hz Outdoor/ Indoor mounting (Synthetic Organic Ester Oil Filled) BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer.		

4685	1000KVA	Each	1350000
4686	1250 KVA	Each	1687500
4687	1600KVA	Each	2160000
4688	2000KVA	Each	2700000
	BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer.		
4689	1000KVA	Each	1620000
4690	1250 KVA	Each	2025000
4691	1600KVA	Each	2592000
4692	2000KVA	Each	3240000
	BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer.		
4693	1000KVA	Each	1890000
4694	1250 KVA	Each	2362500
4695	1600KVA	Each	3024000
4696	2000KVA	Each	3780000
	BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer.		
4697	200KVA	Each	260000
4698	250KVA	Each	325000
4699	315KVA	Each	409500
4700	400KVA	Each	520000
4701	500KVA	Each	650000
4702	630KVA	Each	819000
	BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer.		
4703	200KVA	Each	291200

4704	250KVA	Each	364000
4705	315KVA	Each	458640
4706	400KVA	Each	582400
4707	500KVA	Each	728000
4708	630KVA	Each	917280
	BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer.		
4709	200KVA	Each	364000
4710	250KVA	Each	455000
4711	315KVA	Each	573300
4712	400KVA	Each	728000
4713	500KVA	Each	910000
4714	630KVA	Each	1146600
	BEE 3 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer.		
4715	63KVA	Each	81900
4716	100KVA	Each	130000
4717	160KVA	Each	208000
	BEE 4 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer.		
4718	63KVA	Each	91728
4719	100KVA	Each	145600
4720	160KVA	Each	232960
	BEE 5 Star rated (Corresponding Level as per BIS amended upto date of receipt of tender) , 11/0.433 KV step down, 3 Phase, 50 Hz, Dyn 11 vector group, KNAN [K(K-Class insulating liquid) Natural Air Natural] copper wound transformer.		
4721	63KVA	Each	114660
4722	100KVA	Each	182000

4723	160KVA DRY TYPE 33/0.433 KV, 3 Phase, 50 Hz Indoor mounting 33/0.433 KV Delta/Star,step down, 3 Phase, 50 Hz, Dyn 11 vector group, Cast Resin / VPI (vacuum pressure impregnated) Dry Type, copper wound transformer. Level 3	Each	291200
4724	1000 KVA (losses at 50% loading < 3000watt, losses at 100% loading < 9000watt)	Each	1757500
4725	1250 KVA (losses at 50% loading < 3600watt, losses at 100% loading < 10750watt)	Each	2196875
4726	1600 KVA (losses at 50% loading < 4500watt, losses at 100% loading < 13500watt)	Each	2812000
4727	2000 KVA (losses at 50% loading < 5400watt, losses at 100% loading < 17000watt)	Each	3515000
4728	2500 KVA (losses at 50% loading < 6500watt, losses at 100% loading < 20000watt) Level 4	Each	4393750
4729	1000 KVA (losses at 50% loading < 2790watt, losses at 100% loading < 7700watt)	Each	2021125
4730	1250 KVA (losses at 50% loading < 3300watt, losses at 100% loading < 9200watt)	Each	2526406
4731	1600 KVA (losses at 50% loading < 4200watt, losses at 100% loading < 11800watt)	Each	3233800
4732	2000 KVA (losses at 50% loading < 5050watt, losses at 100% loading < 15000watt)	Each	4042250
4733	2500 KVA (losses at 50% loading < 6150watt, losses at 100% loading < 18500watt) Level 5	Each	5052813
4734	1000 KVA (losses at 50% loading < 2620watt, losses at 100% loading < 7000watt)	Each	2284750
4735	1250 KVA (losses at 50% loading < 3220watt, losses at 100% loading < 8400watt)	Each	2855938
4736	1600 KVA (losses at 50% loading < 3970watt, losses at 100% loading < 11300watt)	Each	3655600
4737	2000 KVA (losses at 50% loading < 4790watt, losses at 100% loading < 14100watt)	Each	4569500
4738	2500 KVA (losses at 50% loading < 5900watt, losses at 100% loading < 17500watt) 33/0.433 KV, 3 Phase, 50 Hz Outdoor mounting 33/0.433 KV Delta/Star,step down, 3 Phase, 50 Hz, Dyn 11 vector group, Cast Resin / VPI (vacuum pressure impregnated) Dry Type, copper wound transformer.	Each	5711875

	Level 3			
4739	1000 KVA (losses at 50% loading < 3000watt, losses at 100% loading < 9000watt)	Each	1850000	
4740	1250 KVA (losses at 50% loading < 3600watt, losses at 100% loading < 10750watt)	Each	2312500	
4741	1600 KVA (losses at 50% loading < 4500watt, losses at 100% loading < 13500watt)	Each	2960000	
4742	2000 KVA (losses at 50% loading < 5400watt, losses at 100% loading < 17000watt)	Each	3700000	
4743	2500 KVA (losses at 50% loading < 6500watt, losses at 100% loading < 20000watt)	Each	4625000	
	Level 4			
4744	1000 KVA (losses at 50% loading < 2790watt, losses at 100% loading < 7700watt)	Each	2127500	
4745	1250 KVA (losses at 50% loading < 3300watt, losses at 100% loading < 9200watt)	Each	2659375	
4746	1600 KVA (losses at 50% loading < 4200watt, losses at 100% loading < 11800watt)	Each	3404000	
4747	2000 KVA (losses at 50% loading < 5050watt, losses at 100% loading < 15000watt)	Each	4255000	
4748	2500 KVA (losses at 50% loading < 6150watt, losses at 100% loading < 18500watt)	Each	5318750	
	Level 5			
4749	1000 KVA (losses at 50% loading < 2620watt, losses at 100% loading < 7000watt)	Each	2405000	
4750	1250 KVA (losses at 50% loading < 3220watt, losses at 100% loading < 8400watt)	Each	3006250	
4751	1600 KVA (losses at 50% loading < 3970watt, losses at 100% loading < 11300watt)	Each	3848000	
4752	2000 KVA (losses at 50% loading < 4790watt, losses at 100% loading < 14100watt)	Each	4810000	
4753	2500 KVA (losses at 50% loading < 5900watt, losses at 100% loading < 17500watt)	Each	6012500	
	11/0.433 KV, 3 Phase, 50 Hz Indoor mounting	Each		
	11/0.433 KV Delta/Star,step down, 3 Phase, 50 Hz, Dyn 11 vector group, Cast Resin / VPI (vacuum pressure impregnated) Dry Type, copper wound transformer.			
	Level 3			
4754	63 KVA (losses at 50% loading < 380watt, losses at 100% loading < 1250watt)	Each	101745	
4755	100 KVA (losses at 50% loading < 520watt, losses at 100% loading < 1800watt)	Each	161500	
4756	160 KVA (losses at 50% loading < 770watt, losses at 100% loading < 2200watt)	Each	258400	

4757	200 KVA (losses at 50% loading < 890watt, losses at 100% loading < 2700watt)	Each	323000
4758	250 KVA (losses at 50% loading < 1050watt, losses at 100% loading < 3150watt)	Each	403750
4759	315 KVA (losses at 50% loading < 1100watt, losses at 100% loading < 3275watt)	Each	508725
4760	400 KVA (losses at 50% loading < 1300watt, losses at 100% loading < 3875watt)	Each	646000
4761	500 KVA (losses at 50% loading < 1600watt, losses at 100% loading < 4750watt)	Each	807500
4762	630 KVA (losses at 50% loading < 2000watt, losses at 100% loading < 5855watt)	Each	1017450
4763	1000 KVA (losses at 50% loading < 3000watt, losses at 100% loading < 9000watt)	Each	1615000
4764	1250 KVA (losses at 50% loading < 3600watt, losses at 100% loading < 10750watt)	Each	2018750
4765	1600 KVA (losses at 50% loading < 4500watt, losses at 100% loading < 13500watt)	Each	2584000
4766	2000 KVA (losses at 50% loading < 5400watt, losses at 100% loading < 17000watt)	Each	3230000
4767	2500 KVA (losses at 50% loading < 6500watt, losses at 100% loading < 20000watt)	Each	4037500
Level 4			
4768	63 KVA (losses at 50% loading < 340watt, losses at 100% loading < 1140watt)	Each	117007
4769	100 KVA (losses at 50% loading < 475watt, losses at 100% loading < 1650watt)	Each	185725
4770	160 KVA (losses at 50% loading < 670watt, losses at 100% loading < 1950watt)	Each	297160
4771	200 KVA (losses at 50% loading < 780watt, losses at 100% loading < 2300watt)	Each	371450
4772	250 KVA (losses at 50% loading < 980watt, losses at 100% loading < 2930watt)	Each	464313
4773	315 KVA (losses at 50% loading < 1025watt, losses at 100% loading < 3100watt)	Each	585034
4774	400 KVA (losses at 50% loading < 1225watt, losses at 100% loading < 3450watt)	Each	742900
4775	500 KVA (losses at 50% loading < 1510watt, losses at 100% loading < 4300watt)	Each	928625
4776	630 KVA (losses at 50% loading < 1860watt, losses at 100% loading < 5300watt)	Each	1170068
4777	1000 KVA (losses at 50% loading < 2790watt, losses at 100% loading < 7700watt)	Each	1857250
4778	1250 KVA (losses at 50% loading < 3300watt, losses at 100% loading < 9200watt)	Each	2321563
4779	1600 KVA (losses at 50% loading < 4200watt, losses at	Each	2971600

	100% loading < 11800watt)		
4780	2000 KVA (losses at 50% loading < 5050watt, losses at 100% loading < 15000watt)	Each	3714500
4781	2500 KVA (losses at 50% loading < 6150watt, losses at 100% loading < 18500watt)	Each	4643125
	Level 5		
4782	63 KVA (losses at 50% loading < 300watt, losses at 100% loading < 1050watt)	Each	132269
4783	100 KVA (losses at 50% loading < 435watt, losses at 100% loading < 1500watt)	Each	209950
4784	160 KVA (losses at 50% loading < 570watt, losses at 100% loading < 1700watt)	Each	335920
4785	200 KVA (losses at 50% loading < 670watt, losses at 100% loading < 2100watt)	Each	419900
4786	250 KVA (losses at 50% loading < 920watt, losses at 100% loading < 2700watt)	Each	524875
4787	315 KVA (losses at 50% loading < 955watt, losses at 100% loading < 2750watt)	Each	661343
4788	400 KVA (losses at 50% loading < 1150watt, losses at 100% loading < 3330watt)	Each	839800
4789	500 KVA (losses at 50% loading < 1430watt, losses at 100% loading < 4100watt)	Each	1049750
4790	630 KVA (losses at 50% loading < 1745watt, losses at 100% loading < 4850watt)	Each	1322685
4791	1000 KVA (losses at 50% loading < 2620watt, losses at 100% loading < 7000watt)	Each	2099500
4792	1250 KVA (losses at 50% loading < 3220watt, losses at 100% loading < 8400watt)	Each	2624375
4793	1600 KVA (losses at 50% loading < 3970watt, losses at 100% loading < 11300watt)	Each	3359200
4794	2000 KVA (losses at 50% loading < 4790watt, losses at 100% loading < 14100watt)	Each	4199000
4795	2500 KVA (losses at 50% loading < 5900watt, losses at 100% loading < 17500watt)	Each	5248750
	11/0.433 KV, 3 Phase, 50 Hz Outdoor mounting	Each	
	11/0.433 KV Delta/Star, step down, 3 Phase, 50 Hz, Dyn 11 vector group, Cast Resin / VPI (vacuum pressure impregnated) Dry Type, copper wound transformer.		
	Level 3		
4796	63 KVA (losses at 50% loading < 380watt, losses at 100% loading < 1250watt)	Each	107100
4797	100 KVA (losses at 50% loading < 520watt, losses at 100% loading < 1800watt)	Each	170000
4798	160 KVA (losses at 50% loading < 770watt, losses at 100%	Each	272000

	loading < 2200watt)		
4799	200 KVA (losses at 50% loading < 890watt, losses at 100% loading < 2700watt)	Each	340000
4800	250 KVA (losses at 50% loading < 1050watt, losses at 100% loading < 3150watt)	Each	425000
4801	315 KVA (losses at 50% loading < 1100watt, losses at 100% loading < 3275watt)	Each	535500
4802	400 KVA (losses at 50% loading < 1300watt, losses at 100% loading < 3875watt)	Each	680000
4803	500 KVA (losses at 50% loading < 1600watt, losses at 100% loading < 4750watt)	Each	850000
4804	630 KVA (losses at 50% loading < 2000watt, losses at 100% loading < 5855watt)	Each	1071000
4805	1000 KVA (losses at 50% loading < 3000watt, losses at 100% loading < 9000watt)	Each	1700000
4806	1250 KVA (losses at 50% loading < 3600watt, losses at 100% loading < 10750watt)	Each	2125000
4807	1600 KVA (losses at 50% loading < 4500watt, losses at 100% loading < 13500watt)	Each	2720000
4808	2000 KVA (losses at 50% loading < 5400watt, losses at 100% loading < 17000watt)	Each	3400000
4809	2500 KVA (losses at 50% loading < 6500watt, losses at 100% loading < 20000watt)	Each	4250000
	Level 4		
4810	63 KVA (losses at 50% loading < 340watt, losses at 100% loading < 1140watt)	Each	123165
4811	100 KVA (losses at 50% loading < 475watt, losses at 100% loading < 1650watt)	Each	195500
4812	160 KVA (losses at 50% loading < 670watt, losses at 100% loading < 1950watt)	Each	312800
4813	200 KVA (losses at 50% loading < 780watt, losses at 100% loading < 2300watt)	Each	391000
4814	250 KVA (losses at 50% loading < 980watt, losses at 100% loading < 2930watt)	Each	488750
4815	315 KVA (losses at 50% loading < 1025watt, losses at 100% loading < 3100watt)	Each	615825
4816	400 KVA (losses at 50% loading < 1225watt, losses at 100% loading < 3450watt)	Each	782000
4817	500 KVA (losses at 50% loading < 1510watt, losses at 100% loading < 4300watt)	Each	977500
4818	630 KVA (losses at 50% loading < 1860watt, losses at 100% loading < 5300watt)	Each	1231650
4819	1000 KVA (losses at 50% loading < 2790watt, losses at 100% loading < 7700watt)	Each	1955000
4820	1250 KVA (losses at 50% loading < 3300watt, losses at 100% loading < 9200watt)	Each	2443750

4821	1600 KVA (losses at 50% loading < 4200watt, losses at 100% loading < 11800watt)	Each	3128000
4822	2000 KVA (losses at 50% loading < 5050watt, losses at 100% loading < 15000watt)	Each	3910000
4823	2500 KVA (losses at 50% loading < 6150watt, losses at 100% loading < 18500watt)	Each	4887500
	Level 5		
4824	63 KVA (losses at 50% loading < 300watt, losses at 100% loading <1050watt)	Each	139230
4825	100 KVA (losses at 50% loading < 435watt, losses at 100% loading < 1500watt)	Each	221000
4826	160 KVA (losses at 50% loading < 570watt, losses at 100% loading <1700watt)	Each	353600
4827	200 KVA(losses at 50% loading <670watt, losses at 100% loading < 2100watt)	Each	442000
4828	250 KVA(losses at 50% loading < 920watt, losses at 100% loading < 2700watt)	Each	552500
4829	315 KVA (losses at 50% loading < 955watt, losses at 100% loading < 2750watt)	Each	696150
4830	400 KVA (losses at 50% loading < 1150watt, losses at 100% loading < 3330watt)	Each	884000
4831	500 KVA (losses at 50% loading < 1430watt, losses at 100% loading < 4100watt)	Each	1105000
4832	630 KVA (losses at 50% loading < 1745watt, losses at 100% loading < 4850watt)	Each	1392300
4833	1000 KVA (losses at 50% loading < 2620watt, losses at 100% loading < 7000watt)	Each	2210000
4834	1250 KVA (losses at 50% loading < 3220watt, losses at 100% loading < 8400watt)	Each	2762500
4835	1600 KVA (losses at 50% loading < 3970watt, losses at 100% loading < 11300watt)	Each	3536000
4836	2000 KVA (losses at 50% loading < 4790watt, losses at 100% loading < 14100watt)	Each	4420000
4837	2500 KVA (losses at 50% loading < 5900watt, losses at 100% loading < 17500watt)	Each	5525000

APFC PANEL

Automatic Power Factor Correction (APFC) System

Automatic Power Factor Correction (APFC) panel, indoor type floor mounted free standing totally enclosed, extendable of following capacity.

4838	50 KVAR	Set	152750
4839	75 KVAR	Set	188500
4840	100 KVAR	Set	206700
4841	125 KVAR	Set	222300

4842	150 KVAR	Set	260325
4843	175 KVAR	Set	279500
4844	200 KVAR	Set	297700
HYBRID Power Factor Correction System			
HYBRID APFC Panel, 3 phase 4 wire, 415 V, 50 Hz AC System for Ambient temperature -5°C to +40°C of following capacity.			
4845	250 KVAR	Set	948350
4846	300 KVAR	Set	1128400
4847	350 KVAR	Set	1262950
4848	400 KVAR	Set	1516450
4849	450 KVAR	Set	1632800
4850	500 KVAR	Set	1849250
4851	550 KVAR	Set	1981850
4852	600 KVAR	Set	2175550
4853	650KVAR	Set	2302300
4854	700KVAR	Set	2477150
4855	750KVAR	Set	2692950
4856	800KVAR	Set	2825550
4857	850KVAR	Set	2927600
4858	900KVAR	Set	3113500
4859	950KVAR	Set	3293550
4860	1000KVAR	Set	3467750

UPS

ON LINE Uninterrupted Power Supply (UPS) system suitable for Single Phase input,Single Phase output AC Supply. with 30 minute backup.			
4861	2KVA	EACH	21604
4862	3KVA	EACH	23511
4863	6KVA	EACH	45439
4864	10 KVA	EACH	75947

Online UPS- Input supply: Three Phase, Output supply:
Three Phase

Online double conversion true sine wave Uninterrupted hot swappable (allow for the replacement or addition of battery modules without shutting down the entire system) modular Power Supply (UPS) system with N+1 modules (N denotes total number of modules required for rated capacity) with 30 minute backup.

4865	10KVA	EACH	104000
4866	20KVA	EACH	155200
4867	30KVA	EACH	228000
4868	40KVA	EACH	252000
4869	60KVA	EACH	388000
4870	80KVA	EACH	452000
4871	100KVA	EACH	476000
4872	120KVA	EACH	524000
4873	160KVA	EACH	744000
4874	200KVA	EACH	828000
4875	300KVA	EACH	1520000
4876	400KVA	EACH	1920000
4877	500KVA	EACH	2240000
Battery type AMF VRLA , ABS container, maintenance free,			
4878	Battery 12V/65 AH	EACH	4080
4879	Battery 12V/120 AH	EACH	7565
4880	Battery 12V/150AH	EACH	9775

GENERATOR CPCB IV +

Supply of 'Silent Type Diesel Generating set as per CPCB IV + or better norms along with having Prime Power Rating of KVA as below, 415 volts at 1500 RPM, 0.8 lagging power factor at 415 V suitable for 50 Hz, 3 phase system with AMF Panel of following capacity.

4881	25KVA	Set	280000
4882	35KVA	Set	310000
4883	40KVA	Set	370000
4884	50KVA	Set	392500

4885	62.5KVA as obsolete (58.5 KVA considered)	Set	415000
4886	82.5KVA	Set	515000
4887	100KVA	Set	600000
4888	125KVA	Set	615000
4889	160KVA	Set	868000
4890	200KVA	Set	1075000
4891	250KVA	Set	1300000
4892	320KVA	Set	1700000
4893	380KVA	Set	1975000
4894	400KVA	Set	2050000
4895	500KVA	Set	2400000
4896	625KVA	Set	3500000
4897	750KVA	Set	4450000

VRV/VRF

OUTDOOR UNIT

Supply of Modular type Variable Refrigerant Flow/Variable Refrigerant Volume air cooled Outdoor units suitable for cooling/heating (For capacity <40 kW_r ISEER 5.4, Capacity $\geq$ 40 and <70 ISEER 5.5, Capacity $\geq$ 70 ISEER 5.6 for ECSBC Building) of following capacity
For Cooling or Heating or both

4898	6 HP to 8 HP	Per HP	14550
4899	10 HP to 12 HP	Per HP	13950
4900	14 HP to 22 HP	Per HP	13275

Supply of Modular type Variable Refrigerant Flow/Variable Refrigerant Volume air cooled Outdoor units suitable for cooling/heating. (For capacity <40 kW_r ISEER 6.4, Capacity $\geq$ 40 and <70 ISEER 6.5, Capacity $\geq$ 70 ISEER 6.6 for ECSBC+ Building) of following capacity
For Cooling or Heating or both

4901	6 HP to 8 HP	Per HP	15520
4902	10 HP to 12 HP	Per HP	14880
4903	14 HP to 22 HP	Per HP	14160

Supply of Modular type Variable Refrigerant Flow/Variable Refrigerant Volume air cooled Outdoor units suitable for cooling/heating (For capacity <40 kW_r ISEER 7.4, Capacity $\geq$ 40 and <70 ISEER 7.5, Capacity $\geq$ 70 ISEER 7.6 for Super ECSBC Building) of following capacity.

	For Cooling or Heating or both		
4904	6 HP to 8 HP	Per HP	16490
4905	10 HP to 12 HP	Per HP	15810
4906	14 HP to 22 HP	Per HP	15045
	INDOOR UNIT		
	4 way Cassette Type Indoor ceiling mounted unit of following capacity.		
4907	0.8 TR	Each	15493
4908	1.0 TR	Each	19366
4909	1.2 TR	Each	26813
4910	1.6 TR	Each	27413
4911	2.0 TR	Each	27600
4912	2.4 TR	Each	28350
4913	2.6 TR	Each	28350
4914	3.6 TR	Each	30750
4915	4.1 TR	Each	31800
4916	4.6TR	Each	36675
	4-way compact VRV/VRF Cassette Type Indoor ceiling mounted unit equipped of following capacity		
4917	0.6 TR	Each	15705
4918	0.8 TR	Each	17450
4919	1.0 TR	Each	27375
4920	1.2 TR	Each	27750
4921	1.6 TR	Each	27750
	Single way wall/corner VRV/VRF Cassette Type Indoor ceiling mounted unit of following capacity		
4922	0.6 TR	Each	21000
4923	0.8 TR	Each	21450
4924	1.0 TR	Each	22500
4925	1.2 TR	Each	29400
4926	1.6 TR	Each	29400
4927	2.0 TR	Each	30300
	Double way VRV/VRF Cassette Type Indoor ceiling mounted unit of following capacity		
4928	0.6 TR	Each	30600

4929	1.0 TR	Each	32550
4930	2.0 TR	Each	39750
4931	3.3 TR	Each	42053
4932	4.2 TR	Each	44576
	High wall type Indoor unit of following capacity		
4933	0.6 TR	Each	11663
4934	0.8 TR	Each	12000
4935	1.0 TR	Each	12375
4936	1.2 TR	Each	15000
4937	1.6 TR	Each	15300
4938	2.0 TR	Each	15750
	External static pressure VRF/VRV ceiling mounted low static ductable type Indoor unit of following capacity		
	Low static ductable units (minimum 19 to 29 pascal external static pressure)		
4939	0.5 TR	Each	21000
4940	0.6 TR	Each	21450
4941	0.8 TR	Each	21675
4942	1.03 TR	Each	21975
4943	1.3 TR	Each	23925
4944	1.6 TR	Each	24300
4945	2.0 TR	Each	25050
	External static pressure VRF/VRV ceiling mounted mid static ductable type Indoor unit of following capacity		
	Mid static ductable units (minimum 30 to 48 pascal external static pressure)		
4946	1.2 TR	Each	23850
4947	1.6 TR	Each	24300
4948	2.0 TR	Each	25050
4949	2.4 TR	Each	26175
4950	3.2 TR	Each	27900
	external static pressure VRF/VRV ceiling mounted mid high static ductable type Indoor unit of following capacity		
	High Static Ductable units (minimum 49 to 77 pascal external static pressure)		
4951	0.8 TR	Each	21887

4952	1.03 TR	Each	22735
4953	1.2 TR	Each	23974
4954	1.6 TR	Each	24300
4955	2.0 TR	Each	25050
4956	2.4 TR	Each	26175
4957	3.2 TR	Each	27900
4958	4.0 TR	Each	31950
4959	4.6 TR	Each	44850
	External static pressure VRF/VRV ceiling mounted high ductable type Indoor unit of following capacity		
	High Static Ductable units (minimum 78 pascal external static pressure)		
4960	5.5 TR	Each	51675
4961	6.6 TR	Each	54225
4962	8.0 TR	Each	59175
	Supply of Y/T/Multi Joints. Joints shall be of same Original Equipment Manufacturer (OEM) make as of ODUs and IDUs		
4963	Indoor Units	Each	3150
4964	Outdoor Multi Joint	Each	5625
	COOPER REFRIGERANT PIPING		
	Soft/hard drawn copper refrigerant piping for VRV/VRF system, complete with fittings of following capacity.		
4965	6.4 mm dia (OD) (Soft drawn) with tube thickness 1.2 mm with 19 mm thick insulation	Meter	178
4966	9.5 mm dia (OD) (Soft drawn) with tube thickness 1.2 mm with 19 mm thick insulation	Meter	240
4967	12.7 mm dia (OD) (Soft drawn) with tube thickness 1.2 mm with 19 mm thick insulation	Meter	338
4968	15.86 mm dia (OD) (Soft drawn) with tube thickness 1.2 mm with 19 mm thick insulation	Meter	427
4969	19 mm dia (OD) (Hard drawn) with tube thickness 1.2 mm with 19 mm thick insulation	Meter	513
4970	22.2 mm dia (OD) (Hard drawn) with tube thickness 1.2 mm with 19 mm thick insulation	Meter	628
4971	25.4 mm dia (OD) (Hard drawn) with tube thickness 1.2 mm with 19 mm thick insulation	Meter	742
4972	28.58 mm dia (OD) (Hard drawn) with tube thickness 1.2 mm with 19 mm thick insulation	Meter	804

4973	31.8 mm dia (OD) (Hard drawn) with tube thickness 1.62 mm with 19 mm thick insulation	Meter	849
4974	34.9 mm dia (OD) (Hard drawn) with tube thickness 1.62 mm with 19 mm thick insulation	Meter	893
4975	38.1 mm dia (OD) (Hard drawn) with tube thickness 1.62 mm with 19 mm thick insulation	Meter	918
4976	41.27 mm dia (OD) (Hard drawn) with tube thickness 1.62 mm with 19 mm thick insulation	Meter	950

UNITARY SYSTEMS

WINDOW AC UNITS

Window type Air conditioners complete with copper power cable upto 3 Mtr, wireless Remote, suitable for working between 180- 260V of following capacity.

Non Inverter Type

4977	1.0 TR with fixed speed 5 Star BEE rating .	Each	22817
4978	1.5 TR with fixed speed 5 Star BEE rating .	Each	25376

Inverter Type

4979	1.0 TR with Inverter 5 Star BEE rating.	Each	23900
4980	1.5 TR with Inverter 5 Star BEE rating .	Each	27836

HI WALL SPLIT SYSTEMS

Air Cooled Hi Wall split type Air conditioners complete with Indoor unit(IDU), Out door unit (ODU) of following capacity.

Inverter Type - Cooling only

4981	0.75 TR with 5 Star BEE Rating .	Each	27156
4982	1.0 TR with 5 Star BEE Rating .	Each	28187
4983	1.5 TR with 5 Star BEE Rating .	Each	30243
4984	2.0 TR with 5 Star BEE Rating .	Each	39868

Inverter Type - Hot & Cold

4985	1.0 TR with 3 Star BEE Rating .	Each	26679
4986	1.5 TR with 3 Star BEE Rating.	Each	29013

DUCTABLE TYPE SPLIT UNITS

Air cooled ducted split type air conditioning machine of following capacity.

Inverter

4987	1.5 TR (BEE 4 Star Rated)	Each	39600
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4988	2.2 TR (BEE 4 Star Rated)	Each	44925
4989	3.0 TR (BEE 4 Star Rated)	Each	61125
4990	3.5 TR (3.2 EER)	Each	88238
4991	4.0 TR (3.2 EER)	Each	99900
	Non Inverter		
4992	1.0 TR (BEE 4 Star Rated)	Each	28950
4993	1.5 TR (BEE 4 Star Rated)	Each	30300
4994	2.0 TR (BEE 4 Star Rated)	Each	39480
4995	2.5 TR (BEE 4 Star Rated)	Each	51450
4996	3.0 TR (BEE 4 Star Rated)	Each	55650
4997	3.5 TR (3.2 EER)	Each	61275
4998	4.0 TR (3.2 EER)	Each	68025
4999	4.5 TR (3.2 EER)	Each	79115
5000	5.5 TR (3.2 EER)	Each	86175
5001	8.5 TR (3.2 EER)	Each	114225
5002	11.0 TR (3.2 EER)	Each	146925
5003	16.7 TR (3.2 EER)	Each	227250

CASSETTE TYPE SPLIT UNITS

Air Cooled Cassette type Air conditioners of following capacity.

Inverter Type- Cooling only

5004	1.5 TR with 5 Star BEE Rating	Each	53775
5005	2.0 TR with 5 Star BEE Rating	Each	59550
5006	2.5 TR with 5 Star BEE Rating	Each	82050
5007	3.0 TR with 5 Star BEE Rating	Each	85913
5008	3.5 TR with 5 Star BEE Rating	Each	96975
5009	4.0 TR with 5 Star BEE Rating	Each	98775

Heating & Cooling

5010	1.5 TR with 3 Star BEE Rating	Each	41775
5011	2.0 TR with 3 Star BEE Rating	Each	46575
5012	2.5 TR with 3 Star BEE Rating	Each	56100

TOWER TYPE SPLIT UNITS

Air Cooled Floor standing Tower type split Air conditioners of following capacity.

	Heat Pump (Heating & Cooling)		
5013	3.3 TR BEE 4 Star Rating	Each	91125
5014	3.8 TR BEE 4 Star Rating	Each	96375
5015	4.6 TR BEE 4 Star Rating	Each	101625
	Cooling Only		
5016	2.4 TR BEE 5 Star Rating	Each	65025
5017	3.3 TR BEE 5 Star Rating	Each	72300
5018	3.8 TR BEE 5 Star Rating	Each	76425
5019	4.6 TR BEE 5 Star Rating	Each	80775
	AIR COOLED PACKAGE UNITS		
	Air cooled ductable type Packaged air-conditioning units complete with Hermetically sealed Scroll compressors fitted inside the indoor unit of following capacity.		
	Inverter		
5020	5.0 TR (2.8 EER)	Each	91260
5021	8.0 TR (2.8 EER)	Each	123370
5022	11.0 TR (2.8 EER)	Each	149305
5023	16.5 TR (2.8 EER)	Each	209105
5024	22.0 TR (2.8 EER)	Each	273390
	CHILLERS		
	AIR COOLED CHILLERS		
	AIR COOLED SCREW CHILLERS COMPLETE WITH Variable Frequency Drive (VFD) of following capacity.		
5025	Upto 74 TR BEE 3 Star rated	Per TR	19500
5026	75 TR - 140 TR BEE 3 Star rated	Per TR	18850
5027	141 TR - 200 TR BEE 3 Star rated	Per TR	17875
	AHRI (Air-Conditioning, Heating, and Refrigeration Institute) Certified Air-Cooled Screw Chiller package complete with VFD (Variable Frequency Drive) of following capacity.		
5028	upto 200 TR - BEE 4 Star Rated	Per TR	20150
5029	201 TR to 250 TR BEE 4 Star Rated	Per TR	18525
5030	251 TR - 300 TR BEE 4 Star Rated	Per TR	17550
5031	301 TR - 350 TR BEE 4 Star Rated	Per TR	16900
5032	351 TR - 400 TR BEE 4 Star Rated	Per TR	16250

AIR COOLED SCROLL CHILLERS

AHRI (Air-Conditioning, Heating, and Refrigeration Institute) Certified Air-Cooled , Scroll Chiller package complete with VFD (Variable Frequency Drive) of following capacity.

5033	Upto 50 TR BEE 4 Star Rated	Per TR	15925
5034	51 TR - 70 TR BEE 4 Star Rated	Per TR	15600
5035	71 TR - 100 TR BEE 4 Star Rated	Per TR	15275

AHRI (Air-Conditioning, Heating, and Refrigeration Institute) Certified Air-Cooled Scroll Chiller package complete with VFD (Variable Frequency Drive) of following capacity.

5036	upto 50 TR BEE 3 Star Rated	Per TR	15275
5037	51 TR - 70 TR BEE 3 Star Rated	Per TR	14300
5038	71 TR - 100 TR BEE 3 Star Rated	Per TR	13975

WATER COOLED CHILLERS

WATER COOLED SCREW CHILLERS

Floor-mounted AHRI (Air-Conditioning, Heating, and Refrigeration Institute) Certified water cooled screw-type chiller machine complete with VFD (Variable Frequency Drive) of following capacity.

5039	Upto 70 TR BEE 3 Star Rated	Per TR	13325
5040	71 TR - 110 TR BEE 3 Star Rated	Per TR	11700
5041	111 TR - 150 TR BEE 3 Star Rated	Per TR	11700
5042	151 TR - 210 TR BEE 3 Star Rated	Per TR	11700
5043	211 TR - 260 TR BEE 3 Star Rated	Per TR	11375
5044	261 TR - 300 TR BEE 3 Star Rated	Per TR	11050
5045	301 TR - 450 TR BEE 3 Star Rated	Per TR	10400
5046	451 TR -600 TR BEE 3 Star Rated	Per TR	10075

Floor-mounted AHRI (Air-Conditioning, Heating, and Refrigeration Institute) Certified VFD (Variable Frequency Drive) Operated water cooled screw-type chiller machine of following capacity

5047	Upto 74 TR BEE 4 Star Rated	Per TR	14300
5048	75 TR - 150 TR BEE 4 Star Rated	Per TR	13650
5049	151TR - 300 TR BEE 4 Star Rated	Per TR	12350
5050	301TR - 450 TR BEE 4 Star Rated	Per TR	11700
5051	451 TR - 525 TR BEE 4 Star Rated	Per TR	10400
5052	526 TR - 600 TR BEE 4 Star Rated	Per TR	9750

	Water cooled scroll-type chiller with Variable Frequency Drive (VFD) of following capacity.		
5053	Upto 40 TR BEE 3 Star Rated	Per TR	15925
5054	41 TR - 75 TR BEE 3 Star Rated	Per TR	15600
5055	76 TR - 150 TR BEE 3 Star Rated	Per TR	15275
	WATER COOLED CENTRIFUGAL CHILLERS		
	Centrifugal Water Cooled Chilling Machine Air-Conditioning, Heating, and Refrigeration Institute (AHRI) certified of following capacity.		
5056	300 TR - 450 TR BEE 3 Star Rated	Per TR	17550
5057	451 TR - 600 TR BEE 3 Star Rated	Per TR	17225
5058	601 TR - 1000 TR BEE 3 Star Rated	Per TR	16900
5059	1001 TR - 1600 TR BEE 3 Star Rated	Per TR	16575
5060	1601 TR - 2000 TR BEE 3 Star Rated	Per TR	16575
	WATER COOLED MAGNETIC CENTRIFUGAL CHILLERS		
	AHRI (Air-Conditioning, Heating, and Refrigeration Institute) Certified water cooled Magnetic centrifugal type chiller machine of following capacity.		
5061	121 TR - 150 TR BEE 3 Star Rated	Per TR	19175
5062	151 TR - 300 TR BEE 3 Star Rated	Per TR	15600
5063	301 TR - 450 TR BEE 3 Star Rated	Per TR	15275
5064	451 TR - 600 TR BEE 3 Star Rated	Per TR	14040
	AHRI (Air-Conditioning, Heating, and Refrigeration Institute) Certified water cooled Magnetic centrifugal type chiller machine of following capacity.		
5065	150 TR - 205 TR BEE 4 Star Rated	Per TR	20800
5066	210 TR - 300 TR BEE 4 Star Rated	Per TR	18850
5067	301 TR - 355 TR BEE 4 Star Rated	Per TR	16575
5068	356 TR - 450 TR BEE 4 Star Rated	Per TR	15275
5069	451 TR - 600 TR BEE 4 Star Rated	Per TR	15210
	Cooling Towers		
	COOLING TOWER		
	Induced Draft counterflow cooling Towers(CTI approved) of following capacity.		
5070	300 GPM	Each	222300
5071	450 GPM	Each	296704
5072	600 GPM	Each	414700

5073	750 GPM	Each	556306
5074	900 GPM	Each	646750
5075	1050 GPM	Each	709459
5076	1200 GPM	Each	819000
5077	1350 GPM	Each	865800
5078	1500 GPM	Each	954200
5079	1800 GPM	Each	1175200
5080	2100 GPM	Each	1331230
5081	2400 GPM	Each	1581080
5082	2700 GPM	Each	1655404
5083	3000 GPM	Each	1770644

AHU & FCU

CEILING SUSPENDED AHU

35.1 Factory built ceiling suspended chilled water double skin type horizontal/vertical air handling units having cooling coil of 4 row deep, of following capacity (as detailed item no. 35.1).

5084	1000 CFM	Each	44915
5085	1600 CFM	Each	52325
5086	2000 CFM	Each	55900
5087	2500 CFM	Each	63700
5088	3000 CFM	Each	70200
5089	4000 CFM	Each	80600
5090	5000 CFM	Each	98800
5091	6000 CFM	Each	111150
5092	8000 CFM	Each	137800
5093	10000 CFM	Each	176800
5094	12000 CFM	Each	212550

FCU (FAN COIL UNIT)

DUCTABLE FAN COIL UNIT

Ceiling Concealed Fan Coil Unit comprising of 3 rows deep chilled water cooling coil of following capacity.

5095	3.0 TR nominal capacity with 1200 Cfm air quantity.	Each	16055
5096	2.5 TR nominal capacity with 1000 Cfm air quantity.	Each	14820

5097	2.0 TR nominal capacity with 800 Cfm air quantity.	Each	12740
5098	1.5 TR nominal capacity with 600 Cfm air quantity.	Each	11115
5099	1.0 TR nominal capacity with 400 Cfm air quantity.	Each	9815

CASSETTE FAN COIL UNIT

Chilled Water Ceiling Suspended Hydronic Cassette type fan coil unit, four(4) way directional flow of following capacity.

5100	4.0 TR nominal capacity with 1600 Cfm air quantity.	Each	32520
5101	3.5TR nominal capacity with 1400 Cfm air quantity.	Each	30224
5102	3.0 TR nominal capacity with 1200 Cfm air quantity.	Each	16705
5103	2.5 TR nominal capacity with 1000 Cfm air quantity.	Each	15210
5104	2.0 TR nominal capacity with 800 Cfm air quantity.	Each	11700
5105	1.5 TR nominal capacity with 600 Cfm air quantity.	Each	10010
5106	1.0TR nominal capacity with 400 Cfm air quantity.	Each	9425

HIGH WALL FAN COIL UNIT

High wall Fan Coil Unit comprising of two 2 rows deep chilled water cooling coil of following capacity.

5107	2.0 TR nominal capacity with 800 Cfm air quantity.	Each	17319
5108	1.5 TR nominal capacity with 600 Cfm air quantity.	Each	15228
5109	1.0 TR nominal capacity with 400 Cfm air quantity.	Each	12426

Evaporative Cooling

EVAPORATIVE COOLING

Factory assembled double skin central evaporative cooling plant of following capacity.

5110	5000 CFM	Each	68900
5111	8000 CFM	Each	91650
5112	10000 CFM	Each	120900
5113	12000 CFM	Each	143325
5114	15000 CFM	Each	170950
5115	20000 CFM	Each	239850
5116	25000 CFM	Each	275600
5117	30000 CFM	Each	336050

AIR COOLED HEAT PUMP FOR HOT WATER

AIR COOLED HEAT PUMP (FOR HOT WATER)

Supply of Heat pumps system for hot water using heat energy source from ambient air to Hot water of following capacity.

5118	200 L	Each	78000
5119	300 L	Each	104000
5120	500 L	Each	136500

SOLAR WATER HEATING SYSTEM

SR.NO.	DESCRIPTION	UNIT	RATE
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SOLAR WATER HEATING SYSTEM (Evacuated Tube Collector)

Evacuated Tube Collector (ETC) Solar Water Heating System comprising of all glass ETC tube absorber of following capacity.

5121	100 LPD	Each	9750
5122	200 LPD	Each	19500
5123	300 LPD	Each	29250
5124	500 LPD	Each	39000

SOLAR WATER HEATER (FLAT PLATE TYPE COLLECTOR)

Flat Plate collector (FPC) Solar Water Heating System comprising of solar flat plate collector ISI Marked of following capacity.

5125	100 LPD	Each	14509
5126	200 LPD	Each	29018
5127	250 LPD	Each	33800
5128	300 LPD	Each	43527
5129	500 LPD	Each	72545

EV CHARGER

Supply of EV charging station
As per specifications and in Compliance to relevant IS codes etc.

5130	Light EV AC Charger (Mode-3) Power : 7 kW, Input power supply: 1phase 230 $\pm$ 10% Volt, output supply: 230 Volt AC, Frequency:50 Hz $\pm$ 3%.	Each	16244
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5131	Light EV DC Charger (Mode 4) Power Level 1: Up to 7 kW, Input power supply: 1phase 230 ±10% Volt/3phase 415 Volt, Frequency: 50 Hz +/-5%.	Each	243750
5132	Parkbay AC Charger (Mode -3) Power Level 2: Normal Power ~11kW/ 22 kW, 3 phase 415VAC(-40% to +20%), Frequency:50 Hz +/-5%.	Each	58500
5133	Parkbay DC Charger (Mode-3) Power Level 2: Normal Power ~24KW and above, 3 phase 415VAC(-40% to +20%), Frequency:50 Hz +/-5%.	Each	617500
Solar Photovoltaic Power Plant			
5134	On-grid Solar Photovoltaic Power Plant conforming to various applicable standards BIS, IEC, MNRE guidelines, the Central Electricity Authority Regulations and CPWD Specifications.	kWp	40000

Façade Lighting

LED floodlight

Cicular/ Rectangular/square LED
RGB/RGBWW/RGBNW/RGBA /RGBW, Facade Floodlight/
Projector luminaire for facade lighting with Pressure Die-
cast aluminium housing, powder-coated finish, with inbuilt
driver, of following wattage.

5135	25W	Each	25025
5136	40W	Each	29705
5137	50W	Each	30485
5138	60W	Each	31785
5139	100W	Each	33777
5140	120W	Each	55705
5141	150W	Each	59995
5142	200W	Each	65942
19.2	LED Linear Profile (AC) LED RGB/RGBWW/RGBNW/RGBA/RGBW, Linear Profile / Wall Washer luminaire for Facade lighting with aluminium extruded/ Die Cast housing with inbuilt driver of following wattage. Wattage per meter		
5143	120W	Each	45760
5144	100W	Each	39182
5145	80W	Each	29718
5146	60W	Each	29387
5147	50W	Each	18486

5148	40W	Each	17804
5149	25W	Each	13694
19.3	LED Linear Profile (DC)		
	LED RGB/RGBWW/RGBNW/RGBA/RGBW, Linear Profile / Wall Washer luminaire for Facade lighting with aluminium extruded/ Die Cast housing with inbuilt driver of following wattage.		
	Wattage per meter		
5150	100W	Each	16783
5151	50W	Each	11365
5152	40W	Each	11466
5153	30W	Each	8344
5154	15W	Each	2380
	RGB LED Strip		
5155	Exterior IP68 LED RGB/RGBWW/RGBNW/RGBA/RGBW. Flexible light strip of 5 mtr standard length with 12-15 Watt per meter.	Each	6730
	Controller		
5156	DMX Controller for the system including address writer and programming, Controller should have Input Power 5-5.5V DC 0.6A and Output Protocol DMX512 with 4 universe, 2 universe etc. as required.	Each	45769
5157	DMX Splitter with Input Voltage range : 190-240V AC, Power Consumption 5W max. Housing is made of CRCA with powder coated finish Mounting complete as required.		4323
5158	RDM-6Wall Splitter, should be able to provide 6- output DMX/RDM splitter with DMX/RDM pass- through supporting bi-directional communications for discovery, addressing and DMX control of Ethernet/DMX products complete as required.	Each	10518
5159	Signal Amplifier to boost the Signal, IP 65 rated module to enhance the depleting DMX Signals, should have IP 66/67 rates inbuilt connectors compatible with Luminaires.	Each	1750
5160	High quality DMX decoder, Power Input : DC 24V etc complete as required.	Each	3750
5161	IP65 "I" power connectors having 3/4 pole cable connector, chuck type strain relief, Current rating 20 A rms continuous.	Each	451
5162	IP65 "I" DMX connectors having 3/5 pole cable connector, Male/Female connector with improved latch, cage type contacts.	Each	451
5163	5 pin XLR Male/Female connectors having 5 pole cable connector, Male/Female connector with improved latch, cage type contacts.	Each	1360

5164	Push button type keypad with 10 buttons. Each button is programmable up to 16 scenerios.	Each	6225
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FOUNTAIN

5165	Nozzle made with gunmetal Grade Geyser jet 100mm with adjustable nozzle for height adjustment of the fountain etc., Complete as required.	Each	7275
5166	40 mm dia Bubbler Jet Nozzles in SS 304 Gr to produce water effect complete as required.	Each	4502
5167	40 mm dia SS Difuser Nozzles in SS 304 Gr to produce water effect complete as required.	Each	2052
5168	Stainless steel 304 gr. Comet effect nozzle with compatible Digitally controlled pump to produce desirable effects with 40 mm dia and 20 mm orifice nozzle. Single Jet Crown Nozzles Complete with ball joint for 15 Degree pivoting made of brass with Nickle plating, flow straightner in following size .	Each	7179
5169	15 mm internal threaded with 6 mm orifice.	Each	1613
5170	15 mm internal threaded with 8 mm orifice.	Each	1763
5171	15 mm internal threaded with 10 mm orifice.	Each	1920
5172	25 mm internal threaded with 12 mm orifice.	Each	2138
5173	25 mm internal threaded with 14 mm orifice.	Each	2588
5174	40 mm internal threaded with 17 mm orifice.	Each	2138
5175	40 mm internal threaded with 20 mm orifice.	Each	5438
5176	Eco Cluster Nozzle made of stainless steel with inlet dia 40 mm and 21 nos Single jets to create 38 mm dia full jet for height 10 mtr water level independent.	Each	25875
5177	Eco Cluster Nozzle made of stainless steel with inlet dia 80 mm and 64 nos Single jets to Create 67 mm dia full jet for fountain height 10 mtr water level independent.	Each	44025
5178	Central hollow jet 65 mm dia for achieving maximum height of 20 mtrs to produce clear full jet stable in wind water level independent Lava bell nozzle for producing transparent water pattern with adjustable ball dia made out of SS 304 Gr./Brass of following Size.	Each	34875
5179	15 mm Dia inlet 290 mm long	Each	2888
5180	25 mm Dia inlet 330 mm long	Each	3638
5181	25 mm Dia inlet 488 mm long	Each	4388
5182	25 mm Dia inlet 636 mm long	Each	5063

	Foam jet High contrast for producing high contrast/foam effect to extensively wind stable, water level independent moc Brass /SS 316L of following size.		
5183	25 mm Dia inlet 32 mm orifice 160 mm long.	Each	12375
5184	25 mm Dia inlet 50 mm orifice 180 mm long.	Each	20700
5185	40 mm Dia inlet 50 mm orifice 180 mm long.	Each	28875
5186	20 mm nylon rope all the accessories Floater Grit Clamping both sides etc., Complete as required.	Mtrs.	48
5187	Suction Screen strainer made of ABS to be fitted in suction to protect the pump from entry of external materials.	Each	1369
5188	100 mm Dia Debris Collection Strainer in Stainless steel 304 gr Complete as reqd. All fixing to be done as per direction instruction and entire satisfaction of Engineer-in-charge.	Each	5338
5189	400 mm Dia wide SS Overflow Skimmer in Stainless steel 304 gr with front openable fitted debris trapping mesh with outlet size of 65mm Dia for fixing in wall.	Each	7238
5190	300x300 mm Dia Main Drain in SS 304 Gr with perforations to disallow foreign material with top openable for maintenance.	Each	5380
5191	ISI Marked openwell submersible pump set contain with stainless steel shaft and cast iron/SS body, suitable for 900LPM at 25 Mtrs Head	Each	34393
5192	ISI Marked vertical Multi Stage submersible pump set contain with stainless steel shaft and cast iron/SS body, suitable for 1200LPM at 14 mtr. head.	Each	47210
5193	Three phase openwell ISI Marked submersible pumpset of 1.5H.P. capacity suitable for operation on 415V(+/-10%), 3-phase AC supply complete as required.	Each	19821
5194	Three phase openwell ISI Marked submersible pumpset of 7.5 H.P. capacity suitable for operation on 415V(+/-10%), 3-phase AC supply complete as required.	Each	39225
	Gun metal gate valve (Screwed type) on the existing pipeline, including testing & Commissioning etc. as required.		
5195	80 mm dia	Each	12146
5196	50 mm dia	Each	5474
	Gun metal Horizontal Non Return valve (screwed type) on the existing pipeline, including testing & Commissioning etc. As required.		
5197	100 mm dia	Each	25584

5198	50 mm dia	Each	6038
5199	Underwater Luminaries surface mounted RGB LED spot light 6 x 3W with IP 68 protection, SS/ aluminium dia casting body with power coated paint, soft rubber gasket.	Each	9488
5200	Underwater Luminaries surface mounted RGB LED spot light 3 x 3W with IP 68 protection, SS/ aluminium dia casting body with power coated paint, soft rubber gasket .	Each	4913
5201	Water proof (IP 68) power supply unit of not less than 200W i/c connection .	Each	5147
5202	RGB controller with remote Kit wattage not less than 200 watt i/c connection.	Each	74366
5203	Signal Amplifier of RGB signal as require i/c connection.	Each	11282
5204	Power Supply/SMPS for suitable to run LED Lights, Body-Metal input Power-220 V AC/ Output Power-12 V DC. PVC insulated PVC sheathed copper conductor submersible flat cable ISI Marked in existing metal/ HDPE / PVC/ pipe/ in bore well/ in sump .	Each	5008
5205	3 Core 1.5 Sq mm Flat Cable	Mtrs.	53
5206	3 Core 2.5 Sq mm Flat Cable	Mtrs.	86
5207	3 Core 4.0 Sq mm Flat Cable	Mtrs.	129
5208	3 Core 6 Sq mm Flat Cable	Mtrs.	185
5209	3 Core 10 Sq mm Flat Cable	Mtrs.	320

Passenger Lifts

Machine room less (MRL) lift with regenrative drive duly compliant with barrier free access as per CPWD General Specification for Electrical Works (Part-III Lift & Escalator) 2003 BIS Codes, NBC 2016, as Amended upto date.

5210	13 passenger (884kg)	Each	1169888
5211	16 passenger (1088kg)	Each	1631103

Goods Lifts

2000kg Machine room less (MRL) goods lift with regenrative drive duly compliant with barrier free access as per CPWD General Specification for Electrical Works (Part-III Lift & Escalator) 2003, BIS Codes, NBC 2016, as Amended upto date.

5212	2000 kg	Each	2356095
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Hospital Lifts (20 passenger (1360 kg))

Machine room less (MRL) Bed lift with regenrative drive duly compliant with barrier free access as per CPWD General Specification for Electrical Works (Part-III Lift & Escalator) 2003, BIS Codes, NBC 2016, as Amended upto date.

5213	20 passenger (1360kg)	Each	1793114
IP-PBX System			
5214	IP based voice communication system with 04 Port Voicemail ,04 Port FXS ,04 Port FXO ,1 PRI Trunk lines (30 Channels) Circuit with CLIP Facility ,min 100 IP users License with provision for further additions ,100 analog users ,01 Nos. IP based Operator Console, 100 Party Conference.	Each	1096741
5214(1)	IP at its core with IP switching technology and 100% non blocking communication system with 04 Port Voicemail, 04 Port FXS, 04 Port FXO, 1 PRI Trunk lines (30 Channels) Circuit with CLIP Facility, min 100 IP users License with provision for further additions, 100 analog users, 01 Number IP based Operator Console, 45 Party Conference.		866149
5215	Type 1 IP Phone 2 VoIP account, 120 x40 DOT matrix screen with backlight , Full HD duplex speaker phone, IPV6, Gigabit Ethernet , PoE complete as required.	Each	8126
5216	Analog phones having standard feautres like 16 digit LCD display , 2 way speaker , caller ID and number storage facility etc complete as required.	Each	1073
5217	20 Pair PVC Telephone cables with Conductor Diameter:05mm Nominal, Insulation Materual: High-Density Polyethylene(HDPE), Insulation Thickness:02 mm Nominal, Certification:BIS(Bureau of Indian Standards) in the existing surface/recessed steel/ PVC conduit as required.	Each	239
LAN SYSTEM			
following capacity 8 port Layer 2 indoor Network Switch having features and specifications etc. as mentioned here under: At least 8 X RJ-45 Gigabit Ethernet Ports and additional 2 X 1G SFP Ports with non-blocking architecture by having Switching capacity of minimum 20Gbps and packet forwarding rate of 14Mpps or higher, 8K MAC table.			
5218	8 port PoE Layer 2 Network Switch with PoE Support of 130W or higher with each copper port supporting 802.3at PoE+.	Each	24375
5219	8 port Layer 2 Non-PoE Network Switch	Each	19500

5220	capacity 8 port PoE Layer 2 Industrial Grade outdoor Network Switch having features and specifications etc. as mentioned here under: At least 8 X RJ-45 PoE/PoE+ Gigabit Ethernet Ports and additional 4 X 1G SFP Ports with non-blocking architecture by having Switching capacity of min. 24Gbps and packet forwarding rate of 17Mpps or higher, console port, 16K MAC table, IPv4 and IPv6 ready, IGMP snooping v1, v2, v3. following capacity 24 port Layer 2 indoor Network Switch having features and specifications etc. as mentioned here under: At least 24 X RJ-45 Gigabit Ethernet Ports and additional 2 X 10G Base -T with 4 X SFP Ports	Each	74100
5221	24 port PoE Layer 2 Network Switch with PoE Support of 370W or higher with each cooper port supporting 802.3at PoE+ min.	Each	87100
5222	24 port Layer 2 Non-PoE Network Switch	Each	59850
5223	following capacity 24 port Layer 3 Network Switch having features and specifications etc. as mentioned here under: At least 20 X RJ-45 Gigabit Ethernet Ports and additional 4 combo 10/100/1000 base-t/SFP Ports with additional 4 SFP+ ports non-blocking architecture by having Switching capacity of min. 128Gbps and packet forwarding rate of 95Mpps or higher, 16K MAC table. Console Port, USB port, Internal AC Power supply	Each	134610
5224	10 Giga Bandwidth Data Centre Network Switch of 48 port Layer 3 having features and specifications etc. as mentioned here under: - ToR/EOR/Core Network Switch 48x 10GbE SFP+, 6x 100GbE QSFP28, RPS, with Console Port, Management Port and 1 x USB 2.0 Type A port, The switch must provide a switching capacity of no less than 2.16 Tbps and a forwarding rate of at least 1600 Mpps. Supplying, Installation, Testing and commissioning of small form-Pluggable (SFP) for Network Switches. Must be hot Pluggable, RoHS Complaint, etc. complete as required.	Each	733500
5225	1G SFP Single Mode/Multi Mode	Each	2925
5226	1G Copper SFP	Each	4875
5227	10 G SFP + Single Mode/Multi Mode	Each	10400
5228	40G SFP	Each	29250
5229	100G SFP	Each	87750

5230	indoor Wireless Access Point for Low Density use having features and specifications etc. as mentioned here under - Dual-band Wi-Fi6 (802.11ax) 574Mbps (2.4GHz) + 1200Mbps (5GHz), WI-FI 6 Certified, 1 x RJ45 console port, 1G LAN POE Port, factory reset, WPA/WPA2/WPA3™ Personal/Enterprise, WEP 64/128-bit, SSID broadcast disable, MAC address access control, Internal RADIUS server.	Each	15795
5231	indoor Wireless Access Point for High density use having features and specifications etc. as mentioned here under- WiFi6 802.11ax 574Mbps (2.4GHz) + 2402Mbps (5GHz), WIFI 6 Certified, 1 x RJ45 console port, MG 2.5G LAN POE Port, factory reset, WPA/WPA2/WPA3™ Personal/Enterprise, WEP 64/128-bit, SSID broadcast disable, MAC address access control, Internal RADIUS server.	Each	24700
5232	Outdoor Wireless Access Point having features and specifications etc. as mentioned here under - WIFI 6 Certified, 1 x RJ45 console port, MG 2.5G LAN POE Port, factory reset, 4 x internal antennas, the transmission coverage of shall be minimum 80 meters in radius, AP should have transmission power minimum of 22 dBm, The AP shall provide a minimum of 29 dBm EIRP for both 2.4 GHz and 5 GHz frequencies. Feild deployment shall be with EIRP as per regulatory guidelines.	Each	32071
5233	indoor Wireless Controller having features and specifications etc. as mentioned here under- The network controller shall be a dedicated hardware appliance designed for centralized management of wireless access points, It must support the management of up to 500 networking devices,	Each	78650
5234	24 port Cat6 Patch Panel loaded. Must be of 1U height with clear label holders and white label with the panel. 24 Ports Cat-6 Patch Panel should have ETL/UL verification program certificate for compliance with ANSI/TIA-568.2-D following CAT6 Patch Cord should have ETL/UL verification program certificate for compliance with ANSI/TIA-568.2-D	Each	3666
5235	Copper Patch Cords of length 1m (3ft)	Each	105
5236	Copper Patch Cords of length 3m (10ft) following CAT6A Patch Cord should have ETL/UL verification program certificate for compliance with ANSI/TIA-568.2-D	Each	158
5237	Copper Patch Cords of length 1m (3ft)	Each	120
5238	Copper Patch Cords of length 3m (10ft)	Each	281

5239	CAT6 Copper Information Outlet (IO) with face plate of color as per site requirement, should have ETL/UL verification program certificate for compliance with ANSI/TIA-568.2-D. All copper Cable and Components should be from same OEM to maintain compatibility and interoperability.	Each	105
5240	CAT6A Copper Information Outlet (IO) with face plate of color as per site requirement, should have ETL/UL verification program certificate for compliance with ANSI/TIA-568.2-D. All copper Cable and Components should be from same OEM to maintain compatibility and interoperability.	Each	210
5241	RJ45 Connector for CAT 6/6A Cables. RJ45 modular plug supports 4 twisted pairs, 8 positions, 8 connectors. Housing: PC, UL94V-2, transparent color. Use for 24- 26 AWG stranded wires. CAT6 UTP LSZH 23AWG Twisted Pair Cable in existing conduit/ on surface, Category 6 Unshielded Twisted Pair.	Each	6
5242	1 Run of cable	meter	25
5243	2 Run of cable	meter	50
5244	3 Run of cable	meter	75
5245	4 Run of cable	meter	100
	Cat6A UTP 4 pair, 23 AWG solid copper cable in existing conduit/ on surface, U/FTP, LSZH, Non-Plenum, Horizontal (solid) Cable suitable for high speed data networking application supporting upto 10Gbps over a 100 meter channel.		
5246	1 Run of cable	meter	35
5247	2 Run of cable	meter	70
5248	3 Run of cable	meter	105
5249	4 Run of cable	meter	140
	Fiber Optic Cable having corrugated steel armoring. The Fiber should be SM Fiber Central -loose tube filled with Thixotropic jelly, duly following Standards: ISO 11801, IEC 60793-1/60794-1-2, ITU-T-REC G.652D and Telecordia GR-20-core.		
5250	Single Mode 6 Core Optical fiber cable	Mtr	28
5251	Single Mode Optical 12 Core fiber cable	Mtr	45
	Multi mode Fiber Optic Cable having corrugated steel armoring, lesser optimize 50/125 μ m with HDPE jacket, LSZH (Low smoke Zero Halogen), Tight-buffered or loose tube, maximum attenuation: ≤ 3.5 dB/km at 850 nm, ≤ 1.5 dB/km at 1300 nm, bandwidth ≥ 2000 MHz-km at 850 nm, ≥ 500 MHz-km at 1300 nm.		

5252	6 Core Multi Mode Optical Fiber Cable	Mtr	44
5253	12 Core Multi Mode Optical Fiber Cable	Mtr	59
	Rack Mount loaded LIU with pigtail, Front-mounted cable saddles for jumper management, suitable to manage both splices and terminations.		
5254	6 Port LIU	Each	3800
5255	12Port LIU	Each	4890
5256	24 Port LIU	Each	6980
5257	Patch Cord of Optical Fiber Multi Mode or Single Mode LC to LC Fiber Duplex having high precision ceramic ferrule with good concentricity, Fiber corning single mode G652D, Cable Type 2mm.	Each	490
	Wall Mount Rack having Fixed Structure with 0.8mm CRCA Sheet, completely knocked-down condition (CKD) Shape, Vertical Mounting Rail 1.6 mm.		
5258	6U Rack	Each	4820
5259	9U Rack	Each	6224
5260	15U Rack	Each	9259
5261	24U Rack with 4 inch Castor wheels and front brake Floor Standing Rack with Main Frame Pillar of 1.25mm CRCA Sheet with removable side panels, L- Shape adjustable Vertical Mounting Rail of 2mm.	Each	14726
5262	42U Rack with 6 inch Castor wheels and front brake Floor Standing Rack with Main Frame Pillar of 1.6mm.	Each	22890
5263	Outdoor Pole/ Wall mount 6U Rack, IP55, Front Door with Filter, Louver & Unique key Lock. 3 Socket PDU 5Amp.	Each	15440
	CCTV CAMERA		
	2 MP IP IR Dome Camera		
5264	2MP @ 25/30fps@1080P (1920×1080) Dome Camera with 2.7mm ~13.5mm Motorized lens, Vari Focal (view Angle for H: 104°–29°; V: 54°–16°; D: 125°–34°)	Each	5591
5265	2MP @ 25/30fps@1080P (1920×1080) Dome Camera with 2.7mm ~13.5mm Motorized lens, Vari Focal (view Angle for H: 109°–30°; V: 56°–17°; D: 131°–35°)	Each	13751
40.2	2 MP IP IR Outdoor Bullet Camera 2MP Fixed Lens IP IR Outdoor Bullet Camera having following specifications and features etc :-		
5266	2MP @ 25/30fps@1080P (1920×1080) Outdoor Bullet Camera with 2.8/ 3.6mm fixed lens.	Each	5390

5267	2MP @ 25/30fps@1080P (1920×1080) Outdoor Bullet Camera with 2.7mm ~13.5mm Motorized lens, Vari Focal (view Angle for H: 97°–28°; V: 52°–16°; D: 117°–32°) 4MP IP IR Dome Camera 4MP IP IR Dome Camera having following specifications and features etc :-	Each	13751
5268	4MP @ 25/30fps@1440P (2560 x 1440) Dome Camera with 2.8/ 3.6/ 6/8/12 mm fixed lens (as per site requirement).	Each	7200
5269	4MP @ 25/30fps@1440P (2560x1440) Dome Camera with 2.7mm ~13.5mm Motorized lens, Vari Focal (view Angle for H: 104°–29°; V: 54°–16°; D: 125°–34°) 4MP IP IR Outdoor Bullet Camera 4MP IP IR Outdoor Bullet Camera having following specifications and features etc :-	Each	16500
5270	4MP @ 25/30fps@1440P (2560 x 1440) Outdoor Bullet Camera with 2.8/ 3.6/6/8/12 mm fixed lens (as per site requirement)	Each	6990
5271	4MP @ 25/30fps@1440P (2560x1440) Outdoor Bullet Camera with 2.7mm ~13.5mm Motorized lens, Vari Focal (view Angle for H: 97°–28°; V: 52°–16°; D: 117°–32°) 5/6MP IP IR Dome Camera 5/6MP IP IR Dome Camera having following specifications and features etc :-	Each	16500
5272	5/6MP ((3200 × 1800)@20 fps or better) IP IR 2.8/ 3.6/6/8/12 mm fixed lens (as per site requirement) Dome Camera.	Each	8891
5273	5/6MP ((3200 × 1800)@20 fps or better) IP IR 2.7mm ~13.5mm Motorized lens, Vari Focal Dome Camera, Angle of View: H: 104°–32°, V: 54°–18°, D: 125°–36° or better 5/6MP IP IR Bullet Camera 5/6 MP IP IR Bullet Camera having following specifications and features etc :-	Each	18750
5274	5/6MP ((3200 × 1800)@20 fps or better) IP IR 2.8/ 3.6/6/8/12 mm fixed lens (as per site requirement) outdoor bullet Camera.	Each	8700
5275	5/6MP ((3200 × 1800)@20 fps or better) IP IR 2.7mm ~13.5mm Motorized lens, Vari Focal outdoor Bullet Camera, Angle of View: H: 104°–32°, V: 54°–18°, D: 125°–36° or better 8 MP IP IR Dome Camera 8 MP IP IR Dome Camera having following specifications and features etc :-	Each	18750

5276	8 MP (3840 × 2160)@20 fps or better IP IR 2.8/ 3.6/6/8/12 mm fixed lens (as per site requirement) Dome Camera,	Each	13500
5277	8 MP (3840 × 2160)@20 fps or better IP IR 2.7mm ~13.5mm Motorized lens, Vari Focal Dome Camera, Angle of View: H: 104°–32°, V: 54°–18°, D: 125°–36° or better	Each	19500
	8 MP IP IR Outdoor Bullet Camera 8 MP IP IR Outdoor Bullet Camera having following specifications and features etc :-		
5278	8 MP (3840 × 2160)@20 fps or better IP IR 2.8/ 3.6/6/8/12 mm fixed lens.	Each	13500
5279	8 MP (3840 × 2160)@20 fps or better IP IR 2.7mm ~13.5mm Motorized lens, Vari Focal outdoor Bullet Camera, Angle of View: H: 104°–32°, V: 54°–18°, D: 125°–36° or better PTZ (Pan, Tilt and Zoom) IP IR Camera PTZ (Pan, Tilt and Zoom) IP IR Camera having following specifications and features etc :-	Each	19500
5280	2 MP IP IR motorized PTZ (1920 × 1080 @ 25/30fps or better) camera, Triple streaming , configurable on resolution Main stream: 1080p@25/30 fps Sub streams :1080p@25/30 fps & D1@25/30 fps or better, varifocal lens 4.3-137 mm or better with Automatic & manual Focus Adjustment provisions, 32x Optical zoom and 12x Digital zoom with angle of view 57°–2.4° , PAN Travel: Pan: 0° ~ 360° endless, Manual Pan: 260° /s, Preset : 300° /s, Tilt, Travel: Tilt: -20° ~ 90°, auto flip 180°, Manual Tilt: 120° /s, Preset : 200° /s	Each	48700
5281	4 MP IP IR motorized PTZ (4MP @25/30fps or better) camera, Triple streaming , configurable on resolution Main stream: 4MP@25/30 fps Sub streams :1080p@25/30 fps & D1@25/30 fps or better, varifocal lens 4.5-135mm or better with Automatic & manual Focus Adjustment provisions, 30x Optical zoom and 12x Digital zoom with angle of view H : 57°–2.4°, m, PAN Travel: Pan: 0° ~ 360° endless; Manual Pan: 300° /s, Preset : 300° /s, Tilt Travel: Tilt: 0-90°, auto flip 180°, Manual Tilt: 200° /s, Preset : 300° /s	Each	54450

5282	5MP IP IR motorized PTZ (2960 x 1668@25/30fps or better) camera, Triple streaming , configurable on resolution:- Main stream: 5MP@25/30 fps Sub streams:1080p@25/30 fps, varifocal lens 3.95mm (±5mm) ~ 177.75mm (±5mm) or better with Automatic & manual Focus Adjustment provisions, 45x Optical zoom and 16x Digital zoom with angle of view H: 65.7°–1.9°V: 39.4°–1.1°D: 73.1°–2.1, PAN Travel: Pan: 0° ~ 360° endless,Manual Pan: 260° /s, Preset : 300° /s, Tilt Travel: Tilt: -20° ~ 90°, auto flip 180°, Manual Tilt: 120° /s, Preset : 200° /s Network Video Recorder (NVR) Following Channel Network Video Recorder (NVR) with camera licenses to record for all channels having specifications and features etc as mentioned below :	Each	75000
5283	08 Channel Network Video Recorder (NVR) having display split :- Main screen: 1/4/8/9, 2nd screen: 1/4/8/9	Each	7001
5284	16 Channel Network Video Recorder (NVR) having display split :- Main screen: 1/4/8/9/16, 2nd screen: 1/4/8/9/16 Following Channel Network Video Recorder (NVR) with camera licenses to record for all channels having specifications and features etc as mentioned below :	Each	8891
5285	32 Channel Network Video Recorder (NVR) having display split :- Main screen: 1/4/8/9/16/25/32(36), 2nd screen: 1/4/8/9/16	Each	45000
5286	64 Channel Network Video Recorder (NVR) having display split :- Main screen: 1/4/8/9/16/25/36/64, 2nd screen: 1/4/8/9/16 Hard Disk Drive Following capacity Surveillance grade Hard Disk with upto 256MB/s Transfer Rate, 256 MB Cache, 7200 RPM Disk Speed, 3.5 inch form factor, SATA Interface, BSMI, ICES-003/NNB-003, CE, FCC, KC, Maghreb, RCM, UKCA, VCCI, CB-Scheme, TUV, UL Certifications.	Each	57375
5287	4TB (Terabytes)	Each	11175
5288	6TB (Terabytes)	Each	13125
5289	8TB (Terabytes)	Each	16125
5290	10TB (Terabytes)	Each	19875

Access Control System
Access Control System with Face recognition

	Access Control System comprising of Face Recognition, Fingerprint, Smart Card, PIN & QR Code based Reader, following sizes of IPS TFT Touch Screens.		
5291	Deleted	Each	0
5292	5" IPS TFT Touch Screen	Each	40500
	Access Control System (without Face Recognition)		
5293	Access Control System comprising of Fingerprint, Smart Card & Password based Reader, 5" Touch Screen LCD Display.	Each	27000
	Access Control System with 2.8" display (without Face Recognition Readers)		
5294	Access Control System comprising of Fingerprint, Card & Password based Reader with Capacitive Touch Keypad, Fake Fingerprint Detection Technology, supports ANSI 378 encryption, AES 128 bit encrypted algorithm to protect important info such as personal info transmitted, 2.8" LCD display.	Each	22500
	Access Control System with 1.77" display (without Face Recognition Readers)		
5295	Access Control System comprising of Fingerprint & Card based Reader with 1.77" LCD Display.	Each	19500
	Software for Web based Access Control System		
5296	Web based Access Control Enrolment and Device Configuration Software.	Each	67500
	Electromagnetic locks		
	Electromagnetic locks (fail safe type) for following type of Doors suitable for operation on 12 V DC having holding force 600 lbs for Single Leaf Door and 1200lbs for Double leaf door with suitable feedback i/c connection complete etc. as required.		
5297	Single leaf door	Each	3356
5298	Double leaf door (Each leaf 600lbs & Total 1200 lbs)	Each	5274
	Exit Push Button		
5299	Heavy Duty SS Finish Exit Push Button for releasing access doors with all necessary installation accessories i/c connection complete etc. as required.	Each	815
	Door Position Sensor		
5300	Door Position Sensor i/c connection complete etc. as required.	Each	432
	Building Management System (BMS)		
Sl.No.	Description	Unit	Rate (₹)
	CENTRAL CONTROL SERVER		

5301	Central Control Server System comprising of Intel® Xeon® CPU E5-2640 x64 (or better) compatible with dual and quad core processors, Windows 11 (Professional, Enterprise, Ultimate, 64 bit). PC System	Each	129500
5302	Client PC System comprising of PC Hardware with minimum 11th Gen Intel® i7 processor CPU E5-2640 x64 (or better), compatible with dual and quad core processors minimum 3.1 Ghz and 8GB or higher RAM. LaserJet printer	Each	63000
5303	LaserJet printer having printing resolution 1200 x 1200 dpi, (dot/inch) or better, 21 to 30 ppm printing speed (LaserJet) complete etc. as required LED Display LED display (LED monitor) industrial grade with 3840x2160 resolution or better, USB playback, bluetooth and miracast connectivity.	Each	15750
5304	32 inch or larger	Each	27500
5305	42 inch or larger	Each	35744
5306	55 inch or larger	Each	56998
	Software for Building Management System : Necessary Software (UL listed) Packages containing Building Management with Capability of Collection, Storage, Processing and Controlling all the Data. Software shall support Open Standard Protocols like XML, BACnet, Modbus, Lon works etc.		
5307	For Upto 500 point included	Each	67640
5308	For Upto 2500 point included	Each	116620
5309	For Upto 5000 point included	Each	171500
	Touchscreen Color portable operator terminal Touchscreen Color portable operator terminal capable of connecting to one of the controllers on the communication bus and view parameters of the complete system.		
5310	4" Touch Screen	Each	43750
5311	7" Touch screen	Each	51727
	DDC CONTROLLERS IP Based DDC controller capable of fully "stand- alone" operation i.e. In the event of loss of communication with other DDC's or Control Station, they shall be able to function on their own.		
5312	DDC Controller-Type-I (Support upto 32 Points)	Each	32550
5313	DDC Controller-Type-II (Support upto 64 Points)	Each	45850
5314	DDC Controller-Type-III (Support upto 96 Points)	Each	49350

5315	DDC Controller-Type-IV (Support upto 128 Points)	Each	51450
5316	DDC Controller-Type-V (Support upto 160 Points)	Each	59920

EXPNSION IO MODULE

Input output (IO) expansion module.

5317	I/O Module with 4 UIO channels	Each	12250
5318	I/O Module with 8 UIO channels	Each	12845
5319	I/O Module with 8 UI channels	Each	13615
5320	I/O Module with 8 DI channels	Each	10822
5321	I/O Module with 16 DI channels	Each	12523
5322	I/O Module with 8 DO channels	Each	11746
5323	I/O Module with 4 DO channels	Each	10920

Enclosure for DDC

Powder coated wall mounted lockable and secure enclosure for DDC panel made of 16 SWG sheet steel IP -55 rated with proper internal mounting accessories.

5324	500X500X200 mm	Each	7350
5325	600mmX600mmX200mm	Each	8792
5326	800mmX1000mmX200mm	Each	12950

Integration Unit

3rd party Integration unit to provide the interface between Ethernet LAN and the 3rd party field control devices or any other devices which need to be integrated.

5327	Integrator for 500 Points	Each	101500
5328	Integrator for 1250 Points	Each	122500
5329	Integrator for 5000 Points	Each	157500

Field Devices

following Field Devices including connection etc. as required

5330	Water Differential Pressure (DP) Switch for Pump Status.	Each	6731
5331	High/Low Level Switch for Tank Level Monitoring.	Each	5490
5332	Air DP Switch for Fan Run Status.	Each	2034
5333	Duct CO2 Sensor, 0-2000ppm, 0-10V/4-20mA Output.	Each	6860
5334	Outside Temp & Humidity Sensor with radiation shield. Measuring Range: Temp:- -15 to 50 Deg C & RH 0-100%, Accuracy: +/- 1 Degc +/- 3%	Each	6475

5335	Immersion Temp Sensor for CHW line. Measuring range: -30 to 110 Deg C, Accuracy: +/- 1.3 Deg C	Each	3185
5336	Water Pressure Sensor . Accuracy: +/-0.3% FSL	Each	6160
5337	Duct type temperature sensor Measuring range: -30 to 110 Deg C, Accuracy: +/- 1.3 Deg C	Each	2492
5338	Current Sensing Relay	Each	2205
5339	Wall Mount Temp. Sensor	Each	4550
5340	Wall/Ceiling Mount CO Sensor	Each	5775
5341	Differential pressure switch for filters & Blowers	Each	1785
5342	Bi-Level Switches (Hi/Low)	Each	3850
5343	Water Level Sensor	Each	4599
5344	Explosion proof level Switch	Each	5292
5345	Water Flow Switch	Each	2625
5346	Indoor Air Quality Sensor (CO2,T,H,PM,TVOC)	Each	22750
5347	Ultrasonic Water BTU Meter including digital display (Clamp-On) suitable for Pipe sizes dia upto 250mm	Each	248500